



Government of the People's Republic of Bangladesh
Ministry of Health and Family Welfare

HEALTH BULLETIN 2018



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Directorate General of Health Services
Mohakhali Dhaka 1212
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ACRONYMS

ACSM	Advocacy, Communication, and Social Mobilization	CFR	Case-fatality Rate
ACT	Artemisinin-based Combination Therapy	CG	Community Group
ADB	Asian Development Bank	CHCP	Community Healthcare Provider
ADP	Annual Development Program	CHT	Chittagong Hill Tracts
AEFI	Adverse Events Following Immunization	CHW	Community Health Worker
AeHIN	Asian eHealth Information Network	CIDD	Control of Iodine Deficiency Disorder
AFP	Acute Flaccid Paralysis	CIN	Cervical Intraepithelial Neoplasia
AHI	Assistant Health Inspector	CMBT	Center for Medical Biotechnology
A2i	Access to information	CPR	Contraceptive Prevalence Rate
AIDS	Acquired Immunodeficiency Syndrome	CR	Civil Registration
ALOS	Average Length of Stay	CRPD	Convention on the Rights of Persons with Disabilities
AMC	Alternative Medical Care	CRVS	Civil Registration and Vital Statistics
AMR	Antimicrobial resistance	CS	Civil Surgeon
ANC	Antenatal Care	CSBA	Community-based Skilled Birth Attendant
API	Annual Parasite Incidence/Application Programming Interfaces	CSG	Community Support Group
APIR	Annual Program Implementation Report	CSO	Civil Society Organization
APR	Annual Program Review	Dev.	Development
ARC	American Red Crescent/Antimicrobial Resistance Containment	D4D	Data for Decision
ARI	Acute Respiratory Infection	DF/DHF	Dengue Fever/Dengue Hemorrhagic Fever
ART	Antiretroviral Treatment/Antiretroviral Therapy	DGDA	Directorate General of Drug Administration
ASD	Autism Spectrum Disorder	DGFP	Directorate General of Family Planning
ASP	AIDS/STD Program	DGHEU	Directorate General of Health Economics Unit
BBS	Bangladesh Bureau of Statistics	DGHS	Directorate General of Health Services
BCC	Behavior Change Communication	DGNM	Directorate General of Nursing and Midwifery
BCG	Bacillus Calmette–Guérin	DHS2	District Health Information System software version 2
BCPS	Bangladesh College of Physicians and Surgeons	DHS	Demographic Health Survey
BCS	Bangladesh Civil Service	DICs	Drop-in-Centers
BDHS	Bangladesh Demographic and Health Survey	DLI	Disbursement Link Indicator
BDT	Bangladeshi Taka	DMCH	Dhaka Medical College Hospital
BEOC	Basic Emergency Obstetric Care	DNA	Deoxyribonucleic Acid
BHFS	Bangladesh Health Workforce Survey	DNCC	Dhaka North City Corporation
BITID	Bangladesh Institute of Tropical and Infectious Disease	DOTs	Directly-observed Treatment-Short Course
BMDC	Bangladesh Medical and Dental Council	DPA	Direct Project Aid
BMRC	Bangladesh Medical Research Council	DR	Disaster Recovery
BNC	Bangladesh Nursing Council	DRPCC	District Rabies Prevention and Control Center
BNHA	Bangladesh National Health Accounts	DSCC	Dhaka South City Corporation
BIRDEM	Bangladesh Institute of Research and Rehabilitation for Diabetes, Endocrine and Metabolic Disorders	DSF	Demand-side Financing
BNNC	Bangladesh National Nutrition Council	EA	Enterprise Architecture
BNSB	Bangladesh National Society of Blind	ECNEC	Executive Committee of National Economic Council
bOPV	bivalent Oral Polio Vaccine	EOC	Emergency Obstetric Care
BOR	Bed-occupancy Ratio/Rate	EPI CES	Expanded Program on Immunization, Coverage Evaluation Survey
BPL	Below Poverty-Line	ERD	Economic Relations Division
BRAC	Bangladesh Rural Advancement Committee	ERP	Emergency Response Preparedness
BRIS	Birth Registration Information System	ESD	Essential Service Delivery
BRN	Birth Registration Number	ETL	Extract Transform-Load
BSL	Biosafety Level	FAO	Food and Agriculture Organization
BSMMU	Bangabandhu Sheikh Mujib Medical University	FBIS	Foodborne Illness Surveillance
BST	British Summer Time	FCPS	Fellow of College of Physicians and Surgeons
BUET	Bangladesh University of Engineering and Technology	FCTC	Framework Convention on Tobacco Control
CBE	Clinical Breast Examination	FD	Finance Division
CBHC	Community-based Healthcare	FETP,B	Field Epidemiology Training Program, Bangladesh
CC	Community Clinic	FP	Family Planning
CDC	Communicable Disease Control	FPI	Family Planning Inspector
CDD	Control of Diarrheal Diseases	FSW	Female Sex Worker
CEmOC	Comprehensive Emergency Obstetric Care	FWA	Family Welfare Assistant
CES	Coverage Evaluation Survey	FWV	Female Welfare Visitor

FY	Fiscal Year
FYP	Five-year Plan
GAVI	Global Alliance for Vaccines and Immunization
GBS	Guillain-Barre Syndrome
GDD	Global Disease Detection
GDP	Gross Domestic Product
GFATM	Global Fund to Fight AIDS, Tuberculosis and Malaria
GHS	Global Health Security
GHSA	Global Health Security Agenda
GNI	Gross National Index
GOARN	Global Outbreak Alert and Response Network
GOB	Government of Bangladesh
GPS	Global Positioning System
GR	Geographical Reconnaissance
HA	Health Assistant
HAV	Hepatitis A Virus
HBV	Hepatitis B Virus
HCV	Hepatitis C Virus
HDI	Human Development Index
HEU	Health Economics Unit
HI	Health Inspector
HIS	Health Information System
HISP,Bd	Health Information System Program, Bangladesh
HIV	Human Immunodeficiency Virus
HMIS	Health Management Information System
HNP	Health and Nutrition Program
HNPSIP	Health, Nutrition and Population Strategic Investment Plan
HPNSDP	Health, Population and Nutrition Sector Development Program
HPNSP	Health, Nutrition and Population Sector Program
HPSP	Health and Population Sector Plan
HR	Human Resource
HRH	Human Resource for Health
HRIS	Health Resource Information System
HRM	Human Resource Management
HWF	Health Workforce
IAEG-SDGs	Inter-agency and Expert Group on SDG Indicator
ICD-10	International Statistical Classification of Disease
ICT	Information and Communication Technology
ICU	Intensive Care Unit
ideSHi	Institute of developing Science and Health initiatives
IEDCR	Institute of Epidemiology, Disease Control and Research
IHR	International Health Regulations
IHT	Institute of Health Technology
IMCI	Integrated Management of Childhood Illness
IMPACT	Improving Public Health Management for Action
INFOSAN	International Network of Food Safety Authority
IPGMR	Institute of Postgraduate Medicine and Research
IPH	Institute of Public Health
IPHN	Institute of Public Health Nutrition
IPV	Inactivated Polio Virus Vaccine
ISDP	Integrated Service Delivery Platform
ISP	Internet Service Provider
IT	Information Technology
IV	Intravenous Fluid
IVM	Integrated Vector Management
JEE	Joint External Evaluation
KA	Kala-azar
KSM	Kala-azar Search Volunteer
LAN	Local Area Networking

LD	Line Director
LEEP	Loop Electrosurgical Excision Procedure
LF	Lymphatic Filariasis
LG	Local Government
LHB	Local Health Bulletin
M&E	Monitoring & Evaluation
MARP	Most-at-risk Population
MAT	Medical Assistance Training
MATS	Medical Assistant Training School
MBBS	Bachelor of Medicine, Bachelor of Surgery
MBDC	Mycobacterium Disease Control
MBT	Medical Biotechnology
MCCoD	Medically Certified Cause of Death
MCH	Medical College Hospital
MCV1	Measles Coverage Vaccine first dose
MCWCs	Maternal and Child Welfare Centers
MDA	Mass Drug Administration
MDGs	Millennium Development Goals
MDR	Multi-drug Resistance
MDT	Multi-drug Therapy
MDV	Mass Dog Vaccination
MIS	Management Information System
MMR	Maternal Mortality Ratio
MNCAH	Maternal, Neonatal, Child, and Adolescent Health
MNHI	Maternal and Newborn Health Initiative
MOCHTA	Ministry of Chittagong Hill Tracts Affairs
MOHA	Ministry of Home Affairs
MOHFW	Ministry of Health and Family Welfare
MOLGRD	Ministry of Local Government, Rural Development and Cooperatives
MPDSR	Maternal and Perinatal Death Surveillance and Response
MR	Measles Rubella Vaccine
MRI	Magnetic Resonance Imaging
MRS	Medical Record System
MSD	Measles Second Dose
MSM	Men having Sex with Men
MSW	Male Sex Worker
MTR	Mid-term Review
NAC	National AIDS Committee
NASP	National AIDS and STD Program
NCD	Non-communicable Disease
NFC	National Fistula Center
NFM	New Funding Model
NHA	National Health Accounts
NIC	National Influenza Center
NID	National ID/National Immunization Day
NINH	National Institute of Neurology & Hospital
NINS	National Institute of Neurosciences & Hospital
NIO	National Institute of Ophthalmology
NIPORT	National Institute of Population Research and Training
NIPSOM	National Institute of Preventive and Social Medicine
NISB	National Influenza Surveillance, Bangladesh
NITOR	National Institute of Traumatology and Orthopedics Rehabilitation
NKEP	National Kala-azar Elimination Program
NLEP	National Leprosy Elimination Program
nm	Nautical Mile
NMCP	National Malaria Control Program
NMSS	National Micronutrients Status Survey

NNP	National Nutrition Program	RT-PCR	Reverse Transcription Polymerase Chain Reaction
NNS	National Nutrition Services	RTRL	Regional TB Reference Laboratory
NPAN2	Second National Plan of Action for Nutrition	SACMO	Sub-Assistant Community Medical Officer
NRPCC	National Rabies Prevention and Control Center	SAP	Strategic Action Plan
NSC	National Steering Committee	SBA	Skilled Birth Attendant
NSP	National Strategic Plan	SBTP	Safe Blood Transfusion Program
NTBB	National Taskforce on Biotechnology of Bangladesh	SCAPAND	Strategic and Convergent Action Plan on Autism and Neurodevelopment Disabilities
NTC	National Technical Committee	SDGs	Sustainable Development Goals
NTDs	Neglected Tropical Diseases	SEARO	Southeast Asian Regional Office
NTP	National TB Program	SHR	Shared Health Record
NTRL	National TB Reference Laboratory	SKKRC	Surya Kanta Kala-azar Research Center
NTV	Nerve Tissue Vaccine	SMoL	Start-up Mortality List
NTWG	National Technical Working Group	SOP	Standard Operating Procedure
OIE	World Organization for Animal Health	SSK	Shashtho Surokhsha Karmasuchi
OMR	Optical Mark Recognition	STD	Sexually Transmitted Disease
OP	Operational Plan	STH	Soil-transmitted Helminthes
OPD	Outpatient Department	SVRS	Sample Vital Registration System
OPV	Oral Polio Vaccine	SWAp	Sector-wide Approach
ORG	Office of Registrar General	TAS	Transmission Assessment Survey
ORS	Oral Rehydration Salt	TAST	Technical Assistance Support Team
ORT	Oral Rehydration Therapy	TB	Tuberculosis
OT	Operation Theater	TCV	Time, Cost, and Visit
PA	Project Aid	TCV	Tissue Culture Vaccine
PCR	Polymerase Chain Reaction	TEMO	Transport & Equipment Maintenance Unit
PDA	Personal Digital Assistant	TFR	Total Fertility Rate
PEP	Post-exposure Prophylaxis	THE	Total Health Expenditure
Pf	Plasmodium falciparum	tOPV	trivalent Oral Polio Vaccine
PH	Public Health	TT	Tetanus Toxoid
PHC	Primary Healthcare	TTGA	Taurocholate-tellurite-gelatin Agar
PHEIC	Public Health Emergency for International Concern	TTI	Transfusion-transmitted Infection
PKDL	Post Kala-azar Dermal Leishmaniasis	TTU	Technical Training Unit
PLHIV	People living with HIV	UAT	User Acceptance Test
PMIS	Personnel Management Information System	UESDS	Utilization of Essential Service Delivery Survey
PMMU	Program Management & Monitoring Unit	UH&FPO	Upazila Health and Family Planning Officer
PMO	Prime Minister Office	UHC	Universal Health Coverage/Upazila Health Complex
PNC	Postnatal Care	UHFWC	Union Health and Family Welfare Center
PoE	Point of Entry	UN	United Nation
PRS	Population Registration System	UNAIDS	Joint United Nations Program on HIV/AIDS
PRSP	Poverty Reduction Strategy Paper	UNDP	United Nations Development Program
PS&	Planning, Supply and ownership management program	UNFPA	United Nations Population Fund
PSM	Preventive and Social Medicine	UNGASS	United Nations General Assembly Special Session
PSTN	Public Switched Telephone Network	UNICEF	United Nations Children's Fund
PWID	People Who Inject Drugs	UPHCP	Urban Primary Healthcare Project
RADP	Revised Annual Development Program	USC	Union Subcenter
RBC	Red Blood Cell	USD	United States Dollar
RCHCIB	Revitalization of Community Healthcare Initiative in Bangladesh	USG	Ultrasonography
RDU	Research and Development Unit	USI	Universal Salt Iodization
REP	Rabies Elimination Program	VA	Verbal Autopsy
Rev.	Revenue	VAC	Vitamin A Capsule
RHC	Rural Health Center	VAD	Vitamin A Deficiency
RHIS	Routine Health Information System	VIA	Visual Inspection with Acetic Acid
RIG	Rabies Immunoglobulin	WASH	Water, Sanitation and Hygiene
RMCS	Revised Malaria Control Strategies	WAZ	Weight-for-age z-score
RPA	Reimbursable Project Aid	WB	World Bank
RRT	Rapid Response Team	WBC	White Blood Cell
rRT-PCR	Real-time reverse transcription (RRT)-polymerase chain reaction	WCBA	Women of Childbearing Age
		WHO	World Health Organization
		WiMAX	Worldwide Inter-operability for Microwave Access

MESSAGE



ZAHID MALEQUE, MP

Honorable Minister

Ministry of Health and Family Welfare

Government of the People's Republic of Bangladesh

It is a matter of great pleasure and pride for me to know that the Directorate General of Health Services (DGHS) is going to publish the Health Bulletin, 2018. They have regularly been publishing the bulletin with exciting and reliable information on the health sector. The world progress is based on appropriate information, and collection of information is one of the toughest job in the context of a developing country. Though Health related information is easily accessible through innovative e-Health and m-Health strategies throughout the world, the bulletin, in fact, accumulates the reports of general health situation of the country as it reflects almost a complete picture of both the public and the most private healthcare service delivery. Therefore, these information are vital for adopting evidence-based public health decisions.

The running 4th sector-wide program called HPNSP has been conducted to achieve the targets of the health-related Sustainable Development Goals (SDGs) within 2030. It is seen as both an opportunity and a challenge for us. I am happy that the Government of Bangladesh is investing their best efforts in the health sector, under the leadership of the most successful Prime Minister of Bangladesh, the Honorable Prime Minister Sheikh Hasina. Her vision guides us to improve every step of our health service delivery from the tertiary hospitals to the community clinic.

I appreciate the hard work of the officials and staff from the Government of Bangladesh and the Development Partners who are involved in the publication of this Health Bulletin. Although the personnel of DGHS are continuously trying for the betterment of this publication by both qualitative and quantitative improvements of data collection, I believe there is still some scope for the development in other areas of interest, especially about collecting data from the private sectors.

I would like to thank Professor Dr. Abul Kalam Azad, the Director General of DGHS and his colleagues who are engaged in collecting data from various sources and analyzing them. These data constitute the heart of the Bulletin. I believe their good work will continue, and Health Bulletin 2018 will fulfill the information needs of all health practitioners and researchers.

Joy Bangla, Joy Bangabandhu.

Long live Bangladesh.

MESSAGE



DR. MD. MURAD HASAN, MP

Honorable State Minister
Ministry of Health and Family Welfare
Government of the People's Republic of Bangladesh

I am pleased to know that the Health Bulletin, 2018 is in the process of printing. This publication of the MIS, DGHS is an authentic document to assess the overall healthcare situation in the country and to identify the priority areas for future attention and actions. I think this bulletin is helping us develop a culture of evidence-based decision-making which is very important for the developing countries. The Directorate General of Health Services along with the Management Information System in particular have contributed much in creating success stories in the health sector that have earned international recognitions in the recent past. The Health Bulletin is like a mirror that reflects our commitments, policies, and efforts for improving the health scenario of the country. It is definitely an immense pleasure that the quality of Health Bulletin is improving year by year, particularly in terms of analysis and authenticity of data. Over the last few years, Bangladesh has made tremendous progress in the health sector under the present government, led by the Honorable Prime Minister Sheikh Hasina. These achievements are the results of the wise planning, careful implementation, good monitoring and evaluation of the country's recent health interventions.

Bangladesh has become a role model for other developing countries in achieving the MDGs. I hope, we will continue to perform better in achieving the targets of SDGs too. By the innovative efforts made by the personnel of MIS, DGHS under the leadership of Professor Dr. Abul Kalam Azad, our country is now equipped with a real time health information system. I congratulate him, and the Director of MIS and his colleagues, partners from DPs, NGOs and to those who have worked hard for bringing out this valuable publication on time.

Joy Bangla, Joy Bangabandhu.

MESSAGE



G M SALEH UDDIN

Secretary

Medical Education and Family Welfare Division

Ministry of Health and Family Welfare

Government of the People's Republic of Bangladesh

Health Bulletin is an annual publication of the Management Information System of the Directorate General of Health Services. This important publication is a vital source of information that is related to the current health care situation. Also, it serves as a reflection of achievements and future challenges for our health department. I feel very happy that now the Health Bulletin 2018 is being published.

Bangladesh has achieved great success by fulfilling the targets of the health-related millennium development Goals. The next target of the Government of Bangladesh is to achieve the health-related sustainable development goals (SDG). To achieve these goals, we need coordinated and relentless effort from everyone, from DGHS personnel at the national level to all stakeholders and health service providers at the grassroots level. The Health Bulletin shows us what activities are now being conducted to achieve the goals and what more is needed to be done.

I would like to extend my special thanks to the Director General, Professor Dr. Abul Kalam Azad and his staff at the DGHS whose noteworthy achievements in the digitalization process have gained international recognitions. The Management information system has been an important contributor to this digital revolution; they deserve special thanks for that.

I hope all the information presented in this bulletin will be very helpful for everyone. Together, we will march forward to create a healthy Bangladesh.

MESSAGE



MD. ASHADUL ISLAM

Secretary

Health Services Division

Ministry of Health and Family Welfare

Government of the People's Republic of Bangladesh

'Health Bulletin' a regular publication of the Directorate General of Health Services, is a precious document for updated information on different health programs, hospital services, academic and public health institutions, and health workforce. This is a source of information used very frequently by the Policy Makers, Researchers, Program Managers and all the stakeholders who are associated with planning and implementation of Health programs.

I am happy to know that the Management Information System (MIS) of the Directorate General of Health Services (DGHS) is going to publish the Health Bulletin. This will help us to analyze the policies and ongoing activities in the health sector and will play an important role in our future health planning.

I appreciate the hard work of personnel in the Directorate General of Health Services for this publication. Their efforts to improve the data collection and data presentation are also praiseworthy. I hope they will continue their good work, despite our limitations and constraints.

Digitalization of the data-collection and record-keeping system has been of enormous help in visiting and revisiting our actual situations of the healthcare services in the country. I would like to thank the Director General Professor Dr. Abul Kalam Azad and the staff at the DGHS whose tremendous effort and noteworthy achievements in the digitalization process have harvested international recognitions.

I wish the best utilization of the Health Bulletin 2018.

MESSAGE



PROFESSOR DR. ABUL KALAM AZAD

Director General

Directorate General of Health Services

Government of the People's Republic of Bangladesh

I am delighted to present Health Bulletin 2018 to our distinguished readers. Health Bulletin, published by the Management Information System (MIS) of the DGHS, has always been a vital and representative source of information on the current healthcare situation in Bangladesh. This annual publication has been recognized as a useful document for all in the health sector, ranging from policy makers to field workers.

The Directorate General of Health Services, especially the MIS has been continuously trying to collect accurate data from the field level and analyze for inclusion in the yearly bulletin. The accuracy of data has a direct impact on understanding of the actual situation, which, in turn, affects the decision-making process. This is an enormous and difficult task but I am very happy that my colleagues at the MIS are doing their jobs efficiently.

Probably, it did not escape the notice of our readers that quality of the Health Bulletin is improving bit-by-bit every year. This has been possible because of the dedication of my colleague Dr. Samir Kanti Sarker and his team from MIS along with Prof. Benazir Ahmed and the members of CSC of DGHS. I extend my heartiest congratulations to all of them for their hard work in bringing out the Bulletin on time.

I am highly grateful to Hon'ble Minister, Hon'ble State Minister, and the Secretary of Health Services division and Secretary of Medical Education and Family Welfare Division of the Ministry of Health and Family Welfare for their kind cooperation, constant support and guidance in all of the activities regarding publication of this bulletin.

I am also thankful to those who contributed at different sector of the publication of this Bulletin including the Development partners. Constructive advice from the readers and multiple stakeholders will help to improve and quality of the Bulletin in future.

MESSAGE



DR. SAMIR KANTI SARKER

Director MIS and
Line Director (HIS and e-Health)

It gives me immense pleasure to finally come to the stage of presenting Health Bulletin 2018. Health Bulletin is the flagship publication of the Directorate General of Health Services under the leadership of Director General Prof. Dr. Abul Kalam Azad. The quality of Health Bulletin is going up year by year, particularly in terms of analysis and authenticity of data. This publication prints technical information on healthcare-related development activities of the Government of Bangladesh each year. Health Bulletin 2018 contains the remarkable achievements in our health sector during 2017 and part of 2018 in some cases. The availability of accurate and timely data is essential to monitor the progress of health development and to evaluate the impact. I am very happy to put forward accomplishments of DGHS, especially our progress in e-Health and m-Health in Bangladesh. Health Bulletin 2018 is rich in quality of contents, accurate in data presentation, and is a high-quality production which will definitely meet the needs of the readers. Now it is well-accepted that updated information is very important for efficient management at all levels. I hope this bulletin enriched with more updated information will be very useful to the implementers, managers, researchers, policy-makers, and to our development partners.

I express special thanks to Hon'ble Minister, Hon'ble State Minister, and the Secretaries of the Ministry of Health and Family Welfare for their cooperation in all the activities, including publication of this bulletin. I would also like to express my heartiest thanks to the respective departments of the DGHS and other sectors for timely sending their data and all the members of the editorial board. I strongly believe and hope that the bulletin will help our policy-makers, donor partners, NGOs health service providers, and other stakeholders to make decisions for improving healthcare services of Bangladesh.

Finally, I would like to extend my heartiest gratefulness to respected Professor Dr. Abul Kalam Azad, Director General, DGHS and all Development Partners and to everyone who worked hard to bring out this publication.

We always welcome the constructive criticisms and suggestions to improve this publication in coming years.

CONTENTS

ii	Editorial board
v	Acronyms
viii	Messages
2	Chapter 01 Bangladesh health situation at a glance
8	Chapter 02 Organizational structure of healthcare System under DGHS, MoHFW
20	Chapter 03 Health rights, acts, policies, strategies, plans and SDG
34	Chapter 04 Public health services
126	Chapter 05 Curative service
150	Chapter 06 Health workforce situation in Bangladesh
158	Chapter 07 Medical education
164	Chapter 08 Health information system and e-health
180	Chapter 09 Health research
184	Chapter 10 Financing healthcare
192	Chapter 11 Development status

For Annex, please visit dghs.gov.bd

BANGLADESH HEALTH SITUATION AT A GLANCE

with demography

BACKGROUND

Over the last 48 years since independence, Bangladesh has made lot of strides in the Health Sector. Undoubtedly there is proliferation in health infrastructures - medical colleges, medical universities, private medical colleges, private clinics, private hospitals, district hospital, rural health centers and community clinics. Many NGOs are also playing a great role and contributing toward health care delivery system. Much progress has been made in the pharmaceutical sector providing affordable medicine, intravenous fluids, anti-cancer drugs etc. There is also increased awareness in the general public on health issues. National and private level campaigns are ongoing to promote mental and child health, vaccination programs, mass deworming programs, use of safe water and latrines, hand washing etc. Family planning awareness is one of the most important successes of Bangladesh. The population growth rate has been brought down. A huge and thriving network of private physicians stretched and spread to all over the country is trying to meet the needs of day to day medical problems. There have been significant gains in terms of polio & small pox eradication. Extensive vaccination, case isolation has resulted in diminishing number of diphtheria and tetanus. Immunization (Valid vaccination coverage) is excellent in Bangladesh. With improvement of overall hygiene and sanitation standard there is a discernible reduction in cholera, typhoid and dysenteries. A great number of tuberculosis satellite clinics now offers free treatment with contribution from NGO's. HIV & AIDS are at a low level of prevalence. Kala Azar and malaria has been significantly curbed down. Malaria and Tuberculosis death rate is totally low at present in Bangladesh. Maternal mortality and neonatal mortality rate has also been reduced.

The number of health care facilities under DGHS is 2257 which is a great contribution for the health sector in Bangladesh. Furthermore, Government hospitals of secondary and tertiary levels, Government hospitals under the DGHS at the upazila and union levels, No. of private registered hospitals and clinics, No. of private registered diagnostic centers, No. of hospital beds, No. of hospital beds in the private-sector (in private hospitals, Total in the DGHS-



DEMOGRAPHY

Estimated population on 1 July 2017: **162.7 million**

Population density (per sq. km): 1,077; in land area 1,236.811 (In 2015; WB 2016)

Population growth rate: 1.37%

Sex ratio (M/F*100): 100.2 /100.0

Health, Population and Nutrition Sector Program

Dependency ratio (percent): Total: 53; Rural: 57, Urban: 47

Child-woman ratio (Per 1,000 women of 15-49 years): Total: 310; Rural: 336, Urban: 279

Crude birth rate (Per 1,000 population): Total: 18.5; Rural: 20.4, Urban: 16.1

Total fertility rate (Per woman of 15-49 years): Total: 2.05, Rural: 2.37, Urban: 1.68

General fertility rate: Total: 68, Rural: 78, Urban: 56

Net reproduction rate (Per woman of 15-49 years): Total:1.00, Rural: 1.09, Urban: 0.80

Crude marriage rate (Per 1,000 population): Total:14.6, Rural: 18.1, Urban: 10.2

Source: SVRS 2017

run hospitals and registered private hospitals has added extra benefits to the health sector. Teaching and training centre for health care and the seats of these institutions have also been increased.

ADMINISTRATIVE UNIT

Division	: 8
City Corporation	: 12
Metropolitan city	: 4
Municipality	: 327
District	: 64
Upazila	: 490
Union	: 4,553
Ward	: 40,977
Village (approx.)	: 87,310

Table 1.1: Population by broad age-group (%) (ref-SVRS 2017)

Age-group (years)	Both sexes	Male	Female
00-14	29.3	29.5	29.2
15-49	54.4	54.1	54.8
50-59	8.3	8.2	8.3
60+	8.0	8.2	7.7

Table 1.2: Age-specific fertility rate (Per 1,000 women of reproductive age)

Age-group (years)	Rate	Age-group (years)	Rate
15-19	75	35-39	26
20-24	134	40-44	7
25-29	105	45-49	3
30-34	59		

Table 1.3: Mean age at marriage (years) (ref-SVRS 2017)

	Male	Female
Total	26.2	18.8
Rural	25.7	18.3
Urban	27.3	19.9

HOUSEHOLD CHARACTERISTICS AND UTILITIES

Household-size (no. of persons): 4.2

Male-headed: 85.8

Female-headed: 14.2

Water and sanitation (% households)

A. Drinking-water: Access to tap and tube well water 98.0

B. Toilet facility: Sanitary: 76.8; Others: 20.6 ; None: 2.6

C. Source of Light: Electricity 85.3, Solar 5.8, Kerosene 8.8, Others 0.1

Source: SVRS 2017

INFORMATION TECHNOLOGY

Internet subscribers (BTRC, 31 August 2018)

Total : 90.501 million

Mobile Internet : 84.685 million

WiMAX : 0.083 million

ISP + PSTN : 5.733 million

Mobile phone users (BTRC, 31 July 2018)

Total : 152.527 million

LITERACY

Literacy rate of population 7+ years (percent):

Both sexes: 72.3 , Female: 70.2 , Male: 74.3

Adult literacy rate of population 15+ years (percent):

Both sexes: 72.9 , Female: 70.1 , Male: 75.7

Source: SVRS 2017

HEALTH SERVICE AND MEDICAL EDUCATION

Total number of government facilities under the DGHS: 2,258

Total primary level facilities (except community clinic): 2,004

Total secondary & tertiary level facilities: 254

Total number of registered private hospitals and clinics: 5,054

Total number of registered private diagnostic centers: 9,529

No. of hospital beds under the DGHS: 52,807

No. of hospital beds in private hospitals registered by the DGHS: 90,587

Total in the DGHS-run facilities and registered private hospitals: 143,394

HEALTH STATUS

Coverage of vitamin A capsule (EPI CES 2016)

Vitamin A coverage:-

Infant (6-11 months): 86.1%

Children (12-59 months): 91.3%

Postpartum women: 37.8%

Family planning

Contraceptive prevalence rate (%): 62.1 (SVRS 2015);

Unmet need for family planning (%): 12.0 (BDHS 2014);

HIV/AIDS

Antiretroviral treatment (ART) coverage among adults needing ART in 2017: 70.03%

HIV prevalence among key populations in 2017: Less than 1%

New HIV infection reported in 2017: 865

Number of ART recipients up to August 2018: 3,265

People living with HIV (PLHIV) in 2017: 5,586

Mortality rate among HIV+ve TB patients/ 100,000 population: 0.11 (WHO 2015)

Table 1.4. Teaching/training institutions for healthcare

Teaching/training institutions	Total	Government	Private
Postgraduate medical teaching institutions	39	22 + autonomous 7	10
Medical colleges	105	36	69
Armed forces & Army Medical Colleges	6		
Dental colleges and dental units	35	9	26
Nursing institutions offering basic BSc Nursing course	60	15	45
Nursing institutions offering post-basic BSc Nursing course	41	4	37
Nursing institutions offering specialized Diploma	4	1	3
Institutions offering Diploma in Nursing & Midwifery	183	43	140
Unani & Ayurvedic Medical College	4	2	2
Homeopathic Medical College	2	1	1
Unani & Ayurvedic Diploma College	23	1	22
Homeopathic Diploma College	61	0	61
Medical assistants training schools	209	9	200
Institutes of health technology (IHT)	108	11	97

Malaria

Malaria positive case/1,000 population (in endemic areas): 1.64 (in 2017)

Malarial mortality/100,000 population (in endemic areas): 0.0007 (in 2017)

Immunization status: EPI Crude coverage 2017

≤12 months old children: BCG 101.3%; OPV1 101.1%; OPV2 99.3%; OPV3 97.9%; Penta1 100.1%; Penta2 99.9%; Penta3 98.5%; PCV1 99.0%; PCV2 98.4%; PCV3 102.5%; MR1 98.8%; MR2 87.2%,

≤23 months old children: BCG 99.5%; OPV1 97.9%; OPV2 97.2%; OPV3 90.4%; Penta1 97.9%; Penta2 97.2%; Penta3 90.4%; MR1 92.3%; full vaccination: 86.8%

Vitamin A coverage: Infant (6-11 months): 86.1%; Children (12-59 months): 91.3%

Immunization (Valid vaccination coverage) (EPI-CES 2016)

≤12 months old children: BCG 99.5%; OPV1 97.8%; OPV2 97.0%; OPV3 90.1%; Penta1 97.8%; Penta2 97.0%; Penta3 90.1%; Measles 87.5%; Full vaccination: 82.3%

≤23 months old children: BCG 99.5%; OPV1 97.9%; OPV2 97.2%; OPV3 90.4%; Penta1 97.9%; Penta2 97.2%; Penta3 90.4%; Measles 92.3%; Full vaccination: 86.8%

Valid tetanus toxoid coverage (%) among women of childbearing age: TT1 98.2%; TT2 96.8%; TT3 89.2%; TT4 73.2%; TT5 52.3%

Life-expectancy at birth (years)

Both sexes: 71.8; Female: 73.1; Male: 70.6 (In 2015; WB 2016)

Maternal and child health, Obstetric care

Antenatal care coverage (at least 4 visits) (%): 31 (In 2014; UNICEF February 2016), 31.2 (BDHS 2014)

Antenatal care coverage (at least one visit by skilled health professional) (%): 63.9 (In 2014; WB 2016)

Births attended by skilled health personnel (%): 42.1 (BDHS 2014)

C-section rate (%): 22.9% (BDHS 2014)

Home delivery rate: 62.2% (BDHS 2014)

Infant mortality rate (per 1,000 livebirths): 24 (SVRS 2017)

Institutional delivery rate (%): 37.4% Public facility: 12.8, Private facility: 22.4, NGO facility: 1.9 (BDHS 2014)

Maternal mortality ratio (per 100,000 livebirths): 172 (SVRS 2017)

Neonatal mortality rate/1,000 livebirths: 17 (SVRS 2017)

Under-5 mortality rate (per 1,000 livebirths:); 31 (SVRS 2017)

Tuberculosis

Incidence of tuberculosis including HIV+ TB: 221 (161-291) per 100,000 population.

Kala-azar

(Source from DHIS2, 2017)

Total number of Kala-azar cases (reported): 380 cases

New Kala-azar (NKA): 210 cases

Relapse Kala-azar (RKA): 61 cases

Kala-azar treatment failure (KATF): 3 cases

Post Kala-azar Dermal Leishmaniasis (PKDL): 102 cases

Death due to Kala-azar: 6 cases

Table 1.5.: WHO estimation of TB burden in Bangladesh 2017

Estimates of TB Burden 2017	Number (thousands)	Rate (per 100,000 population)
Mortality (excludes HIV+TB)	59 (38-85)	36 (23-52)
Mortality (HIV+TB only)	0.17 (0.085-0.29)	0.11 (0.05-0.18)
Incidence (includes HIV+TB)	364 (265-479)	221 (161-291)
Incidence (HIV+TB only)	0.55 (0.27-0.92)	0.33 (0.17-0.56)
Incidence (MDR/RR-TB)	8.4 (3.8-15)	5.1

Table 1.6. Available no. of seats in above mentioned teaching/ training institutions

Name of degree	Total	Government	Private	Foreign
Postgraduate medical degree (CME Survey, 2017) (MD, MS, Diploma, M Phil, MPH, MMed, FCPS, MCPS etc.)	2,486	1,048	1,112	112
Bachelor of Medicine & Surgery (MBBS)	12,082	4,068	6,231	1,783
MBBS offering Armed Forces & Army Medical Colleges	375			
Bachelor of Dental Surgery (BDS)	2,050	532	1,405	113
Bachelor of Unani Medicine & Surgery (BUMS); Bachelor of Ayurvedic Medicine & Surgery (BAMS)	200	100	100	
Bachelor of Homeopathic Medicine & Surgery (BHMS)	150	50	100	
Diploma in Unani Medicine & Surgery (DUMS); Diploma in Ayurvedic Medicine & Surgery (DAMS)	1150	50	1100	
Diploma in Homeopathic Medicine & Surgery (DHMS)	3050	0	3050	
Medical assistants training	14,358	818	13,540	
BSc & Diploma courses in health technology	11,625	2,685	8,940	

Non-Communicable Diseases

Reduction in prevalence of current tobacco use in persons aged over 15 years: 35.3% (2017)

Reduction in prevalence of insufficient physical activity: 15.1%(STEPS 2018)

Relative reduction in prevalence of raised blood pressure: 25.4% (STEPS 2018)

Death rate due to major NCDs (CVDs, cancer, diabetes, and chronic respiratory diseases among population aged 30 to 70 years) per 100000 populations: 398.1-503.7

Mental health situation in Bangladesh as per STEPS, 2018 survey: 16.1% adults and 18.4% child and adolescent have mental diseases.

HEALTH INDICATORS

Health Workforce

Personnel currently working under DGHS of MOHFW (No. of sanctioned posts are given in parentheses)

Table 1.7. Prevalance of Major NCDs (STEPS 2018)

Disease	Prevalence (%)
Ischemic heart disease	3.4
Stroke	0.9
CVD	0.4
Diabetes	6.4
COPD	2.97

Table 1.8. Incidence of five most frequent cancer (STEPS 2018)

Cancer Type	Incidence (%)
Breast	12.10%
Esophagus	11.30%
Cervix uteri	9.70%
Lung	8.80%
Lip and Oral cavity	8.70%

Personnel currently working under DGHS of MOHFW (No. of sanctioned posts are given in parentheses)

No. of personnel: 74,985 (103,743) *

No. of doctors: 20,914 (25,980)

No. of registered physicians: MBBS: 93358; BDS: 9569 (According to BMDC)

No. of Medical technologists:

Total: 5,184 (7,920); Dental: 469 (624); MT (EPI): 483 (499); Lab: 1,488 (2,237), Pharmacy: 1,546 (2,905); Physiotherapy: 109 (296); Radiography: 573 (778); Radiotherapy: 41 (83); Sanitary inspection: 475 (498)

No. of Sub-Assistant Community Medical Officers (SACMO): 3,801 (5,368)

No. of Community Healthcare Providers (CHCPs) for community clinics: 13,507

No. of Domiciliary Staff:

Health Inspectors (HI) 1,047 (1,410); Assistant Health Inspectors (AHI): 3,636 (4,220); Health Assistants (HA) 15,420 (20,908)

*excluding nursing posts

Population-Health Workforce Ratio

(According to SVRS 2017 total population - 162.7 million was considered as the denominator in applicable cases)

Population per registered physician: 1,581

No. of registered physicians per 10,000 populations: 6.33

No. of doctors working under DGHS per 10,000 populations: 1.28

No. of medical technologists working under DGHS per 10,000 populations: 0.32

No. of community and domiciliary health workers working under DGHS per 10,000 people: 2.13

No. of beds in DGHS-run public hospitals per 10,000 populations: 3.24

No. of beds in private hospitals (registered by DGHS) per 10,000 populations: 5.57

ORGANIZATIONAL STRUCTURE OF HEALTHCARE SYSTEM UNDER DGHS, MOHFW

Hierarchy for better health administration

BACKGROUND

Healthcare network in Bangladesh is spread across the country considering the Ministry of Health and Family Welfare (MOHFW) at the leading point and is a complicated network system. The nexus of this web comprises implementing authorities, regulatory bodies and healthcare facilities from national to the community level. Besides, various NGOs and private institutions constitute a large segment of this web along with the public health departments of the Government. The Ministry deliberately handles national-level policy, planning, and decision-making. Various implementing authorities and healthcare delivery systems act promptly to render the national-level policies, plans, and decisions in the provision of healthcare and education. The healthcare systems of the NGOs and the private sector are controlled indirectly by the Ministry and its relevant regulatory bodies.

In March 2017, a new administrative structure of the Ministry of Health and Family Welfare has been created. The Government has divided the Ministry into two divisions. The new divisions under the ministry are: (I) Health Services Division and (II) Medical Education and Family Welfare Division. Health Services Division acts in various way such as making policy regarding health-related matters, management and maintenance of nursing care, health financing and so on. On the other hand, Medical Education and Family Welfare Division acts by making policy regarding family planning matters, medical education, matters relating to medical colleges and medical universities, registration of birth and death etc.

Directorate General of Health Services (DGHS) is the largest implementing authority under the MOHFW with more than one hundred thousand officers and staff members. The DGHS provides technical assistance to the Ministry when new programs and interventions are needed to improve in the existing ones.

Management structure and type of health facilities under the DGHS is very strong. The six tiers of healthcare infrastructure under the DGHS consist of

national, divisional, district, upazila (subdistrict), union, and ward. At the national level, there are institutions both for public health functions as well as for postgraduate medical education/training and specialized treatment to patients.

In the administrative setup of the DGHS there are two Additional Director Generals under The Director General. Additional Director General divides into two sectors: Administration and Planning and Development. Under both personnel there are Directors and Deputy Directors.

A divisional director monitors activities in each division, who are assisted by deputy directors and assistant

directors. There is one infectious disease hospital and one or more medical college(s) with its own hospital at the divisional headquarters.

As the district health manager, the civil surgeon (CS) is responsible for delivering secondary and primary care services. In each district, there is a district hospital and have superintendents to look after the hospital management. In others, civil surgeons look after the district hospitals. Some of the district headquarters have medical colleges with attached hospitals, medical assistants' training schools, and nursing training institutes.

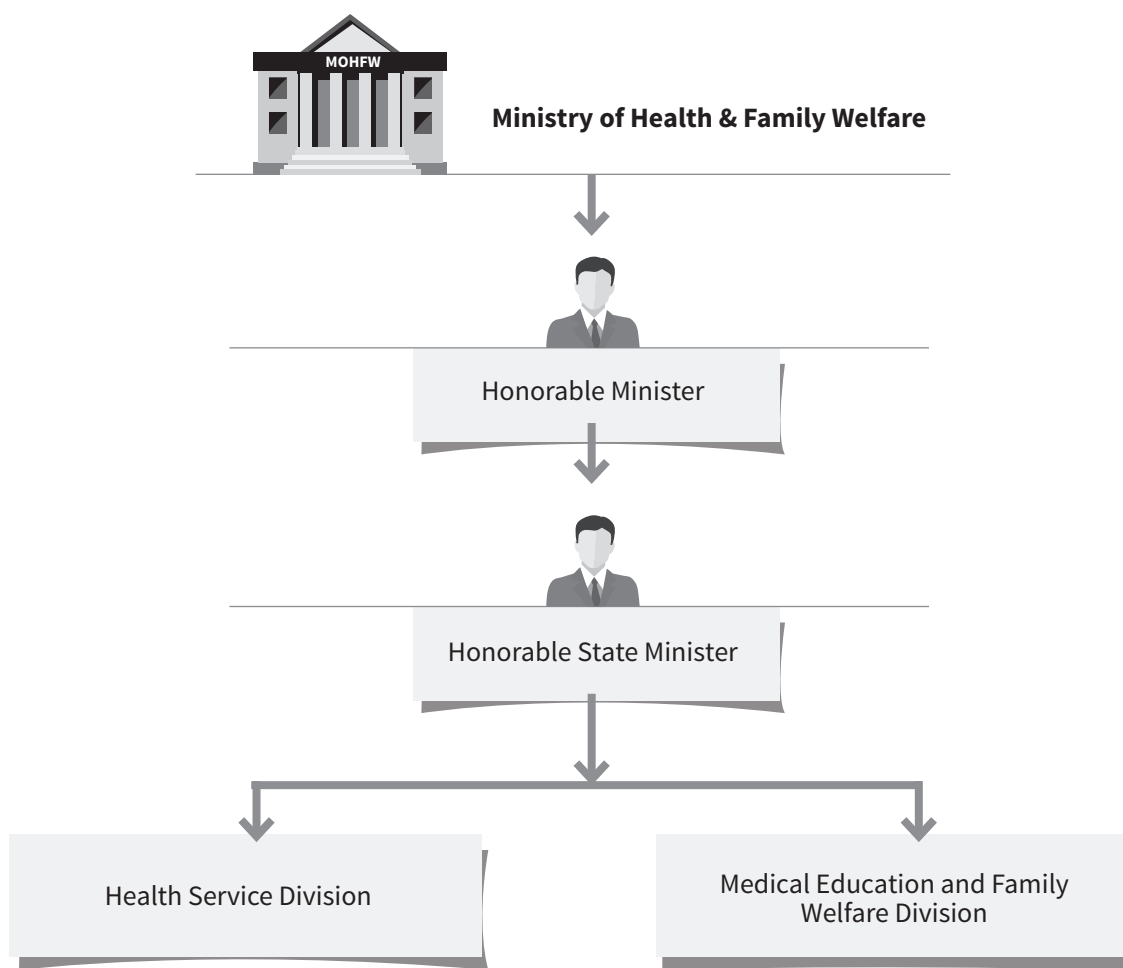


Figure 2.1. Divisions of the MOHFW

HIERARCHY OF PERSONNEL IN THE MINISTRY OF HEALTH AND FAMILY WELFARE (UNDER HEALTH SERVICES DIVISION)

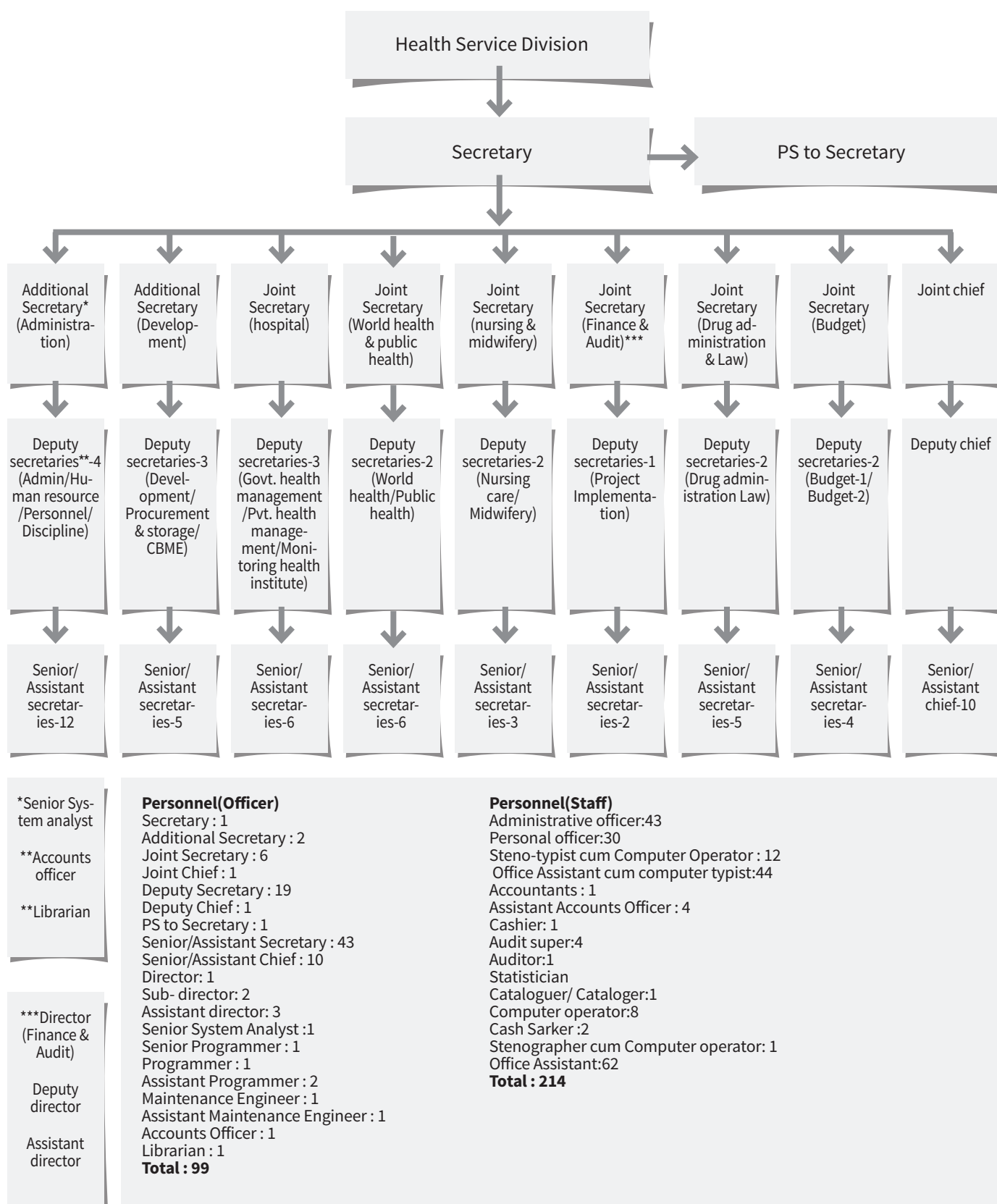


Figure 2.2. Hierarchy of personnel in the MOHFW (ref: MOHFW website, last update april 2017)

HIERARCHY OF PERSONNEL IN THE MINISTRY OF HEALTH AND FAMILY WELFARE (UNDER MEDICAL EDUCATION AND FAMILY WELFARE DIVISION)

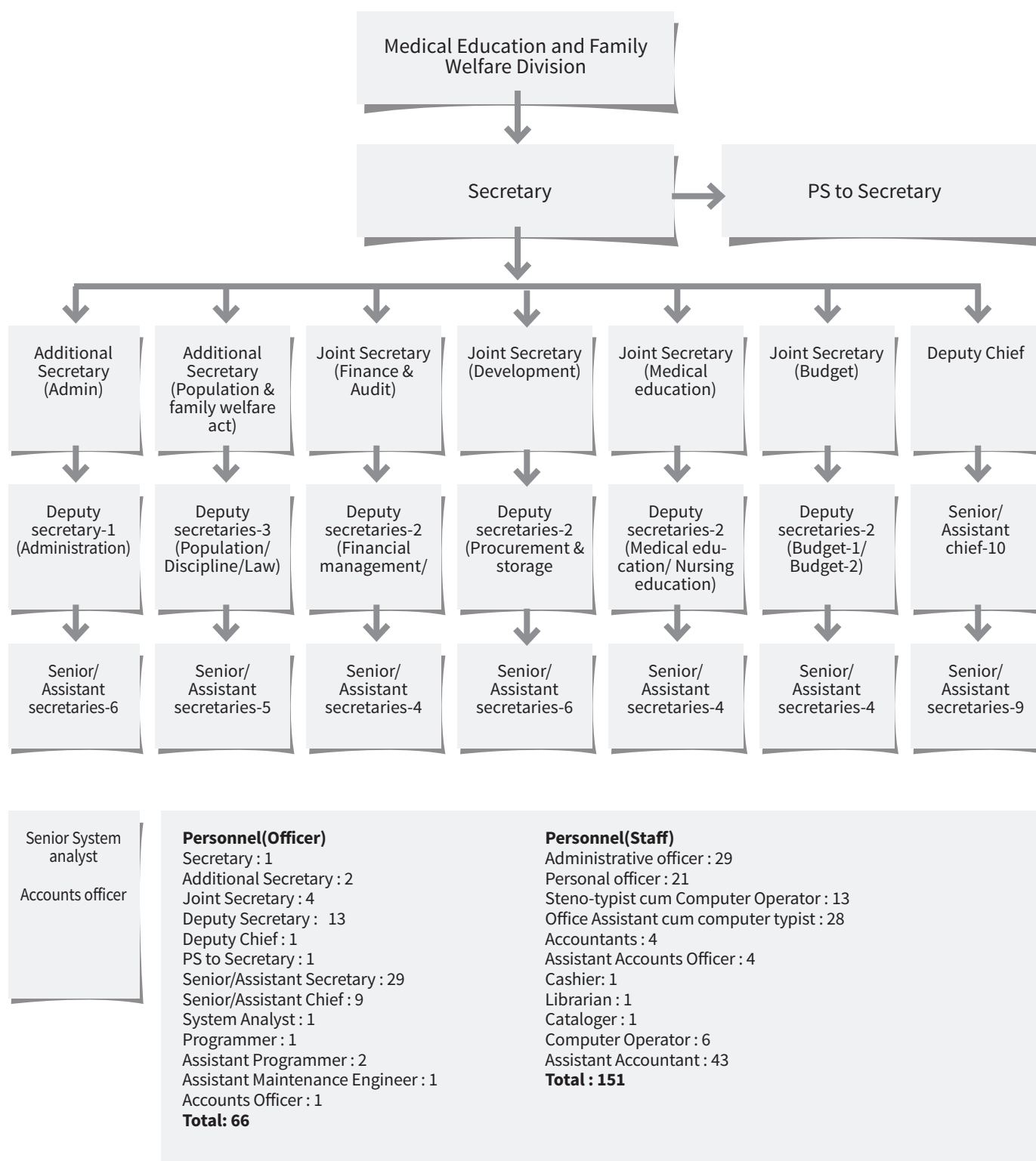


Figure 2.3. Hierarchy of personnel in the MOHFW (ref: MOHFW website, last update april 2017)

IMPLEMENTING AUTHORITIES

There are nine implementing authorities under the MOHFW as follows:

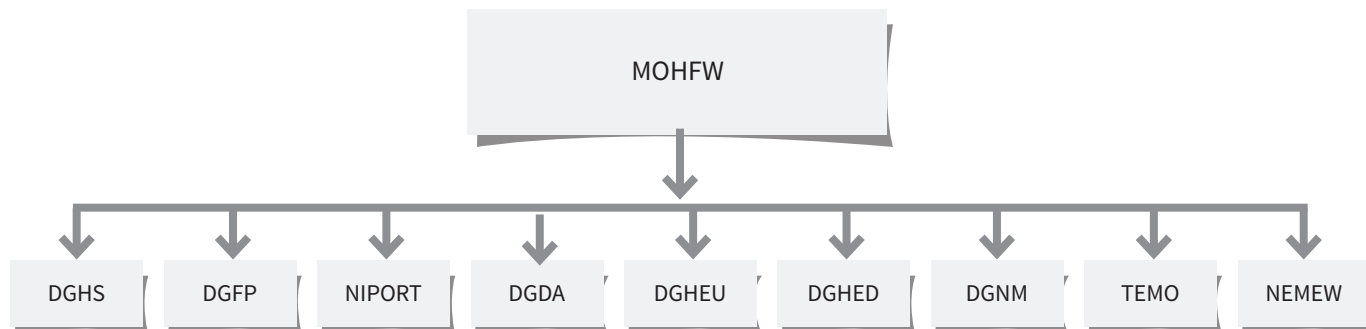


Figure 2.4. Implementing authorities under the Ministry of Health and Family Welfare

REGULATORY BODIES

There are five regulatory bodies under the MOHFW as follows:

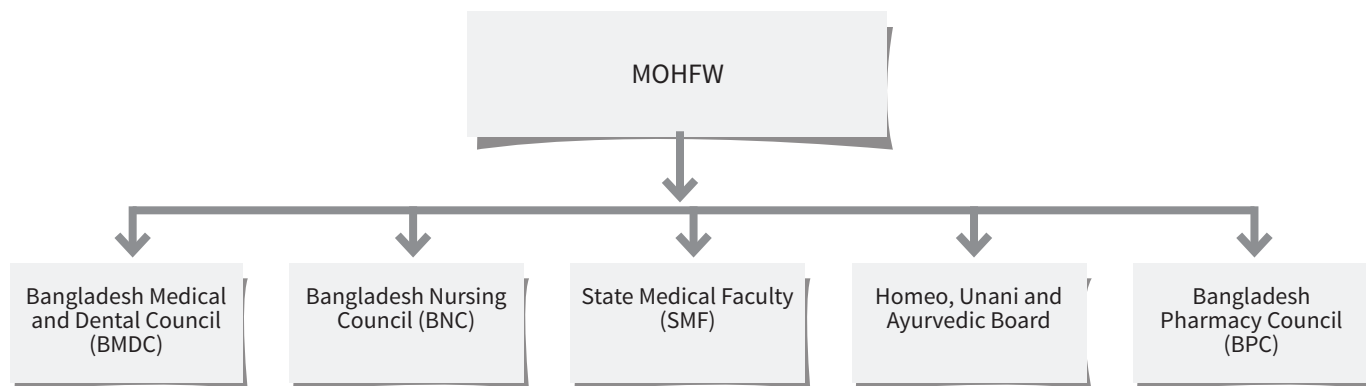


Figure 2.5. Regulatory bodies under the MOHFW

DIRECTORATE GENERAL OF HEALTH SERVICES

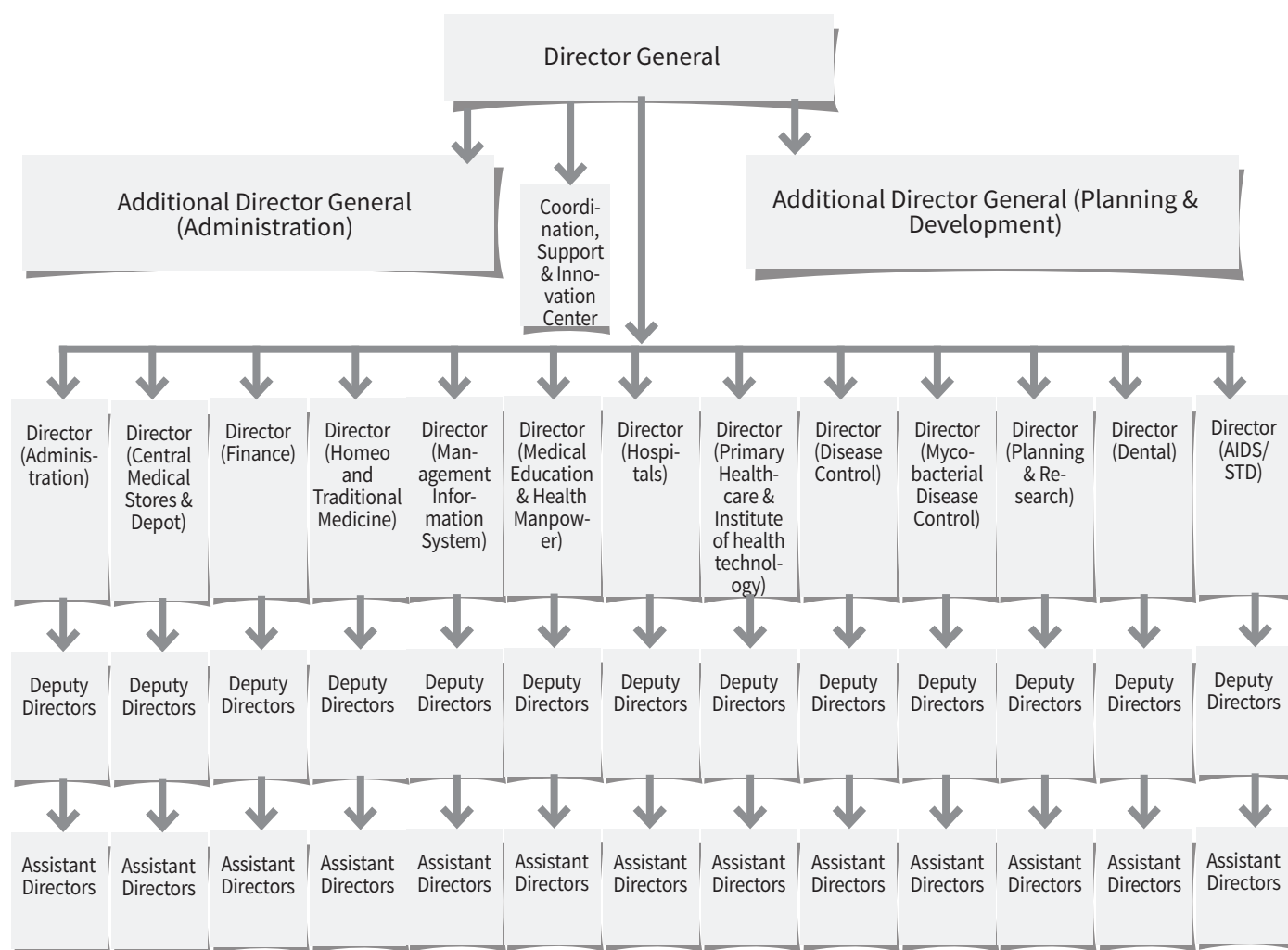


Figure 2.6. Directorate General of Health Services (ref- HRIS,DGHS)

MANAGERIAL HIERARCHY ACCORDING TO TYPES OF FACILITIES FROM NATIONAL TO THE WARD LEVEL

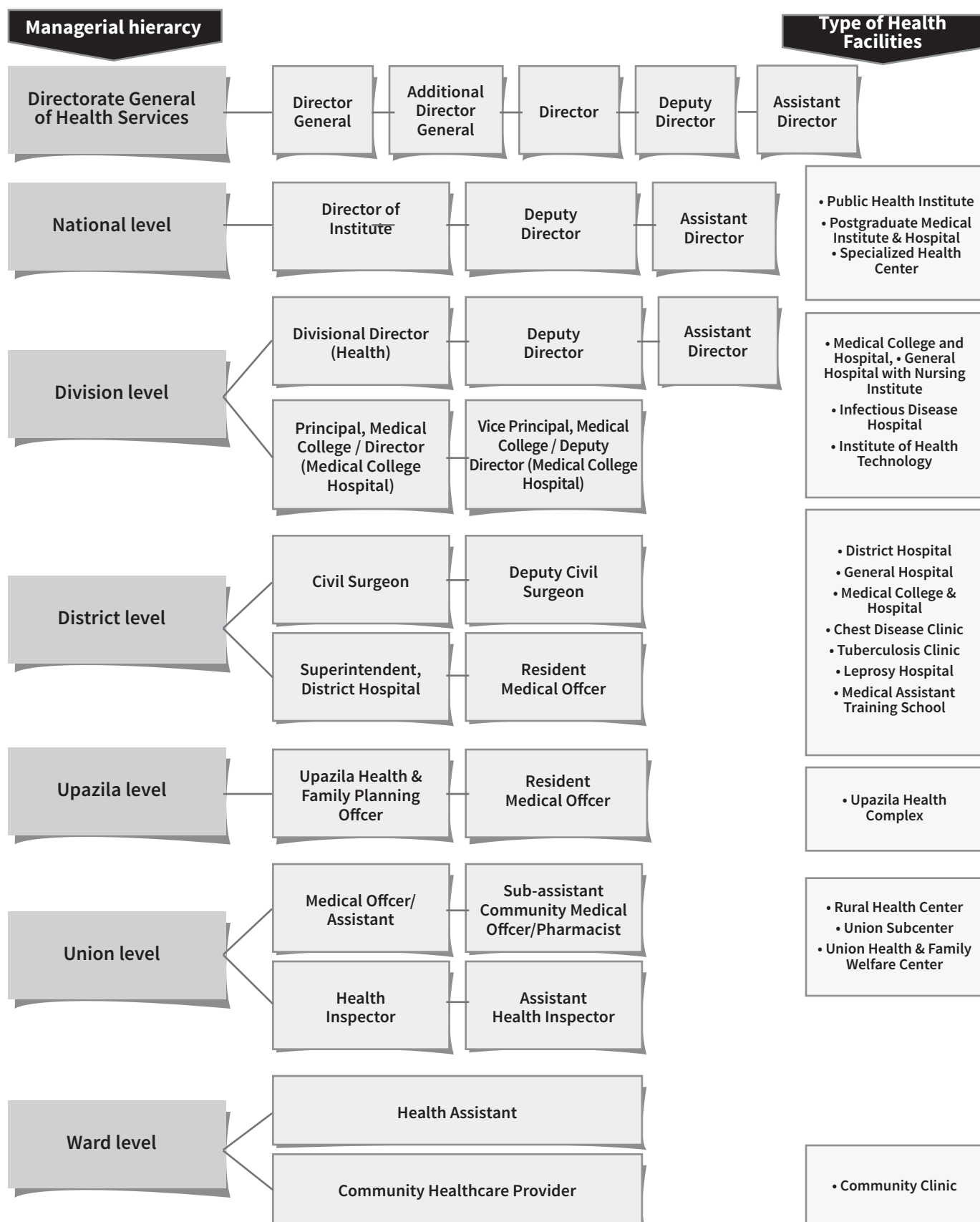


Figure 2.7. Types of facilities from national to the ward level, with managerial hierarchy

ORGANOGRAM OF CIVIL SURGEON OFFICE:

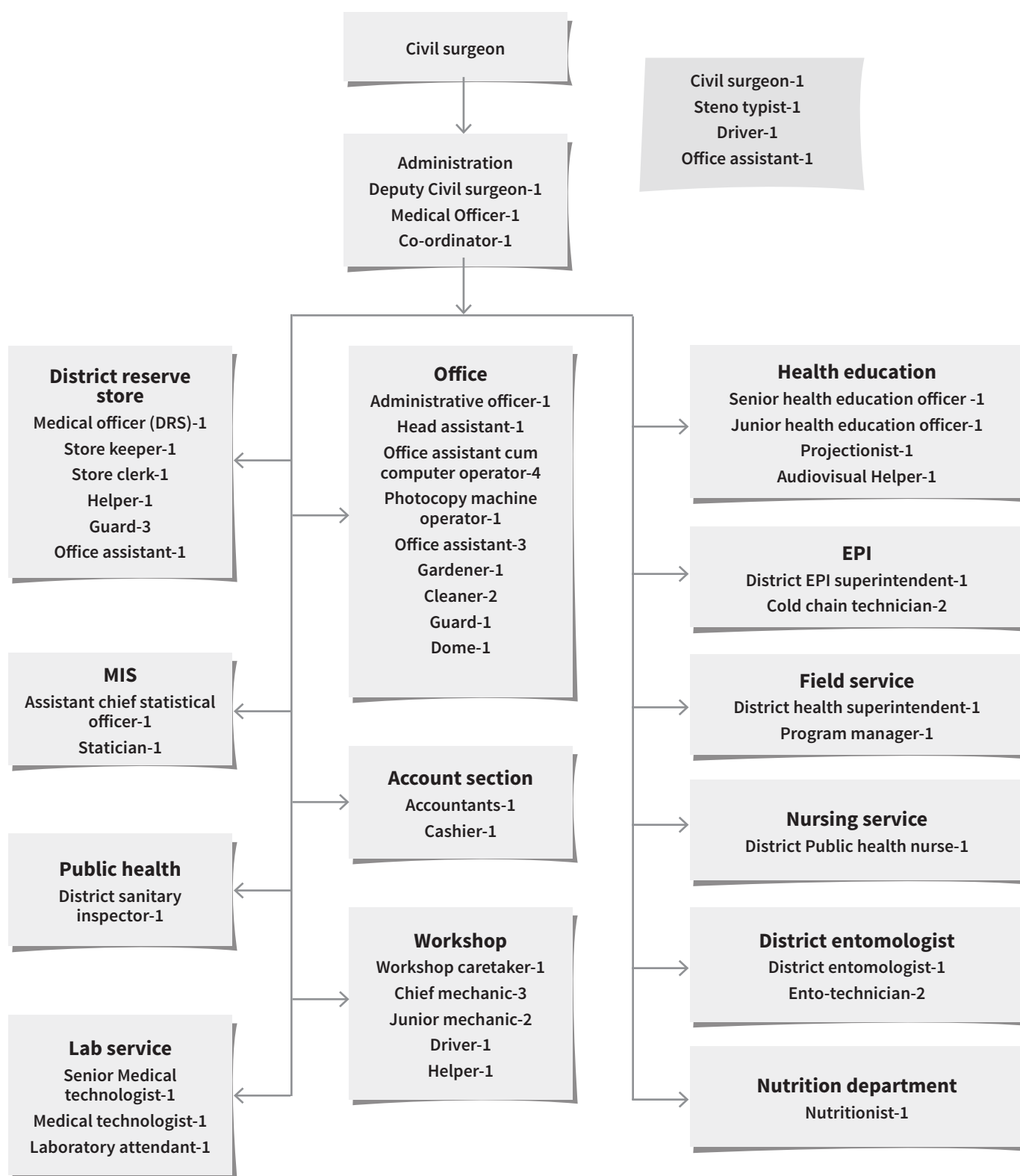


Figure 2.8. Organogram of Civil Surgeon Office (ref- MIS,DGHS)

ORGANOGRAM OF 250 BED HOSPITAL

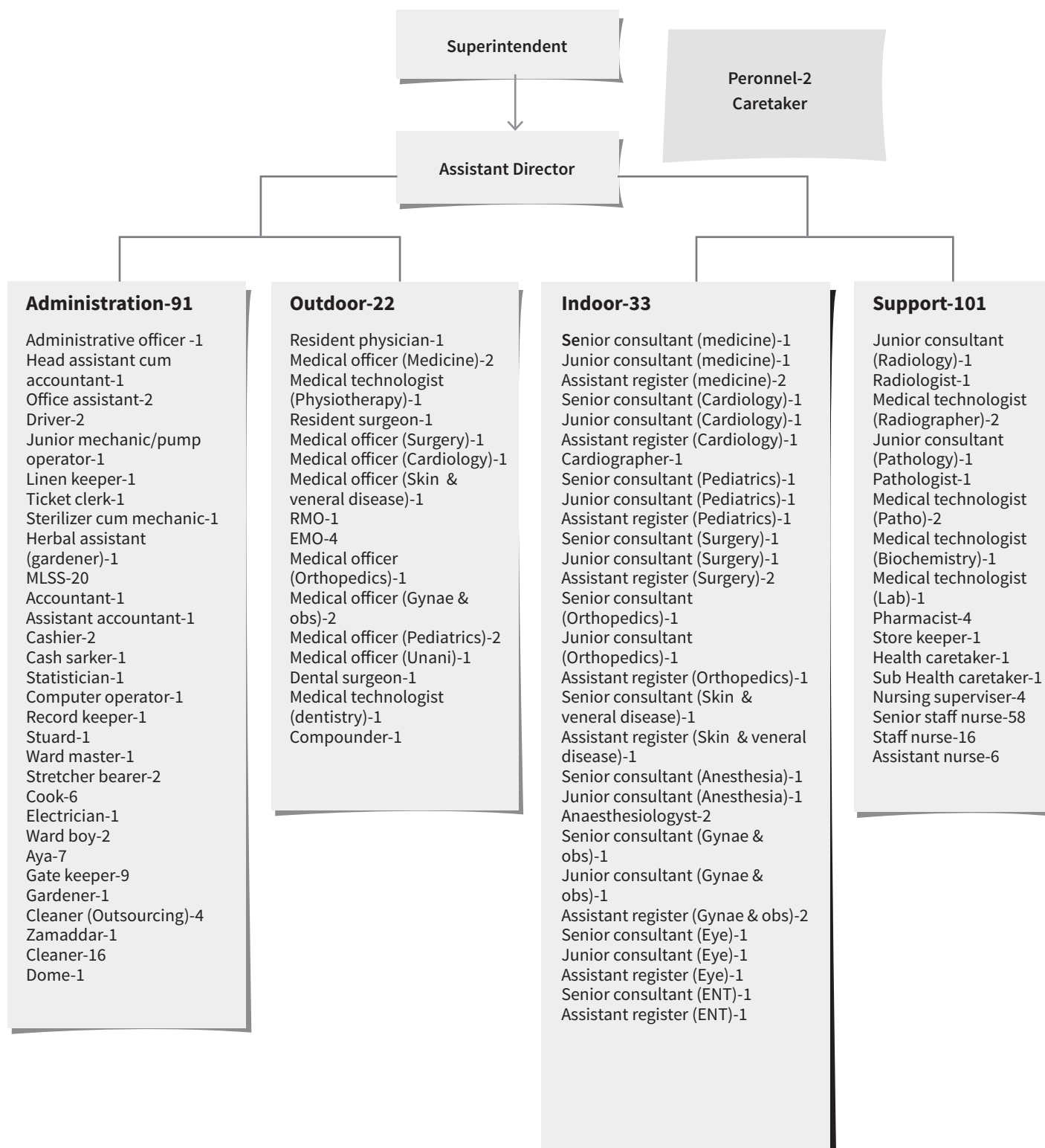


Figure 2.9. Organogram of 250 bedded hospital (ref- MIS,DGHS)

ORGANOGRAM OF 100 BED HOSPITAL

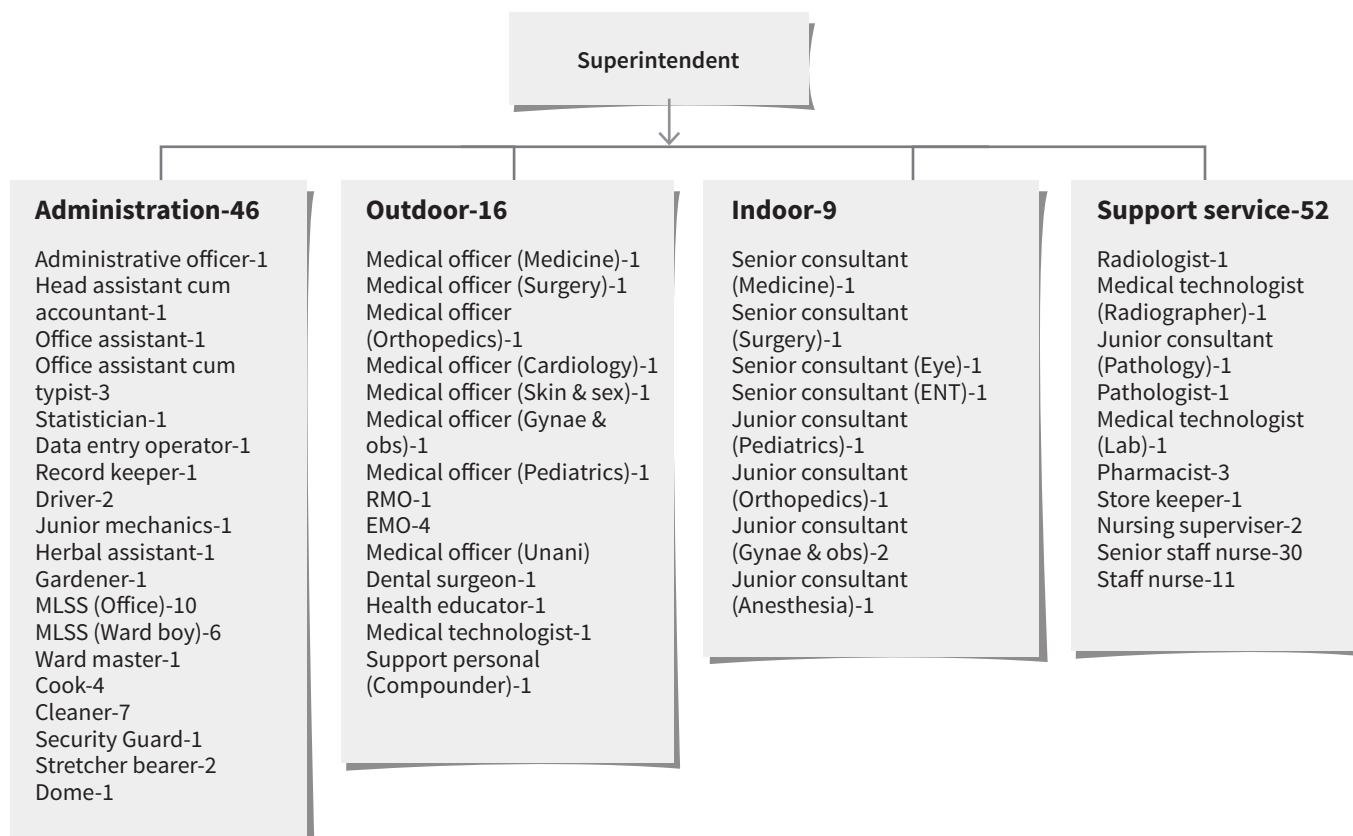


Figure 2.10. Organogram of 100 beded hospital (ref- MIS,DGHS)

ORGANOGRAM OF 50 BED HOSPITAL

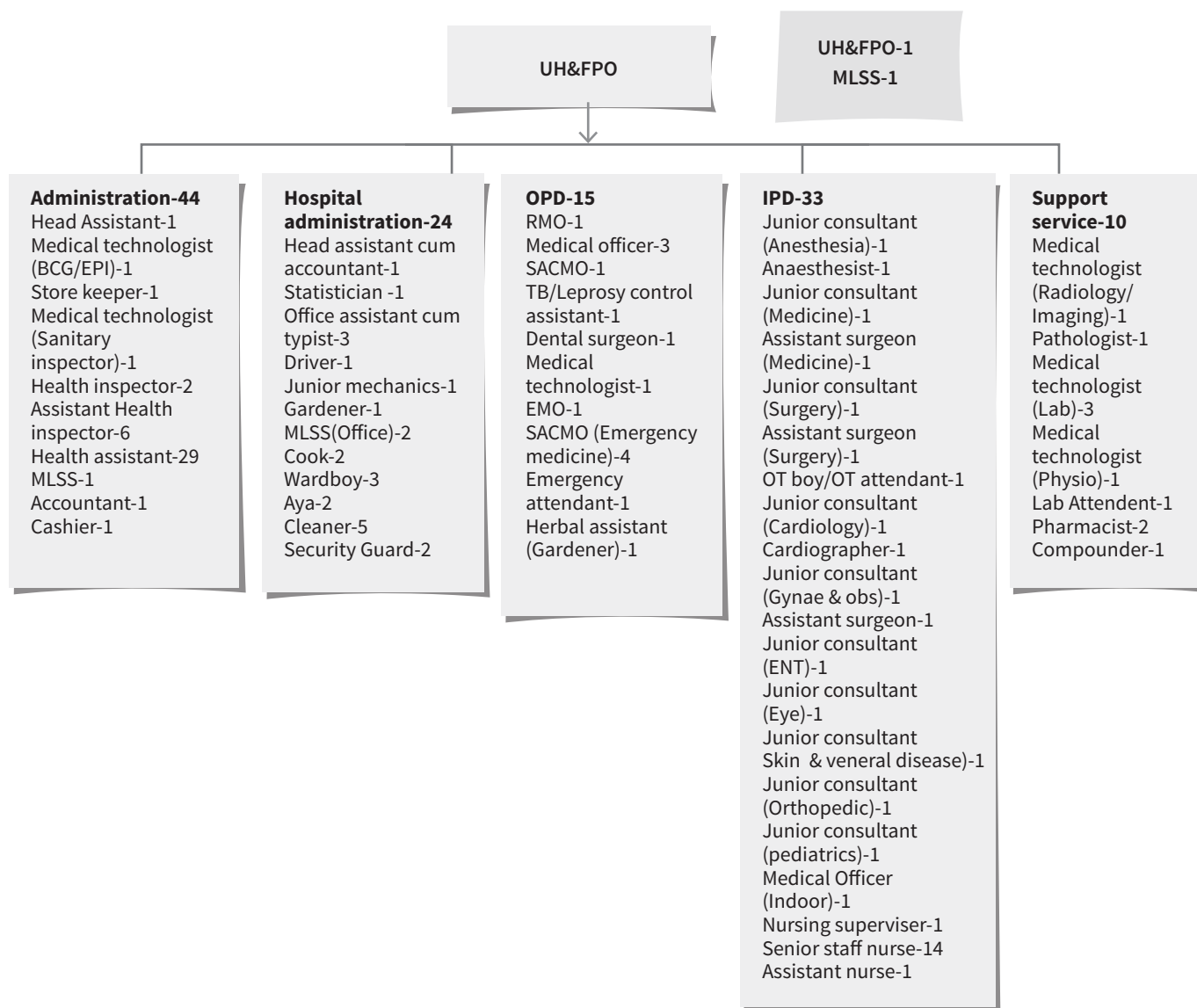


Figure 2.11. Organogram of 50 beded hospital (ref- MIS,DGHS)

ORGANOGRAM OF 31 BED HOSPITAL (Upazila Health Complex)

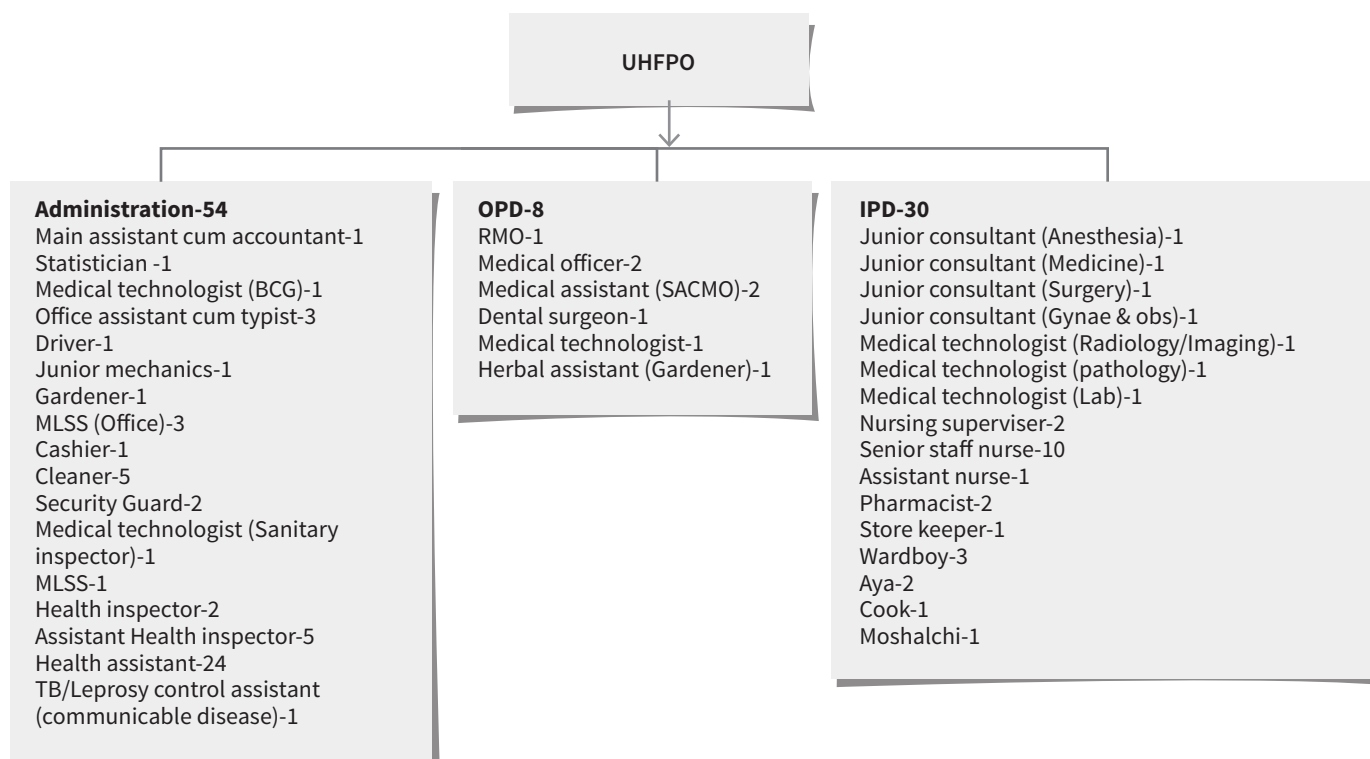


Figure 2.12. Organogram of 31 beded hospital (ref- MIS,DGHS)

ORGANOGRAM OF 10 BED HOSPITAL

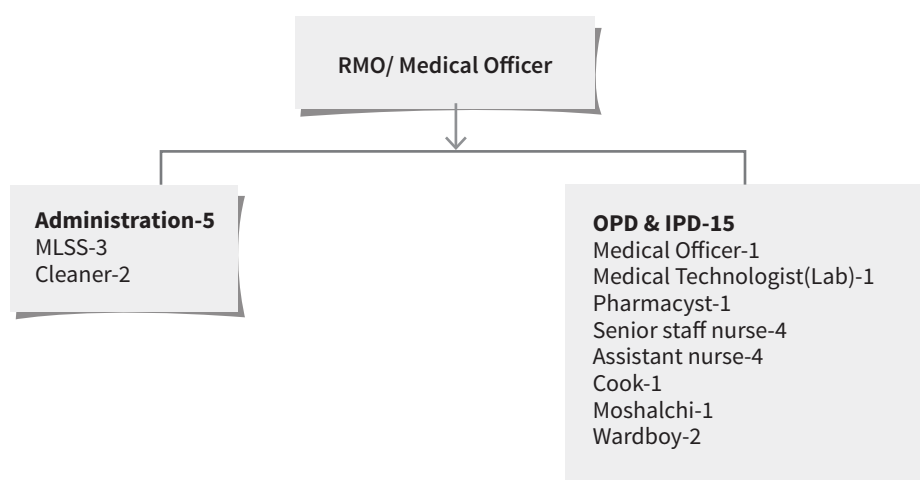


Figure 2.13. Organogram of 10 beded hospital (ref- MIS,DGHS)

HEALTH RIGHTS, ACTS, POLICIES, STRATEGIES, PLANS AND SDG

Progress towards achieving SDG

BACKGROUND

Health is one of the basic human rights. Since the emergence of Bangladesh as an independent country, it has been given utmost priority to ensure human rights and dignity of the population. The Constitution of the People's Republic of Bangladesh guarantees that 'Health is the basic right of every citizen of the Republic'. Article 15 of the constitution says, "It shall be a fundamental responsibility of the state to attain, through planned economic growth, a constant increase of productive forces and a steady improvement in the material and cultural standard of living of the people".

Accordingly, the country's first Five Year Plan (1FYP) was chalked out for the period 1973 - 1978, which was started by the very first President of the country Bangabandhu Sheikh Mujibur Rahman. In that plan, health was a very important sector. Bangladesh is currently implementing its Seventh Five Year Plan (7FYP) for the period 2016-2020. To ensure its progress, Government of Bangladesh adopted a number of policies, acts, rules etc. Recently a number of acts (relating to communicable diseases, mental health, organ transplant, community clinic etc) have either been passed or amended to strengthen the capacity and quality of the health service of the country.

Since the late 1990s, the Government of Bangladesh (GoB) and its Development Partners (DP) have pursued a sector-wide approach (SWAp) in the HNP sector. The Ministry of Health and Family Welfare (MoHFW) has implemented three consecutive Sector Programs in between July 1998 and December 2016. The Ministry of Health and Family Welfare (MoHFW) of the Government of Bangladesh carefully designed the 4th sector-wide approach (SWAp), namely 4th Health, Population and Nutrition Sector Program (4th HPNSP) to initiate its journey towards achieving the health-related SDGs.

The 4th HPNSP's articulation and design have been linked to the 7th Five-year Plan (FYP) of the Government. Other guiding principles are drawn from the national policies on health, nutrition and population (HNP), various HNP sector-related strategies approved by the Government, and the experience

of implementing three successive sector programs in Bangladesh. The 4th HPNSP starts at a time of transition from the Millennium Development Goals (MDGs) to the newly-agreed Sustainable Development Goals (SDGs) to be achieved by 2030. The MDGs had a major focus on specific goals relating to health sector while the SDGs encompass much broader agenda for change. SDGs have been taken into consideration in the 4th sector plan so that the global development agenda can be illustrated in the national plan.

Since August 2017, from Myanmar, about 700,000 forcefully displaced people have crossed into Bangladesh. They are placing an immense strain on an already resource-constrained social service delivery system, including health services. Thus, a fourth component,

beside the original three components, is added with additional financing to enhance the capacity of MoHFW to respond to the crisis. This supports it in extending HNP services to the FDMNs (Forcibly Displaced Myanmar Nationals) in the Cox’s Bazar district.

HEALTH IN THE CONSTITUTION OF BANGLADESH

Bangladesh began its journey as an independent nation in 1971, and the first Constitution was established in 1972. Since the first Constitution, health has historically been a state issue. Under the current Constitution, medical care is included as one of the basic necessities of the citizenry, and it is the fundamental responsibility of the state to ensure health services.

Table 3.1. Health-related statements written in the Constitution of the People’s Republic of Bangladesh

Provision of basic necessities	15. It shall be a fundamental responsibility of the State to attain, through planned economic growth, a constant increase of productive forces and a steady improvement in the material and cultural standard of living of the people, with a view to securing its citizens – (a) the provision of the basic necessities of life, including food, clothing, shelter, education and medical care; (b) the right to work, that is the right to guaranteed employment at a reasonable wage having regard to the quantity and quality of work; (c) the right to reasonable rest, recreation and leisure; and (d) the right to social security, that is to say, to public assistance in cases of undeserved want arising from unemployment, illness or disablement, or suffered by widows or orphans or in old age, or in other such cases.
Rural development and agricultural revolution	16. The State shall adopt effective measures to bring about a radical transformation in the rural areas through the promotion of an agricultural revolution, the provision of rural electrification, the development of cottage and other industries, and the improvement of education, communications and public health, in those areas, so as progressively to remove the disparity in the standards of living between the urban and the rural areas.
Public health and morality	18. (1) The State shall regard the raising of the level of nutrition and the improvement of public health as among its primary duties, and in particular shall adopt effective measures to prevent the consumption, except for medical purposes or for such other purposes as may be prescribed by law, of alcoholic and other intoxicating drinks and of drugs which are injurious to health. (2) The State shall adopt effective measures to prevent prostitution and gambling.

IMPORTANT HEALTH SECTOR POLICIES/ STRATEGIES

- National Health Policy 2011
- Health, Nutrition and Population sector of the 7th Five Year Plan
- National Nutrition Policy 2015
- 4th Health, Population and Nutrition Sector Program (HPNSP) 2017-2022
- Healthcare Financing Strategy 2012-2032: Expanding Social Protection for Health Towards Universal Coverage
- Bangladesh Health Workforce Strategy 2016–2021

NATIONAL HEALTH POLICY 2011

Goal and objectives

To overcome the existing challenges and to create a regulatory framework, the National Health Policy 2011 was implemented. It aims at the following principles:

First: Make necessary basic medical utilities reach people of all upazilas and develop the health and nutrition status of the people.

Second: Develop a proper system to ensure easy and sustained availability of health services.

Third: Ensure optimum quality, acceptance, and availability of primary healthcare and governmental medical services.

Fourth: Reduce the intensity of malnutrition among people and implement effective and integrated programs for improving nutrition status.

Fifth: Undertake programs for reducing the rates of child and maternal mortality.

Sixth: Adopt satisfactory measures for ensuring improved maternal and child health and install facilities for safe and hygienic child delivery.

Seventh: Improve overall reproductive health resources and services.

Eighth: Ensure the presence of full-time doctors, nurses, and other officers/staff, provide and maintain necessary equipment and supplies at each of the upazila health complexes and union health and family welfare centers

(UHFWCs)

Ninth: Devise necessary ways and means to make optimum usage of available opportunities in government hospitals and the health service system.

Tenth: Formulate specific policies for medical colleges and private clinics and to introduce laws and regulations.

Eleventh: Strengthen and expedite the family planning program with the objective of attaining the target of replacement-level of fertility

Twelfth: Explore ways to make the family planning program more acceptable, easily available, and effective among the extremely poor and low-income communities.

Thirteenth: Arrange special health services for the mentally-retarded, the physically-disabled and elderly population.

Fourteenth: Determine ways to make family planning and health management more accountable and cost-effective by equipping it with more skilled manpower.

Fifteenth: Introduce systems for treatment of all types of complicated diseases in the country and minimize the need for foreign travel for medical treatment abroad.

7th FIVE YEAR PLAN (7FYP): 2016 – 2020

Goals, Strategies, and Targets of the 7th Five Year Plan in the Health, Nutrition and Population Sector

Bangladesh will strive to attain a number of targets on the road towards universal health coverage, consistent with the proposed major targets, under the proposed Sustainable Development Goals (SDG) framework of the UN, promoting and sustaining health and nutrition, along with containing population growth as priorities in the human development strategy. Vision 2021 aims at establishing a middle-income Bangladesh, with drastic reduction in poverty and developing conditions that allow individuals to reach and maintain the highest attainable level of health. To realize that vision, the Government has set certain objectives and targets towards achieving Universal Health Coverage (UHC) in the HNP sector. This includes the following:

- Ensure access to and utilization of HNP services for every citizen of the country, with particular emphasis on the elderly, women, children, poor disadvantaged

people, and those living in difficult areas.

- Reduce total fertility rate
- Ensure adolescent and reproductive healthcare
- Strengthen community support and involvement to obtain better results in the implementation of programs
- Improve nutritional status of children and women
- Take effective measures to promote alternative medicines and improve the quality of care
- Meet challenges of emerging, re-emerging, and non-communicable diseases, health hazards due to climate change, and emergency response to catastrophes
- Enhance national capacity for the pre-service education (SBA/nursing, paramedics, midwifery), provide in-service training and better management of human resources
- Improve the quality of hospitals and maternity services and to make these accessible, especially to the women, children, and the poor

Bangladesh has already made great strides towards achieving these objectives. Past and ongoing programs have been promoted, and they are still being promoted for ensuring good health and nutrition. It also concentrates on reducing population growth. In

order to prosecute this progress and to address areas of insufficiency, some of the specific targets for the HNP sector under the 7th FYP are shown in Table 3.2.

These targets can be viewed as ambitious, however these are certainly achievable with timely actions. Bangladesh is now trying to implement these associated policies and institutional reforms. Bangladesh is currently implementing the 4th SWAP in the HNP sector which aligns with the 7th FYP targets for achieving UHC and SDGs by 2030.

4th HEALTH, POPULATION AND NUTRITION SECTOR PROGRAM (4th HPNSP) 2017-2022

The MoHFW is currently implementing its fourth Sector Program titled '4th Health, Population and Nutrition Sector Program (4th HPNSP)' covering a 5.5-year period between January 2017 and June 2022, at an estimated cost of US\$ 14.7 billion.

The 4th HPNSP is guided by Bangladesh's Vision 2021 (transforming the country from a developing country into a middle-income country), which acknowledges that improved health is a necessary and critical condition for the achievement of this vision.

Milestones for Vision 2021 in HNP Sector

- 85% of the population has standard nutritional food

Table 3.2. Targets for the 7th FYP in Health, Nutrition and Population sector

Sl. No.	Indicator	Base year's information (source with year)	7th FYP target FY2020
Impact/Outcome			
1.	Life-expectancy at birth	72.22 (World Bank 2015)	72
2.	Total fertility rate (children per woman)	2.13 (World Bank 2015)	2.0
3.	Under-5 mortality rate (per 1,000 live births)	46 (BDHS 2014)	37
4.	Infant mortality rate (per 1,000 live births)	38 (BDHS 2014)	20
5.	Maternal mortality ratio (per 100,000 live births)	176 (World Bank 2015)	105
6.	Proportion of underweight among under-five children (%)	32.6 (BDHS 2014)	20
7.	Proportion of stunting among under-five children (%)	36.19 (BDHS 2014)	25

Sl. No.	Indicator	Base year's information (source with year)	7th FYP target FY2020
Output			
8.	Proportion of births attended medically- trained care providers (%)	42.1 (BDHS 2014)	65
9.	Contraceptive prevalence rate (%)	62.4 (BDHS 2014)	75
10.	Proportion of children fully-vaccinated by 12 months (%)	78 (BDHS 2014)	75
11.	Proportion of births in health facilities by wealth quintile (ratio of the lowest and the highest quintile)	15: 69.5 (BDHS 2014)	1: 3.5
12.	TB case detection rate (%)	53 (GTR 2014)	75

Source: Ministry of Health and Family Welfare

- Poor people ensure a minimum of 2100 kilo calories of food
- All kinds of contagious diseases are eliminated
- Longevity increases to 70 years
- Infant mortality comes down to 15 from 54 per thousand at present
- Use of birth control methods increases to 80%

4th HPNSP: Strategy and Objectives

Main objectives

The 4th HPNSP is built upon the existing achievements to improve equity, quality, and efficiency with a view to moving gradually towards UHC and achieving health-related Sustainable Development Goals (SDGs) by pursuing the following strategic objectives (SOs), during the next five-and-a half year, starting from January 2017:

- SO 1:** Strengthen governance and stewardship of the public and private health sectors
- SO 2:** Undertake institutional development for improved performance at all levels of the system
- SO 3:** Provide sustainable financing for equitable access to healthcare for the population and accelerated progress towards universal health coverage
- SO 4:** Strengthen the capacity of the MOHFW's core health systems (financial management, procurement, infrastructure development)

SO 5: Establish a high-quality health workforce available to all public and private health service providers

SO 6: Improve health measurement and accountability mechanisms and build a robust evidence-base for decision-making

SO 7: Improve equitable access to and utilization of quality health, nutrition and family planning services

SO 8: Promote healthy lifestyle choices and a healthy environment

Vision, Mission and Goal of the Program

The PIP is based on the vision, mission, goal, development objectives, strategic objectives, and resource envelope as mentioned in the Strategic Investment Plan (SIP) of MOHFW developed in mid-2016.

Vision

“To see the people healthier, happier and economically productive to make Bangladesh a middle income country by 2021” (Vision 2021)

Mission

“To create conditions whereby the people of Bangladesh have the opportunity to reach and

maintain the highest attainable level of health”

Goal

“To ensure that all citizens of Bangladesh enjoy health and wellbeing by expanding access to quality and equitable healthcare in a healthy environment”

Components

The eight strategic objectives of the 4th HPNSP have been categorized under three main components. An additional (fourth) component is added with additional financing to provide health services delivery to the FDMNs and Host Communities in Cox’s Bazar District under Chittagong Division. The 4th component will also be implemented by the MoHFW. The MoHFW will continue to deliver humanitarian aid directly and through the agencies of the United Nations (UN) and local/international NGOs for supply of food, HNP services, sanitation, water and other essential services to these FDMNs.

The components of 4th HPNSP are

1. Governance and Stewardship
2. HNP System’s Strengthening
3. Provision of Quality HNP Services and
4. Development of HNP Services for FDMNs in Cox’s Bazar District

Like the previous three sector programs, it is expected that a significant proportion of DPs support will be channeled through on-budget financing. Since October 2017, the World Bank is implementing its Health Sector Support Project (HSSP), having three components through which the Bank is providing support to the 4th HPNSP. The original HSSP is consistent with the GOB’s program and policies and will play an important role in advancing key result areas with the use of disbursement-linked indicators (DLIs). The original HSSP will use a set of 16 DLIs in responding to the key HNP challenges. RFW (Results Framework) goal-level indicators for the 4th HPNSP are showed in the annex section.

In addition, there are process/output indicators at the OP level covering all the OPs for monitoring OP-wise progress of work on a six-monthly/annual basis. A number of tools will also be used to document and track outputs

and indicators, for regular monitoring of the Program, as illustrated in annexure.

For the implementation of 4th HPNSP, there are now 29 OPs that are distributed across the DGHS (14), DGFP (7), MOHFW (5), and other agencies (one for each of the DGDA, DGNM and NIPORT’s OPs), each of which reflects priority areas of the 4th HPNSP. Each OP is led by a Line Director (LD) and has a functional administrative structure, including staff, budget, and infrastructure.

SUSTAINABLE DEVELOPMENT GOALS

PROGRESS IN 2017: DGHS IN ACHIEVING SDG 3

Bangladesh is currently implementing its Seventh Five Year Plan (7FYP) for the period of 2016-2020. The Ministry of Health and Family Welfare (MoHFW) of the Government of Bangladesh carefully designed the 4th sector-wide approach (SWAp), namely 4th Health, Population and Nutrition Sector Programme (4th HPNSP) -2017/2021 to initiate its journey toward achieving the health-related SDGs. SDGs has taken into consideration in 4th sector plan so that the global development agenda can be illustrated in the national plan. The goal of the 4th sector plan is to achieve sustainable improvement in health, nutrition, and reproductive health, including family planning for all particularly of vulnerable groups – women, children, the elderly and the poor. Conditions will be created in the HNP sector whereby the people of Bangladesh have the opportunity to reach and maintain the highest attainable level of health.

Bangladesh is moving ahead successfully with the action plan to implement SDGs on time. The Perspective Plan stipulates necessary actions to increase coverage of all types of health care and family planning services, to strengthen health administration and to increase the number of skilled professionals in the Health, nutrition and population sector.

Health related SDG remains in the core of SDGs as to ensure quality of life of every citizen throughout life course. MOHFW of Bangladesh is well aware of the fact and is engaged under the leadership honorable minister, honorable state minister and senior officials to pursue achievement of all health related SDGs. So it is believed that Bangladesh as per global commitment of Honorable

Prime Minister Sheikh Hasina will be able to achieve all SDGs targets including the health related SDGs. The SDGs give priority to measurement and accountability, using reliable data generated through the country-led process. The implementing authorities will have to address these requirements. The DGHS has an extensive healthcare network with a proven track-record of accomplishing its tasks. It would use the existing infrastructure and resources in the best possible way to achieve the health-related SDGs.

GOAL 3: HEALTH SDGS

SDG comprises 17 goals, 169 targets including 230 indicators.

Goal 3 of SDGs has been particularly focused on health. The Goal states

“To ensure healthy lives and promote well-being for all at all ages.”

Health sector has the responsibility and stake in all the other 16 goals to apply the principles of SDGs in its own settings as well as to complement achievement of the related goals by other sectors.

Under Goal 3: 13 targets & 25 Indicators

DGHS under ministry of Health has taken number of initiatives to achieve SDG health related goals,

Strengthening of health information system is being done through the use of IT. Deployment of world's DHIS 2 software has been done and telemedicine services has been introduced. DGHS now operates a nationwide call centre 'Shaystho Batayon' for providing health services through mobile network. DGHS also uses routine video conferences to review the health system performance and progress of health related SDG.

To improve maternal health, number of 24 hours emergency obstetric care services has been increased in significant numbers. Electronical registration and tracking of pregnant women to ensure effective maternal health services has also been introduced in this 4th sector plan. Coverage of Antenatal Care (ANC) services for all pregnant women is increasing including TT immunization. Number

of trained Community Skilled Birth Attendant (CSBAs) is also increasing.

The MOHFW strengthened national immunization program to enhance child vaccination coverage. Under 5 mortality rate in 2017 is 31 per 1000 live birth (SVRS 2017) which is 5% less than 2015. Neonatal mortality rate is 17 (SVRS 2017). 4th HNPSP is also strongly addressing NCD prevention programs through raising awareness and mass mobilization, screening, diagnosis and treatment.

Universal Health Coverage is at the center of the health goal. MoHFW is piloting project to implement UHC by 2030 in Bangladesh.

HEALTH INDICATORS IN SDG

According to WHO, 4 main areas of health SDG indicators are

1. Reproductive, maternal, newborn and child health
2. Infectious diseases
3. Non communicable diseases
4. Service capacity, access and health security

BANGLADESH SDG HEALTH DATA GAPS

The monitoring of the health and health-related targets is fairly robust. There are still major data gaps for most health indicators that need to be addressed in order to improve the ability to track global progress.

SDGs data gap analysis was done with the assistance of all data generating agencies including the National Statistical Organization (NSO) of Bangladesh viz. Bangladesh Bureau of Statistics under GED division. The mapping (who to do what in terms of targets) has given responsibility to 40 Ministries/Divisions as lead along with 34 Ministries/Divisions as Co-Lead, and 61 Ministries/Divisions/Independent Institutions have been earmarked as associate in the implementation of 169 targets of 17 SDGs. The mapping also highlighted actions to be taken by leads and co-leads to achieve the SDGs targets during 7th FYP.

Few Challenges to monitor SDG health indicators:

- Data quality and completeness
- Coordination among BBS and other govt office,

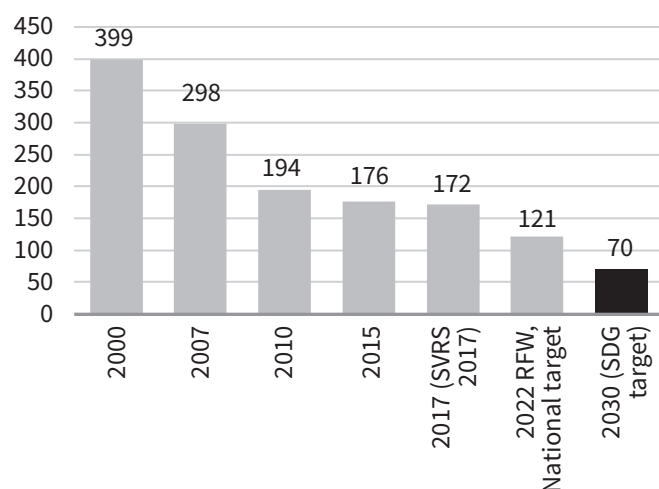


Figure 3.1. Maternal mortality status (SDG 3.1)

- especially for health, DGHS, DGFP and other health organizations.
- Private sector for addressing the methodological gap at this moment in real time data
 - Harmonized and aligned support for strengthening analytical capacity of SDG health indicators
 - Completeness and data quality of routine HMIS system to measure health SDG indicators

Next Initiatives to way forward to measure SDG health indicators at DGHS

- Formation of core group with focal points at DGHS
- Better reporting of (sub)national and global progress on SDG
- Strengthening statistical analytical capacity in the measurement of indicators
- Addressing health data gaps according to GED

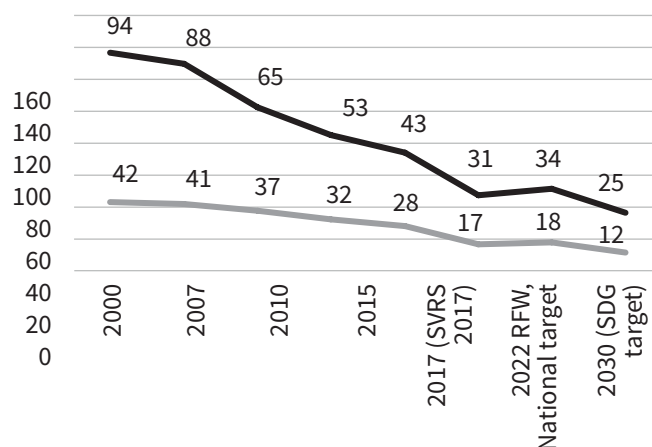


Figure 3.2. Child mortality status (SDG 3.2)

- Improving facility and community data systems
- Enhancing the use and accountability
- Leveraging existing collaborations through technical Working Groups and regional networks where possible

DGHS trying to design more analytic dashboard to provide better insight, focusing on the needs of the expert audiences (ex. health ministry executives/analysts, public health NGOs, academics). Analytic dashboards will suggest data-driven conclusions that expert audiences can interpret and use to determine appropriate responses to achieve SDG health targets.

Bangladesh achieved most of the MDG including health related ones due to strong commitment of the government. Health related SDGs remains in the core of SDGs as they relate to the health wellbeing and the quality of life of every citizen throughout their life course. As per global commitment, hopefully government will be

Table 3.3. 25 SDG 3 indicators plus other relevant health indicators are presented where data is available.

Indicators	SDG target	Baseline data (Year)	2017 Data, DGHS Bangladesh	Milestone by 2020	Target by 2030
Maternal mortality ratio (per 100,000 live births)	3.1.1	181 (SVRS, 2015)	172 (SVRS, 2017)	105	70

Indicators	SDG target	Baseline data (Year)	2017 Data, DGHS Bangladesh	Milestone by 2020	Target by 2030
Proportion of births attended by skilled health personnel	3.1.2	42.1% (BDHS, 2014)	72.3% SVRS, 2017	65%	80%
Under-five mortality rate (per 1,000 live births)	3.2.1	36 (SVRS, 2015)	31 (SVRS, 2017)	34	25
Neonatal mortality rate (per 1,000 live births)	3.2.2	20 (SVRS-2015)	17 (SVRS, 2017)	19	12
Number of new HIV infections per 1,000 uninfected population, by sex, age and key populations	3.3.1	0.04 (Women 15-49 years: <0.1%, Men 15-49 years: <0.1%) (UNAIDS, 2016)	New case 865 Male-639 Female-213 Hijra/Transgender-13	0.03	0.01
Tuberculosis incidence per 100,000 population	3.3.2	287 (NTP, 2016)	221 (Global TB report 2018)	250	150
Malaria incidence per 1,000 population	3.3.3	4.3 (MCP, 2015)	1.64 (From OP)	3	0
Hepatitis B incidence per 100,000 population	3.3.4	-	546 (From OP)	-	-
Number of people requiring interventions against neglected tropical diseases	3.3.5	49,873,889 (WHO, 2016)	47,484,224 (Health SDG Profile: Bangladesh, 2018, WHO)	45,000,000	35,000,000
Mortality rate attributed to (between 30 and 70 years of age from) cardiovascular disease, cancer, diabetes or chronic respiratory disease (%)	3.4.1	18% (WHO, 2016)	Total person 4394 (From OP -% validation is not yet done by NIPSOM)	15%	6%
Suicide mortality rate (per 100,000 population)	3.4.2	7.1 (PSD, 2015)	7.8	5.5	2.4
Coverage of treatment interventions (pharmacological, psychosocial and rehabilitation and aftercare services) for substance use disorders	3.5.1	-	22.212(2016) 16.386(2015) 16.852(2014)	-	-
Death rate due to road traffic injuries (per 100,000 population)	3.6.1	2.49 (PSD, 2015)	347 person (From 8 Division)	2.0	1.2
Proportion of women of reproductive age (aged 15-49 years) who have their need for family planning satisfied with modern methods (%)	3.7.1	72.6% (BDHS, 2014)	72.6% (BDHS, 2014)	75%	100%
Adolescent birth rate (aged 10-14 years; aged 15-19 years) per 1,000 women in that age group	3.7.2	75 (SVRS-2015)	75 (SVRS, 2017)	70	50
Coverage of essential health services	3.8.1	52 (WHO-2016)	50 (Health SDG Profile: Bangladesh, 2018, WHO)	65	100

Indicators	SDG target	Baseline data (Year)	2017 Data, DGHS Bangladesh	Milestone by 2020	Target by 2030
Proportion of population with large household's expenditures on health as a share of total household expenditure or income	3.8.2	-	13.9% (10% threshold) (Health SDG Profile: Bangladesh, 2018, WHO)	-	-
Proportion of the target population covered by all vaccines included in their national programme	3.b.1	78% (BDHS, 2014)		95%	100%
Total net official development assistance to medical research and basic health sectors	3.b.2	177.4 MUS\$ (ERD, FY15)	-	300 MUS\$	500 MUS\$
Proportion of health facilities that have a core set of relevant essential medicines available and affordable on a sustainable basis	3.b.3	-	-		
Health worker density (per 10,000 population)	3.c.1	(a) 7.4 (WHO, 2016) (b) 1: 0.5: 0.2 (HRH Data Sheet, 2014 MoHFW)	“(a) 8.3 (HRD unit, HRH Country profile, 2017, MOHFW) (b) Physician 1: Nurse 0.56: Health Technologist 0.40 (HRD unit, HRH Country profile, 2017, MOHFW)”	(a) 18.9 (b) 1: 1.3: 1.8	(a) 44.5 (b) 1: 3: 5
International Health Regulations (IHR) capacity and health emergency preparedness	3.d.1	87.5 (WHO, 2016)	-	90	100
Mortality rate attributed to household and ambient air pollution (per 100,000 population)	3.9.1	68.2 (WHO, 2012)	-	65	55
Mortality rate attributed to unsafe water, unsafe sanitation and lack of hygiene (per 100,000 population)	3.9.2	5.96 (WHO 2016)	-	5.5	4.5
Mortality rate attributed to unintentional poisoning (per 100,000 population)	3.9.3	5.72 (WHO 2016)	-	5	4.0

Table 3.4. Other Health related SDGs.

Indicators	SDG target	Baseline data (Year)	2017 Data, DGHS Bangladesh	Milestone by 2020	Target by 2030	Remark
Prevalence of stunting among children under 5 years of age	2.2.1	36.1% (BDHS, 2014)	36.1% (BDHS, 2014)	25%	12%	
Prevalence of malnutrition (weight for height) among children under 5 years of age, by type (wasting and overweight)	2.2.2	a) Wasting: 14.3% (BDHS, 2014) b) Overweight: 1.6% (MICS, 2012-13)	a) Wasting: 14.3% (BDHS, 2014) b) Overweight: 1.6% (MICS, 2012-13)	a) 12% b) 1.5%	a) <5% b) 1.0%	a) <5% b) 1.0%
Proportion of women aged 20-24 years who were married or in a union before age 15 and before age 18	5.3.1	Before 15: 23.8% (MICS, 2012-13) Before 18: 58.6% (BDHS, 2014)	-	Before 15: 15% Before 18: 30%	Before 15: 0% Before 18: 10%	
Proportion of women aged 15-49 years who make their own informed decisions regarding sexual relations, contraceptive use and reproductive health care	5.6.1	a) SR: 14% (b) CU: 67% (c) ANC: 65% (BDHS, 2014)	-	a) 20% b) 70% c) 70%	a) 30% b) 80% c) 90%	
Proportion of population with primary reliance on clean fuels and technology	7.1.2	20.8% (SVRS, 2015)	-	25%	35%	
Number of deaths, missing persons and directly affected persons attributed to disasters per 100,000 population	11.5.1	Affected Persons: 12,881 per 100,000 people in 2014 (ICCHL, BBS, 2015)	No data is available	6,500	1,500	
Number of deaths, missing persons and directly affected persons attributed to disasters per 100,000 population	13.1.2	Affected Persons: 12,881 per 100,000 people in 2014 (ICCHL, BBS, 2015)	No data is available	6,500	1,500	
Birth registration coverage	16.9.1		-			
Completeness of cause of death data	17.19.2		-			

Note: (-) implies relevant data are not available

able to achieve all SDG targets including the health related SDGs successfully.

MINISTRY OF HEALTH AND FAMILY WELFARE AND SUCCESS STORIES IN GOVERNANCE

The Ministry of Health and Family Welfare seeks to create conditions whereby the people of Bangladesh have the opportunity to reach and maintain the highest attainable level of health. It is a vision that recognizes health as a fundamental human right and, therefore, the need to promote health and reduce suffering is the spirit of social justice. It performs the following:

Health SDG monitoring activities at DGHS

- Online real time SDG health indicator tracker has been developed and incorporated with central dashboard at the DGHS website
- SDG health indicators information is being used in the national SDG reporting as well as global reporting
- Coordination with Prime minister office is being done to provide SDG health indicator information in the national SDG tracker (16 health indicators)

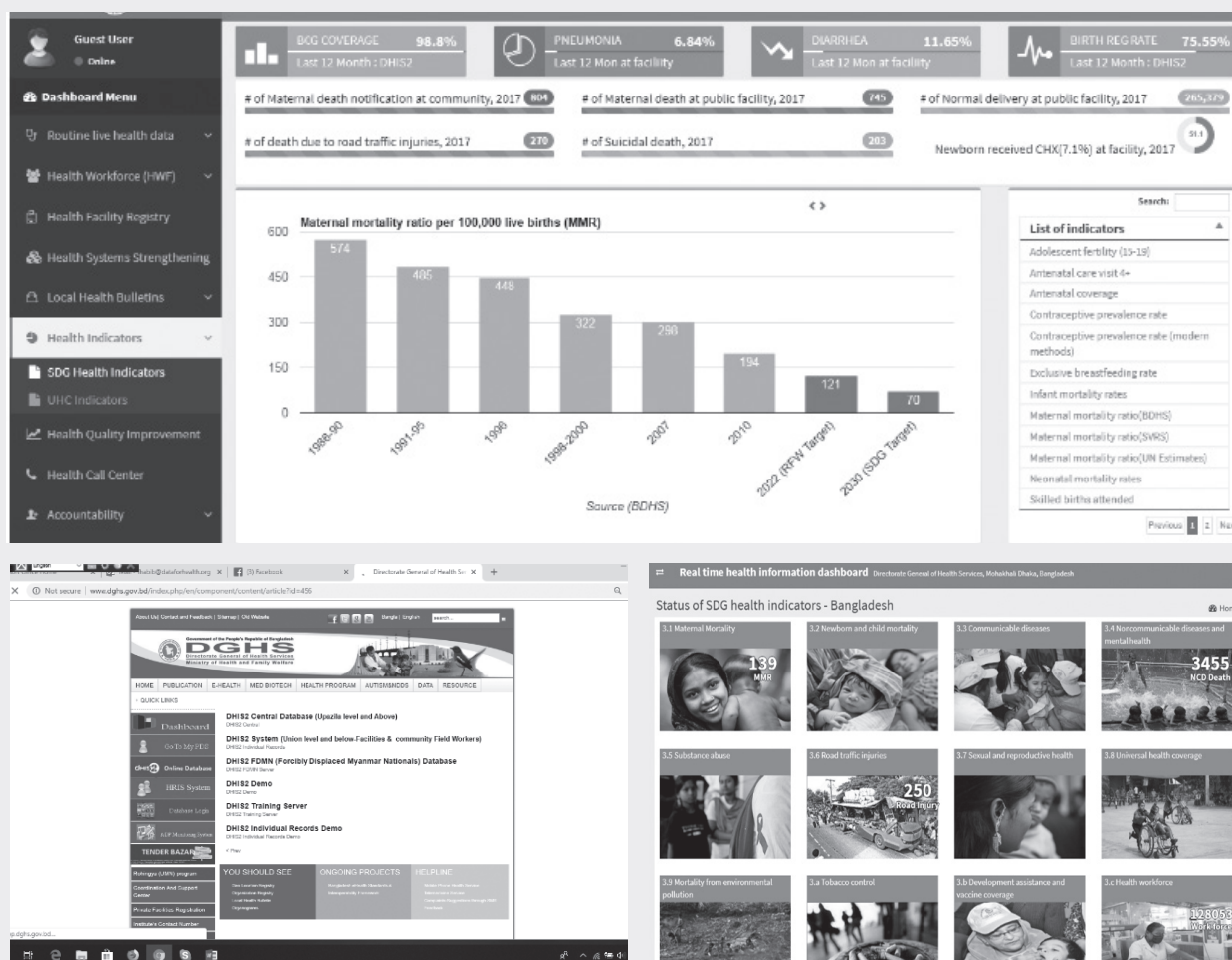


Figure 3.3. Screenshot of the SDG-specific segment of the DGHS dashboard (A click upon any photograph will reveal the relevant indicator)

- Policy regarding health and family planning
- Public health
- Registration of births and deaths.
- Adulteration of foodstuffs and other goods related to health
- Control of epidemics and prevention of infectious and contagious diseases, and quarantine isolation
- Health insurance
- Standardization and quality control of food, water, and other health-related commodities
- Regulations for medical professions and standards
- Administration of medical institutions and coordination
- Determination of standards in institutions for higher medical education or research
- Control of drugs
- Counter-signing of medical bills of the persons holding
- Activities related to maternity and child health centers
- Administration of BCS (Health)
- Administration of BCS (Family Planning)
- Post-mortem examination of dead bodies
- All matters related to administration of morgues
- Secretariat administration, including financial matters
- Administration and control of subordinate offices and organizations under the Ministry
- Liaison with international organizations and matters related to treaties and agreements with other countries and world bodies related to subjects allotted to the Ministry
- All laws on subjects allotted to the Ministry
- Inquiries and statistics on any of the subjects

Table 3.5. The summary of the data availability status of the indicators of Goal 3 in Bangladesh according to mapping documents

SDG Goal	Data Availability of the Indicators		
	Readily Available	Partially Available	Not Available
SDG 3- Ensure healthy lives and promote well-being for all at all ages	3.1.1, 3.1.2, 3.2.1, 3.2.2, 3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5, 3.7.1, 3.7.2, 3.c.1 = 12 Indicators	3.4.1, 3.4.2, 3.5.1, 3.6.1, 3.8.1, 3.8.2, 3.a.1, 3.b.1, 3.b.2, 3.d.1 = 10 Indicators	3.9.1, 3.9.2, 3.9.3 = 3 Indicators

- non-profitable offices
- Reimbursement of customs duty on gifts on non-consumable medical stores received from abroad
- Preparation of schemes, related to family planning, and their submission to the Prime Minister or the Cabinet through the Planning Commission
- Coordination and evaluation of all executive functions to related projects and programs
- Preparation and coordination of activities related to family planning through other ministries/divisions and offices
- Survey, monitoring, evaluation, and compilation of statistics of field activities in matters related to family planning

- allotted to this Ministry
- Fees in respect of any of the subjects allotted to this Ministry except fees taken in courts

ACTS/ LAWS IN HEALTH SECTOR

Health is a basic human right of every citizen. To ensure this basic right to every citizen, there must be timely, modern legislation. The Government of Bangladesh adopted a number of plans, policies, acts etc to develop, maintain and improve the capacity and quality of healthcare system of Bangladesh. Some recent and important acts/ laws in the health sector is mentioned below:

- The Communicable Diseases Prevention, Control and Eradication Act, 2018

- Mental Health Act, 2018
- Community Clinic Trust Act, 2018
- Bangladesh College of Physicians and Surgeons Act, 2018
- Transplantation of Human Organ Act, 2018
- Bangladesh Nursing and Midwifery Council Act, 2016
- Medical Education-1: Chittagong Medical Law, 2016
- Medical Education-1: Rajshahi Medical Law, 2016
- The Smoking and Using of Tobacco Products (Control) (Amendment) Act, 2013
- Bangladesh Medical & Dental Council Act, 2010
- The Drugs (Control) (Amendment) Act, Ordinance, 2006
- Institute of Child and Maternal Health Law, 2002
- Safe Blood Transfusion Law, 2002
- Bangabandhu Sheikh Mujib Medical University Act, 1998
- The Medical Practice Private Clinics and Laboratories (Regulation) (Amendment) Ordinance, 1984
- The Allopathic System (Prevention of Misuse) Ordinance
- The Bangladesh Unani and Ayurvedic Practitioner Ordinance, 1983
- The Bangladesh Homeopathic Practitioner Ordinance, 1983
- The International Centre for Diarrhoeal Disease Research Ordinance, 1978
- International Centre for Diarrhoeal Disease Research, Bangladesh (Amendment) Act, 1978
- The Pharmacy Ordinance, 1976
- The Medical Colleges (Government Bodies) Ordinance, 1961
- The Medical Degrees Act, 1916
- The Vaccination Act, 1880

PUBLIC HEALTH SERVICES

SUB-CHAPTER 4.1

Maternal, Newborn, Child and Adolescent Health

MATERNAL HEALTHCARE

The Ministry of Health and Family Welfare, Bangladesh in collaboration with UNICEF, has been running the facility-based Emergency Obstetric Care (EOC) Program, in all districts of Bangladesh, to improve the maternal health situation. All the government medical college hospitals, district hospitals, upazila hospitals, and maternal and child welfare centers (MCWCs) provide obstetric care services, inclusive of emergency obstetric care. A number of private clinics or hospitals and health-related NGOs are also serving as partners in this program. Obstetric care is classified into two categories in this program; they are: Comprehensive Emergency Obstetric Care (CEmOC) and Basic Emergency Obstetric Care (BEOC). Currently, all medical college hospitals, 59 district hospitals, 3 general hospitals, 132 upazila health complexes, and 71 MCWCs provide CEmOC, and rest of the upazila health complexes provide BEOC. The list also includes NGOs and private care providers from a number of districts. Under a program, jointly operated by the Management Information System (MIS) of the DGHS and UNICEF, data are collected from the EOC facilities.

Table 4.1.1 Sources of data on emergency obstetric care which have been analyzed for translating them into a format called United Nations Process Indicators.

Type of hospital/ facility	Number	Percentage (%)
Medical college hospital	15	2%
District and general hospital	62	9%
Upazila health complex	416	62%
Maternal child and welfare center	71	10%
Private care providers	64	9%
NGO care providers	51	8%
Total	679	100

A network for improving
safe, effective,
timely, efficient,
equitable, and
people-centered
health care
for maternal,
newborn, child and
adolescent health

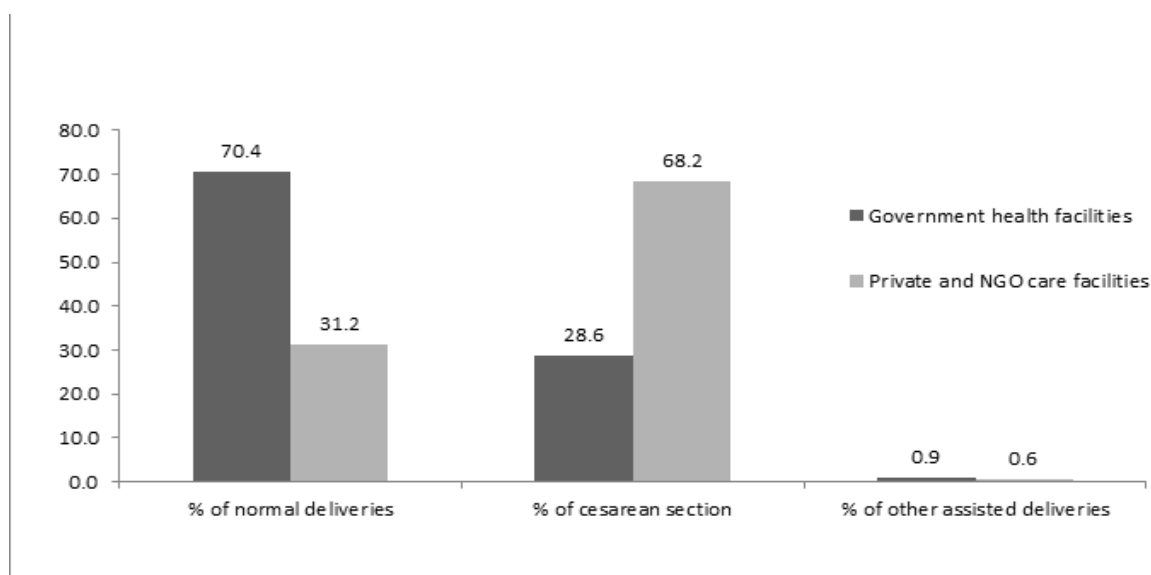


Figure 4.1.1 Percentage of deliveries by type between government and non-government EmONC facilities in 2017

Table 4.1.2 Number of deliveries in the government, private and NGO health care EmONC facilities in Bangladesh in 2017

Type of delivery	Government					Non-government		
	MCH	DH/GH	UHC	MC&WC	Total	Private facilities	NGO facilities	Total
No. of normal deliveries	48901	67363	142137	29298	287699	95588	37549	133137
% of normal deliveries	46.3%	61.2%	90.0%	84.0%	70.4%	26.4%	57.7%	31.2%
No. of cesarean section	54830	41938	14796	5419	116983	264140	27069	291209
% of cesarean section	51.9%	38.1%	9.4%	15.5%	28.6%	73.0%	41.6%	68.2%
No. of forceps/ vacuum/ breech deliveries	1842	797	964	154	3757	2078	482	2560
% of forceps/ vacuum/ breech deliveries	1.7%	0.7%	0.6%	0.4%	0.9%	0.6%	0.7%	0.6%
Total number of deliveries in EmONC facilities	105573	110098	157897	34871	408439	361806	65100	426906

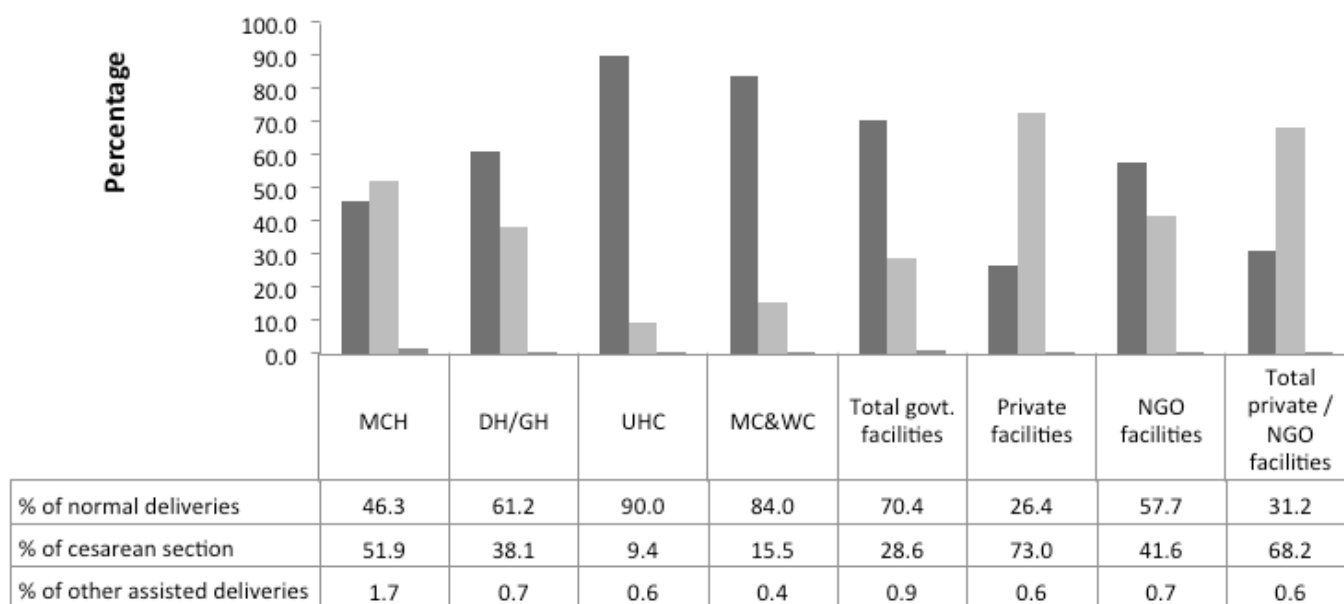


Figure 4.1.2. Percentage of normal, assisted and cesarean deliveries (obstetric care service) provided by government and non-government emergency health care facilities in 2017.

Demand-side Financing (DSF): Maternal Health Voucher Scheme

- The Ministry of Health and Family Welfare, Bangladesh, in collaboration with WHO, introduced in 2007 an innovative Maternal Health Voucher Scheme, a demand-side financing (DSF) initiative, to improve access to and use of quality maternal health services for the poor pregnant women.
- The number of DSF upazilas was increased in phases. Currently, this program is being implemented in 55 upazilas of 41 districts.
- Poor women are defined by specific selective criteria (approximately 35% of all the pregnant woman of an upazila) and pregnant woman are selected by local government representatives, health managers and other stakeholders.
- Registration was done by field workers. The total number of cumulative beneficiaries reached 14, 64, 078 (From July, 2006 to June, 2018).
- All the voucher holders received health services free of charge (ANC, PNC, Safe delivery and treatment of complications, including cesarean section, transportation cost, and laboratory test).
- If the voucher holder mother has her delivery in Health facility or by SBA at home, she will get cash incentive for nutritious food.
- Both public and NGO as well as private facilities participate in the DSF scheme.
- Under the current sector program, there is a target to scale-up the program every year.

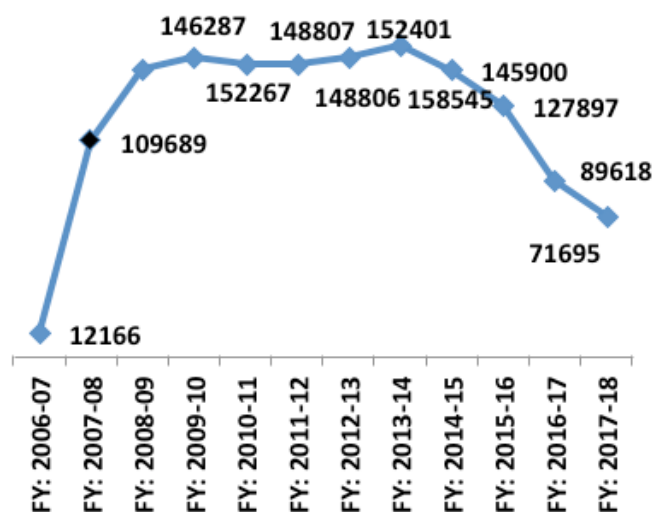


Figure 4.1.3. Number of DSF (demand-side financing) beneficiary pregnant women by fiscal year (total 14,64,078)

Maternal and Newborn Health Initiative

- The Maternal and Newborn Health Initiative (MNHI) is being implemented by the Director of Primary Healthcare of the DGHS in 11 districts of Bangladesh, with the assistance of UNFPA, UNICEF, and WHO. Some of them are funded by DFATD Canada.
- The districts are Thakurgaon, Jamalpur, Narail, Maulvibazar, Panchagarh, Sirajganj, Patuakhali, Barguna, Rangamati, Sunamganj, and Bagerhat.
- The program focuses on saving maternal and newborn lives through creating need-based demand and priority-based actions.
- The broad principle of this program is to find the bottlenecks through data analysis.
- Finally, the health managers develop Evidence-based Planning and Budgeting (DEPB) for every upazila and hospital. Around 25 districts have been covered under DEPB by UNICEF.
- The civil surgeon and deputy directors of family planning of the respective districts serve as the local focal points for the program.
- UNICEF has designed a comprehensive model to improve health of neonates, mothers, and young children in hard-to-reach areas of Bandarban, Cox's Bazar, and Netrakona district. It has introduced a default-tracking system to track every mother and child.
- Under health system strengthening program, 10 consultants are working to improve the information system by using data for planning.
- Innovative dashboard has been created for all health managers.
- As a part of improving quality, Maternal and Perinatal Death Surveillance and Response (MPDSR) program has been introduced in 10 districts. This has shown a good impact in reducing maternal and neonatal deaths.

Community-based skilled birth attendants and midwives

- The MoHFW undertook a short-term measure to tackle shortage of skilled manpower by providing training courses.
- The Directorate General of Health Services has also been implementing community-based skilled birth attendant (CSBA) training program since 2003, to train and educate the family welfare assistants/female health assistants, on midwifery skills.
- The CSBAs are trained to conduct normal safe deliveries at home and to identify the risks and complicated cases so that they can motivate the women and their family members to refer to the nearby health facilities where comprehensive EOC services are available. The CSBA training program is now organized in 465 upazilas of 64 districts.
- The total number of CSBA in Bangladesh (up to December, 2017) were 11509. In November 2018, this number has increased to 11555.
- Training is given to female CHCPs and female HAS
- Training was imparted to 54 participants from Sher-E-Bangla MCH, Adhunik Sadar Hospital of Brahmanbaria, Moulvibazar and Civil Surgeon Office of Feni in 2017-2018 FY.
- Training will be provided to 80 more participants in 2018-2019 FY

Table 4.1.3. Represents levels of training institution in the whole country

Training institution at divisional level	Training institution at district level
• Medical College Hospital • Nearest Family welfare center	• Sadar Hospital • Nursing institution • Family welfare center

Obstetric fistula program

- In Bangladesh, obstetric fistula and other maternal morbidities affect thousands of women. Currently, it is estimated that approximately 50,000 women in the country are living with obstetric fistula.
- UNFPA has been assisting the Government of Bangladesh in strengthening the quality of service delivery and capacity development of service care providers at 10 government medical college hospitals.
- Bangladesh endorses the UN declaration to end obstetric fistula within a generation. In the new HPNSP (2017-2022) government has made lots of references for the care and prevention of fistula.
- UNFPA and other partners has been providing both technical and financial support to GoB to make Bangladesh fistula-free.

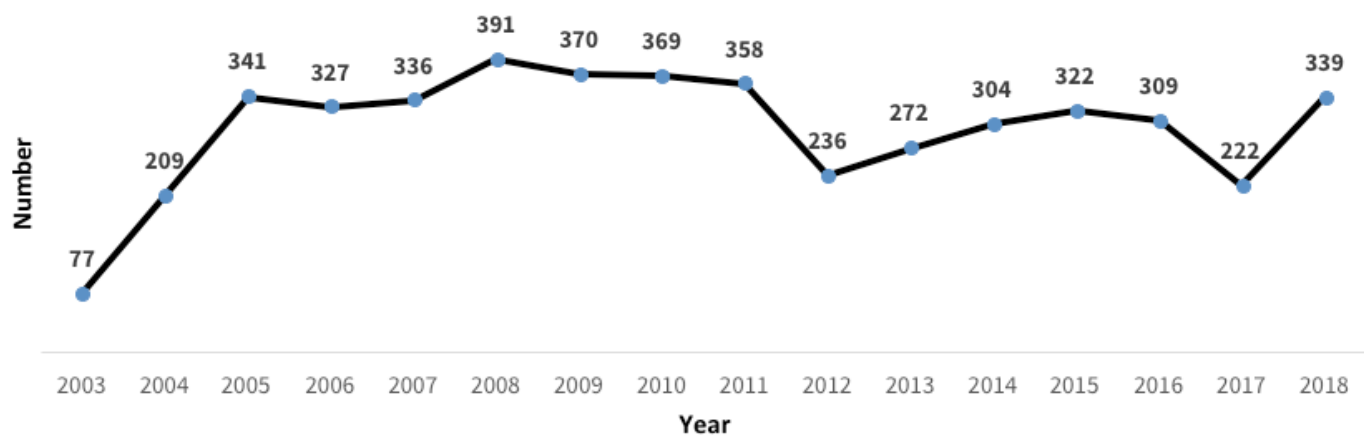


Figure 4.1.4. Number of fistula repair surgeries (N=4782) at the government medical college hospitals during the period 2003-2018

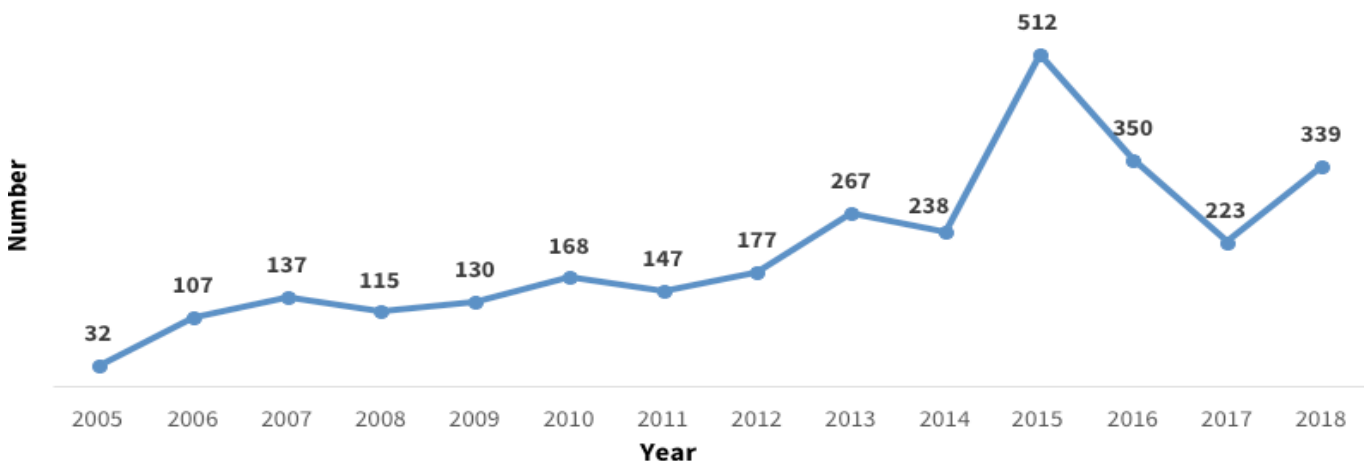


Figure 4.1.5. Number of fistula repair surgeries (N=2942) at private hospitals during the period 2005-2018

Cervical and Breast Cancer Screening Programs in Bangladesh

- Cervical and breast cancers are the most common cancers among Bangladeshi women, and they contribute to a significant disease burden of Bangladesh. Routine screening and early detection of cervical and breast cancer can reduce the death rates and can improve life-expectancy of cancer patients.
- The Government of Bangladesh (GOB) has developed a cervical and breast cancer screening program in all districts of Bangladesh with the technical assistance of Bangabandhu Sheikh Mujib Medical University (BSMMU) and UNFPA.
- Since 2005, a total of 417 centers [Table-1] have been established throughout the country and about 2036 service providers (Doctors = 386, Nurses/FWs/ Paramedics=1650) from 64 districts were trained on cervical and breast cancer screening, based on visual inspection with acetic acid (VIA) and clinical breast examination (CBE).
- Fourteen government medical college hospitals, along with BSMMU, have been developed as referral centers through development of colposcopy clinics with facilities for colposcopy, histopathology, and management of precancerous condition of the cervix.

Table 4.1.4. Total number of VIA and CBE centers established at various institutions (2005-2017).

Name of Locations	Number of Centers
District hospitals	34
BSMMU, NICRH, ICMH medical college hospitals	34
MCHTI, MFSTC, MCWC	61
Upazila health complexes	248 out of 482
Union health & family welfare centers	30 out of 3725
BGB,CMH/ Private/NGOs	10
Total	417

- A total of 1,816,802 VIA tests were performed from 2005 to June, 2018 at different service centers. Of the tested women, 85,079 (4.68%) were found VIA-positive. (figure 4.1.7)

The coverage of the screening tests has been increasing every year. All the VIA+ve cases were referred to the colposcopy clinic at BSMMU and also to different medical college hospitals.

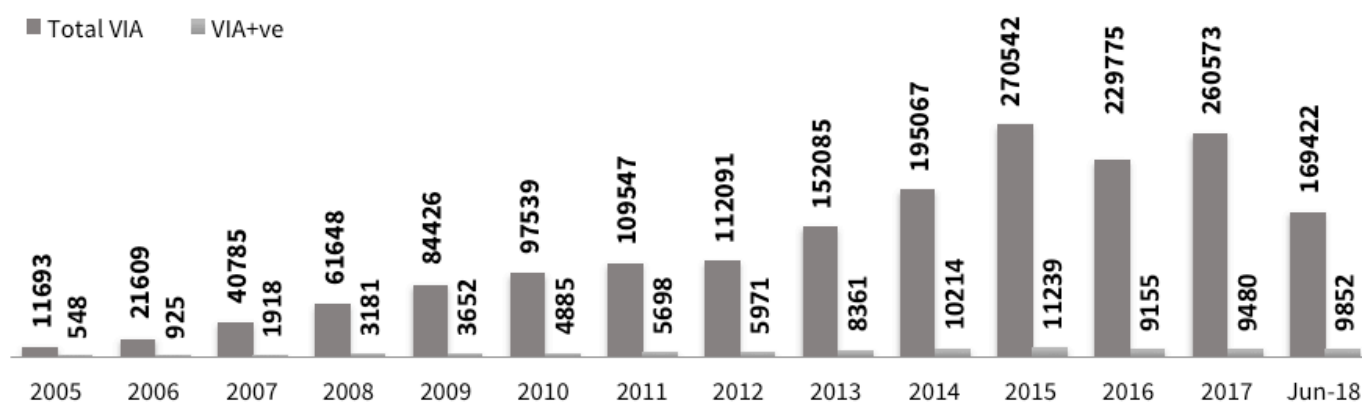


Figure 4.1.6. Number of VIA tests performed and VIA+ve results (2005-June 2018)

Table 4.1.5. Total number of colposcopy tests done in 2017 at different institutions.

Name of institution	Number (%) of colposcopy tests
Bangabandhu Sheikh Mujib Medical University	2812 (29.6)
Rajshahi Medical College Hospital	1192 (12.60)
Chittagong Medical College Hospital	829 (8.70)
Khulna Medical College Hospital	901 (9.60)
Sher-E-Bangla Medical College Hospital, Barisal	269 (2.80)
Faridpur Medical College Hospital	134 (1.50)
MAG Osmani Medical College Hospital, Sylhet	291 (3.00)
Mymensingh Medical College Hospital	511 (5.50)
Dhaka Medical College Hospital	749 (7.90)
Dinajpur Medical College Hospital	423 (4.40)
Comilla Medical College Hospital	370 (3.90)
Rangpur Medical College Hospital	282 (2.90)
Shaheed Suhrawardy Medical College Hospital	122 (1.4)
Sir Salimullah Medical College & Mitford Hospital	157 (1.60)
Shaheed Ziaur Rahman Medical College Hospital	138 (1.40)
Failure to attend	300 (3.20)
Total	9480 (100)

Table 4.1.6. Management of referred patients at BSMMU (January 2005 to December 2017)

Management of Women	Number
Follow-up after 3 years	10334 (43.80%)
Follow-up after 1 year	3053 (12.90%)
LEEP Biopsy (Loop Electrosurgical Excision Procedure)	3332 (14.10%)
Cryotherapy	131 (0.60%)
Cold Coagulation (local ablative method)	2428 (10.30%)
Referred to Oncology due to Cancer	1413 (6%)
Failure of treatment and follow-up	2897 (12.30%)
Total	23588 (100.00%)

- Figure 4.1.8 shows that from 2007 to 2017, a total of 15,44,286 CBE tests were performed at service centers; among them, a total of 21,250 (1.38%) women were CBE-positive. The coverage of CBE screening is increasing every year.
- In 2017, a total of 261788 screening tests were conducted. Of them, 1.67% were CBE-positive. All these CBE-positive cases were referred to the Department of Surgery at BSMMU/MCHs and DHs.
- In Bangladesh, LEEP is acquired acceptability as a commonly-used outpatient treatment procedure for CIN (Cervical Intraepithelial Neoplasia) under local anesthesia and thermo-coagulation without local anesthesia (Basu et al 2010).

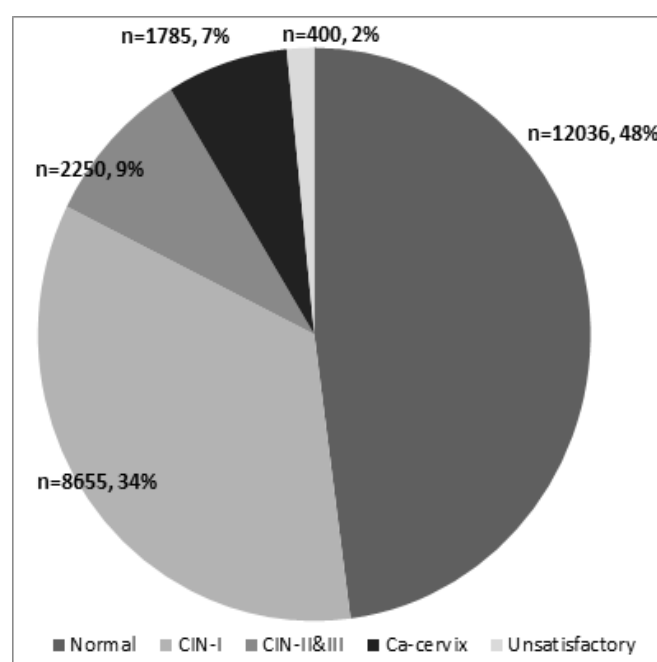


Figure 4.1.7. Colposcopy findings of VIA+ve women attending BSMMU (from January 2005 to December 2017, N=25,126).

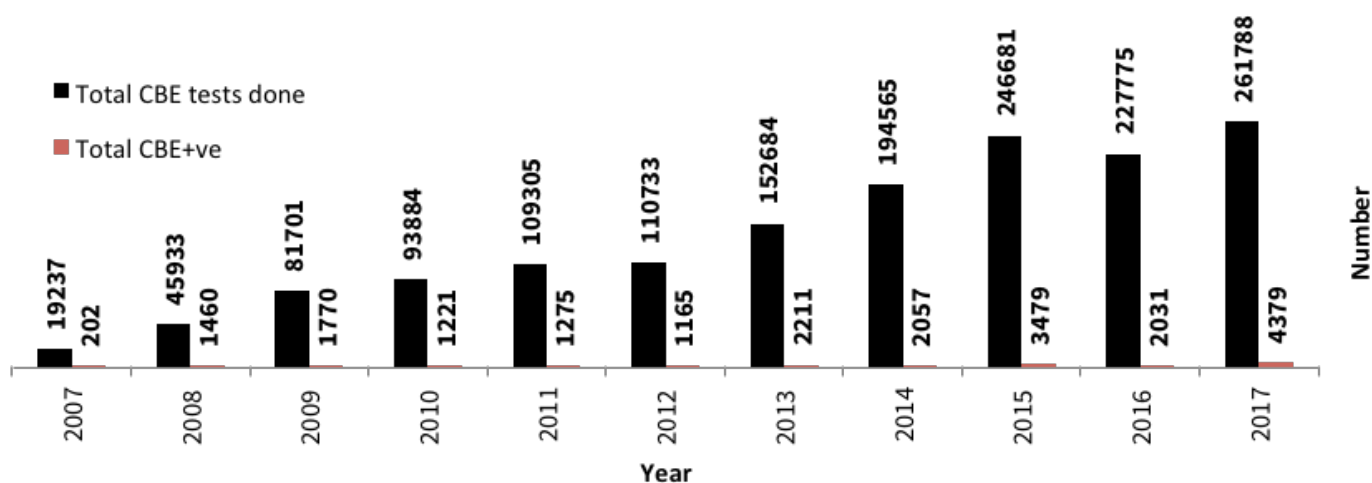


Figure 4.1.8. Number of CBE tests done, with results (2007-2017)

CHILD HEALTH SITUATION IN BANGLADESH

Bangladesh has been able to reduce the under-five mortality, well ahead of the MDG 4 target (from 144/1000 live births in 1990 to 38/1000 live births in 2015), reducing by 72% since 1990, with the annual rate of reduction being 5.4.

There has been a significant shift in the epidemiological pattern of under-five mortality. Neonatal deaths still account for almost 61% of all under-five mortality (U5MR).

Diarrheal deaths among under-five have decreased substantially over the last few years (2%). However, ARI/ pneumonia continues to be the single most cause (21%) of U5MR.

The care seeking from trained providers, for pneumonia and diarrhea, has increased remarkably over the last few years. ORT use rate for diarrhea management has increased to more than 80%, vitamin A supplementation coverage among under-five children has increased from 85% in 2005 to 92% in 2011, while the exclusive breast feeding practices have stood at 54% (in 2014).

Progress on EPI and introduction of new vaccines have progressed well and scaling up of C-IMCI and IMCI, and nutrition corners is taking place. About 81% of children aged 12-23 months received all scheduled vaccines by 12 months of their age.

Under the Reach Every District Community Strategy routine EPI has further been strengthened in low performing

32 districts and 4 City Corporations. NGO/Private sector partnership exists for EPI program in a limited scale, especially in urban areas.

Challenges

- Despite the reduction in mortality over time, continued disparities have been observed between the richest and the poorest quintiles, between urban and rural areas and between divisions (e.g. Khulna, Sylhet).
- Mothers' education was found to be associated with U5MR. With the highest mortality rate of 82/1,000, noted among mothers with no education. Compared to them, among the mothers with secondary education or higher, the rate is 39/1000.
- As for under-five deaths pneumonia, neonatal causes and drowning are the major causes. The number of diarrheal deaths among under-five children has decreased substantially over the last few years (2%). However pneumonia remains the single most important cause (21%) of under-five mortality followed by neonatal causes.
- Although deaths due to preventable communicable diseases continued to decrease over times, the share of deaths due to drowning has resulted in a sharp rise from 26% in 2007 to 42% in 2011, among the age group of 1-4 years. Currently, a clear multi-sectoral strategy to address child drowning is absent.
- The current child health interventions mainly focus on mortality, with limited emphasis on child development.

Table 4.1.7. Relevant Result Frame Work Indicators (RFW)

Indicators	Unit of Measurement	Base line (Year and Data Source)	Projected Target(2022)
Under 5 Mortality Rate	No.of/1000 live birth	31 (2017 SVRS)	34
Neonatal Mortality Rate	No. of /1000 live birth	17 (2017 SVRS)	18
Maternal Mortality Ratio	No. of /100000 live birth	172 (2017 SVRS)	121

Sustaining, the already achieved high coverage for EPI, and reaching the unreached to further increase the coverage, remain a challenge.

National Newborn Health Program (NNHP) and IMCI

- To improve newborn and child survival, a National Newborn Health Program (NNHP) will be launched through this sector program and the IMCI program will be strengthened.
- The NNHP will focus on achieving the equitable and effective coverage of high impact newborn services and practices at scale, through quality implementation of a comprehensive newborn health care package by the public health system.
- The activities under the IMCI program will be implemented at community and at all level of facilities in line with national policy and strategy for achieving high coverage and quality of facility and community based IMCI services.

Program Management

- The existing IMCI unit under Line Director MNCAH will be strengthened and will be renamed as National Newborn Health Program and IMCI. The unit will have one Program Manager (PM-NNHP & IMCI), five Deputy Program Manager (DPM-Admin & Finance; DPM- Newborn Health; DPM- Training and Child Injury; DPM-Field Monitoring and Data Quality; DPM- Coordination and Logistics Management).
- Along with strong leadership and ownership of MOH&FW officials' at all level, functional coordination among different departments and facilitation role from development partners will be crucial to ensure successful implementation on NNHP and IMCI.

- A District MNCH Focal Person will work in district level to strengthen supervision and monitoring of all maternal newborn and child health related activities of the district level and below. With the support of development partners, as group of consultants, they will work from district and division level to facilitate the implementation of NNHP throughout the country.

Activities

1. Promotion of birth preparedness, newborn care preparedness and proper care seeking through a comprehensive SBCC approach
2. Immediate and essential newborn care (ENC), including application of 7.1% Chlorhexidine to the umbilical cord and postnatal care (PNC) in the facility and in the home by visiting community health workers
3. Management of sick newborn resuscitation using HBB protocol
4. Kangaroo Mother Care (KMC) for management of preterm/LBW babies
5. Judicial use of antenatal corticosteroid in threatened premature deliveries
6. Care of critically ill newborn at Special Care Newborn Unit (SCANU)
7. Strengthen effective referral systems, including reliable and affordable transportation
8. Development of NNHP management toolkit and training of health managers and MNCH focal persons on Comprehensive Newborn Care Package (CNCP) Implementation
9. Capacity building of the health care providers and field workers of different levels on CNCP for the management

of healthy and sick newborn

10. KMC training for doctors and nurses to manage preterm low birth weight babies at upazila health complex and higher level facilities
11. ETAT and sick newborn care training of doctors and nurses for management of sick newborn at SCANU/NSU
12. Ensure logistics commodities and medicines at union, upazila and district level facilities under DGHS for the management of sick newborn as per national guidelines
13. Establishment of standard KMC corner in all upazila and district level facilities and ensuring commodities and logistic supplies, and functional newborn resuscitation devices (Bag & Mask) in all facilities. Provision of a district MNCH focal person to coordinate, supervise and

monitoring of all maternal newborn and child health related activities at the district level and below and reportable to Line Director MNCAH

16. Revision of National Neonatal Health Strategy and Standard Operating Procedure (SOP) for newborn health services. Community engagement by using community group (CG) and community support groups (CSG) of community clinics in coordination with Community Based Health Care Service (CBHCS)
18. Strengthen societal engagement by orienting school teachers, local representatives, multi sartorial officials, NGOs through different existing platforms
19. Organization of national newborn campaign

ADOLESCENT AND SCHOOL HEALTH PROGRAM

BACKGROUND

Adolescent health has been highlighted frequently during HPNSP, as a priority area that needs further and focused attention. The Bangladesh national ARSH strategy was formulated in 2006 and related action plan was developed in 2013. Two major operational plans included plans to carry out selected health activities of ARSH strategy.

Training modules were developed by DGHS. Training was provided to the service providers, school teachers and adolescent peers in 10 districts and, adolescent friendly health services (AFHS) have been established in 6 districts and 2 city corporations. A guideline on AFHS (2015) and strategy and guideline on IV/STI/AIDS/MR/PAC are in place.

Recently (2017), Bangladesh National Adolescent health strategy for 2017 to 2030 has been developed and approved. Related action plan for Adolescent health is also developed and soon will be approved.

Emphasis is given on the following 3 thematic areas: as (i) strengthen adolescent, sexual and reproductive health issues (ii) nutrition (iii) violence against adolescent and mental health. All the activities and strategies have been planned according to these 3 thematic areas.

Adolescent pregnancy and child bearing entail a high risk of maternal death among adolescents, and the children of the young mother have higher level of morbidity and mortality rate of 5. There are estimated 569,000 annual births among 15-19 year adolescents girls in Bangladesh. They face a number of important health risks arising out of early pregnancies, violence and inadequate nutrition. Almost twice as many newborn deaths occur among them.

OBJECTIVES

- Improve knowledge of adolescents on sexual and reproductive health (SRH), Nutrition, violence against adolescent and mental health and others issues.
- Promotion and ensuring access of adolescent friendly health services at the facility and community level
- Creation of positive changes in the behavior and attitude of the gatekeepers of the adolescents towards reproductive health.

Target of the program / Target population: 10-19 year age group population (about 36 million).

MAJOR ACTIVITIES IN 2017

- Upgrading of health facilities (all UHCs and UH & FWCs / USC) in phases and to provide adolescent friendly SRH services
- Counseling and developing awareness among adolescents, especially senior girl students on personal hygiene practices, including menstrual hygiene management, nutrition and ASRH.
- Formation of a coordination committee among the adolescent sexual and reproductive health (ASRH) implementing partners.
- Incorporation of ASRH component into the existing MIS of DGHS
- Community mobilization around ASRH issues through courtyard meetings, interpersonal communication and workshops
- Development and printing of training manuals, guidelines, and booklets regarding adolescent health
- IEC activities: Short film show, TVC, poster, calendar, folder, wall-painting, bill-board, AFHS logo, Brochure, Leaflet, Class routine with SRHR information etc.
- Development and dissemination of key messages and materials in all available and appropriate channels
- Development of supervision and monitoring tools and establishment of supervision and monitoring system

MAJOR ACTIVITIES IN 2017

- Training of 690 trainers (TOT) of district and upazilla level managers of DGHS, DGFP and DG of education sector on adolescent health care in Chandpur and Sunamgonj districts
- Training of 1050 secondary school and colleges teachers on adolescent health care in Chittagong, Sunamgonj and Chandpur districts
- Training of 510 secondary school students as Peer Educators from the adolescent groups in Moulvibazar, Sunamgonj, Chandpur and Barisal districts
- Orientation of 750 persons as Gatekeepers (Chairman, Member, Imam, Kazi, School management committee and local population) in Moulvibazar, Sunamgonj, Chandpur districts
- Orientation of 44100 field level workers
- Workshop for capacity development, coordination and performance Appraisal: 40 persons
- Special program was undertaken in the low perform areas of Chittagong and Sylhet Division.
- Operational Research on adolescent health
- Developed strategy on Social and Behavior Change Communication for adolescent health
- Developed dashboard for adolescent health indicators and strengthen HMIS system for adolescent health
- Training of trainers (TOT) of secondary school teachers of Dhaka City Corporation on adolescent health, with the support of UNICEF and WHO
- Training of frontline basic health workers of Tangail district on adolescent health with the support of UNICEF and Embassy of the Kingdom of the Netherlands.
- Assessment of current school-based HPN services. in Chittagong and Sylhet division. An action plan developed for school based HPN services, with the support of UNFPA

Expanded Program on Immunization

BRIEF HISTORY OF EPI IN BANGLADESH

- EPI in Bangladesh was launched on 7 April, 1979
- Intensified both in rural and urban areas as Universal Child Immunization (UCI) from 1985
- EPI service was made available to all the target groups in phases by the year 1990
- Polio eradication and elimination of maternal and neonatal tetanus (MNT) activities started in 1995
- AFP, measles and NT surveillance started in 1997
- Introduced Hepatitis B vaccination in 2003
- Introduced AD syringes for vaccination from 2004
- Introduced Hib vaccine in 2009
- Incorporated MR vaccine and measles second dose from 2012
- Introduced PCV and IPV from 2015
- Introduced fractional dose of IPV (fIPV) from 2017

OBJECTIVES OF EPI

- Improve immunization coverage among children under one and child bearing age women
 - a. At least 95% fully immunization coverage among under one year children at national level and at least 85% full immunization coverage in all districts
 - b. TT5 coverage among women of childbearing age reached at least 80% at national level and at least 75% in all districts
- Maintain polio free status
- Maintain Maternal and Neonatal Tetanus Elimination Status
- Elimination of Measles & Rubella
- Prevention of diseases protected by new and underused vaccines

TETANUS TOXOID (TT) FOR WOMEN OF CHILDBEARING AGE

- Like the childhood vaccination coverage, TT vaccination coverage is assessed as crude and valid coverages. The valid TT coverage is assessed in terms of the valid doses that a woman receives. And, the crude TT coverage is assessed in terms of all TT doses - both valid and invalid - that a woman receives.

Table 4.2.1. Valid vaccination coverage of ≤12 months old children as found in EPI administrative data 2017

Age Group	BCG	OPV1	OPV2	OPV3	Penta1	Penta2	Penta3	MR1	MR2	PCV1	PCV2	PCV3
≤12 months	101.3	100.1	99.3	97.9	100.1	99.9	98.5	98.8	87.2	99.0	98.4	102.5

- A TT dose administered before the recommended interval is considered to be invalid. Thus, a TT3 dose given earlier than the recommended 6-month interval after a valid TT2 is enumerated as an invalid TT3 dose. The information of TT vaccination is obtained from a woman's TT card (if available). If it is not available, the information is collected from the woman's vaccination history reported by her.
- The country is maintaining the maternal and neonatal tetanus-free status since 2008. The immunization program of Bangladesh aims to immunize the women of childbearing age by administering TT before the age of 18 years.
- A period of 2 years and 7 months is required to complete all the 5 doses of TT vaccines. If a woman starts TT vaccination at the age of 15 years and maintains the exact interval, she would be able to complete all the doses before she reaches the age of marriage, ensuring protection for her entire reproductive life..
- However, the coverage gradually falls for the subsequent TT doses for pregnant women and is remarkably lower for the third and fourth doses. TT coverage does not increase as expected considering the number of target child bearing age women. This aspect needs attention to ensure effective coverage.
- The urban-rural comparison shows that valid doses of TT2 to TT5 coverage were higher in rural areas than in urban areas. Five doses of valid TT vaccine give protection to a woman against tetanus throughout her reproductive life. The findings suggest that more than half of the women in all divisions received 5 doses of valid TT vaccine.

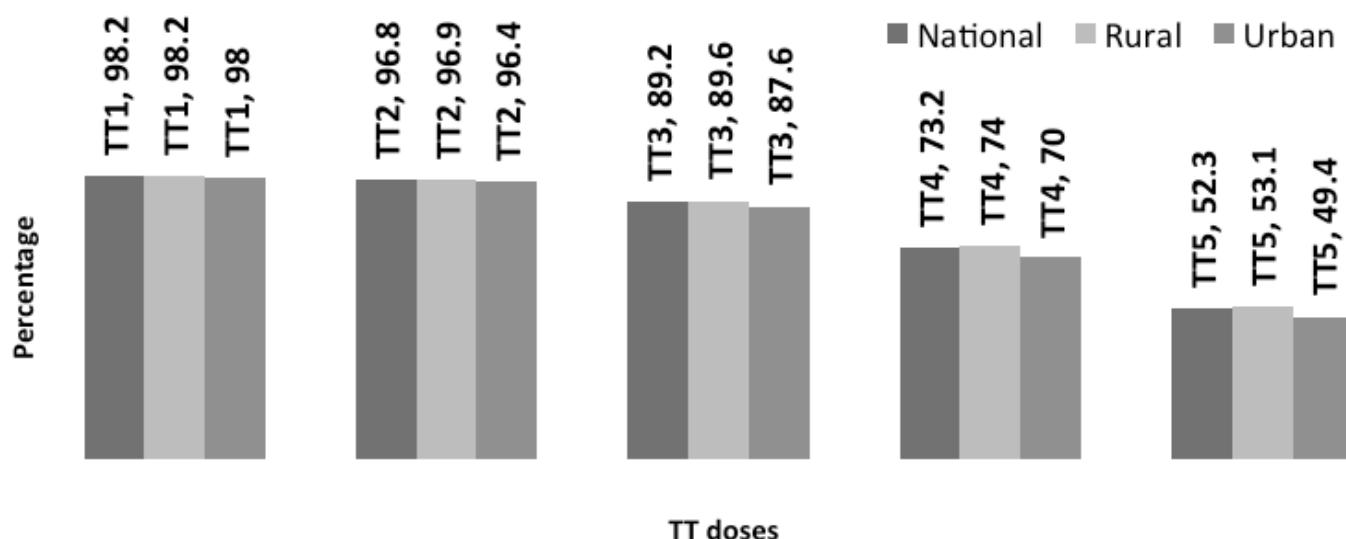


Figure 4.2.1. Levels of valid TT vaccination coverage in rural and urban areas, with national average in 2016

UNIVERSAL ROUTINE CHILD IMMUNIZATION

The EPI CES 2016 validated the immunization coverage rates by cross-checking EPI cards, with history taken from mothers and caregivers of children.

Table 4.2.2. Valid vaccination coverage of ≤ 23 and ≤ 12 months old children as found in EPI CES 2016

Age Group	BCG	OPV1	OPV2	OPV3	Penta1	Penta2	Penta3	MR1	MR2
≤ 12 months	99.5	97.8	97.0	90.1	97.8	97.0	90.1	87.5	82.3
≤ 23 months	99.5	97.9	97.2	90.4	97.9	97.2	90.4	92.3	86.8

- Bangladesh showed a success story on polio eradication. The country is polio-free since 2000, with 18 exceptional cases of wild polio virus imported from neighboring India in 2006. National Immunization Day is observed every year.
- The polio eradication program in Bangladesh illustrates Government's commitment through providing 100% cost of routine polio immunization and 95% cost of supplementary polio immunization activities. Bangladesh, despite being free from polio for a long time, could not achieve polio-free certification as one country of the WHO South-East Asia Region, viz. India, could not eradicate the last traces of polio. However, in India, last case of polio was detected in 2011.
- After being polio-free for more than 36 months, Bangladesh, along with other 10 member countries of the South-Asia Region, obtained the polio-free certification in March 2014. As per the global polio eradication end-game strategic plan (2013-2018), Bangladesh introduced inactivated polio virus vaccine (IPV) in March 2015 for prevention of vaccine-derived polio viruses due to type two component of polio vaccine.
- As per country's plan, Bangladesh switches from trivalent oral polio vaccine (tOPV) to bivalent oral polio vaccine (bOPV) in April 2016 for prevention of outbreak due to type 2 component of OPV.
- The country is also satisfactorily progressing toward achieving the measles elimination goal in the WHO's South-East Asia Region by 2020. National measles control activities have been accelerated since 2004 and already implemented all recommended strategies for measles elimination and rubella/congenital rubella syndrome control.
- Valid measles-rubella vaccination coverage was 87.5% and 92.3% among ≤ 12 and ≤ 23 months old children respectively according to EPI CES 2016. Coverage of the first dose (MCV1) of measles vaccine was estimated to increase from 64% in 2001 to 87.5% in 2016 among ≤ 12 months old children. Coverage of the first dose (MCV1) of measles vaccine was estimated to increase from 78% in 2005 to 92.3 in 2016 among ≤ 23 months old children.

Table 4.2.3. Valid vaccination coverage of ≤ 23 and ≤ 12 months old children by demography as found in EPI CES 2016

Age Group	Sex		Residence		Division						
	Male	Female	Rural	Urban	Barisal	Chittagong	Dhaka	Khulna	Rajshahi	Rangpur	Sylhet
≤ 12 months	82.2	82.5	83.5	77.1	87.5	83.5	77.9	84.5	84.9	82.5	79.2
≤ 23 months	86.8	86.9	87.6	83.5	97.8	90.7	84.0	88.1	88.5	87.2	84.8

VALID VACCINATION COVERAGE WITHIN THE AGE OF 12 MONTHS

- Improvement in valid vaccination coverage within the age of 12 months is the ultimate goal of the EPI. The program has set a target of achieving 95% vaccination coverage nationally and at least 90% in each district.
- Figure 4.2.2 indicates the increasing trend in the coverage

from 2001 to 2016. A remarkable increase in vaccination coverage has occurred in the last 15 years. It increased up to 30.3 percentage points: from 52% in 2001 to 82.3% in 2016.

- The analysis of trend indicates that gradual improvement in BCG, Penta3, and MR coverage attributed to the continuous improvement in the fully valid vaccination coverage.

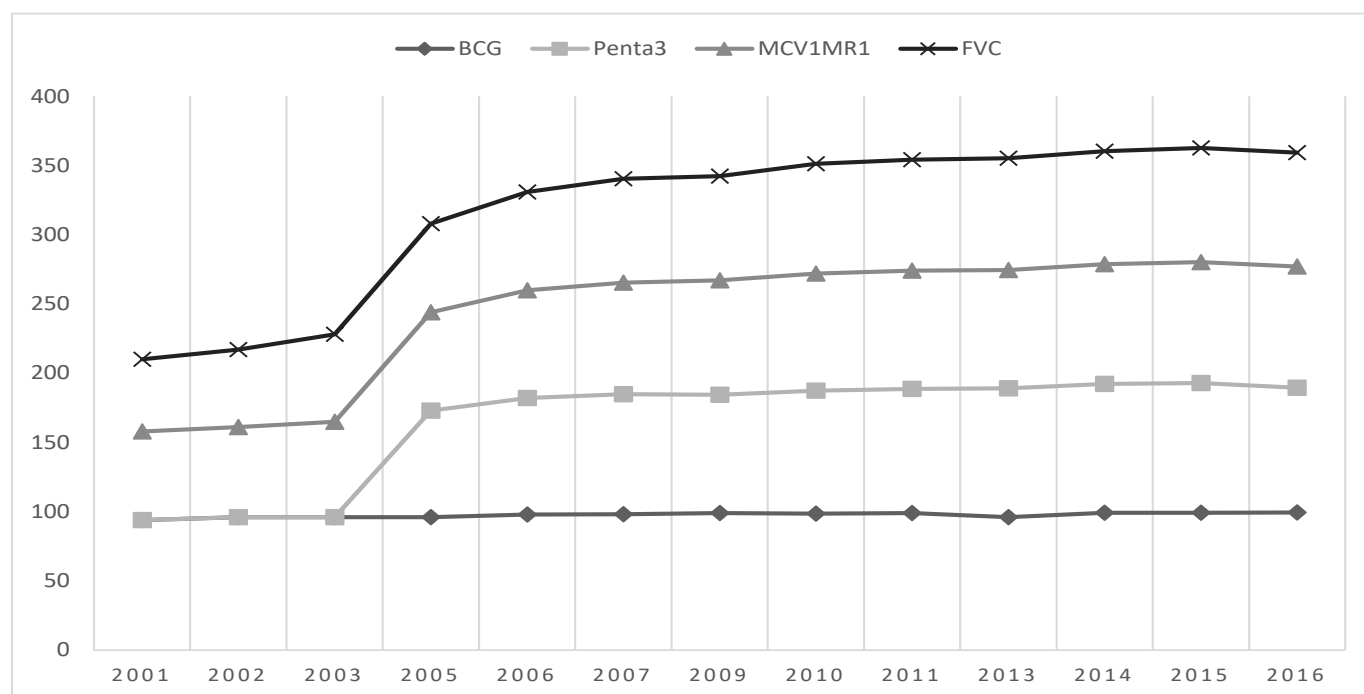


Figure 4.2.2. Annual trend of national-level valid vaccination coverage among children of ≤12 months (2001-2016)

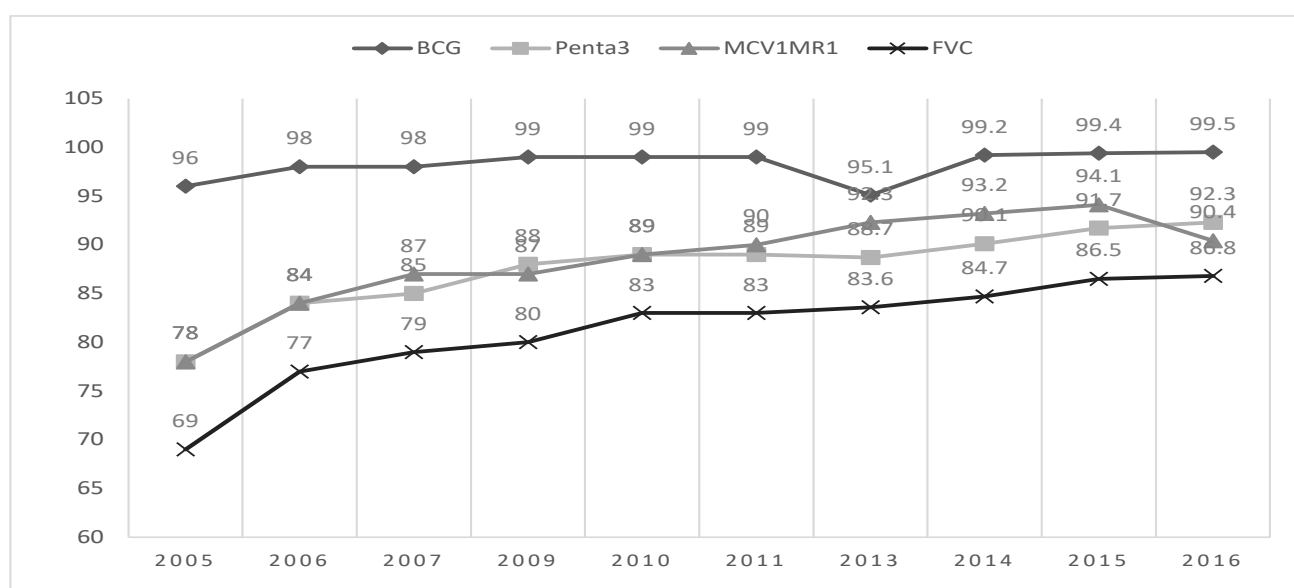


Figure 4.2.3. Annual trend of national-level valid vaccination coverage among children aged ≤23 months, from 2005 to 2016

VALID VACCINATION COVERAGE WITHIN THE AGE OF 23 MONTHS

- Valid coverage was defined as vaccines received by following the EPI-recommended age and dose interval for each antigen.
- Figure 4.2.3. shows the valid vaccination coverage among children aged ≤ 23 months since 2005. The Figure portrays a gradual improvement in valid vaccination coverage, which increased by 17.8 percentage points: from 69% in 2005 to 86.8% in 2016.
- Nationally, 86.8% of children received all the scheduled doses of all antigens, with BCG coverage being at 99.5%. Penta1 coverage was 97.9%, Penta2 97.2%, and Penta3 90.4%.

Moreover, MR1 coverage (92.3%) was revealed to be 7.2 percentage points lower than BCG (99.5%).

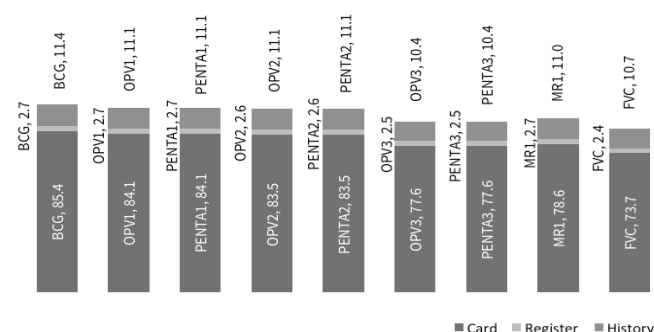


Figure 4.2.4. National-level valid vaccination coverage within the age of 23 months as revealed from cards, registers, and history

- It is evident from the Figure that, nationally, 82.3% of children received all the scheduled vaccines within the age of 12 months, following the EPI-recommended age and dose intervals for each antigen.
- The drop from BCG coverage (99.5%) to the 3rd dose of pentavalent administrations, Penta1 (97.8%), Penta2 (97%), and Penta3 (90.1%) was 9.4 percentage points. Valid MR1 coverage was 12 percentage points lower than for the BCG coverage.
- Administering vaccines without following the EPI-recommended minimum age and intervals caused invalid doses as well as drop-outs from BCG; and the subsequent dose of OPV and Penta vaccines attributed to lower MR1 coverage.

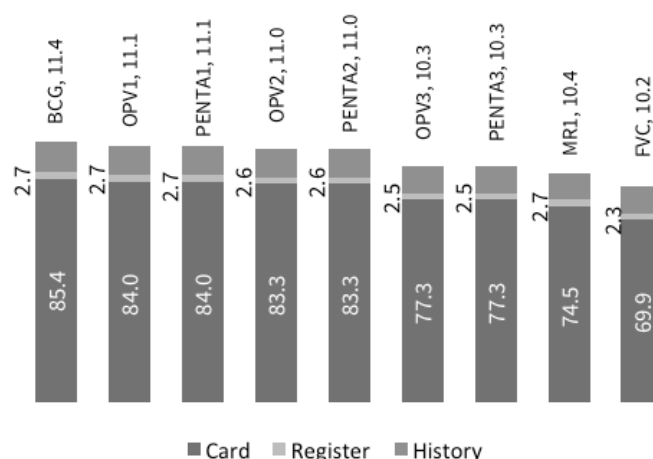


Figure 4.2.5. National-level valid vaccination coverage within the age of 12 months as revealed from cards, registers, and history in 2016

VITAMIN A SUPPLEMENTATION COVERAGE

- CES 2016 found that overall 86.3% of infants aged 6-11 months and 90.7% of children aged 12-59 months received vitamin A capsules, with a slight variation in proportions between urban and rural areas.

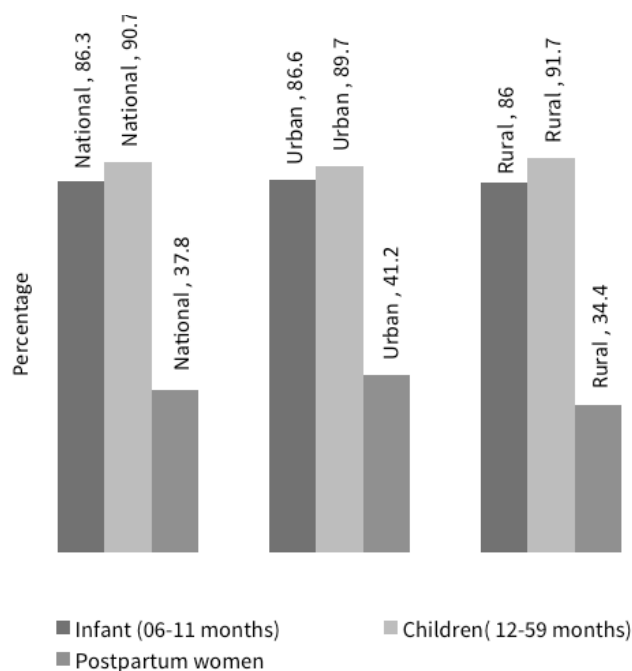


Figure 4.2.6. Vitamin A supplementation coverage among 6-11 months old infants, 12-59 months old children, and postpartum women in rural and urban area, with national average, in 2016

Communicable Disease Control Programs

INTRODUCTION

Bangladesh has been conducting a number of programs for prevention, management and elimination of major communicable diseases of public health importance. Some of these programs have successfully been able to eradicate, prevent and eliminate a considerable number of communicable diseases at the country level. Also, these programs have played a significant role in eradication of Polio at the global level. Bangladesh is now able to eliminate leprosy, visceral *leishmaniasis* (Kala-azar), lymphatic filariasis and tetanus among mothers and their neonates. The country has set targets to control and eliminate a number of other communicable diseases in the near future; these targets are specially aligned with SDG goals. During 2017 and 2018, there were outbreaks of a few communicable diseases: one Chikungunya and two measles outbreaks in Dhaka and Chittagong, and one Diphtheria outbreak among Forcibly Displaced Myanmar Nationals (FDMN) and the nearby Bangladeshi-host community of Cox's Bazaar. These outbreaks induced huge morbidity and a few deaths. However, due to prompt and constructive actions from DGHS, it has been possible to successfully control and manage all these outbreaks. The following section will present some of the important programs of these recent communicable disease control programs.

MALARIA ELIMINATION PROGRAM

- The incidence of malaria has been in decline for the last couple of years in Bangladesh since the introduction of malaria control program in the 1960s. This has been strengthened and reprogrammed over time.
- Out of 64 districts, malaria is now endemic in 13 districts with variable transmission potentials (high, moderate, and low).
- Historically, over 90% of the total cases are reported from three Chittagong Hill Tract districts (Rangamati, Khagrachhari, and Bandarban) and the coastal district of Cox's Bazaar, with low to very low endemicity in other districts.
- There has been consistent decrease in the number of cases and deaths due to malaria over the last one decade, other than an upsurge in 2014, due to an intensified EDPT, LLIN use and additional vector control measures through 'Hot Spot Approach' in the high endemic hilly districts.
- In 2017 and especially in 2018, there has been a significant reduction of malaria cases. Around 29 thousand cases in 2017, and around 10 thousand cases up to

Table 4.3.1 Malaria Situation, 2017

Status	Estimates
Risk Population	17.80 million
Endemic Districs	13
Total Cases	29,247
PF	80%
Pv	15%
Mixed	5%
Death	13
API	1.64/1000 populaiton
Mortality	0.07/10000 population
ABEr	5.54%
Test Positivity Rate	2.96%
Endemic Districs	13

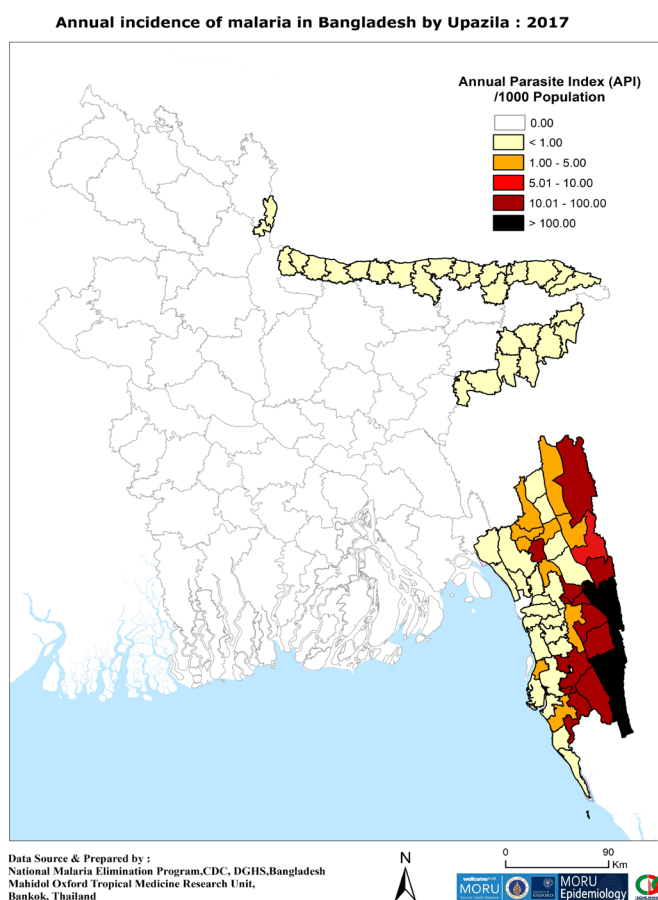


Figure 4.3.1 Malaria Endemcity

November 2018 have been reported .

- There are 65% reduction in morbidity and 92% reduction in mortality in 2017, compared to 2008. During the same period, 98.6% case reduction was achieved in 8 elimination-targeted districts.
- Over 85% of the total malaria cases were diagnosed and treated in the community.
- Only 2.3% of the total positive cases were severe in 2017.

DECLINE OF MALARIA DISTRIBUTION

- Malaria is becoming an increasingly focal disease in Bangladesh.
- In 2017, 3 of the 13 endemic districts had an Annual Parasite Incidence (API) greater than 1, compared to the year 2008.
- The three Chittagong Hill Tract districts (Khagrachhari, Rangamati and Bandarban) together accounted for ~93% of total confirmed malaria cases that have been detected in the 13 endemic districts.

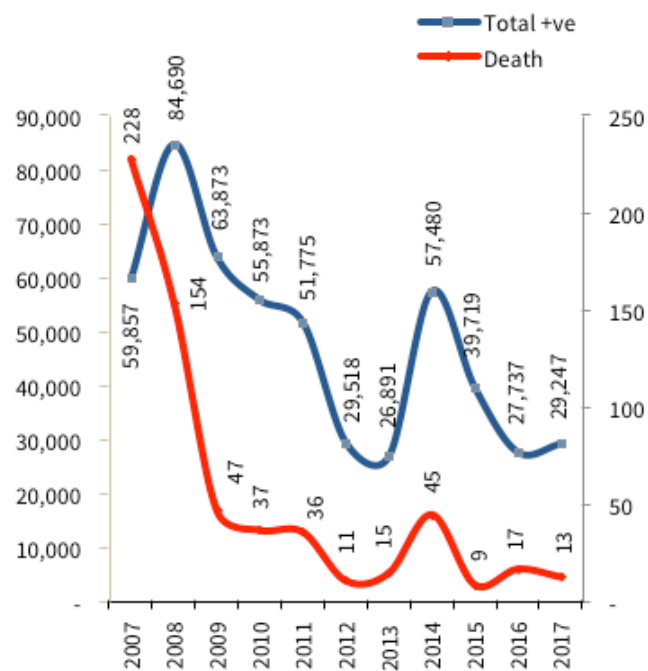


Figure 4.3.2 Decreasing trend of malaria cases and death tolls over the last decade

- The population of these three districts constitutes only around 12% of the total population at risk. Eight elimination-targeted districts reported only 126 cases in 2017.
- All those cases were investigated to identify the indigenous cases for further 'Foci Investigation and Responses'.
- Out of 13 malaria-endemic districts, four most endemic districts (Bandarban, Khagrachhari, and Rangamati and Cox's Bazar) have reported ~99% of total cases. In eight districts, out of total 13, death cases were reported among the populations.

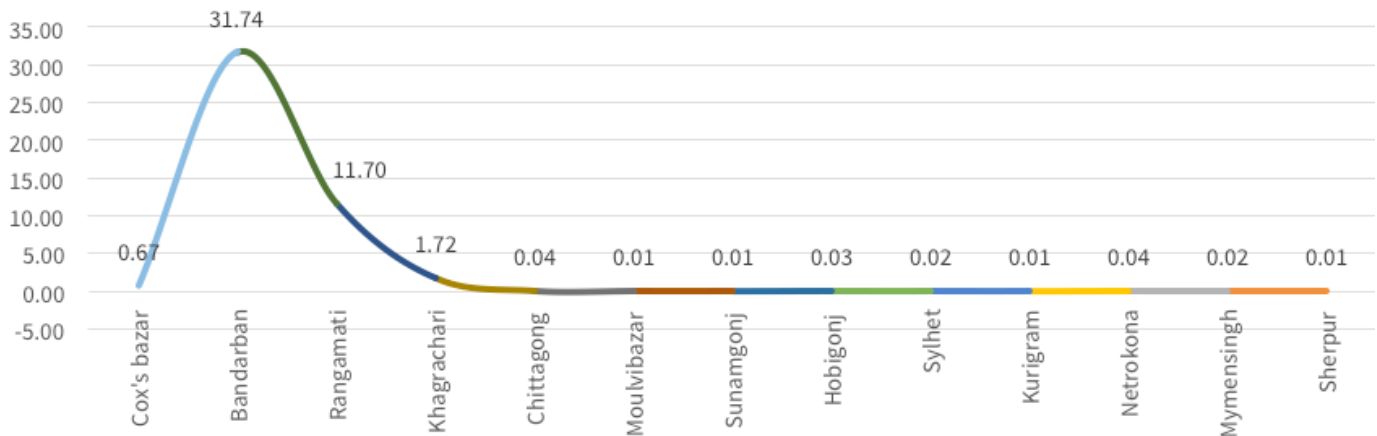


Figure 4.3.3 Concentrated endemicity in hilly districts

SPECIES DISTRIBUTION

- The large majority of malaria cases are caused by *Plasmodium falciparum*.
- However, a steady increase in *Plasmodium vivax* cases has been observed in the last few years.
- A total of 29,247 malaria cases were reported in 2017, and *falciparum* malaria accounted for 85% (including mixed cases) among them.

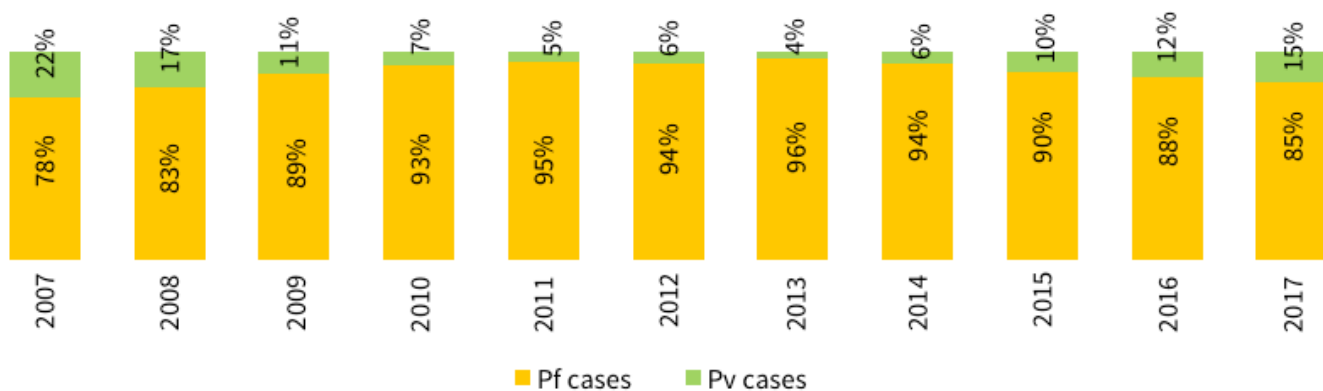


Figure 4.3.4 Proportion of Pf and Pv cases from 2007 to 2017

SEASONAL PATTERN

- The monsoon (May–September) is the peak period for malaria transmission.
- The following figure illustrates the seasonality of malaria transmission. Number of cases rises rapidly in June and it peaks between June and August.

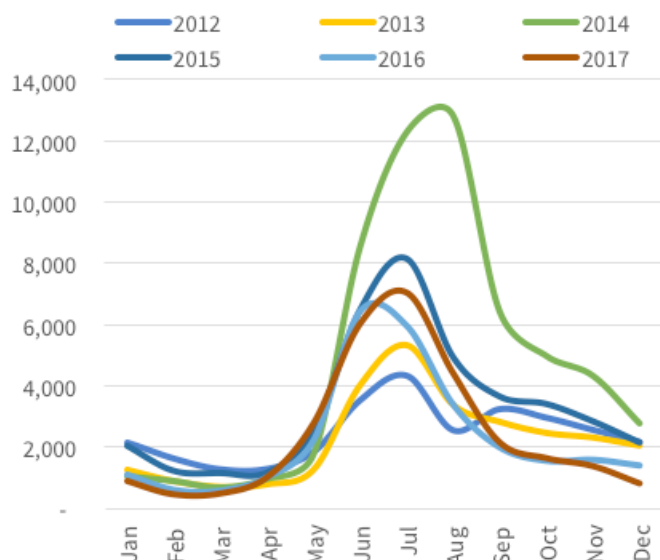


Figure 4.3.5 Seasonality of Malaria Transmission

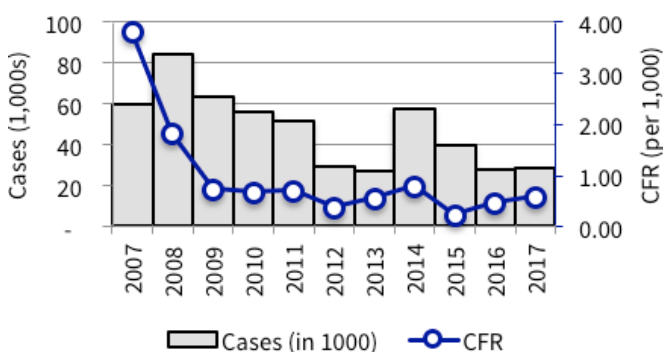


Figure 4.3.6 Case Fatality Rate

MALARIA CONTROL MEASURES FOR FDMN

- During the ongoing catastrophic humanitarian crisis in Cox's Bazar, the national program has been highly vigilant in implementing malaria control and prevention activities among Forcibly Displaced Myanmar Nationals (FDMN).

- The national program executed its operation in Health Camps for FDMN through mainstream health operation of Directorate General of Health Services, in coordination with partner NGOs and other non-governmental health agencies.
- Suspected malaria cases among FDMN populations are being diagnosed and 14 positive cases have been treated accordingly. Adequate LLIN, antimalarial and diagnostic kits are ensured for them.
- Regular entomological surveys are conducted in camp areas.
- Additional human resources have been recruited through partners, and awareness messages are disseminated in their native language.

‘MALARIA FREE BANGLADESH’ INITIATIVES

In consideration of the prevailing situations and keeping the target achievements in mind, the National Strategic Plan for Malaria Elimination 2017-2021 has been drafted by setting the vision of “A Malaria Free Bangladesh by 2030”, aligned with the “Global Technical Strategy for Malaria 2016-2030”.

OBJECTIVES

1. To reduce Annual Parasite Incidence in the 13 ‘endemic’ districts, from current status to less than 0.46, by 2021.
2. To interrupt the transmission of malaria in eight of the country's 13 ‘endemic’ districts by 2021.
3. To ensure that the remaining 51 districts are free from malaria transmission by 2021.
4. To prevent the re-establishment of malaria infection in districts where transmission has been interrupted.
5. To prevent the emergence of ACT resistant *Plasmodium falciparum* in Bangladesh

Over the next five years, Bangladesh aims to eliminate malaria in ‘low endemic areas’, while accelerating control efforts in ‘high endemic areas’. After 2025, it is expected that all areas will either be targeted for elimination, or for prevention of reintroduction, so that by 2030 Bangladesh will be malaria free.

LLIN DISTRIBUTION

- Distribution of Long-Lasting Insecticidal Nets (LLIN) among the risk population is one of the key interventions in malaria prevention; between 2008 to 2017, ~9.13 million LLINs have been distributed at free of cost among the beneficiaries.
- Among them, 2.24 million LLINs was distributed in 2017 only.
- A new initiative of distributing additional LLINs among high risk groups (jhum cultivators, armed forces, pregnant women, etc.) has been started in 2017.

ACSM

- As part of Advocacy, Communication and Social Mobilization (ACSM) activities, various initiatives has been

undertaken by the program to increase awareness among the mass population.

- Awareness programs are broadcasted from local radio stations in Bangla, Chakma and Marma languages.
- Awareness raising messages on malaria prevention and treatment are telecast (in the form of message scrolling) from local satellite channels, in 3 CHT districts.
- Regular community drives and “uthanboithoks” (court yard meetings where the local people meet and greet to share their opinions) are conducted. Advocacy meetings are conducted at all levels.
- World Malaria Day is observed on 25th April each year. A malaria hotline number (01787-691370) has been introduced in 2017.

AEDES TRANSMITTED DISEASE CONTROL PROGRAM

BACKGROUND

- Three diseases have been included under Aedes transmitted disease control program. These are: Dengue, Chikungunya and Zika.
- Dengue occurred sporadically in Bangladesh from 1964, until a large epidemic occurred in 2000 which established the virus.
- A large number of cases were notified in 2002. Since then, reported outbreaks have generally declined with increased notifications in subsequent years.
- However, in 2018, dengue has appeared all over the country in an epidemic form, causing serious health hazards. This happened because of the outbreak of a new mutated strain.
- There was a huge outbreak of Chikungunya in Dhaka city in 2017. Therefore, it has been added as another Aedes transmitted disease in Bangladesh.
- There is no indigenous transmission of Zika virus in Bangladesh. Nonetheless, one imported case was reported in 2016.

- In the future, the inadequate active interventions, unplanned urbanization, environmental deterioration, increasing population mobility, and economic factors may heighten the risk of these Aedes transmitted diseases, especially Dengue.

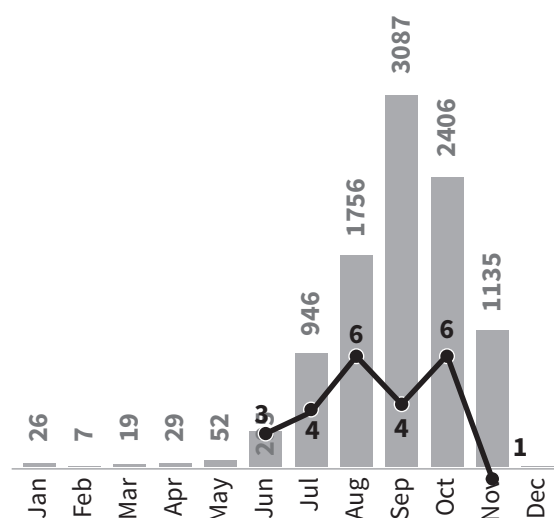


Figure 4.3.7 Monthly Trend of Dengue Cases and Deaths in 2018 up to November

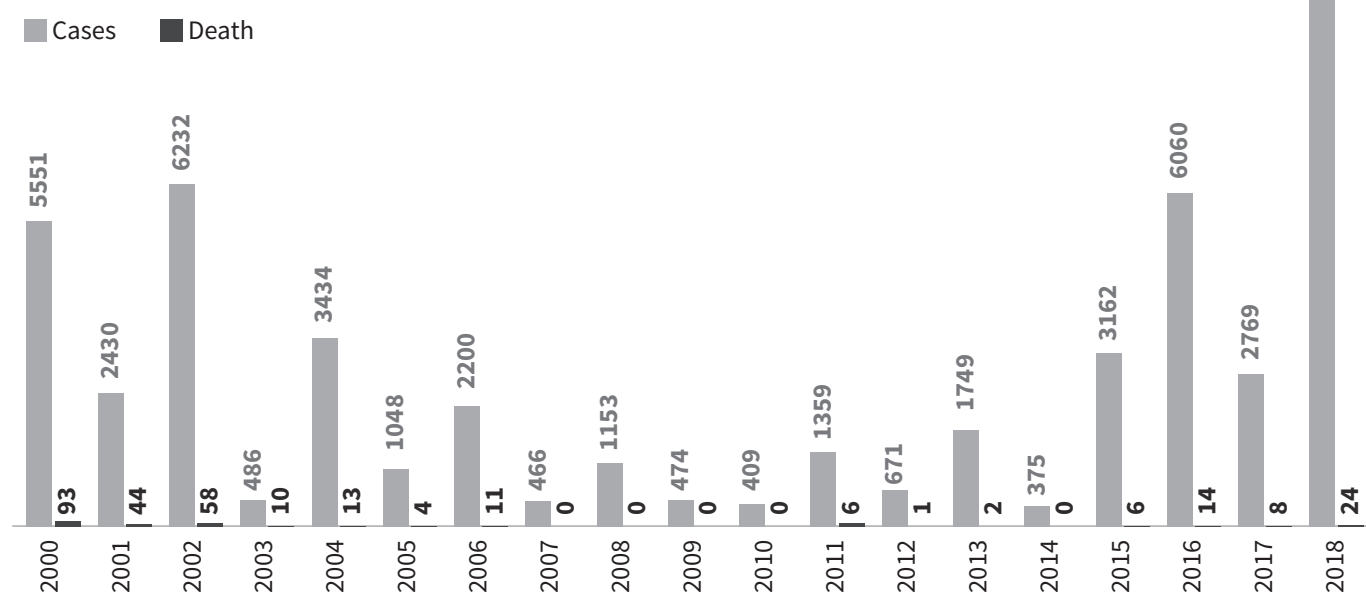


Figure 4.3.8 Dengue Cases & Deaths by year

PREVENTION AND CONTROL PROGRAM FOR DENGUE

At present, the only method of controlling or preventing dengue and Dengue Hemorrhagic Fever is to combat the vector mosquitoes. *Aedes aegypti* and *Aedes albopictus* mosquitoes are the two vectors responsible for dengue. Aedes mosquito breeds in clean water. Awareness is the key to control this disease outbreak.

CAPACITY BUILDING

- Training must be provided to all doctors at all levels, starting from consultants, registers to MOs and also to senior staff nurses who are engaged in clinical management of dengue and chikungunya.
- The guidelines on clinical management of dengue & chikungunya have been prepared

LAB CAPACITY DEVELOPMENT:

- Distribution of rapid diagnostic kits (both dengue & chikungunya) is done

EPIDEMIOLOGICAL AND ENTOMOLOGICAL SURVEILLANCE

- The surveillance is carried out throughout the year
- Breeding sources of mosquitoes are reduced through Community Drives and engagements

ADVOCACY, COMMUNICATION AND SOCIAL MOBILIZATION (ACSM)

- During the initial stage of the dengue epidemic, awareness building programs of the community was undertaken, in a large scale. .
- Renowned personalities of the society came forward, distributed leaflets, organized rallies
- CDC, DGHS also organized rallies and distributed leaflets among the city dwellers.
- CDC, DGHS coordinated awareness activities with the city corporations, NGOs and other relevant stakeholders.
- A technical group was formed to update the guideline, based on evidences and field experiences.
- National Guideline for Clinical Management of Dengue Syndrome (4th edition.) was updated in 2018

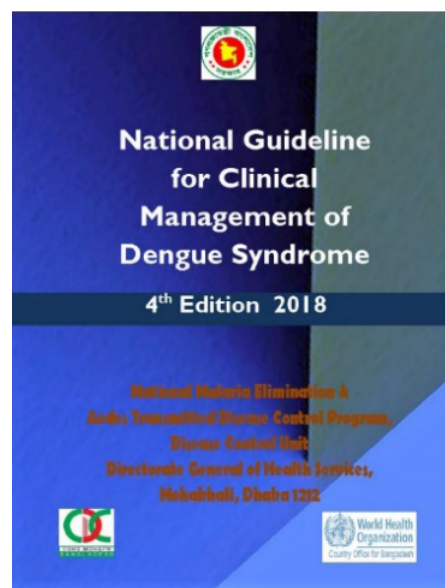


Figure 4.3.9 Guideline for the clinical management of Dengue

CHIKUNGUNYA CONTROL MEASURES

BACKGROUND

- Chikungunya is a viral disease caused by a virus transmitted by Aedes mosquitoes (*Ae. Aegypti* & *Ae. albopictus*).
- Since this disease is caused by Aedes mosquito, its epidemiology is very much similar to dengue fever.
- *Aedes aegypti* is the main vector of transmission of chikungunya in Bangladesh.
- The first outbreak of chikungunya in Bangladesh was identified in 2008.

PUBLIC HEALTH MEASURES

Infected patients can spread the infection by spreading the virus through mosquitoes. It is important to break this transmission by minimizing the vector density by community participation, and by taking appropriate control measures in the hospital setting by adopting the following measures:

- To conduct risk communication among the household members
- To minimize the vector population



Figure 4.3.10 Vector control and public awareness activity for chikungunya

- To minimize the vector-patient contact (Aedes mosquitoes bite during day time, mostly in the mornings and late afternoons)
- To report the infection to the nearest public health authority.

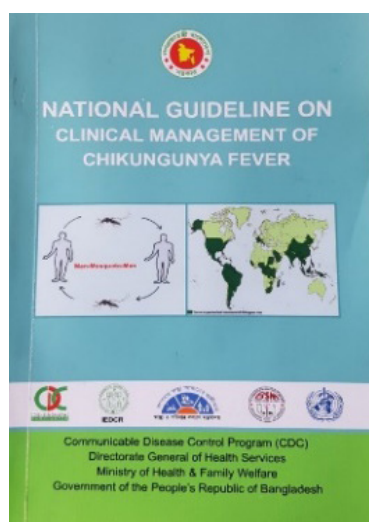


Figure 4.3.11 National guideline for the management of chikungunya

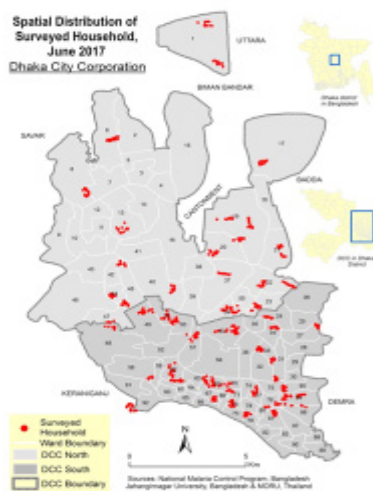
RISK COMMUNICATION TO THE HOUSEHOLD MEMBERS

- Chikungunya is a disease that is transmitted by mosquitoes.
- House hold members may also be infected by chikungunya as they share the same environment.
- Hence, there is no need to isolate the patient or to segregate the patient.
- Rather it is important to reduce the vector population inside the infected house.

VECTOR SURVEY

- The Entomological Department of CDC, DGHS and Students of the Zoology Department (Entomology), Jahangir Nagar University, has jointly done pre and post monsoon vector density surveys in Dhaka North and South City Corporations and also in Chittagong City Corporation.
- Also, they have distributed the awareness leaflets to house owners and helped them to learn about the mosquito breeding sites management.

751 households, 01 - 5 June, 2017



1957 households, 19-28 July, 2017

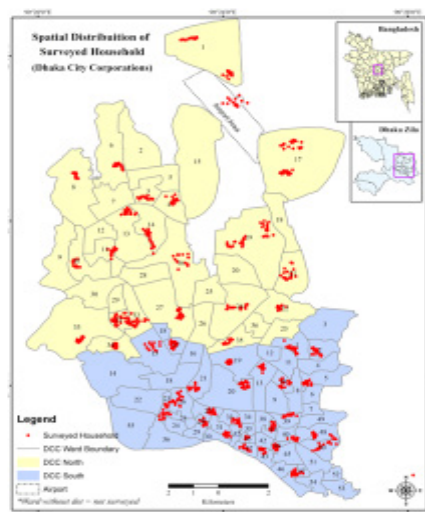


Figure 4.3. 12 Household survey in 2017

Area	BI	HI	Area	BI	HI
Banani	100	60	Monipuripara	70	35
Mohammadpur	90	45	Pirerbagh	65	35
Dhanmondi 32	80	60	Dhakeshori Road	60	50
Kathalbagan	80	55	Dhanmondi 1	60	55
Noyatola	80	65	Shantinagar	60	45
Tolarbagh	80	55	Shakhari Bazar	55	40
Bakshi Bazar	75	55	Siddik Bazar	55	30
Mayakanon	75	50	Lalmatia	50	70
Mintu Road	75	35	Nakhalpara	50	35
Uttara-4	75	55			
Baridhara DOHS	70	50			
Elephant Road	70	50			

Figure 4.3.13 Breteau and house indices in different areas of DCC

CONCLUSIONS

- The spatial distribution and density of *Aedes aegypti* was far higher than that of *Aedes albopictus*.
- Monsoon larval indices were variable across many areas of Dhaka, however they were relatively high everywhere.
- The relationship among larval, pupal and adult stages of the mosquito vector and the factors influencing *Aedes* survival rates are poorly understood.

RECOMMENDATIONS

- A continuous national integrated entomological surveillance system should be established to have two outcomes. One is to monitor longitudinal trends more effectively, and the other one is to yield evidence to effectively target delivery of vector control interventions (under IVM) to reduce arboviral transmission.
- Combined with case-based surveillance and meteorological data, this system should provide early warning alerts in order to prevent outbreaks through the use of vector control rapid response teams.
- A national Communication for Behavioral Impact (COMBI) program needs to be established. This can be multi-faceted in nature, targeting different sub-groups (e.g. primary/secondary school children through educational activities) through a range of media channels.

Zone	HC	+H	WC	+WC	BI	CI	HI	PI
North City Corporation	779	215	1850	275	35.30	14.86	27.60	10.65
01	80	22	104	26	32.50	25.00	27.50	0.00
02	160	31	435	40	25.00	9.20	19.38	0.00
03	220	70	582	85	38.64	14.60	31.82	23.18
04	139	42	357	55	39.57	15.41	30.22	21.58
05	180	50	372	69	38.33	18.55	27.78	1.11
South City Corporation	1178	275	3081	361	30.65	11.72	23.34	22.24
01	200	65	654	89	44.50	13.61	32.50	22.00
02	207	53	554	67	32.37	12.09	25.60	53.62
03	260	57	700	74	28.46	10.57	21.92	13.08
04	221	44	486	67	30.32	13.79	19.91	22.62
05	290	56	687	64	22.07	9.32	19.31	7.93
Total	1957	490	4931	636	32.50	12.90	25.04	17.63

Figure 4.3.14 Different indices by zones of North and South Dhaka City Corporation

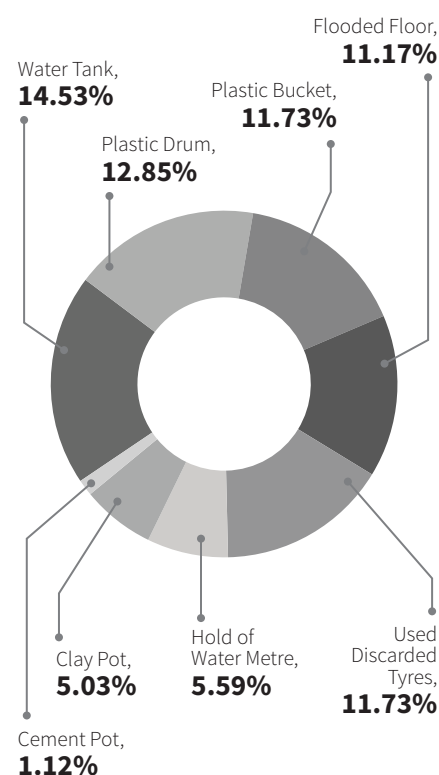


Figure 4.3.15 Breeding Sources of aedes mosquito

FILARIASIS ELIMINATION PROGRAM

BACKGROUND

- Wuchereria bancrofti is the most common parasite, and Culex mosquitoes are the main vectors for transmission of lymphatic filariasis (LF) in Bangladesh.
- Lymphatic Filariasis is a leading cause of permanent and long-term disability worldwide; hence, WHO has targeted it as one of the seven communicable diseases for elimination by the year 2020.
- The target of Bangladesh was to eliminate the disease by 2015 through mass drug administration (MDA) and morbidity control.
- Out of 64 districts of the country, it was endemic in 34 (based on ICT survey); 19 out of 34 districts were eligible for mass drug administration during 2001-2004.
- MDA was launched in November 2001 (Round I) at Panchagarh district; afterwards, it was scaled up in 19 districts by 2008, following the baseline survey of the area.
- According to the WHO protocol, at least five years' surveillance, after "Stop MDA" with no sign of transmission or sign limited to cut off value, is required to declare the elimination of the disease.
- MDA has been stopped in all the 19 eligible endemic regions.

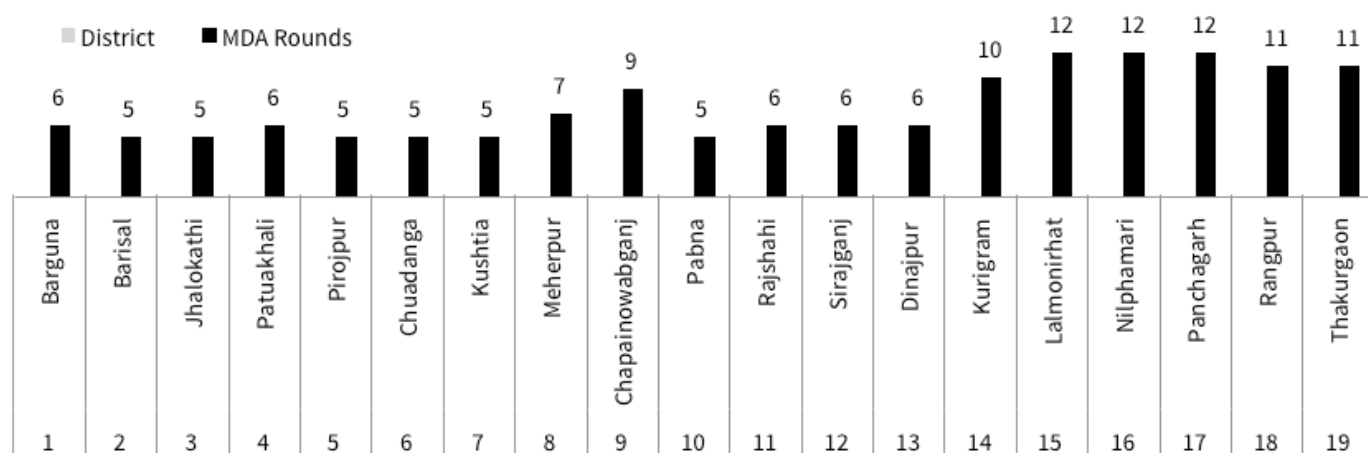


Figure 4.3.16 MDA rounds in 19 endemic districts during 2001-2015

GOAL, OBJECTIVES, AND STRATEGIES

- The goal of the program is elimination of Filariasis as public health problem by 2021, and the objectives are to reduce microfilaria prevalence to <1% as well as to give relief to the patients of lymphedema caused by filariasis.
- Strategies adopted to achieve this goal and objectives are: mass drug administration
- (MDA) is conducted among the at-risk population, once a year, for successive five years through door-to-door household registration. Apart from, pregnant women, children aged < 2 years, and severely-ill patients, this is done to all for the alleviation of sufferings of lymphedema patients by community-based morbidity control.

Table 4.3.2 MDA rounds from 2001 to 2015

Year	District/ Implementation unit (IU)	Total population (Million)	Coverage (%)		
			Reported	Survey	Actual (Eligible)
2001	1	0.81	95.5	93.0	ND
2002	4	5.18	93.6	83.2	87.3
2003	6	8.73	93.3	77.9	81.9
2004	10	11.75	98.6	ND	ND
2005	12	20.16	90.3	78.0	82.2
2006	13	23.92	92.2	78.2	82.2
2007	17	31.0	91.5	82.4	84.3
2008	20	42.0	90.53	79.38	83.06
2009	19	35.0	96.87	83.33	85.76
2010	19	35.0	92.47	60.23	62.98
2011	14	29.70	97.14	92.35	94.90
2012	09	16.67	98.11	89.76	92.78

Year	District/ Implementation unit (IU)	Total population (Million)	Coverage (%)		
			Reported	Survey	Actual (Eligible)
2013	04	8.66	98.86	88.9	93.26
2014	01	3.01	81.72	69.5	90.33
2015	01	3.40	79.83	67.3	93.93

TRANSMISSION ASSESSMENT SURVEY (TAS)

- Transmission Assessment Survey (TAS) is the WHO-recommended survey protocol to assess the status of elimination to stop MDA.
- As per WHO protocol, MDA has been stopped in 19 districts.
- The outstanding question for Bangladesh's LF Program was how to assess the 15 endemic districts that were found to have low prevalence (<1%) and not eligible for MDA.
- The results of the TAS conducted to date showed the promising signs that the National LF Program would be able to 'shrink the LF map' by approximately 38 million people, and in this way, it can start to move one step closer to the elimination goal.

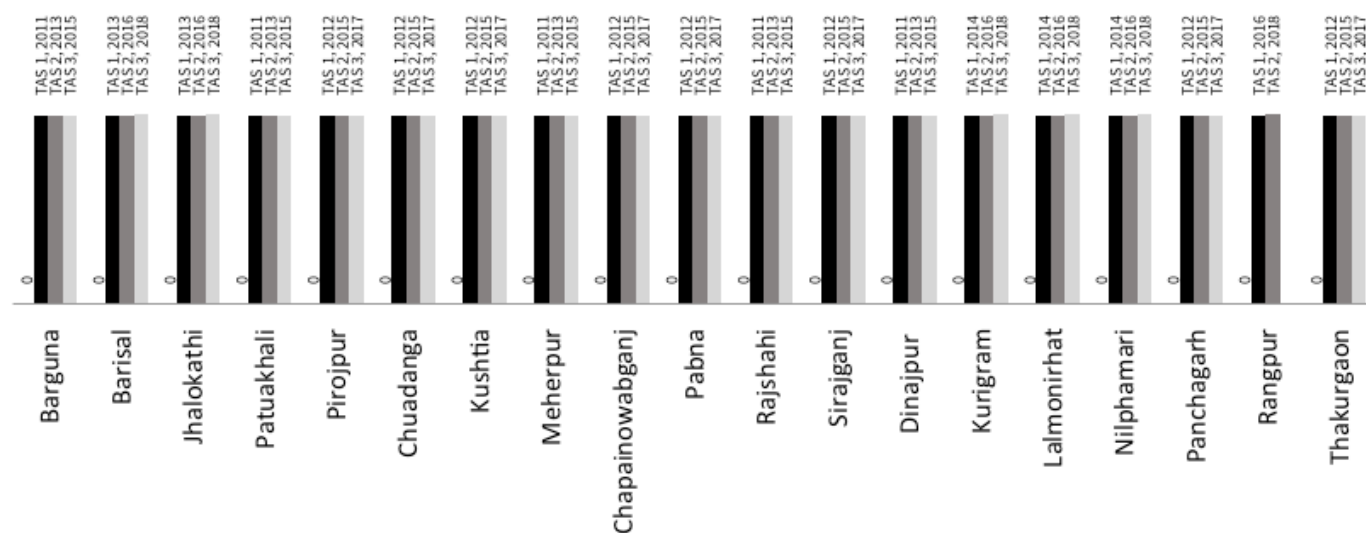


Figure 4.3.17 TAS status in 19 districts

MORBIDITY MANAGEMENT AND DISABILITY PREVENTION (MMDP)

- In the 19 highly-endemic districts, 8,145 staff members of the community clinics (CC) were trained to search for LF cases, during 2013-2016, by visiting each household in their catchment areas and registering all cases.
- In the 15 low-endemic districts, a team of 10 trained field assistants conducted active case finding. By utilizing health facility records and by engaging the local health workers and community informants, they tried to identify

and medically verify the cases. These also included the cases that were reported via SMS to mHealth database.

- Morbidity Management and Disability Prevention (MMDP) services for filariasis patients have been established in 19 endemic districts of the country.
- A total of 120 surgeons and medical officers have been trained on hydrocele surgery during the year.
- A total of 4384 CHCP and HA have been trained on MMDP services from 2192 community clinics of 12 endemic districts.

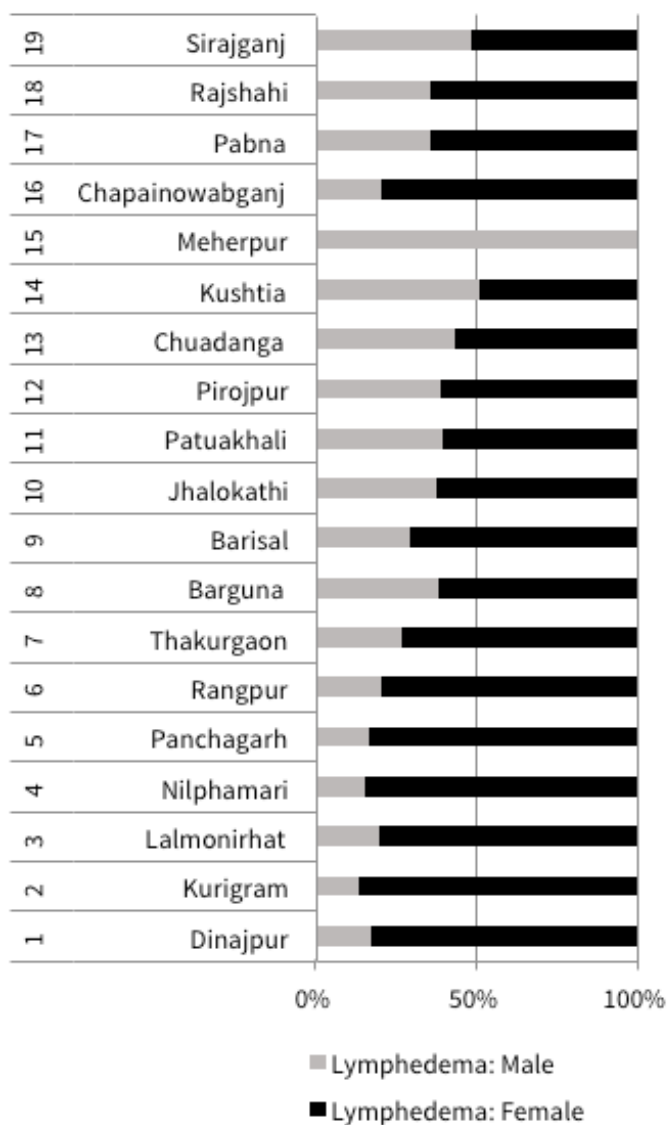


Figure 4.3.18 Lymphedema patients showing district-wise proportions of males and females

ACHIEVEMENTS

- During 2004-2012, reportedly 8,500 hydrocele surgeries were done by the program
- At present, the estimated number of hydrocele sufferers in 19 districts is approximately 12,824
- In 2016, one hundred forty-three hydrocele surgeries were performed in Panchagarh district
- In 2017, the program aimed to perform 575 hydrocele surgeries in Panchagarh and Nilphamari districts
- In 2018, 1075 hydrocele surgeries have been performed

Table 4.3.3 Distribution of patients by districts

District	Lymphedema	Hydrocele	Other*	Total patients
Dinajpur	2,044	1,021	2	3,067
Kurigram	1,239	699	20	1,958
Lalmonirhat	5,383	1,937	-	7,320
Nilphamari	5,475	1,715	50	7,240
Panchagarh	1,341	1,634	5	2,980
Rangpur	8,532	2,654	13	11,199
Thakurgaon	2,767	2,001	50	4,818
Barguna	542	466	15	1,023
Barisal	427	50	1	478
Jhalokathi	592	49	-	641
Patuakhali	423	202	11	636
Pirojpur	314	31	6	351
Chuadanga	53	5	1	59
Kushtia	145	7	7	159
Meherpur	0	0	0	0
Chapainowabganj	693	201	18	912
Pabna	229	67	20	316
Rajshahi	209	26	12	247
Sirajganj	208	59	7	274
Total	30,616	12,824	238	43,678

*Others include breast lymphedema and female genital swelling

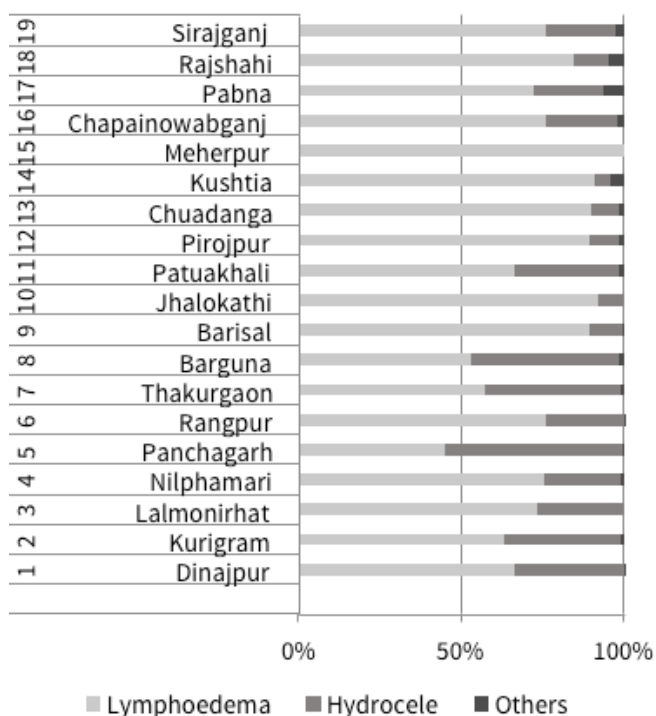


Figure 4.3.19 Lymphedema, hydrocele and other patients in endemic districts

SOIL-TRANSMITTED HELMINTHS (STH) CONTROL PROGRAM

BACKGROUND

- Soil-transmitted helminths (STH) control is an important component of the Filariasis Elimination Program of the Communicable Disease Control Unit of the DGHS.
- The STH Control Program has been integrated with Filariasis Elimination Program, with the aim to minimize its operation cost.
- The nationwide school-based deworming program has been started in 2008, with an aim of regular deworming among 75-100% of school-age children (World Health Assembly Resolution 54.19 of 2001).
- The 'Little Doctor' program (involving school students) is an initiative under STH Control Program.
- It has been introduced nationwide through all primary and secondary level of educational institutions for maintaining peer-education in STH control and developing health-related well-being, followed by regular and proper hygiene practices.
- There are three 'Little Doctors' (selected students) in each team, and a team is assigned for each class or section of the institutions.
- It is estimated that there will be a total of 22,50,000 'Little Doctors' each year from about 1,50,000 primary and secondary level institutions of the country.

GOAL OF THE SCHOOL-BASED DEWORMING PROGRAM

To control intestinal helminths among school-aged children of 5-12 years.

GOAL 2017-2021

- To eliminate intestinal helminths among children (reducing moderate to heavy intensity to <1%)

OBJECTIVE

- To reach 75%-100% school-aged children under the deworming program (according to World Health Assembly Resolution 54.19)
- To educate school children about health and hygiene practices

ACHIEVEMENTS

- Every primary-level institution in the country is covered under the school-based deworming program to deworm all school-aged children aged 5-12 years, twice a year (April and October), till the year of 2016, thereafter children aged 5-16 years.
- Table 4.3.4 shows the reports of deworming coverage as sent by the civil surgeons' offices of the concerned districts.



Figure 4.3.20 Little Doctors



- STH prevalence was found higher in the survey report of 2017-2018. Previous survey was conducted among the only school children of grade five.
- However, this survey of 2017-2018 was conducted in the community among 9 districts to cover three different age groups: PSAC- 1- 4 years, SAC – 5 – 14 years and adult 14 years and above.

Table 4.3.4 Reported coverage of school-based deworming program (2008–2016)

Year	Round	No. of districts covered	No. of children		Reported coverage (%)
			Targeted	Treated	
2008	November	64	15,743,159	15,482,778	94
2009	May	64	19,303,404	19,101,496	98
	November	64	19,303,404	18,782,212	97
2010	May	64	19,837,612	19,440,860	98
	November	64	2,19,71,611	2,17,45,757	98.97
2011	May	64	2,20,70,512	2,17,35,040	98.48
	November	64	2,20,82,923	2,19,92,383	99.59
2012	May	64	2,22,63,213	2,20,40,581	99
	November	64	2,22,63,192	2,20,38,334	98.99
2013	April	64	2,49,86,323	2,47,99,113	99.25
	October	64	2,50,89,864	2,48,98,332	99.23
2014	April	64	2,48,86,323	2,46,98,576	99.25
	October	64	2,50,90,960	2,49,93,205	99.61
2015	April	64	3,12,81,837	2,67,04,597	85.37
	October	64	3,13,46,057	2,66,00,439	84.86
2016	April	64	3,12,81,837	2,63,96,187	84.38
	October	64	3,12,81,837	2,68,34,605	85.78
2017	April	64	4,05,56,232	4,02,87,201	99.33
	October	64	4,05,56,232	4,04,96,342	99.85
2018	April	64	4,06,45,956	4,03,35,323	99.23
	October	64	4,06,45,956	4,03,42,521	99.25

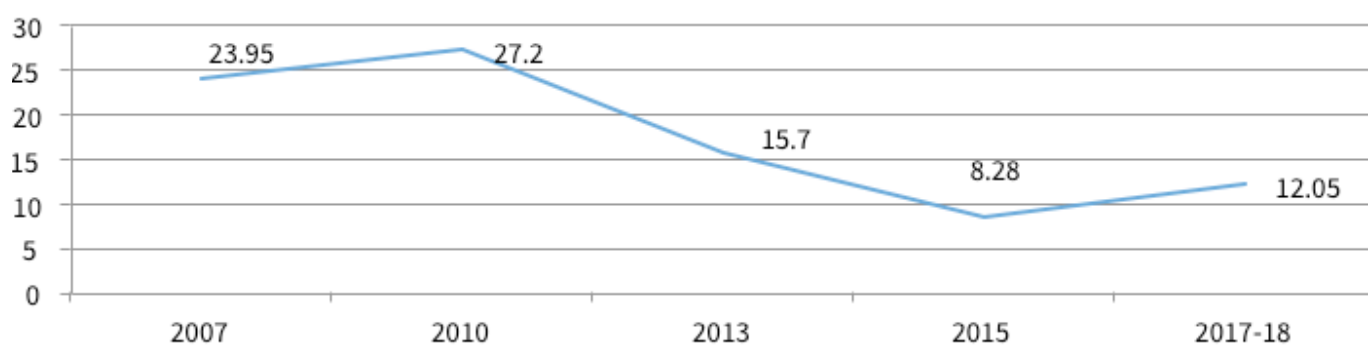


Figure 4.3.21 STH prevalence among school children in different years

ANTIMICROBIAL RESISTANCE CONTAINMENT (ARC)

BACKGROUND

- Antimicrobial Resistance (AMR) is a growing global public health threat.
- In concordance with the global and WHO activities on ARC, the Ministry of Health and Family Welfare (MoHFW) in Bangladesh has come forward and formulated different committees with representation from multi-sectoral stakeholders, including inter-ministerial National Steering Committee (NSC), headed by honorable Minister for MoHFW, and National Technical Committee (NTC) headed by Director General of Health Services.
- Subsequently, National Strategy for Antimicrobial Resistance Containment in Bangladesh was developed and approved by the NSC and the NTC, with further recommendation of developing a National Action Plan (NAP).

OBJECTIVES OF THE NATIONAL STRATEGY FOR ARC:

1. To establish multi-sectoral approach for planning, coordination and implementation of antimicrobial resistance containment (ARC) activities
2. To promote and ensure rational use of antimicrobial agents in human health, livestock and fisheries sectors
3. To promote and strengthen infection prevention and control measures to minimize the emergence and spread of AMR
4. Promoting and strengthening biosafety and biosecurity principles and practices as well as containment measures
5. To review, update and strengthen regulatory provisions
6. To strengthen surveillance system for AMR containment
7. To promote operational research and education in the area of AMR
8. To establish Advocacy, Communication and Social Mobilization (ACSM) for ARC activities

TARGET OF THE PROGRAM

1. To ensure rational use of AMs by
 - a. Developing a standard treatment guideline (STG) to ensure adherence to STG
 - b. Establishing and Strengthening the national reference laboratory and regional microbiology laboratories
 - c. Ensuring GLP at all microbiology laboratories
- d. Updating STG annually based on AST data

- e. Developing antibiotic policy
 - f. Ensuring availability of AMs at all healthcare facilities (HCFs)
 - g. Enforcing and enhancing regulatory provisions for the use of AMs
 - h. Ensuring good pharmacy practice (GPP)
2. To ensure good manufacturing practice (GMP) at pharmaceutical industries
 3. To ensure infection prevention and control (IPC) activities at HCF & community level
 4. To establish an efficient surveillance system by setting up
 - a. A web-based surveillance and laboratory networking for AMR
 - b. A surveillance system for AMs use
 5. To conduct ACSM for healthcare providers (HCPs), Drug sellers, Community members, Animal and fisheries farm owners
 6. To ensure adequate funding for basic and operational research

MAJOR ACTIVITIES IN 2017

1. To plan a draft of five years National Action Plan (NAP) on ARC, User Guideline and Laboratory SOP and Training Module
2. To conduct seminars on Anti-microbial Resistance Containment (ARC) with stakeholders, in the presence of the honorable State Minister of MOHFW
3. To include Anti-microbial Resistance Containment (ARC) as different programs in new Operational Plan (OP)
4. To conduct a laboratory Based Surveillance on AMR pattern. under the agreement of Global Health Security Agenda (GHS), CDC, USA
5. To conduct hands on training on “Anti-microbial Susceptibility Test” on December 2015
6. To provide laboratory instrument to operate the Laboratory Based Surveillance on AMR pattern
7. To ensure the involvement of mass media to increase awareness among the mass people about Anti-microbial Resistance

MAJOR ACHIEVEMENTS

First situation analysis on AMR of Bangladesh is done with the help of GARP, Bangladesh.

VIRAL HEPATITIS

BACKGROUND

- Globally viral hepatitis is the seventh leading cause of death, accounting for 1.4 million deaths per year.
- It has been estimated that more than 10 million people are living with viral hepatitis in Bangladesh.
- Liver diseases account for 8-12% admissions in medicine wards and remain the third leading cause of death in tertiary medical college hospitals of the country, which is approximately more than 20,000 deaths per year.
- Hepatocellular carcinoma is the third leading cause of cancer related deaths in Bangladesh, next only to cancers of lungs and stomach.
- By 2030, globally, 20 million people are expected to die from hepatitis B, and the number of people living with hepatitis C is increasing, despite an existing and effective cure.

GOAL

To prevent and control viral hepatitis, in order to reduce 90% of viral hepatitis related morbidity and mortality from Bangladesh by 2030.

OBJECTIVES

1. To establish multi-sectoral approach for planning, coordination and implementation of viral Hepatitis prevention and control activities
2. To promote and strengthen viral hepatitis prevention and control measures
3. To promote and strengthen bio-safety and bio-security principles and practices and containment measures
4. To strengthen national surveillance system for viral hepatitis containment
5. To promote and ensure rational use of anti-viral agents
6. To review, update and strengthen regulatory provisions
7. To establish Advocacy, Communication and Social Mobilization (ACSM) for viral hepatitis prevention and control activities
8. To promote operational research and education in the area of viral hepatitis prevention and control
9. To adapt special care system for viral hepatitis in

pregnant and pediatric population

10. To adapt special care system for viral hepatitis related advanced liver disease patients
11. To adapt special measures for people who inject drugs (PWID)
12. To develop training module and standard guideline for management of viral hepatitis
13. To develop international cooperation and collaboration with experts, professional bodies and development partners
14. To introduce adult vaccination program for hepatitis B and E patients

TARGET OF THE PROGRAM

1. To develop a National Guideline
2. One of the principal concerns in tackling the issue of viral hepatitis is irrational use of anti-viral drugs
3. To ensure Good Manufacturing Practice (GMP) of antivirals. Also, the quality of anti-viral is crucial for the control of viral hepatitis.
4. To set up Prevention and Control (IPC) Measures for Nosocomial Viral Hepatitis
 - The IPC measures in healthcare facilities reduce the risk of nosocomial transmission of viral hepatitis.
 - These measures include hand hygiene, use of personal protective equipment (PPE), maintaining a clean environment, cleansing and sterilizing (or high level disinfection) of instruments, improving safety in risky areas of the health facility, safe use and disposal of sharps medical waste, appropriate waste disposal and vaccination.
5. Surveillance

Surveillance for viral hepatitis will provide information on the magnitude and trends in viral hepatitis and for monitoring the effect of interventions. It is necessary to formulate policy decisions, update national formularies, inclusion of antivirals in the lists of essentials and ensuring availability of appropriate antivirals.
6. To monitor the use of Viral Hepatitis
7. To monitor the use of antivirals

8. To monitor the manufacture of quality assured antivirals
9. To conduct research and education
10. Researches on use of antivirals, viral hepatitis and hepatitis viruses and IPC are essential for prevention and control of viral hepatitis. Educational works are needed to alleviate the irrational use of antivirals.
11. To sustain Advocacy, Communication and Social Mobilization (ACSM)
12. Antiviral use is influenced by the interplay of knowledge, attitude and interactions of prescribers, patients and promoters of manufacturers as well as the characteristics of the health system(s) and the regulatory environment. In this context, coordinated interventions are needed to encourage rational use of antivirals.

MAJOR ACTIVITIES IN 2017

- Establishing a multi-sectoral approach for planning, coordination and implementation of Viral Hepatitis

FOOD AND WATER BORNE DISEASES (F&WBD)

BACKGROUND

- Food and water borne diseases are an important cause of morbidity and mortality worldwide. An estimated 600 million or almost 1 in 10 people in the world fall ill after eating contaminated food and 420,000 die annually. This results in the loss of 33 million daily adjusted life years (DALYs).
- Forty percent of the food borne disease burden is carried by children under 5 years of age, with 125,000 deaths every year (WHO, 2017)
- About 663 million people in the world are still surviving without an access to clean drinking water, despite the MDG target for clean water being met in 2010
- Waterborne diarrheal diseases, for example, are responsible for 2 million deaths each year, with the majority of the ailment occurring in children under 5 (WHO, 2017)

GOAL

Reduction of the burden of food and waterborne diseases to less than 50% from the baseline (2017) by the year 2022.

activities

- Training of more than 1000 Doctors on viral hepatitis of different districts was given.
- Training of 100 Doctors and Nurses, on Advance management of viral hepatitis of Chittagong, was conducted.
- Updating National Strategy and budgeted National Action Plan for Viral Hepatitis
- Building awareness through advocacy, communication and social mobilization
- Conducting Hepatitis B diagnosis and providing vaccines among 250 health care providers at free of cost
- Distributing 8000 drugs for Hepatitis C among the patients at free of cost.
- Supplying water purification tablets according to the needs of the people all over the country
- Developing national strategic plans of Viral Hepatitis

Advocacy on Hepatitis at community level and distribution of WPT were done at Haliashahar in Chittagong.

OBJECTIVES:

1. Reduction of hospitalization from diarrheal diseases to less than 30% from the baseline (2017) by 2022, in Bangladesh.
2. Reduction of HEV related maternal mortality by 30% of the baseline (2017) by 2022, in Bangladesh.
3. Reduction of hospitalization rate of fever, due to typhoid, to less than 30% from the baseline (2017) by 2022, in Bangladesh.

TARGET OF THE PROGRAM

1. Case Management
 - To ensure proper and timely case management for F&WBDs by using standard treatment guideline
 - To provide specific training for proper and uniform case management, including avoidance of nosocomial infections
 - To ensure sufficient pre-positioned medical supplies for case management

- To create a list of essentials for management

2. Surveillance

- To establish and strengthen hospital and lab based web based surveillance through MIS using DHIS-2 form for early detection of disease outbreak.
- To facilitate early detection of outbreak of diseases and immediate response

3. Vaccination

- To ensure reactive vaccination at community level to prevent spreading of the disease or outbreak
- To provide preemptive vaccination at community level with an identification of hotspots through surveillance in Bangladesh
- To provide vaccines in humanitarian crisis, during natural disasters like flood, earthquake etc.

4. Water, sanitation and hygiene Intervention

- Creating an improved access to safe water, proper sanitation and waste management and food safety for preventing F&WBDs
- Spreading health education among various communities by adopting preventive behavior for averting contamination
- Setting up an improved communication and public information system

MAJOR ACTIVITIES

- National Action Plan for Food and Water borne Diseases has been done
- Cholera saline, WPT has been distributed countrywide according to need
- Oral Cholera Vaccine has been distributed in adult population

NATIONAL RABIES ELIMINATION PROGRAM

BACKGROUND

Rabies, a dreadful viral zoonotic disease, is one of the major public health concerns in Bangladesh.

Before 2010, more than 2000 rabies related deaths were estimated to occur in the country. >95% cases of dogs were found to be the single most important cause of transmitting rabies, followed by cats, jackals and other animals.

Considering the public health impact of human rabies, Communicable Disease Control (CDC) unit of Directorate General of Health Services (DGHS), has been implementing the National Rabies Control Program since 2010.

In 2017, in the 4th HPNSP, the program has been established as a vertical program under CDC operational plan and shifted the target from control to elimination of rabies by 2022, in collaboration with the Department of Livestock (DLS), Local Government, non-Government organizations (NGOs), and development partners.

Rabies: Bangladesh
Area – 147,570 sq.km
Human Population – >160 million
Dog Population . – 1.6 million (estd.)
Stray Dogs: 83%
Annual Human Dog bites:>0.3 million
Annual Human rabies: >2000 (before 2010)
Rural burden: 85%
Mostly affected (>45%): children <15 yrs
Annual animal Rabies: approx. 25,000

Figure 4.3.22 Rabies situation in Bangladesh

GOAL

- To eliminate dog mediated human rabies (one of the public health problems in Bangladesh) by 2022

OBJECTIVE

- To reduce dog mediated human rabies mortality to less than 1 per million populations by 2022
- To reduce dog mediated animal rabies mortality to less than 10 per million animals by 2022
- To ensure regular reporting and strengthen active surveillance of human and animal rabies cases

TARGET

- Ensuring continuous supply of human anti-rabies vaccine and rabies immunoglobulin (RIG) in all rabies prevention and control centers by 2020
- Completing three rounds of mass dog vaccination throughout the country by 2022
- Establishing a referral laboratory for rabies for diagnosis of suspected rabies cases (human and animal) by 2019
- Strengthening a country wide rabies surveillance for animal and human by 2019
- Building awareness among community people to receive PEP and encouraging them for being concerned about the dogs around them
- Promoting responsible dog ownership campaigns to prevent dog bites by the year 2022

STRATEGIC ACTIVITIES

- Advocacy, Communication and Social Mobilization
- Modern Management for dog bite cases
- Mass Dog Vaccination (MDV)
- Dog population management (DPM)
- Rabies surveillance and Research

Important Activities and achievements in 2017-18

INTER-SECTORIAL COORDINATION

- A nationally coordinated and supervised comprehensive program through one health approach is being conducted for achieving rabies elimination. Ministry of Health and Family Welfare along with the Ministry of Fisheries and Livestock and Ministry of LGRD are playing a key role in eliminating rabies.
- Different national and international organizations as well as

development partners including WHO, WAP, OIE, FAO, CDC, Atlanta, are also being involved in different capacities to closely work with Rabies Elimination program in Bangladesh.

- Following committees are reformed for conducting the program
 - National Steering Committee (NSC)
 - Joint Coordination Committee (JCC)
 - Technical Working Group (TWG)
 - District Coordination Committee (DCC)
- Review of national strategy and national MDV action plans are designed
- Implementation of project for evaluation of mass dog vaccination in Bangladesh, in collaboration with CDC, Atlanta, DLS and others, is introduced.
- Reports with stakeholders are being shared

Animal bite management (to save human Lives)

THREE APPROACHES:

1. Disinfection by washing the wound with soap and water for 15 minutes
2. Active immunization with intra-dermal Inj. ARV
3. Local infiltration of RIG at the wound

ACTIVITIES AND ACHIEVEMENTS

- Functioning of 67 Rabies Prevention and Control Centre, one at the national level, at Infectious Disease Hospital, Dhaka (NRPCC) and at least one at each of the district Sadar Hospital (DRPCCs)
 - Standard management of dog/other animal bite cases following national Guideline and WHO protocol
 - Provision of modern tissue culture vaccine (Inj. Anti-rabies vaccine) in intra-dermal route (ID-RV) in all the centers and RIG at NRPCC for free of cost
 - Provision of training and refresher's training for physicians and nurses every year
 - Provision of vaccination cards, forms and registers for documentation
 - Documentation and reporting of animal bite cases through DHIS-2 and direct reporting to CDC, DGHS
 - Observation of national and district rabies prevention and control centers

MASS DOG VACCINATION

Mass dog vaccination within one week is one of the main strategies for rabies elimination by stopping the transmission of rabies from dog to dog/other animal and human beings.

GOAL

Completion of 3 rounds of MDV by following the international standard with >70% coverage in the smallest geographical area, throughout the country by the year 2021

ACHIEVEMENTS

Scaling up MDV to all the district municipality/City Corporation, 15 districts and vaccination of 6.5 lac dogs by June 2018

SURVEILLANCE

- Verbal autopsy of all reported rabies cases
- Field investigations and surveillance of 5 unusual animal bite events in 2018
- Development of draft guidelines for laboratory for human rabies

ADVOCACY, COMMUNICATION AND SOCIAL MOBILIZATION (ACSM)

- Promotion of mass awareness for timely and appropriate animal bite management, including disinfecting the affected area by washing it with soap and water, community awareness for dog vaccination and dog ownership promotion throughout the country, during MDV campaign
- Awareness activities at different medical colleges

OBSERVATION OF WORLD RABIES DAY

“Share a message, save a life” has been the slogan of World Rabies day in 2018. With Platform-(organization for medical and dental society of Bangladesh), World Rabies Day has been celebrated in 38 medical college hospitals in Bangladesh to aware the mass population. Other important facilities include:

- Multi-sectoral involvement for observing World Rabies Day
- Programs at national level, districts and sub-district levels
- Rally and discussion at Primary schools through **Little Doctor Campaign**



World Rabies Day celebration

- Mass media disseminating the activities of World Rabies Day with special importance
- Various steps are taken for Mass awareness building, regarding rabies control

OPERATIONAL RESEARCH:

- Evaluation of Mass Dog Vaccination in Bangladesh (jointly conducted by CDC, DGHS, DLS and CDC, Atlanta)
- Evaluation of knowledge, attitude and practice regarding rabies, dogs and treatment seeking behavior in dog bite events.

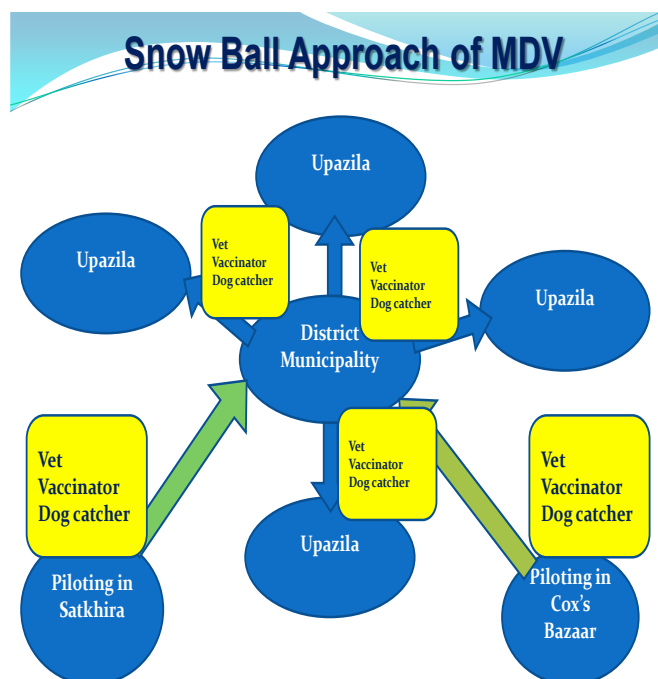


Figure 4.3.23 Snow ball capacity building in MDV in Bangladesh

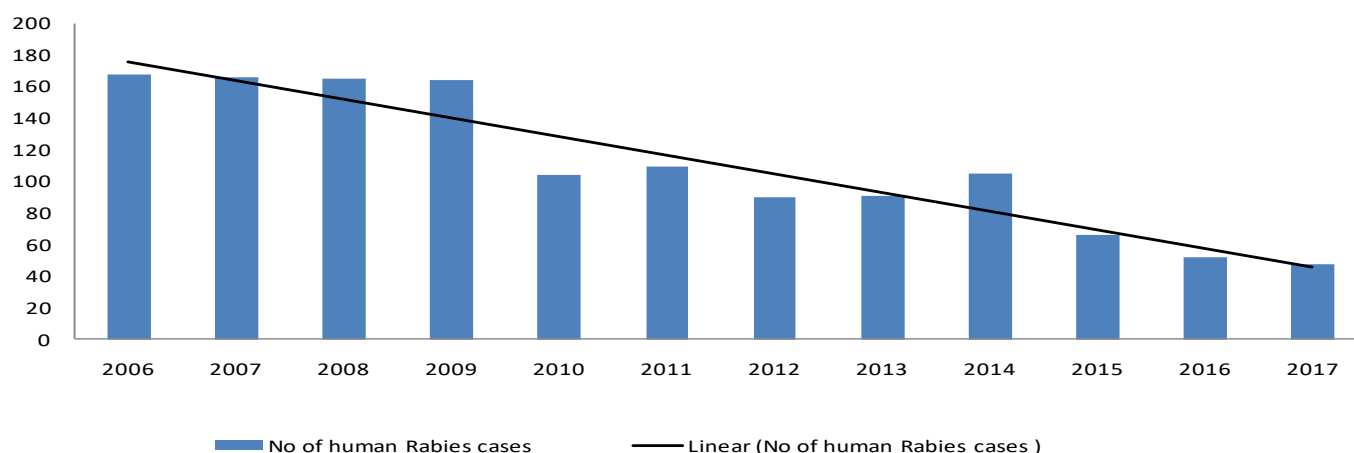


Figure 4.3.24 Declining trend of human rabies cases in Bangladesh from 2006 to 2017

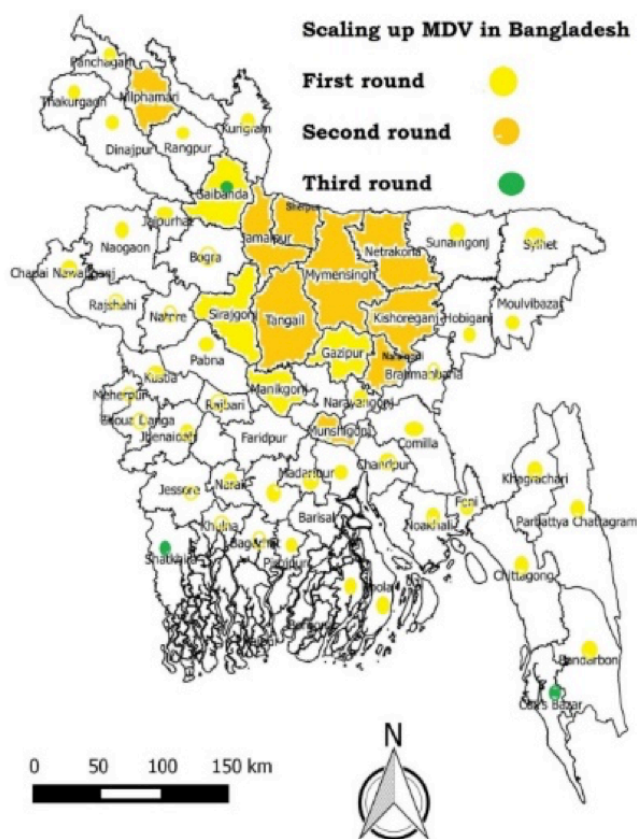


Figure 4.3.25 Areas covered by MDV

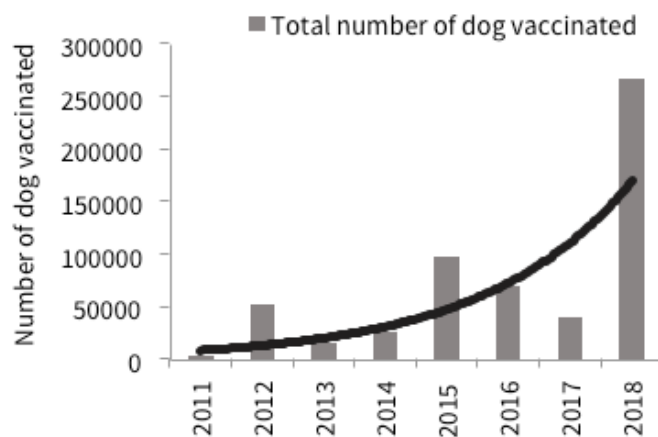


Figure 4.3.26 MDV Coverage in Bangladesh

Table 4.3.5 comparison of program features between 2011-16 and 2017-2022

Areas	2011-16	2017 onwards
National strategy	Yes	Updating
Committees	NSC, JCC, NTWG	NSC, JCC, NTWG
Multi-sectoral collaboration	Health, Livestock, LG, WHO, FAO, OIE, World Animal Protection, HSI	Strengthening
Fund allocation	7.2 million USD	320 million USD
Vaccine	TCV and RIG	TCV and RIG
Accessibility to vaccine per year	TCV: 200 thousand	TCV: 350 thousand
RPCCs	67 (all districts)	Better functioning
Trained HCPs	<1000	>3000
MDV	<3 lac dogs	>3 lac dogs
Lab diagnostic capacity and Surveillance	No	Establishing

FUTURE PLANS

- To complete at least 3 rounds of MDV throughout the country
- To ensure availability of human vaccines and RIG with free of cost
- To establish a country wide rabies surveillance system and laboratory diagnostic facilities
- To improve rabies case reporting
- To strengthen and promote responsible dog ownership

CONCLUSION

- Bangladesh is leading a national multipronged rabies elimination program towards its goal of eliminating rabies by 2022, and serving as a role model for others in the region with similar economic and sociocultural characteristics.
- Despite of various challenges, the national REP of Bangladesh has already been achieved with strong political commitment, adequate allocation, strategy guided activities and 'one health approach' programs through multi-sectoral collaboration to combat the disease.

NATIONAL KALA-AZAR ELIMINATION PROGRAM (NKEP)

BACKGROUND

- Kala-azar (KA) or visceral leishmaniasis (VL) is a neglected tropical disease affecting the poor and marginalized rural population.
- Around 31 million people, residing in 100 upazilas in 26 districts where Kala-azar is endemic, are at risk in Bangladesh.
- The National Kala-azar Elimination Program (NKEP) of Bangladesh set the target of elimination of kala-azar by 2017 with the goal to reduce the annual incidence of kala-azar to less than 1 patient per 10,000 population.
- At the beginning of 2014 “No kala-azar transmission” initiative was adopted and implemented in moderate and hyper-endemic upazilas.
- Kala-azar patients are detected and treated mainly through primary healthcare centers (upazila health complexes) and referral centers, especially at Surya Kanta Kala-azar Research Center (SKKRC) and medical college hospitals.
- In the WHO-supported VL Elimination Program in Bangladesh, single-dose AmBisome (Liposomal

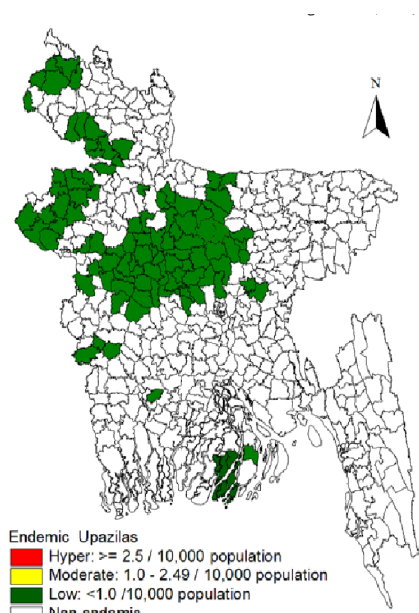
amphotericin B) has been introduced in the treatment for kala-azar since 2013 which is now available in all upazilas.

- WHO Bangladesh, icddr, b, MSF have been providing technical assistance to strengthen this program. Government provides huge amount of Deltamethrin instead of DDT i.e. Integrated Vector Management (IVM) is done successfully.
- NKEP was successful in reducing the number of hyper-endemic Upazila to zero, in 2016, with no upazilas having incidence rate more than > 01 case per 10,000 Population.

OBJECTIVES

- Reducing the incidence of New Kala-azar and PKDL to less than 1 case per 10 000 populations at the upazila level in Bangladesh by the end of 2017
- Reducing incidences of KA in the poor, vulnerable and remote endemic areas
- Reducing case fatality rates from high to negligible level
- Reducing cases of PKDL to interrupt transmission of kala-

Endemic areas of Bangladesh (2017)



In 2017, no hyper- or moderate endemic Upazila

In all endemic Upazilas, **NKEP has achieved the elimination target** of < 1.0 / 10,000 population in 2017

Figure 4.3.27 Kala-azar Endemicity

azar

- Preventing and treating Kala-azar-HIV-TB co-infections in the endemic areas

ACTIVITIES

- From the very beginning of the National Kala-azar Elimination Program, multisectoral approach has been adopted to achieve the targets in time.
- Different actions were taken for sensitization, capacity building, vector control measures, proper case detection and timely reporting, and dissemination of information.
- A large scale capacity building activity was conducted by NKEP in year 2017
- A total 20,751 people were trained and sensitized in different capacities on Kala-azar related issues. National Kala-azar elimination program adopted web-based reporting in 2016

SOME KEY IMPORTANT STRATEGIC DOCUMENTS (PREPARED BY NKEP)

- National Kala-azar Case Management Guideline (Updated in 2017)
- Kala-azar Outbreak Management Guideline in Bangladesh, 2016
- Monitoring tools and Supervisory Checklist (draft developed)
- Integrated Vector Management Guideline (IVM)
- National strategy on Kala-azar Elimination in Bangladesh

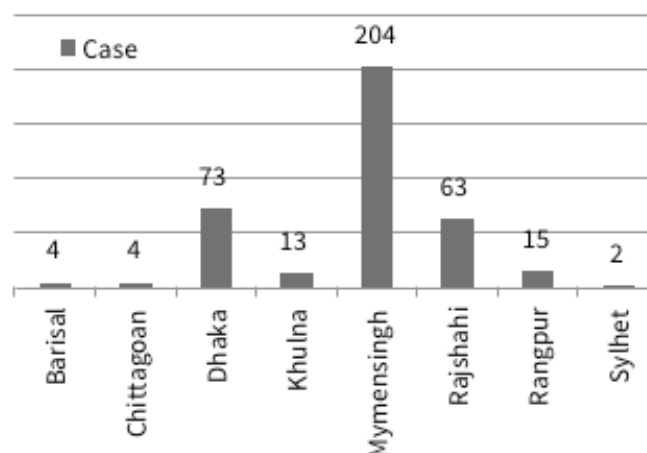


Figure 4.3.28 Divisionwise Kala-azar Data, 2017

Table 4.3.6 Age and sex distribution of Kala-azar cases by Disease category in 2017

Sex by age group	Disease type				
	KATF	NKA	PKDL	RKA	Total
Female					
< 5 yrs	0	1	2	0	3
5-14 yrs	0	15	4	2	21
> 14 yrs	1	68	23	12	104
Male					
< 5 yrs	0	6	1	0	7
5-14 yrs	2	17	6	5	30
> 14 yrs	1	87	64	39	191
Total	4	194	100	58	356

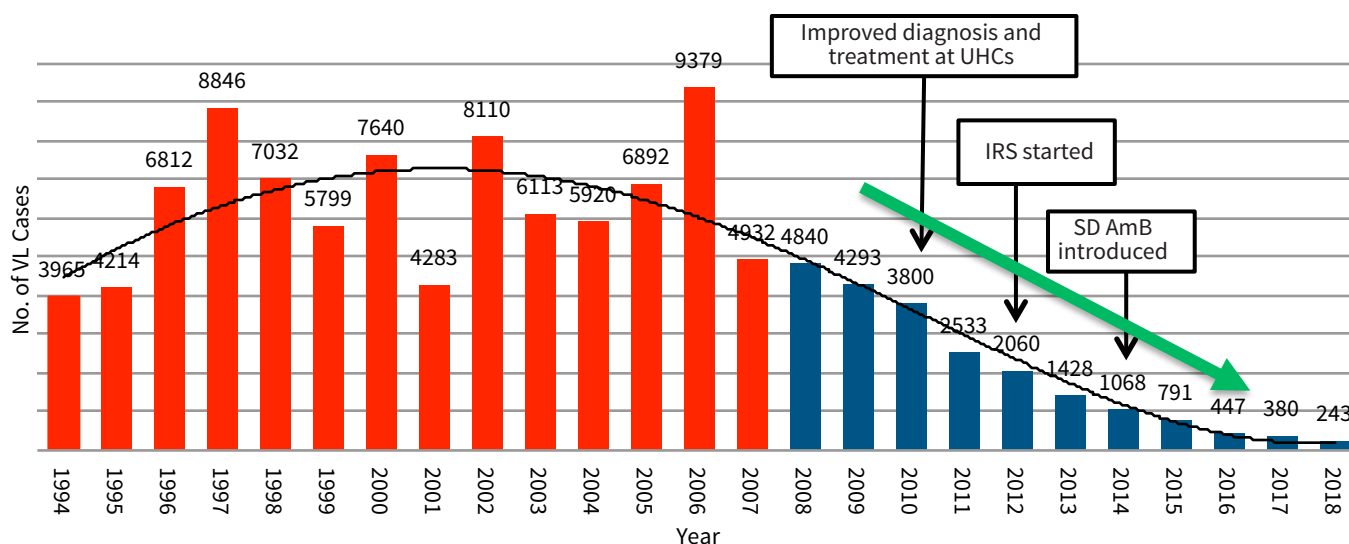


Figure 4.3.29 How Kala-azar Programs achieved elimination of the disease

(draft updated)

- NKEP is about to reach elimination target with the assistance from Govt. WHO, icddr,b, MSF

There is a significant reduction of total number of cases in 2018, in comparison to the previous year, although death rate was similar

- There is 36% reduction in incidence rate of VL in 2018, in comparison to 2017 (figure 4.3.29), indicating effectiveness of the strategies implemented by the NKEP

NATIONAL TUBERCULOSIS PROGRAM

BACKGROUND

- Bangladesh is one of the world's most densely populated countries. It has a population of 165 million. This huge population poses some serious health challenges, and tuberculosis is one of them.
- TB remains the leading cause of death from communicable diseases, and it is the second highest cause of disability among the productive age group.
- National Tuberculosis Control Program (NTP), under the Mycobacterial Disease Control (MBDC) unit of the Directorate General of Health Services (DGHS), is working with a mission of eliminating TB from Bangladesh.

GOAL

To end the tuberculosis epidemic.

GOVERNMENT POLICY AND STRATEGIES

- Bangladesh has been implementing sector-wide approaches (SWAp) in the health sector since 1998. Currently 4th Health, Population and Nutrition Sector Program (HPNSP) is being implemented. This is being covered from January 2017 to June 2022.
- In all the sector programs, tuberculosis control has been

recognized as one of the priority programs.

- Government of Bangladesh is committed to provide TB diagnosis and treatment completely free of cost to all the citizens of the country.
- The Government of the People's Republic of Bangladesh is also procuring first line anti-TB drugs with domestic funding from 2018. It has increased its overall GOB budget of TB Control in the operational plan (2017-2022) to US\$ 51.89 Million.
- Tuberculosis was declared as a mandatory notifiable disease by the government in 2014.
- NTP policies and strategies to date have been informed by international standards, formulated by the World Health Organization, such as DOTS strategy launched in 1993, and the Stop TB Strategy that underpinned the Global Plan to Stop TB 2006–2015.
- New multisectoral strategic approaches and new international targets of END TB strategy has also been adopted in 2016 with a vision of 'A Bangladesh Free of Tuberculosis'.
- The National Strategic Plan for TB Control was updated in line with WHO END TB strategy for 2018-22.

OBJECTIVES OF THE NATIONAL STRATEGIC PLAN 2018-2022

Pillar 1: Integrated, Patient-Centred Care And Prevention

Objective 1

To increase annual case detection of all forms of TB to more than 90% of all incident cases by 2022 (from baseline of 57% in 2015) with childhood TB contribution of 8% by 2022 (from baseline of 4% in 2015)

Objective 2

To further increase the treatment success rate at levels of more than 90% in all forms of detected non-MDR TB cases and ensure quality-controlled treatment services at all implementation sites

Objective 3

To increase annual case detection of MDR- TB to 4,100 cases (from baseline of 880 in 2015), and Child MDR case detection to 112 cases by 2022 (from baseline of 0 in 2015), and improve management of MDR-TB cases through country-wide implementation of the shorter MDR-TB treatment regimen

Pillar 2: Bold Policies And Supportive Systems

Objective 4

To ensure that no TB-affected families face catastrophic costs due to tuberculosis by 2022

Objective 5

To ensure that 100% of TB service facilities receive regular supervision and monitoring with appropriate feedback, resulting in remedial actions by 2022

Objective 6

To ensure the long-term availability of 100% of required funding for activities at all program levels, through effective planning and partner coordination, and the continuing increase of GOB contributions to the NTP budget

Pillar 3: Intensified Research And Innovation

Objective 7

To ensure adequate support for operational research to foster innovation

TB EPIDEMIOLOGY IN BANGLADESH

WHO estimated the incidence and mortality for all ages and all forms of TB in 2017, in Bangladesh to be 221 and 36 per 100,000 (Table 4.3.7). These WHO estimates will be updated, based on the results from a recently completed TB prevalence survey, for which the results are included here.

Table 4.3.7 WHO estimates of TB burden, 2017

Estimates of TB Burden 2017	Number in thousands (95% confident interval)	Rate per 100,000 population (95% confident interval)
Mortality (excludes HIV+TB)	59 (38-85)	36 (23-52)
Mortality (HIV+TB only)	0.17 (0.085-0.29)	0.11 (0.05-0.18)
Incidence (includes HIV+TB)	364 (265-479)	221 (161-291)
Incidence (HIV+TB only)	0.55 (0.27-0.92)	0.33 (0.17-0.56)
Incidence (MDR/RR-TB)	8.4 (3.8-15)	5.1 (2.3-9)

Source: Draft Global TB Report 2018

NATIONAL TUBERCULOSIS PREVALENCE SURVEY IN BANGLADESH, 2015-16

- The recent survey of 98,710 participants, conducted between 2015 and 2016, is designed in accordance with WHO standards.
- The estimated prevalence of bacteriologically confirmed pulmonary TB cases, and smear-positive bacteriologically confirmed TB cases among the population aged ≥15 years respectively were 287 per 100,000 population (95% CI: 244-

330) and 113 per 100,000 population (95% CI: 87-139) .

- The prevalence was higher among males compared to females and was higher in urban compared to rural clusters. Prevalence increased with age and was much higher among ≥65 years age group.
- Overall estimated prevalence (all ages, all forms of TB) was 260 per 100,000 population (95% CI 220-301).

MAJOR ACTIVITIES OF THE NTP

To achieve the objectives, the main activities adopted by the NTP are:

- Developing policies, strategies and guidelines for TB control
- Ensuring proper planning and budgeting for TB control activities through GoB OP and develop human resources capacity for TB control including training
- Promoting early detection of smear-positive patients at all levels of the health services
- Implementing quality assurance system for smear microscopy
- Diagnosing smear-negative, extra-pulmonary and childhood TB
- Ensuring Directly Observed Treatment (DOT) through community
- Ensuring participation and involvement of government and non-government health care providers.
- Maintaining a high treatment success rate (>90%) among diagnosed new smear-positive patients
- Ensuring uninterrupted supply of drugs, laboratory equipment and consumables as well as other logistics
- Implementing standardized recording and reporting system
- Involving academic medical institutes and hospitals, private practitioners, special services like prisons, defense, industries and other corporate sectors in the NTP activities
- Strengthening cooperation and collaboration between the government of Bangladesh and Non-Government Organizations (NGOs) who are involved in controlling tuberculosis
- Conducting regular supervision, monitoring and evaluation of the NTP thus measuring program performance and impact
- Ensuring programmatic management of drug-resistant TB. The following measures can be taken for this purpose:
- Establishing linkage for management of TB-HIV co-infection

Table 4.3.8 TB case notification and treatment success in 2017

Total TB case notified	244,201
TB treatment coverage (notified/estimated incidence)	67%
Notified TB/HIV co-infection	89
Laboratory-confirmed MDR/RR-TB cases	944
MDR/RR-TB patients started on treatment	920
Laboratory-confirmed XDR-TB cases	6
XDR-TB patients started on treatment	6
Treatment success of New and relapse cases registered in 2016	94%
Treatment success HIV-positive TB cases registered in 2016	49%
Treatment success MDR/RR-TB cases started on second-line treatment in 2015	78%

Source: Draft Global TB Report 2018

- Maintaining liaison with development partners and establishing inter-sectoral and inter-ministerial collaboration
- Carrying out operational research related to TB control

ACHIEVEMENTS OF TB CONTROL IN BANGLADESH

Since the introduction of DOTS in Bangladesh in 1993, remarkable progress in TB control has been achieved in terms of DOTS coverage, diagnosis and treatment of TB cases.

DOTS COVERAGE

Bangladesh adopted the internationally recommended DOTS strategy in 1993 and by 2007 NTP reached the 100% DOTS coverage. Over 2 million people with symptoms of TB are yearly being tested. Also, 1 million lives were approximately saved in the last decade.

TB CASE NOTIFICATION:

- After the introduction of the DOTS strategy in 1993, the overall progress was slow and steady in case finding, until 2001. In 2001, case notification rate for new smear positive cases were 59/100,000 population.
- Through social support for the ultra-poor group and additional effort for addressing detection of smear negative

and child TB cases with special attention in remote areas, this case notification increased to 149/100 000 population in 2017.

- As a result, the gap increased between incidence and notification narrowing (Figure 4.3.20). In 2017, a total number of 244,201 TB cases were notified to NTP and they were subsequently put on DOTS.
- Up to September 2018, a total number of 199625 TB cases were notified to NTP; they were immediately put under treatment. Therefore, 73 % case is detected and 27 % case is missing.

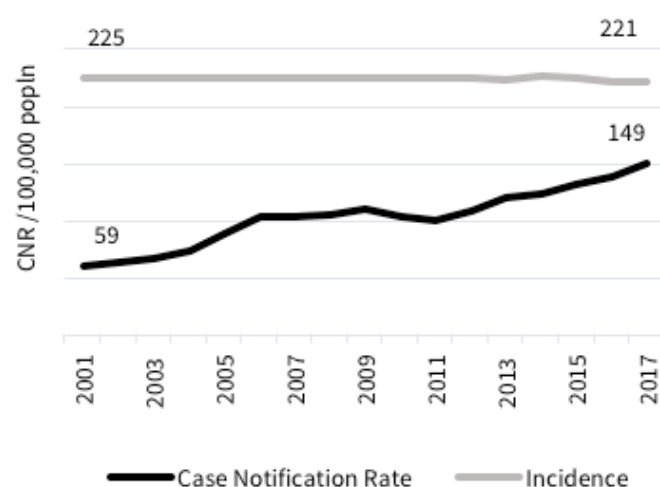


Figure 4.3.30 Trends of Incidence and Notification rate in Bangladesh from 2001 to 2017

TB TREATMENT COVERAGE

Bangladesh is steadily progressing towards universal health coverage. 67% of the estimated TB cases were detected and put under treatment in 2017.

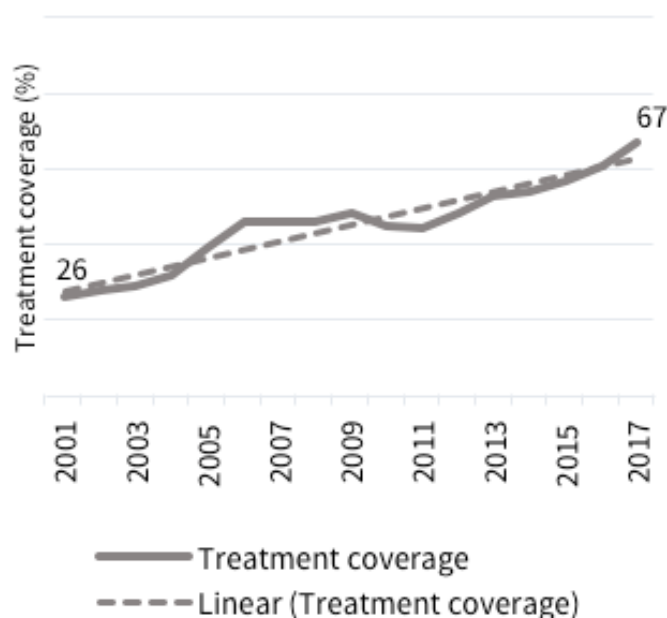


Figure 4.3.31 Trend of TB treatment coverage in Bangladesh from 2001 to 2017

TB TREATMENT OUTCOMES

- Treatment success rates under DOTS have been consistently high from the beginning, and it crossed the global target of 85% in 2003.
- After strengthening DOTS and ACSM activities, NTP has been maintaining over 91% treatment success rates since 2005.

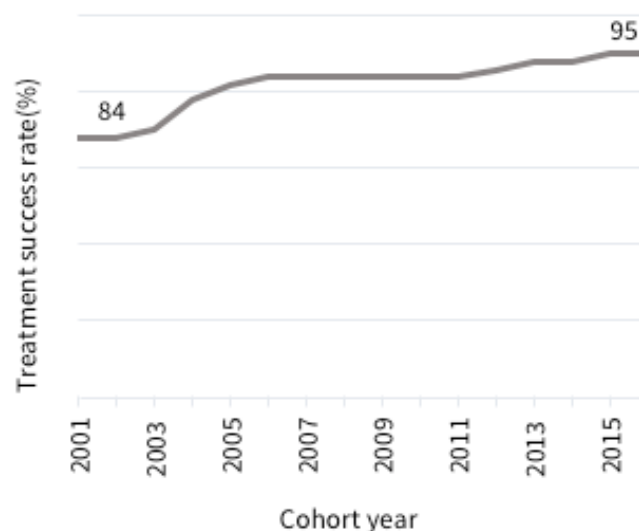


Figure 4.3.32 Trends of treatment success rates (Bacteriologically confirmed TB) from 2001 to 2015

TB MORTALITY

- TB deaths in Bangladesh is in a declining trend. From 2001, there is a 50% reduction in TB mortality while SDG target has been 90%.
- There is 19% reduction in TB death, in two years, from 2015 to 2017.

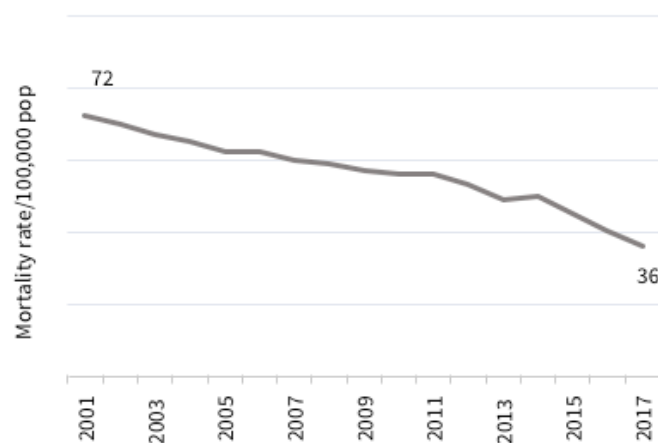


Figure 4.3.33 Trend of TB mortality in Bangladesh

DRUG-RESISTANT TB

- The MDR-TB shorter treatment regimen research conducted in Bangladesh, famously known as “Bangladesh Regimen”, has significantly reduced the MDR-TB treatment duration, from 20 months to 9 months, with a better treatment success.
- Bangladesh has started implementing this shorter treatment regimen from 01 April 2017, under programmatic condition. This has been scaled up all over the country by 2017.
- By the end of 2017, a total number of 6,240 MDR-TB patients were treated with second line anti-TB drugs.
- Bangladesh is maintaining high treatment success rate in MDR-TB treatment over the years with a gradual switch to 9-month regimen.
- NTP Bangladesh introduced new anti-TB drugs i.e. Bedaquiline and Delamanid in 2016. The country has successfully been treating XRD, Pre-XDR and complicated MDR-TB cases since.
- A total number of 26 XDR-TB cases were diagnosed and put on DOTS till December 2017.

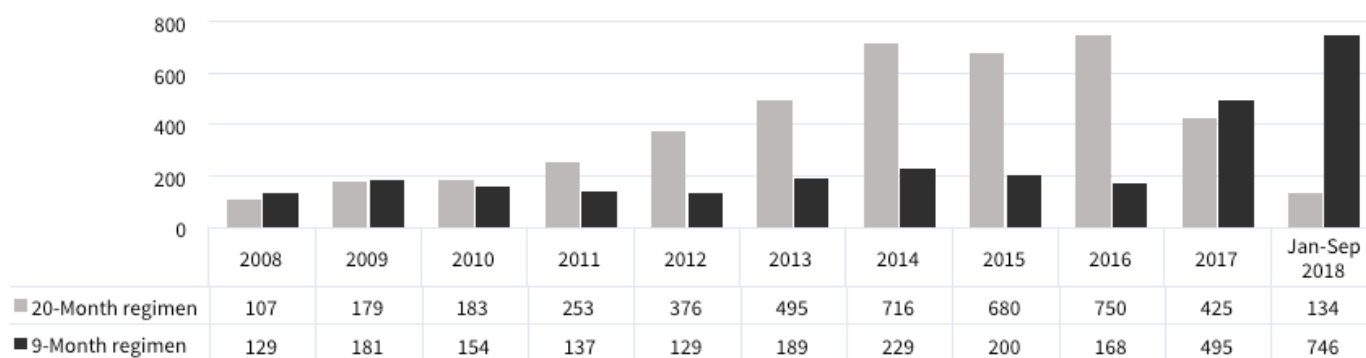


Figure 4.3.34 Number of MDR-TB cases put on second-line treatment under different regimen

CHILDHOOD TB

In 2017, 4.3% of new TB cases were children under 15 years of age. This is up from 3.3% in 2014 and 2.8% in 2013. Therefore, progress is being made in identifying new pediatric cases. 10,189 childhood TB cases were identified in 2017 which increased to double in comparison to 2013 (figure 4.3.35).

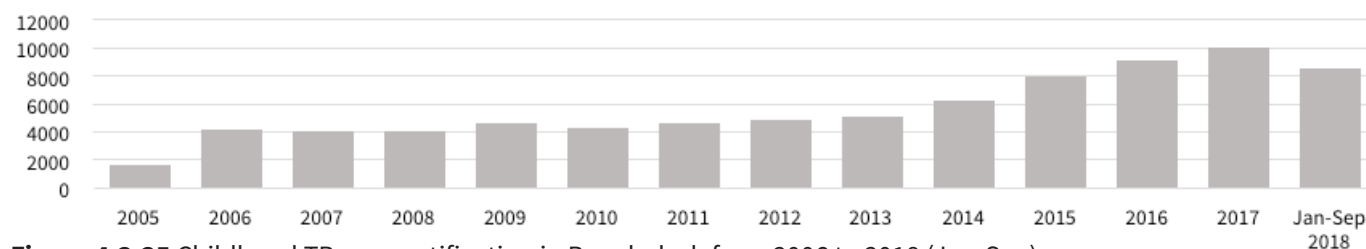


Figure 4.3.35 Childhood TB case notification in Bangladesh from 2006 to 2018 (Jan-Sep)

New child friendly formulation of anti-TB drug has been introduced in Bangladesh from April 2017.

TB LABORATORY SERVICES:

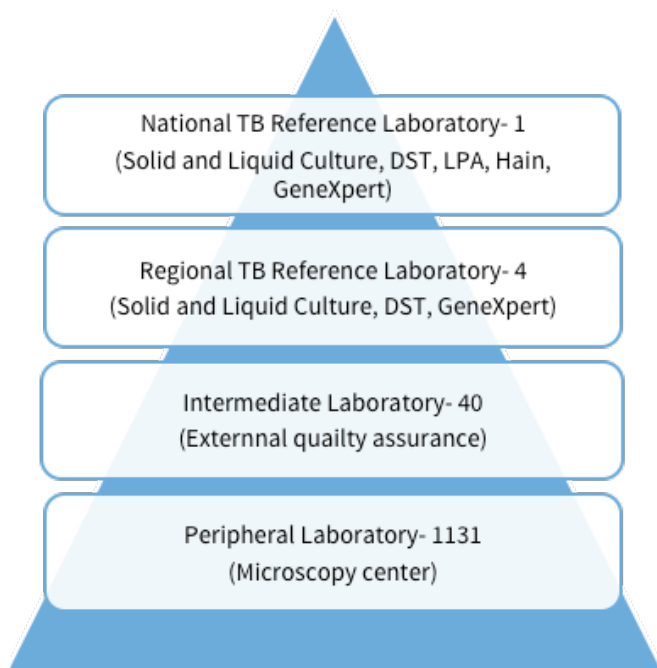


Figure 4.3.42 TB Laboratory Services

- Apart from the aforementioned activities, NTP is currently running 193 GeneXpert machines in 163 sites all over the country.
- Bangladesh has introduced new diagnostic algorithm for testing all presumptive TB cases by GeneXpert (a more sensitive tool for diagnosis TB cases) from January 2018.
- In 2017, a total number of 77,560 presumptive cases were tested by GeneXpert.

PUBLIC-PRIVATE MIX (PPM)

- The multisectoral response including Public-Private Mix (PPM) is one of the strengths of NTP to combat TB in Bangladesh.
- Currently, NTP is working with 28 NGOs with a vision to END TB. The partners include BRAC, Damien Foundation, LEpra, HEED, RDRS, TLMI, LAMB hospital, PIME Sisters, NATAB, ICDDRDB and BGMEA.

TB/HIV RESPONSE

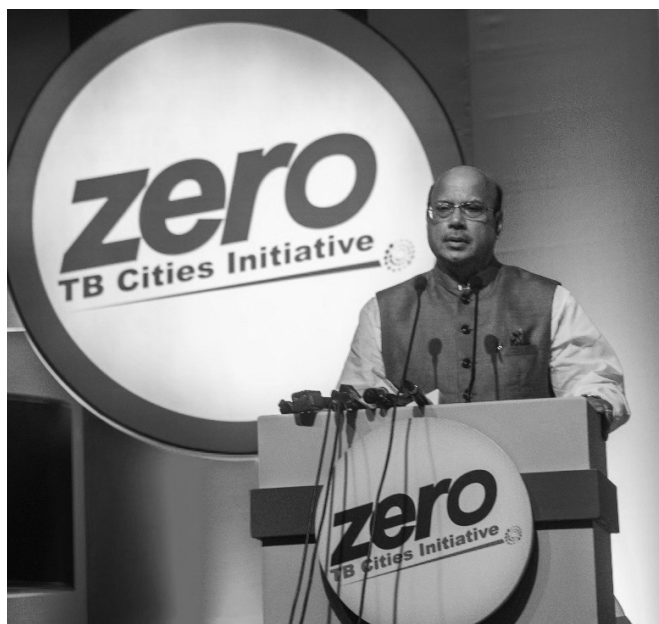
- Bangladesh is maintaining a low prevalence and a low incidence rate (below 0.1%) through the national response to HIV for the last two decades.
- Among most of the key populations, the prevalence is still less than 1% (MSM, Transgender, PWID, FSW etc.) and it is less than 0.01% in the total population.
- In 2017, 3995 TB cases were tested for HIV, and 89 TB/HIV co-infected cases were identified. Among them, 86 cases received ART.
- NTP has recently published the 2nd edition guidelines for TB/HIV collaborative activities.

MONITORING AND EVALUATION

- NTP is collecting quarterly TB reports through DHIS-2, an integrated government electronic reporting system.
- Besides paper-based recording and reporting, an electronic individual case based recording and reporting tool has been introduced in Bangladesh. This e-TB manager is functional in 246 of the 482 upazilas.

POLITICAL COMMITMENT

- The Government expresses strong commitment towards the TB program. This commitment is reflected in their current operational plan as it has organized significant increase in the domestic funding.
- The Honorable Minister of the Ministry of Health and Family Welfare attended the “Ministerial Meeting Towards Ending TB in the South-East Asia, 15-16 March 2017” held in New Delhi and signed the “Delhi Call for Action” to End TB in the WHO South-East Asia Region by 2030.
- The Honorable Minister of the Ministry of Health and Family Welfare attended the first WHO global ministerial conference held in Moscow, Russian Federation from 16 -17 November 2017.
- In October 2017, the Honorable Minister of the Ministry of Health and Family Welfare declared “Zero TB cities initiatives” and called upon all stakeholders in urban areas to focus on local government participation in a comprehensive set of activities to treat, diagnose, and prevent TB.



Mohammed Nasim, former Minister of the Ministry of Health and Family Welfare, Bangladesh declaring “Zero TB cities initiatives”.

TB PROGRAM FOR FDMN

- More than 8 million refugees have recently been infiltrated in the southern part of Bangladesh from Myanmar.
- All kinds of health services, along with specially designed TB programs, are being provided in the areas where these refugees are living.
- NTP is providing additional human resources to provide support in the camps. Fifteen additional microscopy laboratories have been established in that area and one GeneXpert has been set up.

PROGRAM SUPPORTED BY DP (DEVELOPMENT PARTNERS)

- Bangladesh TB Control Program funding sources are done by GoB through the MoHFW budget, the Global Fund and USAID.
- Apart from this, WHO is also extending financial and technical support in some area as a technical partner. Some major highlights of the roles being played by these development partners in TB control are:

1. USAID:

Challenge TB Bangladesh (CTB) is part of a 5-year cooperative agreement funded by USAID. CTB provides Technical Assistance (TA) to the National Tuberculosis Control program

(NTP) with a countrywide coverage.

2. WHO:

WHO is providing support to increase efforts for detection of TB cases, as well as diagnostics and laboratory strengthening. It is also conducting several other activities, such as:

- Maintaining high cure rates
- Improving the quality of the TB control services
- Strengthening major critical components of the service delivery system
- Addressing the issue of drug resistance
- Setting up norms and standards
- Assisting to take evidence based policy decisions
- Mobilizing partnerships for TB control
- Supporting research, monitoring and development.

3. icddr,b:

icddr,b is the only Sub – Recipient (SR) of Global Fund that is implementing the TB Control Program in direct partnership with NTP. Also, it is providing cutting-edge research which is scalable in resource-limited settings.

4. IOM Bangladesh Providing Migration Health Assessments since 2006:

IOM, affiliated by NTP, is working in Dhaka and Sylhet for the diagnosis and treatment of TB.

PROGRAM IMPLEMENTED BY OTHER PARTNERS (NGOS, PRIVATE SECTORS)

Bangladesh represents a unique example of close collaboration between NTP, NGOs and TB Control Programs; there are clear geographic assignments from the central level to avoid overlaps.

The following NGOs are implementing TB Control Programs in partnership with NTP:

1. BRAC

BRAC is the principal recipient of The Global Fund to strengthen health system, and it has expanded DOTS services nationwide. Other NGOs are implementing their actions for TB through BRAC.

2 Damien Foundation

Damien Foundation, a Belgian NGO, is meaningfully contributing to tuberculosis control and leprosy elimination in Bangladesh.

3. HEED Bangladesh

HEED Bangladesh (Health, Education and Economic Development) is working to control TB in 24 Upazillas under Moulvibazar District.

4. LAMB Hospital

LAMB (Lutheran Aid to Medicine in Bangladesh) TB Control Program is working in 4 Upazilas. Parbatipur (one poursova and one union), Chirirbandar and Khansama upazilla under Dinajpur district and Saidpur upazilla under Nilphamari district covering approximately 788,529 people.

5. LEPRO Bangladesh

Lepra is covering 7,512,080 number of people for the TB control program in Sirajgonj, Pabna, and Natore. It is receiving its funding mainly from GFATM NFM with additional support from Lepra UK.

6. NATAB

NATAB has been working as a civil society advocacy agency to identify different groups by vocation, profession, religion, ethnicity and other possible classification and to turn the variation into strength.

7. NGO Health Service Delivery Project (NHSDP):

NHSDP contributes to DOTS support to National Tuberculosis Control Program through Surjer Hashi (SH) clinics in Dhaka, Chittagong, Rajshahi and Khulna City Corporations

8. PIME Sisters:

PIME Sisters joined the NTP to implement its activity in Khulna City Corporation area.

9. Rangpur-Dinajpur Rural Service

Rangpur-Dinajpur Rural Service (RDRS) Bangladesh provides services to TB patients in 5 Upazilas of Lalmonirhat and 9 Upazilas of Kurigram District through 47 clinics.

10. Salvation Army

The Salvation Army is mainly responsible for supporting TB control activities in Mirpur, Dhaka. This area is mostly inhabited by Bihari (Urdu speaking) refugees living in unhygienic slum conditions.

11. The Leprosy Mission Bangladesh

TLM is strengthening the health system by integrating its Leprosy and TB control services in Government Health facilities in ten upazilas of Thakurgaon and Panchagarh districts. TLMB also fosters networking between the Government service providers and other stake holders in this regard.

LEPROSY ELIMINATION PROGRAM

BACKGROUND

- *Mycobacterium leprae*, an acid fast bacillus, causes leprosy. It is a chronic infectious disease that primarily affects peripheral nerves. Secondly, it affects the skin, mucous membrane, and other tissues of the body, like muscles, eyes, bones, testes, and various internal organs.
- Bangladesh has achieved the elimination of leprosy at the national level by the end of December, 1998.
- It was 2 years ahead of WHO-targeted date. The 'elimination' as defined by the WHO is to reduce registered prevalence to less than 1 case per 10,000 people.
- After achieving elimination at the national level, the National Leprosy Elimination Program (NLEP) is consolidating its effort to achieve subnational (district-level) elimination.
- There is 25% decrease in leprosy prevalence in 2018 than the previous year.

Table 4.3.9 Registered prevalence rate of leprosy (per 10,000 people) in the last 10 years

Year	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Prevalence Rate	0.29	0.25	0.22	0.22	0.2	0.21	0.23	0.2	0.2	0.15

MDT COMPLETION

- The MDT completion rate is more than 85 % in the last few years; it is the highest among SEARO countries.
- Some years MDT completion rate is more than 100% because of the cases identified in the previous year and the MDT completed in the following year.

Table 4.3.10 MDT completion in the last 10 years

Year	2011	2012	2013	2014	2015	2016	2017	2018 (Up to 3rd quarter)	Total
New Cases	3990	3590	3156	3621	3976	3000	3754	2857	27944
MDT Completion	4051	3328	3437	3107	3442	3420	3332	2509	26626
%	101.5	92.7	108.9	85.8	86.56	113.33	88.75	87.81	95.28

Source: 1) TB MIS, 2) DHIS2 and 3) WHO Global TB Report

AIDS AND SEXUALLY TRANSMITTED DISEASE (ASP)

BACKGROUND

- Prevalence of HIV/AIDS is still low in Bangladesh, but the country remains vulnerable to an HIV epidemic because of the high prevalence of the disease in neighboring countries and the high mobility of people within and beyond the country.

- The most important factors that may contribute to a potential HIV epidemic include: high rate of needle-sharing among people who inject drugs (PWIDs), low rate of condom-use, and high prevalence of sexually transmitted infections (STIs) among the key population groups.
- The Government of Bangladesh responded to HIV and AIDS from the first case detected in 1989, by forming the National AIDS Committee (NAC) and developing the first AIDS policy.

- The National AIDS/STD Control Program (NASP) under Directorate General of Health Services (DGHS) is responsible for coordinating all stakeholders and development partners who are involved in HIV/AIDS control activities throughout the country.

4th NATIONAL STRATEGIC PLAN FOR HIV AND AIDS RESPONSE 2018-2022

- The strategic plan is developed with an aim to guide the national response to HIV and AIDS to achieve the global targets on 'Ending AIDS by 2030' and treatment target of '90-90-90' by 2020.

PROGRAM OBJECTIVE

1. To implement services to prevent new HIV infections by increasing program coverage and case detection
2. To provide universal access to treatment, care and support services for the people living with HIV
3. To strengthen the coordination mechanisms and management capacity at different levels to ensure an effective national multi-sector HIV/AIDS response
4. To strengthen strategic information systems and research for an evidence based response

ACTIVITIES IN 2017

- In total, 865 new HIV (63 cases among the Rohingya refugees) infections have been detected in 2017 (Figure 10.34). Until October 2017, the total number of detected cases was 5,586, of whom 924 people living with HIV (PLHIV)

have died, leaving 4,662 known people living with HIV.

- However, most of the cases are likely to remain undetected; the total national estimate of PLHIV is around 13,000 (source: National HIV/AIDS Estimate and Projection Using Spectrum, 2017).

SURVEILLANCE

- The national surveillance system for HIV was set up by the Government of Bangladesh in 1998. Based on UNAIDS/WHO guidelines, a flexible system was formed so that it can be easily adapted to the state of epidemic. It includes both serological surveillance and a behavioral surveillance survey (BSS).

MSM, MSW AND TG

- During May to September 2015, a cross-sectional survey was conducted to assess changes in risk behaviors and prevalence of HIV and active syphilis among MSM, MSW, and Transgender (TG) in selected sites in Bangladesh.
- In Dhaka, both behavioral and serological surveillance were conducted among all groups but, in Hili, only serological surveillance was conducted.
- The overall prevalence of HIV was 0.2% (1/531) for MSM, 0.6% (3/497) for MSW, and for TG 1.4% (4/279). The overall prevalence of active syphilis form 1.1% (6/531) for MSM, 1% (5/497) for MSW, and for TG was 1.8% (5/279).

PEOPLE WHO INJECT DRUGS (PWID)

- The HIV risk behaviors were found to be higher among male PWID in two neighborhoods in Dhaka, designated as A1 and



Figure 4.3.43 Distribution of new HIV/AIDS-positive cases and deaths, due to HIV, reported by year in Bangladesh (1998-2017)

A2, especially in regard to drug-injecting practices.

- Those in A1, had injected drugs for a longer time, took more injections, shared their needles/syringes more often, and had more injection-sharing partners.
- When all male PWID were considered, trends over years (since 2002) showed improvements in almost all key indicators of risk. However, in 2016, more than 50% were still found to be sharing their used needles/syringes
- The 197 HIV-positive male PWID, >60% borrowed or lent last week, >30% were married or had non-transactional sex partners or bought sex from FSWs; some had multiple sex partners, and <40% consistently has used condoms last year while having physical relationships with FSWs.
- Structural factors can also impact the HIV epidemic, such as living conditions; 54.7% of HIV-positive PWID, in A1, lived on the streets. Such chaotic lifestyles impact negatively on risk-taking behaviors.
- In this population group, all individuals with structural risks and vulnerabilities assist in further spread of HIV. Population, beyond this group, is not devoid of this risk either.

FEMALE SEX WORKER (FSW)

- Condom use among the FSWs has increased 60-80%, in comparison to the last study done in 2006, as condoms were made available through NGOs.
- FSWs selling sex in hotels of Dhaka were the most vulnerable of all the FSW groups sampled.
- These FSWs had large numbers of clients, little knowledge about HIV and STIs, and of HIV testing; only 4.7% were tested,

counseled, and received their HIV test results last year.

- In the hotels of Dhaka, there were approximately equal numbers of FSWs (between 15-24 years and 25-49 years) while, in other places, fewer younger FSWs were found.
- Differences in risk behaviors, between the two age-groups, varied in different areas. Nonetheless, a few indicators highlighted the greater vulnerability of younger FSWs compared to the older ones.

HIV PREVALENCE OVER THE ROUNDS

- Over the rounds of surveillance, the overall HIV prevalence has remained at <1%, irrespective of whether the total population is considered or segregated for the key (most at-risk) and bridging populations (Figure 4.3.44).
- It is to be noted that bridging population groups (e.g. mainly truck-drivers, dockyard workers, etc.) were not sampled since Round VI of the surveillance.

CROSS-BORDER MOBILITY IN RECENT YEARS AMONG FSW, MSM, MSW, AND HIJRA

- Two areas where more than one population group with HIV were detected include: Benapole (female PWID and residence-based FSWs) and Hili (TG and casual FSWs); both sites share borders with West Bengal, India.
- Figures show the percentages of FSWs, MSM, MSWs, and TG, living in border areas, who crossed the border in the year

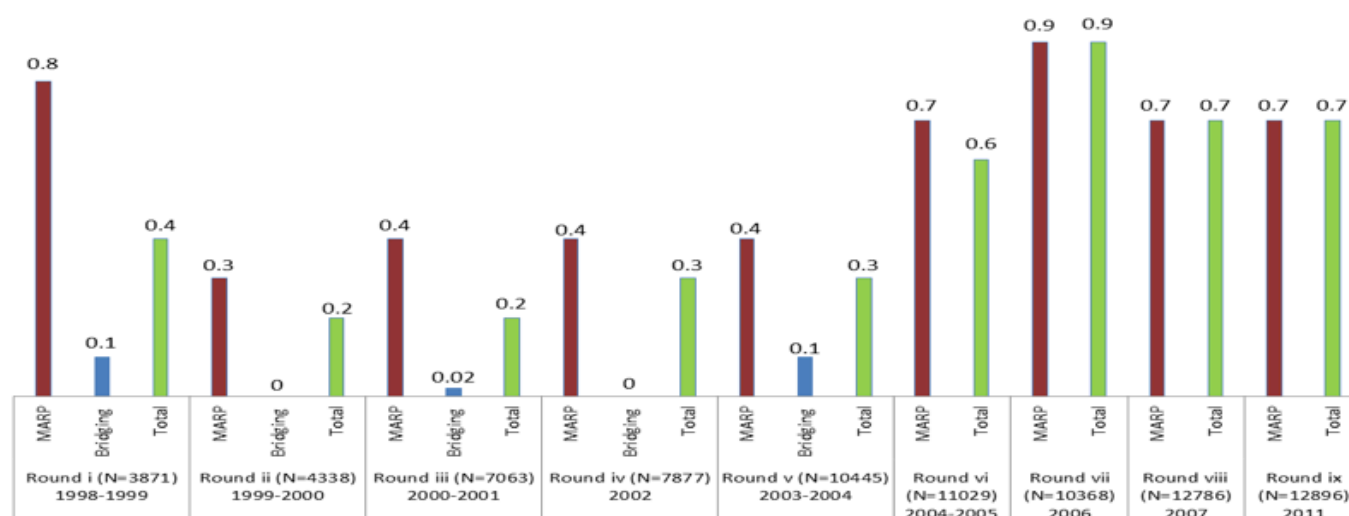


Figure 4.3.44 HIV prevalence over the years

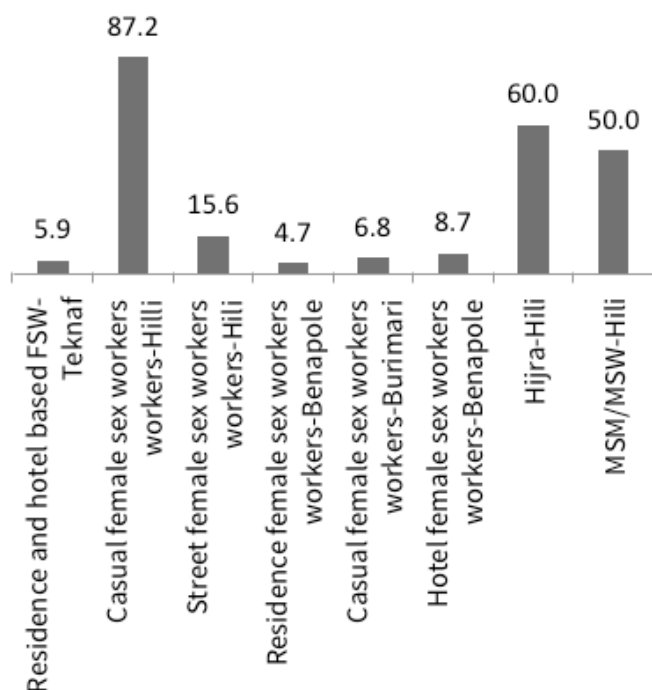


Figure 4.3.45 Cross-border mobility of different key populations, 2011

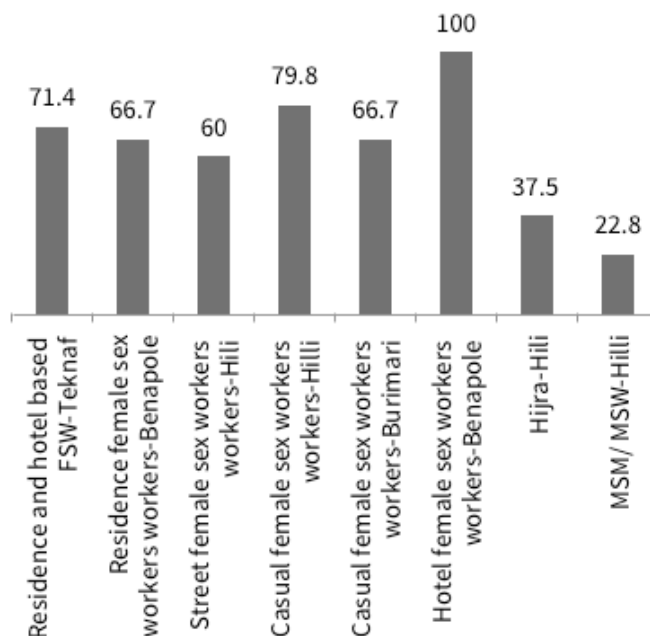


Figure 4.3.46 Selling sex while being abroad in 2016 (by those who crossed the border in 2011)

2011 and sold sex while being abroad. It is clear that cross-border mobility is more common in Hilli.

GEOGRAPHICAL AND OCCUPATIONAL DISTRIBUTION

- Regular case reporting showed highest number of PLHIV in Dhaka, but Chittagong and Sylhet have the highest concentration of PLHIV. Maps show the division-wise case reports.

Map (1) shows the division-wise HIV-positive cases in 2017. In total, 865 cases were identified while most of the cases (418) were identified in Dhaka division.

Map (2) shows the division-wise commutative HIV-positive cases from 1989 to 2017. The highest numbers of HIV-positive cases (1,851) were identified in Dhaka division.

Map (3) shows that 3,265 HIV-positive cases received ART from different service centers in the country. The government facilities provided ART drugs to 2,717 HIV

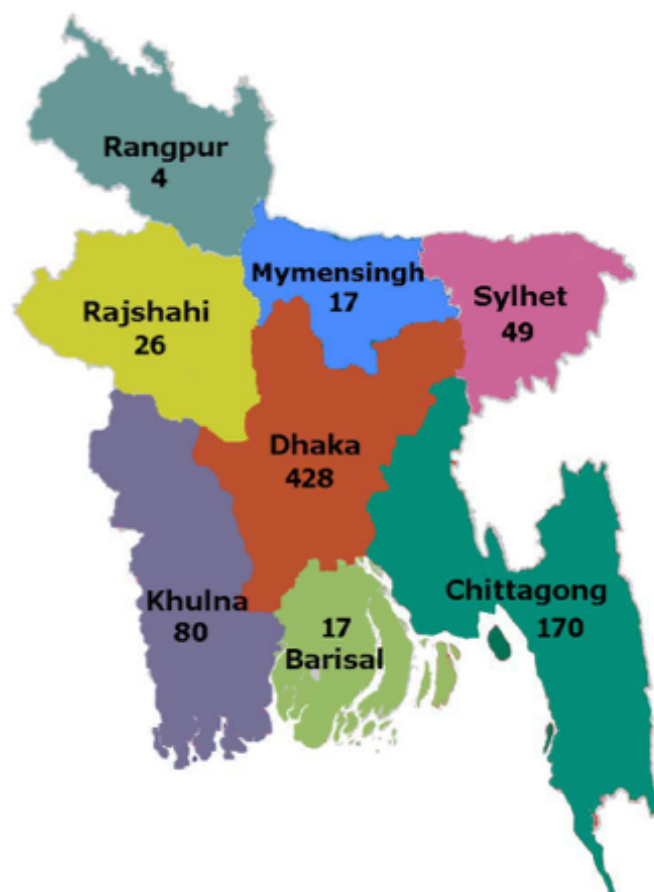


Figure 4.3.47 Geographical distribution of PLHIV (2017)

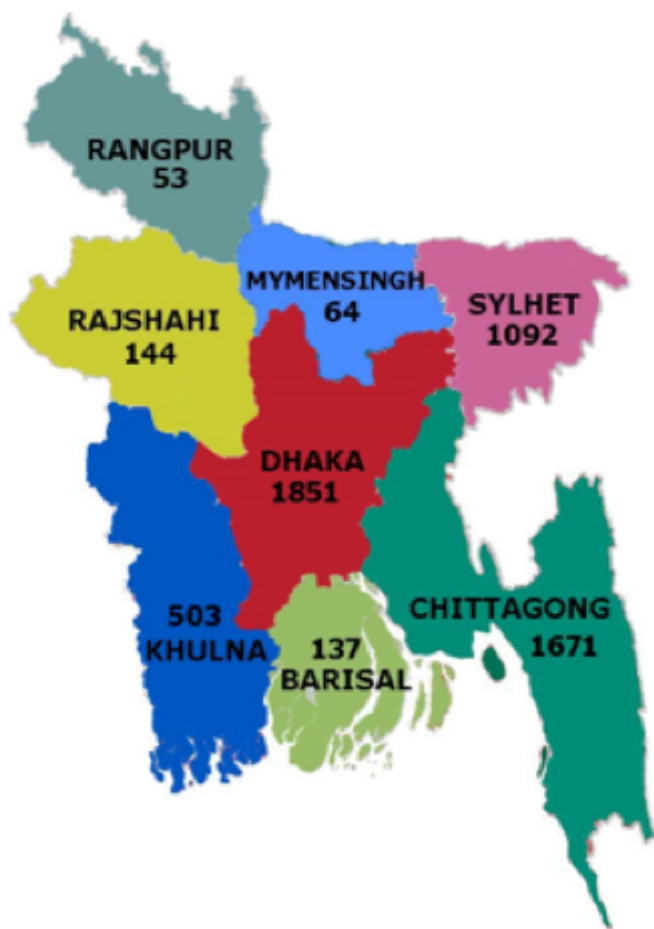


Figure 4.3.48 Geographical distribution of PLHIV (1989-2017)

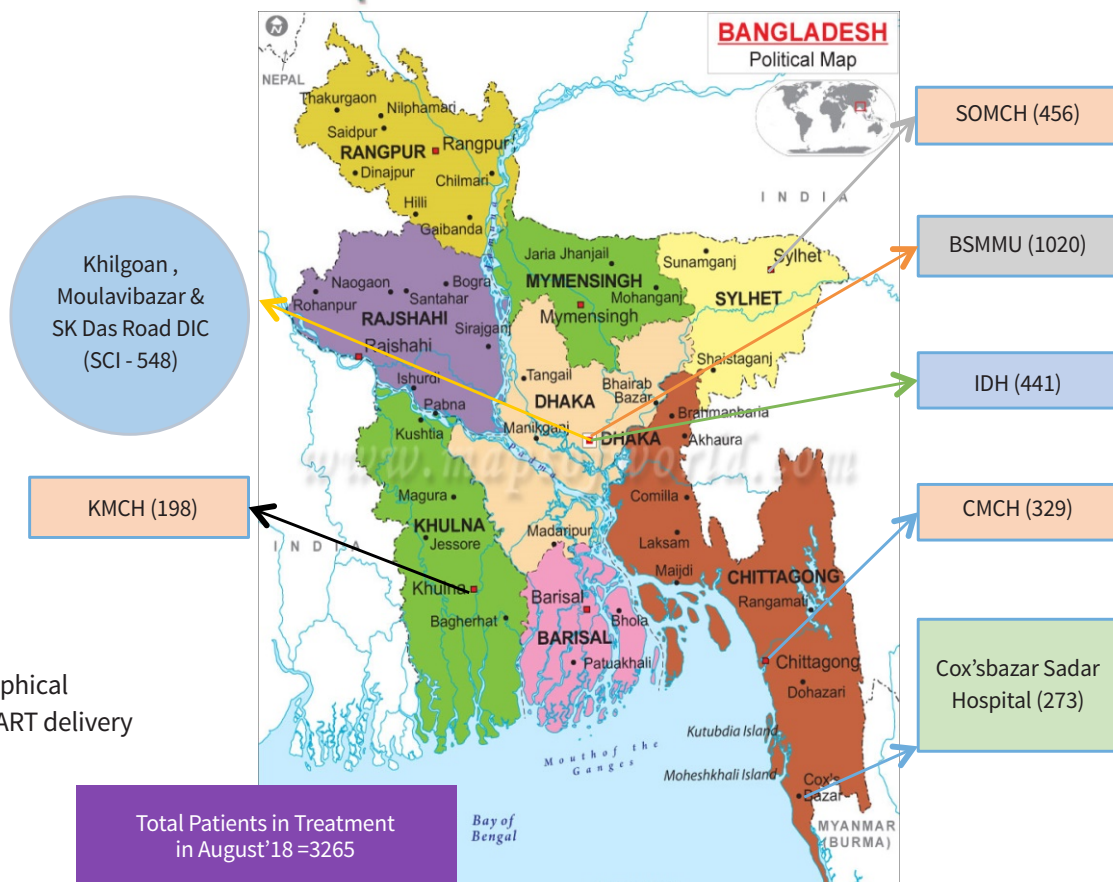


Figure 4.3.49 Geographical distribution showing ART delivery

Positive patients, and only 548 patients received ART from 3 comprehensive DIC, implemented by Save the Children with Government support.

Table 4.3.11 Key populations coverage on size estimation, 2015

Sl	Key population	Estimated size (2015-2016)
1	Street-based FSWs	41,350
2	Residence-based FSWs	39,078
3	Hotel-based FSW	17,976
4	Brothel-based FSWs	3,856
Total FSWs		102,260
5	MSM	101,695
6	MSWs	29,777
Total MSM		131,472
7	PWID (Male)	32,021
8	PWID (Female)	1,045
Total PWID		33,066
9	TG/Hijra	10,199

Table 4.3.12 Health service coverage among the vulnerable population groups

Key population	Size estimation 2015	% Covered by GF	% Covered by 4th OP
Female sex worker (FSW)	102,260	25%	13%
MSM and MSW	131,472	23.6%	3%
TG/Hijra	10,199	39.8%	10%
People who inject drugs (PWID)	33,067	35%	30%

Estimated size of key populations

A new estimation of the size of the key populations was planned in Bangladesh. As per existing information, the estimated sizes of the different key populations are shown in Table 4.3.11.

HEALTH SERVICE COVERAGE AMONG THE VULNERABLE POPULATION GROUPS

- Health services coverage was provided 25% to 39.8% of the estimated key population in 2015.

INVESTMENT CASE STUDY

- An initiative, relating to HIV/AIDS was taken in January 2015, to conduct an investment case study to explore and maximize the impact of HIV programs with limited resources, in a sustainable manner, to make substantial progress to the global goal of “Ending AIDS by 2030.” Update on “Investment Case 2018” is ongoing.

CARE, SUPPORT, AND TREATMENT SERVICES

- The Government has taken an initiative to provide optimum care, support, and treatment (CST) services at the government and NGO facilities for key populations vulnerable to HIV/AIDS and PLHIV.
- The Government had a target to provide optimum care to 65% PLHIV by 2014 under the Millennium Development Goal 6 (MDG 6).
- Therefore, the package of services aimed to take initiative for early detection of HIV, through HIV testing and counseling (HTC), to detect more cases and increase CST coverage.

GOVERNMENT INITIATIVE FOR COMPREHENSIVE CARE, SUPPORT, AND TREATMENT TO PLHIV

- ASP is procuring 100% ARV drugs from November 2012 onwards
- ARV drugs are dispensed through 6 government health

facilities

- HIV testing service is provided, with free of cost, from 23 government health facilities around the country
- 103 NGO facilities are providing BCC, community sensitization, drug adherence, opportunistic infections (OIs) management, and capacity-building of health service providers
- Five tertiary-level health institutions are supporting PMTCT (Prevention of mother-to-child transmission) among ANC attendees.
- In addition to the abovementioned treatment, care and support package, the Government of Bangladesh recognized the need of the key populations: FSWs, MSM, TG, and PWID and is providing prevention services for them. The major services covered STI management, BCC

and advocacy, HIV testing and counselling (HTC), and community sensitization.

NATIONAL REPORTING SYSTEM FOR HIV AND AIDS

- In 2013, a unified online national reporting system for HIV and AIDS was established. This was a collaborative initiative among ASP, ICDDR'B, MIS of the Directorate General of Health Services, and UNAIDS.
- Through the MIS database, data on HIV and AIDS program, relating to key populations, are now being collected on output/coverage indicators, every three months, from all drop-in-centers (DICs) and service delivery points, including HTC centers for the general population.

Non-communicable Diseases

Significant disease burden: increasing multi-Sectoral approaches

6.4%

of adults have **diabetes**

31%

of the population have raised total **cholesterol**

25.4% of the population have raised **blood pressure**

16.1%

of adults and **18.4%** of children and adolescent population suffer from any form of **mental disorder**

8%

reduction in **tobacco use**

All the primary health care settings are equipped with **basic technologies** and supplied with **medicines**, required to manage **HTN and diabetes**

Bangladesh is currently going through an epidemiological transitions. Like most countries, the burden of non-communicable diseases (NCDs) in Bangladesh is emerging as a public health challenge. Non-Communicable Disease Control (NCDC), the operational plan (OP), is taking the stewardship role on behalf of the Bangladesh Government. To overcome the increasing burden of NCDs, total allocation for this OP has been increased to more than three folds in the 4th HPNSP than that of the previous one.

Bangladesh is now going towards the successful achievements of the 2025 NCD targets and 2030 SDG targets. Overall mortality due to cardiovascular diseases, cancers, diabetes or chronic respiratory diseases has relatively reduced to 25% (WHO) from 2014 to 2017; Relative reduction in the prevalence of raised blood pressure is 25% from 2010 to 2018

The Government approved the 'Strategic Plan for Surveillance and Prevention of NCDs in Bangladesh' for five years in 2011. Very recently it has endorsed 'Multi-Sectoral Action Plan for Prevention and Control of NCDs 2018-2025' with national NCD targets, A National Multi-sectoral NCDC Coordination Committee (NMNCC) is already approved. NMNCC secretariat has been established and equipped with necessary logistics.

The Government has started dialogue with stakeholders, including WHO, on measures to improve diets, including awareness campaigns on dietary salt reduction. The NCDC program is piloting community based NCD management model (preferably hypertension and diabetes) in some selected sub districts following WHO PEN (Package of Essential Non-communicable Disease intervention) protocol. The cancer registry protocol and formats are on trial in eight government medical college hospitals under the guidance of NICRH and NCDC.

- 6.4% adults have diabetes
- 31% population have raised total cholesterol
- 25.4% population have raised blood pressure
- 16.1% of adult and 18.4% of child and adolescent population suffer from any form of mental disorder
- 8% reduction in tobacco use
- All the primary health care settings are equipped with basic technologies and supplied with medicines, required to manage HTN and diabetes

Source: STEPS 2018

COMPONENTS OF NCDC OP

- Major NCD components such as cardiovascular diseases (CVDs), cerebrovascular disease (stroke), cancer, diabetes, chronic obstructive pulmonary disease (COPD).
- The mental health components such as mental health, autism, neuro developmental disorders (NDDs), substance–abuse and harmful use of alcohol
- Other components such as injury prevention including poisoning and snakebite, violence against women and children, climate change, occupational health, environmental health hazards (water, arsenicosis, air, and soil), The prevention and treatment for disability through physical therapy/physiotherapy (especially among elderly people/senior citizens), palliative care, ear care, oral health, prevention of thalassaemia emergency preparedness and response (EPR) and post-disaster health management.

GOAL OF NCDC PROGRAM:

To reduce mortality and morbidity of NCDs in Bangladesh through control of risk factors and improving health service delivery.

OBJECTIVES OF NCDC PROGRAM:

- To promote development and implementation of effective, integrated, sustainable, and evidence-based public policies for non communicable diseases, their risk factors, and determinants
- To foster, support, and promote social and economic conditions that address the determinants of chronic non communicable diseases and empower people, to increase control over their health and to adopt healthy behaviors.
- To strengthen the capacity and competencies of the health system for the integrated early detection, management and control of the risk factors of non communicable diseases.

NON COMMUNICABLE DISEASE CONTROL TARGETS

In alignment with the UN High Level Political Declaration of 2011, Bangladesh is committed towards achieving the 2025 NCD targets and 2030 SDG targets. Potential indicators for the 2025 targets are listed in Annexure. Through the implementation of the Multisectoral NCD Control and Prevention of NCDs (2018–2025), Bangladesh aims to achieve the following proposed 2025 targets:

Table 4.4.1 Health indicators and service coverage among the vulnerable population groups

	Baseline	2018	2025 targets
Overall mortality from cardiovascular diseases, cancers, diabetes or chronic respiratory diseases	59% (STEPS 2010)	67%	25% relative reduction
Reduction in the harmful use of alcohol	0.9% (STEPS 2010)	1.3 (STEPS 2018)	10% relative reduction
Reduction in the prevalence of current tobacco use in persons aged over 15 years	43.3% (2009)	35.3% (2017)	30% relative reduction
Reduction in the prevalence of insufficient physical activity	27% (STEPS 2010)	15.1%(STEPS 2018)	10% relative reduction
Reduction in mean population intake of salt/sodium	10.6 gm/day (95% CI 9.6-11.6)	9 gm/day (95% CI 8.7-9.3)	30%relative reduction
Relative reduction in prevalence of raised blood pressure	17.9% (STEPS 2010)	25.4% (STEPS 2018)	25% relative reduction

	Baseline	2018	2025 targets
Halt rise in obesity and diabetes	Overweight: 14.3% =>BMI 25-29.9 kg/M2, obese: 3.6%=>BMI30kg/ M2 (STEPS 2010) Diabetes: 6.3 % of the adults (IDF) 5.5% (STEPS 2006)	Overweight: 22.6% =>BMI25- 29.9 kg/M2 obese:6.7%=>BMI30kg/M2 (STEPS 2018) Diabetes: 6.4 % of the adults (STEPS 2018)	0%
Reduction in the proportion of households using solid fuels (wood, crop residue, dried dung, coal and charcoal) as the primary source of cooking	41.4% straw & leave 4.8% brans 42 % wood, bamboo 10.3 Natural Gas (BBS 2005) COPD 4.3% (2007, Bangladesh Lung Foundation) 2.97 % (WHO survey)	30.2% straw & leave 3.5% brans 41.3% wood, bamboo 23.1% Natural Gas (BBS 2017) COPD 1.9% (2018, DGHS, NCDC)	50%
Increase the number of eligible people receiving drug therapy and counseling (including glycaemic control) to prevent heart attacks and strokes		20 Upazila Health Complexes /200 community clinics	50%
Improve the availability of affordable basic technologies and essential medicines, including generics, required to treat major NCDs in both public and private facilities	NCD corner piloting in every Upazila Health Complex	All primary healthcare settings are equipped with basic technologies and supplied with medicines, required to manage HTN and diabetes	80%

NATIONAL INTEGRATED NCD ACTION PLAN

Responding to the commitment made in 2014 Outcome Document for developing national multi-sectoral policies and plans to achieve the national targets by 2025, The Ministry of Health and Family Welfare adopted the “Multi-sectoral Action Plan for Prevention and Control of Non-communicable Diseases (MAP)” in May 2018, with a vision to reduce premature deaths and disabilities due to non-communicable diseases (NCDs). The plan is a priority blueprint for key-stakeholders and includes a three-year operational plan from 2018 to 2021 in alignment with the 7th Five Year Plan and the 4th Health, Nutrition and Population Strategic Investment Plan (HNPSIP) of the Government. Implementation of the action plan employs a “health in all policies” approach, engaging actors outside the health sector to influence public policies on shared risk factors, such as tobacco use, unhealthy diet, physical inactivity, harmful use of alcohol, and exposure to poor indoor air quality.

The objectives of MAP are to scale up responses to NCDs through effective multi-sectoral partnerships; improve the capacity of individuals, families and communities to live a healthy life and reduce the risk of developing NCDs by increasing health literacy and creating healthy and safe environments; strengthen the health system by improving access to health care services for primary prevention, early detection and treatment of NCDs; and establish a sound surveillance, monitoring and evaluation system that generates data for evidence-based policy and program development. In alignment with the UN High Level Political Declaration of 2011, Bangladesh is also committed towards achieving the 2025 NCD targets and 2030 SDG targets.

The MAP is based on the four strategic priority action areas in alignment with the Action plan for the prevention and control of NCDs in South-East Asia. Actions and activities that are potentially implementable, low cost, and have a high health impact are included in the plan.

Action area 1:

Advocacy, leadership and partnerships aimed at increase advocacy, promote multi sectoral partnerships and strengthen capacity to accelerate and scale-up the national response to the NCD epidemic by establishing a National Multi-sectoral NCD Coordination Committee (NMNCC).

Action area 2:

Health promotion and risk reduction for promoting the development of population-wide interventions to reduce exposure to key risk factors. Actions include: fully implementing tobacco control laws; placing restrictions on availability of illicit alcohol; developing collaborative efforts to reduce trans-fats, saturated fats and salt intake; encouraging consumption of adequate servings of fruits and vegetables; encouraging physical activity; and establishing healthy settings in cities, schools and work places.

Action area 3:

Health systems strengthening for early detection and management of NCDs and their risk factors aimed at improving the efficiency and coverage of NCD services to achieve universal health coverage (UHC), particularly through the primary health care system. Key activities include: screening for NCDs and its risk factors; increasing capacity for prevention and management of NCDs through the essential service package in primary health care settings; reviewing the essential medicines list and making basic NCD medicines available at the primary health care level; integrating healthy lifestyle education in all health facilities; incorporating NCD prevention and management into the health workforce curriculum, and identifying sustainable health financing options to cover basic NCD services to protect the poor from financial risks.

Action area 4:

Surveillance, monitoring and evaluation, and research which includes key actions for strengthening surveillance, and monitoring and research in NCD control. Key activities include: conducting surveys on tobacco and NCDs at regular intervals; strengthening national cancer registration; generating NCD mortality data through civil registration and vital statistics; strengthening the management information system (MIS) to capture hospital-based NCD data; developing a national priority research agenda for NCDs; ensuring implementation evaluation of the NCD operational framework; and evaluating compliance with tobacco laws, and food safety regulations policies.

MAJOR NCDs (CVD, CANCER, DIABETES, COPD)

- In Bangladesh the burden of non-communicable disease is an emerging public health challenge. Cardiovascular disease, diabetes, cancer and chronic respiratory disease are responsible for 67% of all death in Bangladesh.
- Total number of annual NCD death is over 500000. Among them about 22% are premature deaths. 6.4% of adult population is diabetic and 25% is hypertensive
- According to Global Burden of Disease Study 2016 report, age-standardized death rate due to NCDs among all population in Bangladesh is 369.78 (335.79-406.84) per 100000 population.
- Death rate due to major NCDs (CVDs, cancer, diabetes, and chronic respiratory diseases among population aged 30 to 70 years in (398.1-503.7) per 100000 population.

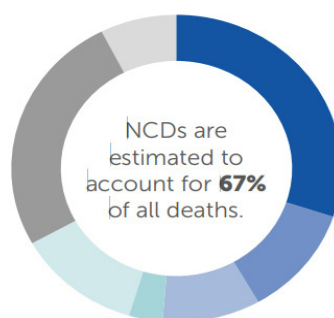
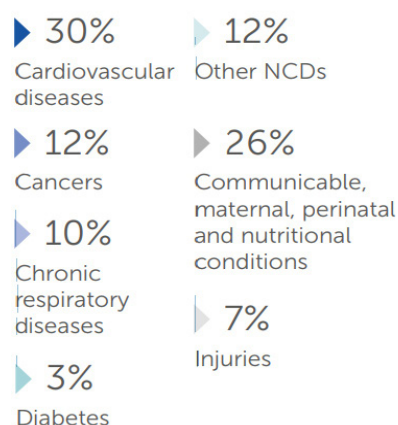


Figure 4.4.1 Proportional mortality in Bangladesh in 2016 (according to WHO)

ONGOING ACTIVITIES ON MAJOR NCDS IN NON-COMMUNICABLE DISEASE CONTROL PROGRAM, DGHS

1. Early detection for prevention and control of non-communicable diseases through community clinic.
2. Setting up of NCD corners in the sub district and district level health facilities are going on.
3. Essential medicines, diagnostics and equipment are made available at each tier of the healthcare systems including both in public and private sector. National Institute of Cardiovascular Diseases has recently been expanded.
4. Inclusion of NCD risk factor variables in DHIS2.
5. Life style modifications and physical activities promotion programs are being strengthened and expanded.
6. Developed and adopted guidelines to manage NCDs, especially for Primary Health Care (PHC) settings, based on the WHO Pack- age of Essential Non-communicable Disease Interventions (WHO PEN).
7. Agreements with different private sectors/NGOs/INGOs to develop innovative sustainable community based NCD management model (preferably hypertension and diabetes) in some selected sub districts.
8. Drugs for NCDs have been included in national essential drug lists and national drug procurement systems. Overall, around 11% of the MOHFW budget is allocated to procure medicines and medical supplies.

ONGOING ACTIVITIES ON CANCER

1. The cancer registry protocol and formats are on trial in eight government medical college hospitals under the guidance of NICRH and NCDC.
2. Developed and adopted guidelines to manage NCDs, especially for Primary Health Care (PHC) settings, based on the WHO Pack- age of Essential Non-communicable Disease Interventions (WHO PEN).
3. A multi-sectoral action plan has been adopted defining specific roles for each NCD stakeholder.

MENTAL HEALTH

Table 4.4.2 Mental health situation in Bangladesh as per STEPS 2018 Survey

Common mental disorder among	Percentage of all population suffering from any mental disease
Adults - Anxiety disorder - Depression - Schizophrenia - Substance abuse	16.1
Children and adolescents - Mental retardation - Autism	18.4

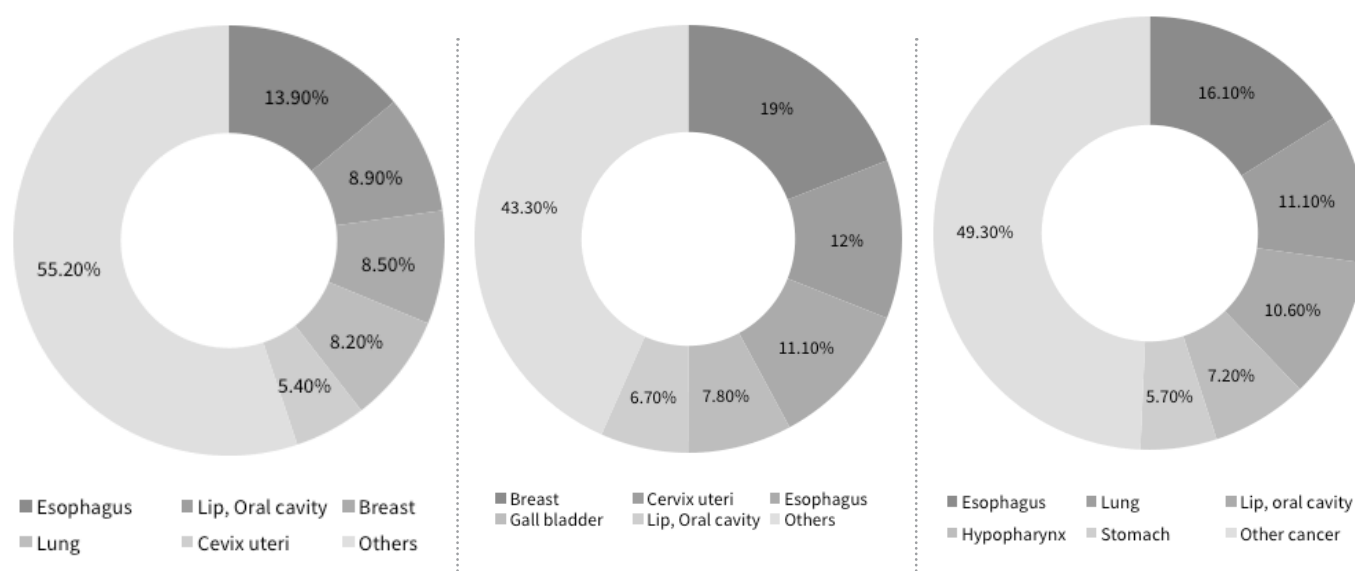


Figure 4.4.2 Number of new cases of cancer in 2018 in both sexes, all ages (GLOBOCAN 2018)

Source: (GLOBOCAN 2018)

TOBACCO CONTROL:

- Bangladesh, one of the first countries to sign the WHO FCTC, has accomplished the Global Adult Tobacco Survey (GATS). This is a key step of Tobacco Control Framework (FCTC)
- The number of smokers is decreased at an encouraging rate. In 2009, the percentage of average smokers was 43.3% whereas in 2017 it decreased to 35.3%, a reduction by 18.5% which is a remarkable achievement.
- Passive smoking also reduced remarkably. Smoking in restaurants reduced from 79.7% to 49.7%, indoor smoking from 62.2% to 42.7%, and in public transport from 53.6% to 44%.
- Advice to 'Quit Smoking' by health staffs increased up to 65.8% from 52.9%.
- Tobacco advertising, promotion and sponsorship drastically have fallen to 5.5% from 43.6%

SHISHU BIKASH KENDRA OR CHILD DEVELOPMENT CENTRE

Child Development Centre (CDC), commonly known as Shishu Bikash Kendra (SBK), are multidisciplinary specialized service centers located in different medical college hospitals in Bangladesh catering the need for providing the diagnostic and therapeutic services for children with ASD and other NDDs. There are 15 CDCs in the country at present in medical college hospitals.

25 more CDCs are planned to be established in the next 4 years in medical college and district hospitals. The clinics emphasizes on development and psychological assessments of the children with ASD and NDDs and the follow-up visits for ensuring functional improvement in motor, vision, hearing speech communication skills, cognitive and behavioral domains.

Each CDC is run by a multidisciplinary team comprising a Child Physician, a Child Psychologist and a development therapist. The team is dedicated to ensuring that every child came to the centre is undergone a comprehensive assessment, diagnostic work-up and selection of best suited intervention technique. They work closely with the parents and family of the child to optimize his/her development potential, improve the quality of life and prevent disability.

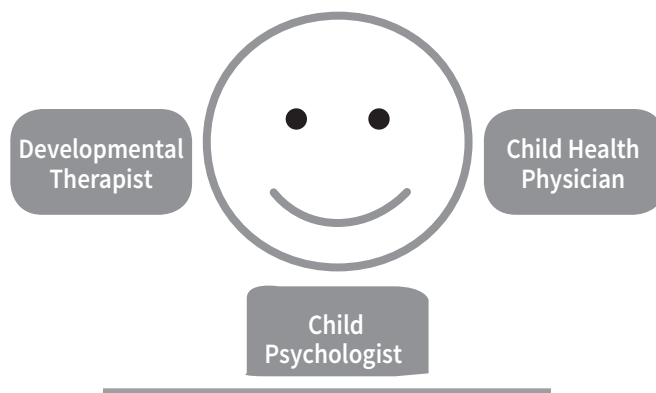


Figure 4.4.3 Multidisciplinary team at Shishu Bikash Kendra (SBK)

The journey began with the establishment of 5 SBKs in August 2009 in Dhaka Medical College Hospital, Shaheed Suhrawardi Medical College Hospital, Sir Salimullah Medical College & Mitford Hospital, Sher-e-Bangla Medical College Hospital, Barisal and MAG Osmani Medical College Hospital, Sylhet. Following this success 5 more centers were established in the following year that included Chittagong, Khulna, Rajshahi, Mymensingh and Rangpur Medical College Hospitals. In February 2014, another 5 centres started providing services in Dinajpur, Faridpur, Comilla, Cox's Bazar and Shaheed Ziaur Rahman Medical College Hospital, Bogra.

Under the 4th HPNSP Hospital Services Management OP will establish 25 more SBKs in the next three years. There are several other CDCs working in autonomous and private hospitals. They include Dhaka Shishu (Children) Hospital, BIRDEM, Shishu Swasthyo Foundation Hospital, ICMH, Matuail, CMH, Dhaka and Apollo Hospitals, Dhaka.

Since the inception of services in 2009, 124649 children (till June, 2018) were served in the 15 CDCs of them 15839 were served during July 2017 - June 2018 period. In addition to diagnosis and intervention of ASD and NDDs, the CDCs have been running weekly clinics for epilepsy, mental health, and well-baby clinic. In total 294432 follow-up visits were recorded during the same period. Rapid Neurodevelopmental Assessment (RNDA), Autism Diagnostic Observation Schedule (ADOS), Independent Behaviour Assessment Scale (IBAS), Bayley Scales for Infant Development (BSID), Wechsler Intelligence Scale for Children (WISC IV), DSM-V, ICD-10 are used among other scales and methods for assessment of children here. The

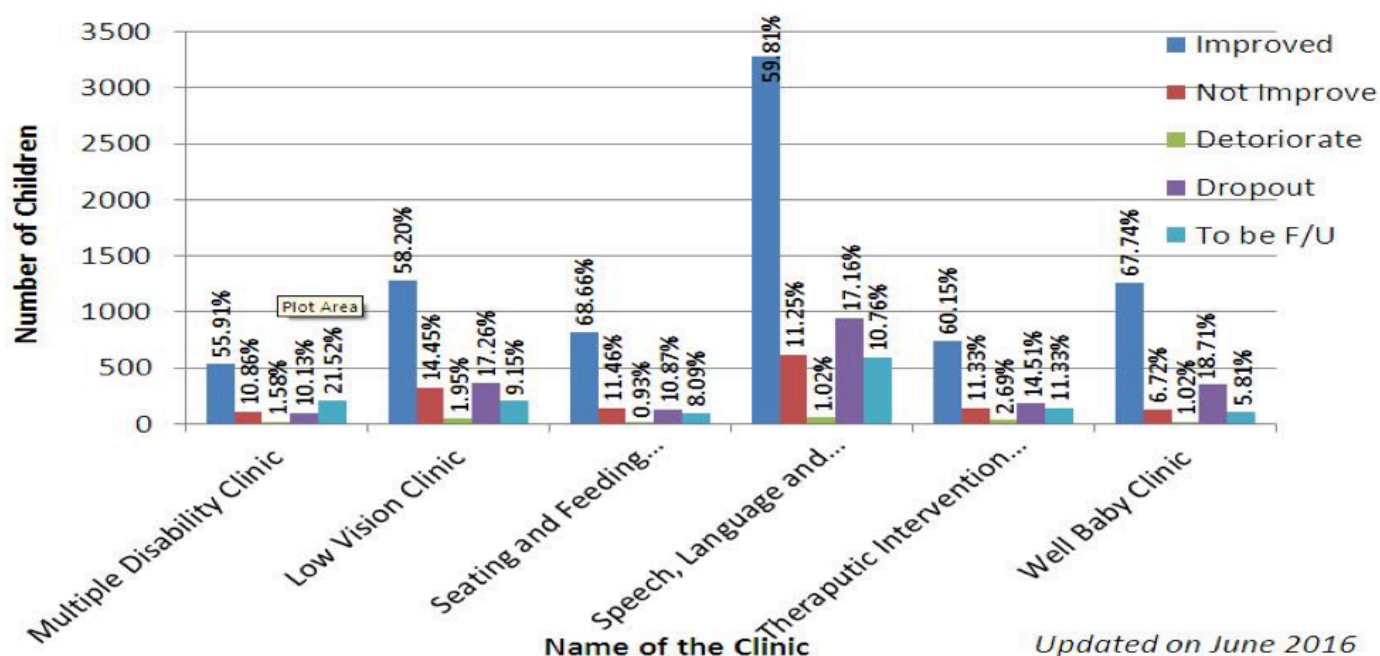


Figure 4.4.4 percentage of children prognosis according to type of clinic

CDCs are providing technical assistance to various Child development programme in Bhutan, Guatemala, Haiti and Zambia in recent years. The CDCs are continuously engaged in research activities and conducted three major epidemiological surveys on Autism and NDDs.

INJURY SITUATION IN BANGLADESH

Injury is also the leading cause of disability. The WHO estimates that approximately 70,000 yearly deaths occur in Bangladesh due to various types of injury (burn, drowning, acid and accidents at work), estimated 18,000 yearly death due to road traffic injury. Poisoning is a leading cause of morbidity and mortality across the country with an approximate annual admission of 64,000 cases, (2% of all admission) injury and poisoning is responsible for 9% of all hospitalized deaths. Snakebite is a rural health problem among the poor farming community causing more than 700,000 episodes and more than 6000 deaths.

Prevention and emergency management of various injuries will be addressed in the Component-5 of NCDC. These include: road traffic injury, drowning, prevention and treatment of poisoning, burns, prevention and treatment of snakebite, occupational injury, intentional injuries, and violence against women (GBV).

ROAD TRAFFIC INJURY

The National Committee to Protect Shipping, Roads and Railways (NCPSRR), a non-government organisation, revealed an information on road traffic accident in this yearly survey and observation report. According to the report, the number of road accidents and casualties increased in 2017 compared to 2016. The number of accidents increased by 15.82% and death toll by 25.56% in 2017 compared to that of 2016. In 2017, the total number of accidents was 3472 in which total number of death was 4,284 (including 516 women and 539 children) and number of injured person was 9,112.

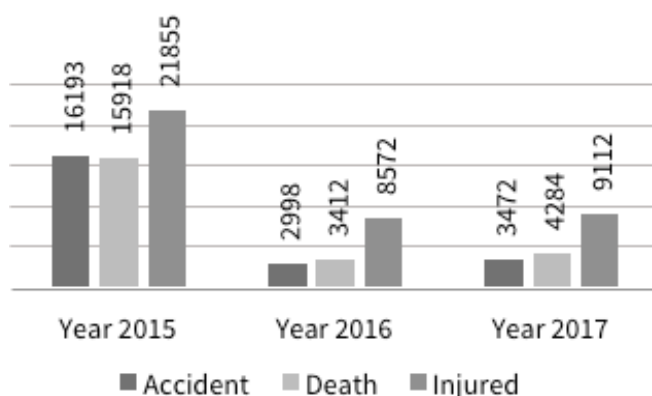


Figure 4.4.5 Number of accidents, injuries and death in last 3 years found in NCPSRR 2017

On going activities on Road Traffic Injury

1. Education and publicity to improved knowledge and behavior of road users, and to raise awareness about traffic rules and safety. Multi-sectoral co-ordination (non-health interventions and stewardship)
2. Education of pedestrian regarding the use of road, foot path and over bridge and signals through school, community safety program and media. Multi-sectoral coordination (non-health interventions, and stewardship).
3. Driver recruitment, training and testing – Ensuring minimum standards for drivers such as ensuring eligibility according to age, physical fitness and provide adequate training on safe driving practices. Multi-sectoral co-ordination (non-health interventions, stewardship and regulatory).
4. Training on pre-hospital care by the first responder and comprehensive training on trauma case management to the health care providers. Cross cutting issue: Linked to Essential Services Delivery (ESD), Secondary and Tertiary-level Hospital Management System (HSM).
5. Engineering and environmental approach multisectoral co-ordination (non-health interventions, stewardship and regulatory).
6. Legislation and standards multisectoral co-ordination (non-health interventions, stewardship and regulatory).
7. Vehicle design and safety equipment multisectoral co-ordination (non-health interventions, stewardship and regulatory).
8. Emergency medical care Cross cutting issue: Linked to Essential Services Delivery (ESD), Secondary and Tertiary-level Hospital Management System (HSM). (i) Ensure emergency assistance for rescue and transportation of the crash victims within golden hour. (ii) Establish emergency medical service system – Pre-hospital care and hospital care. (iii) Ensure treatment of the road crash victims in public and private hospitals.
9. Collaboration and coordination multisectoral co-ordination (non-health interventions and stewardship). (i) Effective co-ordination and collaboration of road safety council which is responsible for planning, implementation and monitoring of road safety activities, formulating road safety policy, facilitating funding,

dissemination of information, organizing road safety education and publicity and training. (ii) Collaboration between relevant sectors/departments. /agencies for injury information and reporting and also in providing emergency medical services. (iii) Effective collaboration with vehicle owners association and workers association.

Snake bite

- Snake bite become important health hazard in Bangladesh. There are more than 6000 deaths and 700000 bites annually imposing a great burden on health care system and also economy of the country.
- Medically important snakes include Cobra, Krait and Viper. Most common victims are young, working rural people engaged in farming and cultivation making snakebite an important occupational health hazard. The incidence is also likely to increase during natural disasters.
- The majority of the bite (71%) took place in the lower extremities. Eighty six percent of the victims receive some form of management within two hours of snakebite, although only 3% of the victims went directly to either a medical doctor or a hospital (According to a nationwide community-based epidemiological study of snakebite and its socioeconomic consequences in Bangladesh-2010)

Activities for Snake Bite Prevention

1. A national level guideline on snake bite management has developed by 3 consultative workshops and approved by MOHFW. Guideline is in the printing process.
2. Establishment of Venom Research Centre in the Zoology Department of Chattogram University to produce Anti Snake Venom (Made in Bangladesh) for the very first time.

CLIMATE CHANGE AND RESPONSE

- The Non-communicable Disease Control Unit of the DGHS, with technical support from WHO, has an action plan which includes raising awareness and building capacities of medical professionals towards understanding the potential impact of climate change on human health and adaptation.
- Accordingly, training modules—one for doctors and

another for nurses, paramedics, health assistants, and family welfare assistants—were developed and published. Several training and orientation sessions have been organized at the central, district and upazila levels. Around 50000 field level personnel have been trained on climate change, its health impact, and management. After receiving training, the field-level health workers are now organizing courtyard sessions at the communities. They are also registering climate-sensitive diseases in their respective areas. Necessary collaborations are being made to strengthen the activities of Climate Change Health Promotion Unit (CCHPU) of the Ministry of Health and Family Welfare. The WHO-IEDCR has jointly conducted the ‘Health Vulnerability and Adaptation’ assessment of the current and future scenarios. Later, another regional assessment of ‘Vulnerability and Adaptation’ was carried out in the coastal and drought-prone areas of Bangladesh. The assessment generated primary qualitative and quantitative information regarding the health, including the ‘WASH’ components (water, sanitation and hygiene) vulnerabilities of the study population, which will support informed decision to define policy and strategy to addressing the impact of climate change. Development process of the Health-National Adaptation Plan (H-NAP) may also be benefited from findings of this assessment.

OCCUPATIONAL HEALTH

- Occupational Health is a neglected but very important is our country perspective. In a developing country like Bangladesh, occupational health and related hazards greatly affect the productivity of the country and may hamper development. But often our workers as well as the health care professionals are reluctant about occupational safety measures. Even the health care workers don't know that in how much virulent environment they are working in, what routine safety measures are to be taken, what the potential hazards of their job are. So, in our country perspective, this is the time to focus on this sector to ensure a optimal working environment, so that Bangladesh will be able to achieve the 2030 sustainable development goal.
- Ongoing activities: Capacity building of Doctors, Nurses and Para health professionals at district level for management and prevention of common occupational

health problems through Training.

ENVIRONMENTAL HEALTH HAZARD

- Ongoing activities: Sensitization of Doctor, Nurse and para health professionals regarding harmful effects of environmental health hazards by Seminars at District level

ARSENICOSIS

- Arsenicosis is a major public health problem. Groundwater contamination by arsenic resulted in chronic arsenicosis in a large area of the country. Various health problems including skin problems, peripheral nerve problem even cancer can occur due to arsenic contamination.
- Collaboration between DGHS and DPHE was ongoing to address the problem under a National Arsenic Mitigation Policy (2004) for providing safe water and providing care to the patients.
- Arsenicosis Scenario in Bangladesh 2017: In a survey titled ‘Prevalence Of Of Arsenicosis And Their Treatment Availability’ was conducted in 2017-2018 Financial Year funded by NCDC operation plan. The key findings are-
 - i. Heavily contaminated districts are Bagerhat (100%), Chuadanga (100%), Lakshmipur (92%), Jashore (75%), Magura (75%), Feni (67%), Chattagram (58%), Satkhira (50%)
 - ii. Frank clinically diagnosed patient found in Laxmipur and Noakhali, Chattagram, Jaipurhat
 - iii. Urinary excretion of arsenic without any clinical manifestation seen in Chandpur (2psn), Netrokona (1psn), Jashore (1psn), Narayanganj (1psn), Sylhet (1psn) and Moulvibazar (1psn)
- Apart from this study, a provisional report was collected from 42 districts gathered from the arsenicosis patient registrar book. It revealed a total of 18862 patients in October 2017.

Ongoing Activities

- Workshop on updating Arsenicosis detection, screening tools, management protocol,
- Updating Arsenic mitigation policy

- Training on Arsenicosis screening, management and treatment protocol among Doctors at selected district level for capacity building.

THALASSEMIA

- Thalassaemia is an autosomal recessive disease, which means that if two carriers get married to each other then they will have a 1 in 4 (25%) chance of having an offspring with B-Thalassaemia Major in every pregnancy.
- B- Thalassaemia (Hb E disease) is the most common genetic disorder in Bangladesh with a gene prevalence rate of around 6%. This translates that 6 out of every 100 Bangladesh carry this mutant gene. With a population of about 160 million, around 10 million people are healthy carriers of this gene. Over 5000 affected children are born.
- These children require regular monthly blood transfusions and halation therapy to remove the excess iron from their bodies in order to just stay alive. A large amount of money is required for each child per annum. Currently, majority of the patients in Bangladesh do not get adequate treatment, especially the transfusion of safe blood has been a big problem and the majority of the Thalassaemics are positive for infections such as Hepatitis B, C and HIV.
- Since there is no easily available cure for this disorder, the only hope lies in adopting a preventive program on the line pursued by countries like Italy, Cyprus and Iran, which has resulted in either complete control or significant reduction in the births of new Thalassaemia Major Children in these countries. Considering the gravity of the issue, the Government of Bangladesh should take the lead and initiative Bangladesh Thalassaemia Prevention Programme (BTPPP), which will be unique and can be compared with any other preventive program of the world.
- Ongoing activities on Thalassemia: Day Celebration, International Conference on Prevention & Management of Thalassemia
- For Future there is plan of Awareness programme, Module development, Iron chelating drug, hemoglobin electrophoresis machine at district level for Thalassemia prevention

DISABILITY

- Disability refers to a situation where any person who is affected with long-term or permanent physical, psychological, intellectual, developmental and sensory-related damages or obstacles which in interaction with attitudinal and environmental barriers toward the person restricts the person's equal and effective participation in society.
- The Household Income and Expenditure Survey (HIES) of Bangladesh (2010) puts the prevalence rate of disability at 9.1%
- The DGHS's effort to 'demystifying disability' for primary health care (PHC) workforce, particularly medical doctors is a way forward to strengthen health care systems and structures by making it more responsive, equipped and ready to cater people with disabilities.
- Common Disabilities in Bangladesh includes NDDs, Autism Spectrum Disorder (ASD), Cerebral Palsy (CP), Down Syndrome, Intellectual Disability, Mental Disability, Physical Disability, Visual Disability, Hearing & Speech Disability, Deaf-Blindness Disability, Multiple Disability and Others Disability.

ONGOING ACTIVITIES ON DISABILITY IMPROVEMENT

- Development of guideline- Defining disability for health workforce in Bangladesh (by 06 workshops for one year 2017-18)
- Development of Sign Language Training Manual
- Sign Language Training in Upazila.
- Survey done: Service Availability and Readiness Assessment (SARA) survey for NCDC and disability Service delivery system in Bangladesh
- Priority based circulation for management of disability is given at all level
- Shibalay, Manikganj Pilot project is ongoing (04 Sensitization workshop for doctors already done & 03 field worker training program ongoing)
- According to guideline development of module preparation activities ongoing
- 01-03 Days Compulsory training for development of at least one doctor for identification of disability at upazila level.

THE STEPS SURVEY

The STEPS survey for Non-communicable disease (NCD) risk factors in Bangladesh was carried out from January to June 2018. Bangladesh followed all three steps (step—1, Step—2 and Step—3) in this survey. A total of 8185 adults participated in the survey. The overall response rate was 82.67%.

STEP—1

(Socio demographic and behavioral information)

1. Population characteristics

Among the 8185 respondents, 53% were female and 47% were male. The average age of the respondents was around 40 year.

2. Unhealthy diet

a. Fruits and vegetables consumption

82% of population don't eat recommended amount of fruit or vegetables. (men 83%, women 80%).

b. Extra salt intake

About half (47%) of the population take extra salt while eating (women 49%, men 44%).

2010 vs 2018: Consumption of required amount of fruit

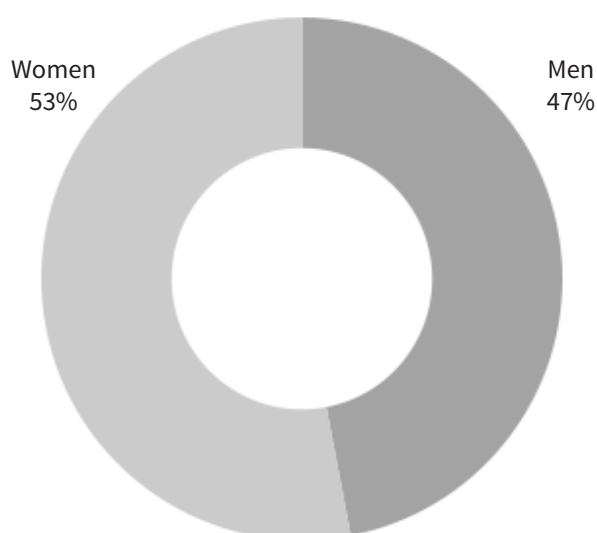


Figure 4.4.6 Characteristics of the survey respondents

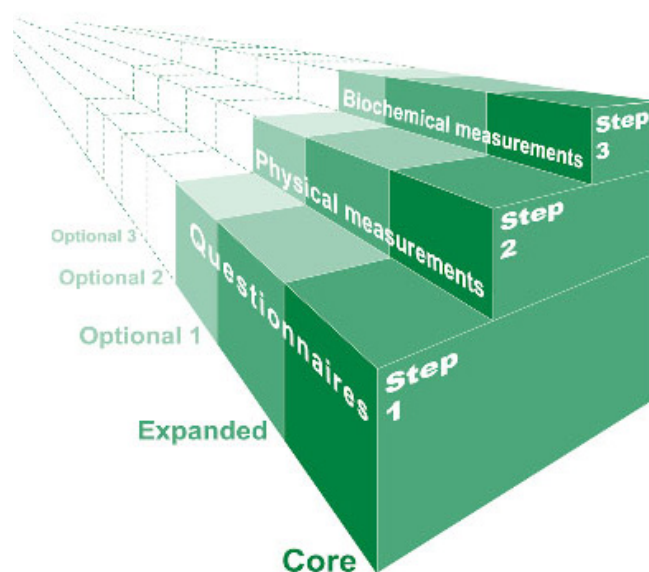


Figure 4.4.7 Steps of STEPS survey

and vegetables has increased from 2.3% (2010) to 18.2% in (2018)



3. Physical inactivity

2010 vs 2018: Insufficient physical activity has dropped down to 15.1% (2018) from 27% (2010). That means the people are becoming more physically active than the previous time.



4. Harmful use of alcohol

2010 vs 2018: Lifetime abstainer has decreased from 94% (2010) to 92% (2018). The percentage of current drinker has increased to 1.3% (2018) from 0.9% (2010).



5. Tobacco use

Tobacco data are almost same as GATS 2017 data.



6. Cervical cancer

Majority of the respondents never had screening for cervical cancer. Only 7.5% women age between 30—49 years had

cervical cancer screening test in their lifetime.



7. Oral health

About 39.5% of the respondents have pain or discomfort caused by their teeth or mouth during the past 12 months, however only 5.8% had seen a dentist within the past 6-12 months. On the other hand, 69.0% never received any dental care.

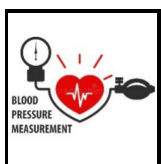
STEP—2

(Physical measurement)



8. Body mass index (bmi)

Study found 13% were underweight (BMI <18.5 kg/m²) and 22.6% were overweight (BMI ≥25—29.9 kg/m²), 6.7% were obese (BMI ≥ 30 kg/m²).



9. Blood pressure (bp)

2010 vs 2018: The trend of getting raised blood pressure shows in increment from 2010 (17.9%) to 2018 (25.4%).

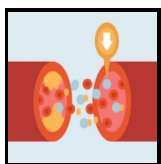
STEP—3

(Biochemical measurement)



10. Diabetes

Prevalence of diabetes was 6.4% among the sample population. Again, 6.4% of the respondents were pre-diabetic. 45.2 % of those having raised blood glucose, were taking medication.



11. Cholesterol

About 31% of the study population had raised total cholesterol. Among them 4.7% were taking medication. The survey revealed about 28.6% had raised cholesterol level: they never checked their blood cholesterol previously; this survey noticed their raised blood cholesterol for the first time.

National Eye Care

General objective: To improve eye care service delivery at all levels of health facilities in Bangladesh.

SPECIFIC OBJECTIVES:

- To improve eye care facilities at the primary and secondary care levels with needed infrastructure development, appropriate equipment and MSR.
- To improve capacity of HR related to eye care services (Ophthalmologists, Nurses, Paramedics).
- To expand & sustain Demand Side Financing (DSF) / Vouchering scheme to increase accessibility to the poor, marginalized and disadvantaged population.
- To control childhood blindness and increase awareness of mass population on blindness prevention.
- To reduce avoidable blindness due to cataract, refractive error, diabetic retinopathy, glaucoma, ocular injury and others.
- To strengthen coordination among GO-NGO & private eye care providers

COMPONENT/ACTIVITY-WISE PHYSICAL PROGRESS

- 24 cataract screening & surgical camp has been successfully held in this year.
- Total cataract surgery: 6320 (4118 under DSF activities)
- Total OPD service: 73995
- 18312 pieces reading glasses supplied in free of cost
- 30 ophthalmologists have been Trained in 8 batches on microsurgery
- 30 nurses have been Trained in 5 batches on Eye OT & Ward Management
- 22 ophthalmologists have been Trained in 4 batches on glaucoma, retina & Paediatric Ophthalmology.
- 300 health assistants of have been trained for Primary Eye Care training different District.
- 5312 children had given treatment and 300 pairs of glass had delivered for refractive error as part of school sight testing program.
- One diabetic retinopathy screening camp has been held in Gopalganj District.
- One glaucoma screening camp has been successfully held in this year in Narayanganj District.

Disease Surveillance and response of IEDCR

BACKGROUND:

Institute of Epidemiology, Disease Control and Research (IEDCR) is the focal institute for conducting disease surveillances and outbreak investigations in Bangladesh. IEDCR has been engaged in controlling diseases and conducting research on events of public health importance and developing public health workforce. The IEDCR comprises eight departments; Biostatistics, Epidemiology, Medical Entomology, Medical Social Science, Microbiology, Parasitology, Virology, and Zoonosis.

The IEDCR has five laboratories: (i) Virology, (ii) Parasitology, (iii) Microbiology, (iv) Entomology, and (v) One Health laboratory. IEDCR has the national reference laboratories for influenza, Nipah, and HIV. These laboratories play a significant role in the diagnosis of emerging infections, conduct and assist in training programs for laboratory personnel, and take part in quality assurance program.

MAJOR ACTIVITIES OF IEDCR IN 2017

• One Health

- *Revision of One Health Strategic Framework*

Strategic framework for One Health approach in Bangladesh 2022 was developed and later endorsed by Ministry of Health & Family Welfare, Ministry of Fisheries & Livestock, Ministry of Forest & Environment to infectious diseases is in the process of government approval.

- *Zoonotic Disease Prioritization*

In July, 2017, the One Health secretariat arranged a workshop to prioritize the zoonotic diseases in. The following six diseases have been prioritized- Anthrax, Brucellosis, Nipah, Rabies, Zoonotic Influenza and Zoonotic Tuberculosis.

- *Coordinate Outbreak Investigations and Responses*

One Health secretariat has coordinated zoonotic disease outbreak investigations and responses particularly on anthrax and avian influenza.

- *One Health Conference*

IEDCR with its partner arranged 9th One Health Bangladesh Conference on 17-18 September, 2017 at Radisson Blue Dhaka Water Garden. The theme of the conference was 'Achieving Sustainable Development Goals through One Health Approach'.

• Public Health Emergency Operation Centre

In Bangladesh, the Public Health Emergency Operations Centre (PHEOC) was formally launched for the first time at IEDCR in June, 2017. The US CDC has been providing formal trainings to IEDCR officials. The PHEOC of IEDCR coordinates the public health emergency responses in Bangladesh. PHEOC remains in

‘watch mode’ in searching of public health emergencies. When any such event is likely to occur, the PHEOC moves to ‘alert mode’ where not only communicate with stakeholders but also take preparations to control those emergencies. During a large scale public health emergency, the PHEOC moves to ‘response phase’.

- **Chikungunya Outbreak, 2017**

In July 2017, PHEOC was activated for the first time in response to the chikungunya outbreak in Dhaka metropolitan city. An Incident Action Plan (IAP) was developed for the control and management of the outbreak. During the outbreak, IEDCR confirmed the diagnosis of suspected patients, provided health advice by hotlines, published daily news bulletin and conducted regular media briefing, laboratory validation of different diagnostic methods, risk communication to members of the national rapid response team, civil surgeons and journalists of print and electronic media, engagement of civil society for social mobilization and expert consultation on vector control in urban and rural areas.

- **Activities among Forcefully Displaced Myanmar Nationals (FDMNs) in Cox’s Bazar Bangladesh**

The PHEOC of IEDCR has been participating in the coordinated national and international efforts to overcome the health related challenges of these huge number of influx people from the neighboring country.

- **Rapid Baseline Risk and Need Assessment Survey**

The PHEOC conducted a rapid baseline risk and need assessment survey to provide valuable information for

taking appropriate public health measures.

- **Expanding Cholera Surveillance**

The PHEOC, IEDCR also conducted a feasibility study for the expansion of the ongoing diarrhoeal disease (Cholera) surveillance among the FDMNs in collaboration with icddr,b. It is already started in 7 sentinel sites which will be expanded to 10 sites.

- **Diphtheria Outbreak in FDMNs**

In response to a diphtheria outbreak among FDMNs, an Outbreak investigation team was deployed from the PHEOC, IEDCR in early December, 2017. The team worked together with concerned national and international agencies to control and manage the outbreak. IEDCR provided laboratory support during outbreak to confirm the diagnosis of diphtheria among FDMNs.

- **Integrated Disease Surveillance among FDMNs**

Foreseeing the threats of impending outbreaks of other diseases among this vulnerable population, PHEOC of IEDCR has planned to launch IDS for the early warning, alert and response to prevent, control and manage if there any outbreak.

- **Training on Incident Management System (IMS)**

PHEOC also conducted an orientation training on IMS for the health officials in Cox’s Bazar.

- **Biosafety and Biosecurity**

To safeguard the people along with the healthcare professional from pathogens, Institutional Biosafety and



Biosecurity Committee (IBC) formed at the IEDCR in 2017.

• Entomological Activities

The laboratory of medical entomology department is carrying out the biological efficacy tests of different types of public health insecticides supplied by different departments. The insecticides are tested against mosquito, housefly, cockroaches (American & German), and bed bugs. This department is providing regular services at free of cost to many organizations. In 2016-17, medical entomology department investigated the bioefficacy of over 500 insecticides.

SURVEILLANCE ACTIVITIES OF 2017

• National Influenza Surveillance, Bangladesh (NISB)

Through NISB, IEDCR collects nasopharyngeal and throat swab from patients with Influenza Like Illness (ILI) and Severe Acute Respiratory Infection (SARI). NISB has ten sentinel sites throughout the country. The sentinel sites includes eight district hospitals and two medical college hospitals.

In total 2,748 samples were collected from SARI and ILI patients in 2017, out of which 336 (12.22%) were influenza, of them, 190 (57%) were influenza A and 140 (42%) were influenza B. Common sub-types of influenza A was AH1pdm09 (64%) and influenza B was B/Vic.

• Hospital Based Influenza Surveillance (HBIS)

Nasopharyngeal and throat swabs are collected from each enrolled SARI case for molecular diagnosis of influenza virus infection. The platform started subtyping of influenza B from 2016.

Of 2,726 samples collected from SARI patients (age ≥ 5 years), 4613 (22%) had influenza, of 845 ILI samples, 187 (22%) had influenza and 658 samples SARI patients (age < 5 years), 47 (7%) had influenza.

• Hospital Based Dengue Surveillance

IEDCR restarted hospital based dengue surveillance in May

2017 in four hospitals – Dhaka Medical College Hospital, Uttara Adhunik Medical College Hospital, Khulna Medical College Hospital and Chittagong Medical College Hospital. So far, 274 blood samples were collected through the surveillance and among them 91 samples have been tested and 10 had a positive result for dengue IgM.

• Web Based Dengue Surveillance

IEDCR receives reports of dengue patients through passive reporting using District Health Information System 2 (DHIS 2). Thirty two hospitals (both government and private) of Dhaka city are participating in the reporting. In 2017, a total of 1,061 dengue cases were reported through this system.

• Web Based Integrated Disease Surveillance System

Bangladesh and SEARO WHO developed a strategy for promoting an integrated approach. In this context, Integrated Diseases Surveillance (IDS) strategy called for coordinated approach to data collection, analysis, interpretation, use and dissemination of surveillance information for decision making and implementation of public health interventions with a goal to control and prevent communicable disease.

• Cell Phone Based Disease Surveillance System (CPBDSS)

CPBDSS is a newer platform of IEDCR where surveillance data were collected using cell phone. Total 10,358 completed interviews were made through this platform. Among the completed call, 6,839 (66.03%) were male and 3,518 (32.07%) were female. The interviewees included 5,579 (54%) urban and 4,752 (46.0%) rural respondents. The surveillance found, 27% were current smoker, 36% take smokeless tobaccos, 5% drink alcohols, 15% of respondents exercised regularly at least for 10 minutes.

• Sexual and Reproductive Health Surveillance of women aged between 15 to 49 years in Bangladesh

Out of 28,230 calls 7,694 (27.3%) were complete calls. The mean age of the respondents were 29.5 ± 16.9 years. More than half (57.5%) respondents were married before the

Table 4.5.1. List of diseases reported through web based integrated disease surveillance system

Acute watery diarrhoea	Cutaneous anthrax	Pulmonary tuberculosis (PTB+)
Blood dysentery	Malaria	PTB-
Pneumonia	Enteric fever	Extrapulmonary tuberculosis (EPTB)
Severe pneumonia	Probable rabies	Tuberculosis in children
Very severe disease	Kala azar	Acute hepatitis
Severe acute respiratory infections	Dengue fever	Acute meningitis-encephalitis syndrome

age of 18 years. About 59% were below the age of 20 years during the first pregnancy. About 95% of current pregnant women have knowledge about ANC visits. About 50% of the respondents heard about breast cancer and about 77% heard about breast cancer.

• **Child Health and Mortality Prevention Surveillance Network**

Child Health and Mortality Prevention Surveillance (CHAMPS) is a network of disease surveillance sites in developing countries with high under-five mortality rates. IEDCR, icddr,b and Bangabandhu Sheikh Mujib Medical University (BSMMU) are jointly implementing the surveillance in Bangladesh. Field sites are Faridpur Medical College Hospital and Baliakandi upazila of Rajbari district, situated in the west-central part of Bangladesh.

The Bangladesh site will establish epidemiologic surveillance of under-5 deaths and stillbirths utilizing verbal autopsy and Minimal Invasive Tissue Sampling (MITS). The tissue samples will be tested by both conventional and advanced histopathology and molecular screening for various pathogens.

• **Food Borne Illness Surveillance System Bangladesh**

The project duration was 22 months. In the project period, all of the 10 sentinel sites established, and capacity developed in terms of logistics and personnel and total of 30 food safety emergencies were responded.

• **Cholera Surveillance**

Cholera surveillance is a joint collaboration between IEDCR and icddr,b, funded by the Bill and Melinda Gates Foundation (BMGF). Now all the functioning 22 sites are line listing the diarrhea patients, conducting Rapid Diagnostic Test (RDT) at the site and collecting epidemiological data with samples for culture. In addition to that, stool samples from previous 10 FBIS sites were also tested for Salmonella and Shigella in the laboratories of IEDCR and icddr,b.

• **Antimicrobial Resistance (AMR) Surveillance in Bangladesh**

Five infectious diseases (urinary tract infection, diarrhoeal diseases, wound infection, pneumonia and septicemia) have been selected and specific samples (urine, stool, wound swab, sputum, blood and endotracheal aspirate) are being collected. In 2017, five sentinel sites started their reporting. In total 2,619 samples were tested for antimicrobial resistance in these sites.

• **Anthrax Surveillance and Response**

Along with culture and isolation technique molecular diagnostic methods are being used at IEDCR for diagnosis. A high level stakeholder meeting on anthrax surveillance was held at DLS for prioritizing anthrax disease prevention and control in Bangladesh. Molecular training on anthrax was conducted in Manipal University, Karnataka, India as well as at IEDCR. A joint epidemiological and laboratory capacity assessment of anthrax endemic area was performed for anthrax surveillance.

• **Media Surveillance / Monitoring**

OUTBREAK INVESTIGATIONS AND RESPONSES

As the National Rapid Response Team (NRRT), IEDCR responds any unusual health events or diseases on an emergency basis. Names and the number of outbreaks are listed below.

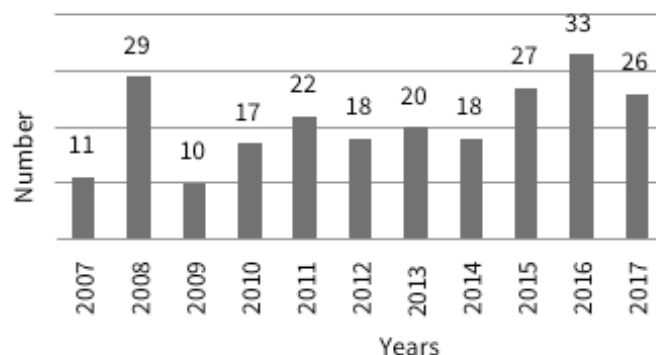


Figure 4.5.1 Number of outbreaks responded by IEDCR from 2007-2017

OUTBREAKS IN 2017

- Avian Influenza
- Atypical Pneumonia
- Chikungunya
- Cutaneous Anthrax
- Diarrheal Diseases
- Diphtheria
- Encephalitis
- Hepatitis: A and E
- Hand, Foot and Mouth Disease
- Measles
- Respiratory Tract Infection

CASE INVESTIGATIONS IN 2016-2017

- Atypical pneumonia
- Chikungunya
- Severe respiratory illness following Hajj
- Zika Like symptoms following return from endemic countries
- IHR 2005 reported case investigations

PROGRAM FOR IHR, MIGRATION HEALTH & EMERGING REEMERGING DISEASES

BACKGROUND

The International Health Regulations (IHR) are an international legal instrument that is binding on 196 countries across globe, including Bangladesh. In May 2005, the fifty-eighth World Health Assembly (WHA) adopted the International Health Regulation (IHR (2005) which subsequently entered into force on 15 June 2007, with 196 member countries signing the document. The member countries are required by the IHR (2005) to develop minimum core public health capacities. Attaining the minimum public health core capacity as per IHR (2005) will significantly contribute to the SDG goal for 2, 3 and 6.

GOAL

The Goal of the IHR (2005) was “to prevent, protect against, control and provide a public health response to the international spread of diseases in ways that are commensurate with and restricted to public health risks, and which avoid unnecessary interference with international traffic and trade.”

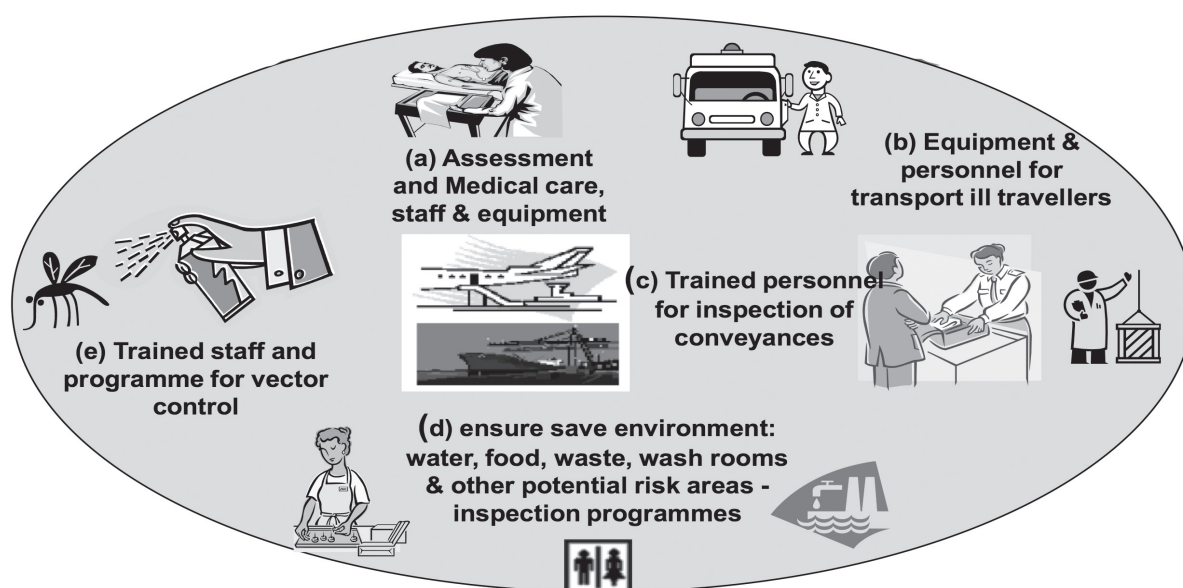
TARGET

To strengthen the capacity of Bangladesh, along with other 195 countries, for early warning, risk reduction and management of national and global health by 2019

OBJECTIVES

- Enactment of adequate legal framework and laws for implementation of IHR 2005
- Implementation of Multisectoral and Multi-disciplinary coordination and communication mechanism
- Establishment of cross border surveillance of priority diseases among neighboring countries (with India, Myanmar)
- Enactment of Establishment of cross border surveillance
- Inclusion of component of IHR in curriculum of medical and public health, and courses in related fields and in all related training
- Adoption of mechanisms and documented procedures among all relevant sectors, particularly those responsible for human health and animal health to minimize the transmission of zoonotic diseases from animal to human
- Strengthening laboratory capacities, infection prevention and control
- Developing human resource for IHR 2005 implementation
- To Ensure Plan of action for Risk Communication and strategy development
- To ensure Plan of action for strengthening core capacities of Points of Entry (PoE)
- Strengthening capacities to respond to potential hazards

PoE Core capacity requirements at all times (routine)



IHR COORDINATION PROGRAMME



World Health Organization

Figure 4.6.1 IHR coordination programme

- To detect, assess and respond to food safety issues and related hazards
- Implementation of mechanism for timely detection and effective response to potential radio-nuclear hazards/ events/emergencies in collaboration with other sectors responsible for radio nuclear managements
- To ensure timely detection and effective response to potential chemical risks and /or events in collaboration with other sectors responsible for chemical safety, industries, transportation and safe disposal.

MAJOR ACTIVITIES IN 2017

1. Revision and Finalization of the 3rd National Avian and Pandemic Influenza Preparedness and Response Plan, Bangladesh: 2016-2020 started in the context of IHR (2005).
2. Prioritization of important zoonotic diseases in collaboration with USAIDs started and will be finalize soon.
3. Joint external evaluation was successfully completed for International Health Regulation (2005) to assess the country capacity to prevent, detect and rapidly respond to public health threats. The JEE is based on peers assessing peers and uses a Multisectoral One Health Approach, which is imperative to combat global public health emergency threats.

4. Activities were undertaken to contain Middle East Respiratory Syndrome Corona virus (MERS-CoV) infection in returning Hajjees during the Hajj of 2017.
5. A consultative workshop was carried out in July 2017 by CDC on communicable diseases for doctors, nurses and health related personals.

MAJOR ACHIEVEMENTS IN 2017

1. Diphtheria outbreak was successfully contained in the settlements of forcibly displaced Myanmar nationals, Cox's Bazar, Bangladesh. A campaign was carried out to immunize the children aged 0-6 years with PENTA vaccines during 13-27 December 2017. After completion of the first round children ages 7-15 years were also vaccinated.
2. Disease Control Unit of Directorate General of Health Services (CDC) procured 200 vial Azithromycin for management of diphtheria cases and are planning to procure Azithromycin Tablet as per requirement for proper case management.
3. Table Talk exercise was organized during Annual Retreat in Coxsbazar on Standard Operating Procedure (SOP) for Public Health Emergencies for International Concern (PHEIC), Strategy and guideline for management of PHEIC

at Points of Entry (PoE)s and reporting of PHEIC in PoE.

4. Completed the finalized version of National Management Guideline for Chikungunya Fever in collaboration with the Medicine society of Bangladesh.
5. Draft National Strategy was also finalized for Zika virus although there is no active case of Zika identified in Bangladesh till now.
6. Health screening of passengers coming from affected countries is done regularly in Hazrat Shahjalal International Airport (HSIA), Chittagong Sea port and Benapole Land port. A total of 6 doctors and 22 Sanitary Inspectors and Nurses worked in shift at Dhaka Airport. During emergency, immigration officials of Hazrat Shahjalal International Airport (HSIA) were trained to screen passengers at the airport properly. In an aim to prevent the fatal disease Ebola from spreading and contaminating people in Bangladesh, the government has set up seven archway thermal scanners at different PoEs, to check body temperatures of the arriving passengers.

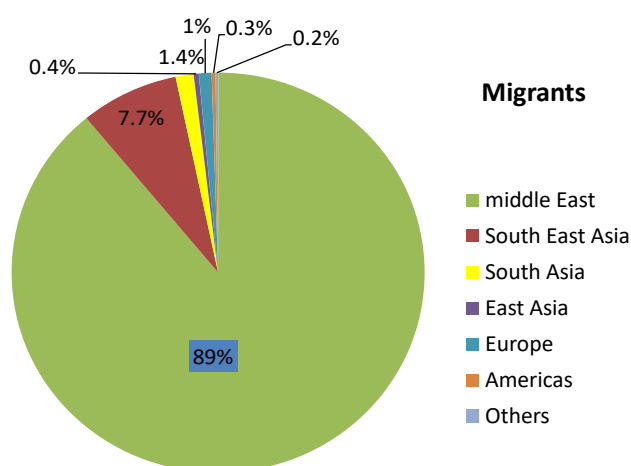
CHALLENGES AND RECOMMENDATIONS FOR WAY FORWARD

- Achieving international public health security is one of the main challenges arising from the emerging and mosaic landscape of public health. Shared vulnerability implies shared responsibility. Strengthening countries' disease surveillance and response systems is central to improving public health security in each country as well as globally. Meeting the requirements of IHR (2005) compliance is a big challenge.
- Member States including Bangladesh have to comply with developing core capacities for surveillance and response which covers surveillance and emergency threats from infectious diseases, as well as core capacities for designated airports, ports and ground crossings to help ensure global, regional and national health security, and the capacity to detect and combat diseases of radio-nuclear, chemical and food origin in the community.

MIGRATION HEALTH

Migration health addresses the physical, mental and social needs of migrants and their families, and the public health needs of communities of the countries of origin and destination, through policies and practices corresponding to the emerging challenges facing mobile population today. Migrants constitutes a large number of population moving out of the country and generating huge remittances. Despite

of the migrant's great contribution to the economy of Bangladesh, it is unfortunate that they are often exposed to occupational health hazards and have very limited access to health benefits. Health of migrants has become more prominent concern of the government. The honorable Prime minister Sheikh Hasina declared in the 64th World Health Assembly about the country's commitment to ensure the welfare of migrants especially by migrant inclusive health policies. The Director General of Health Services has already developed a national strategic action plan on



Source: Health Vulnerabilities of Migrants from Bangladesh, Baseline assessment IOM, Dhaka, August 2015

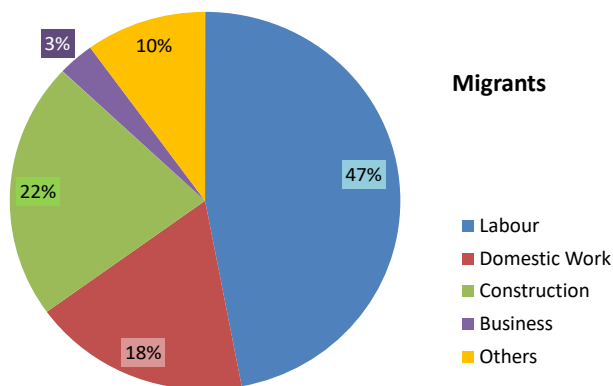
Figure 4.6.2 Intended or past country of work for departing and returnee migrants

migration health for the tenure 2015 to 2018.

Strategy

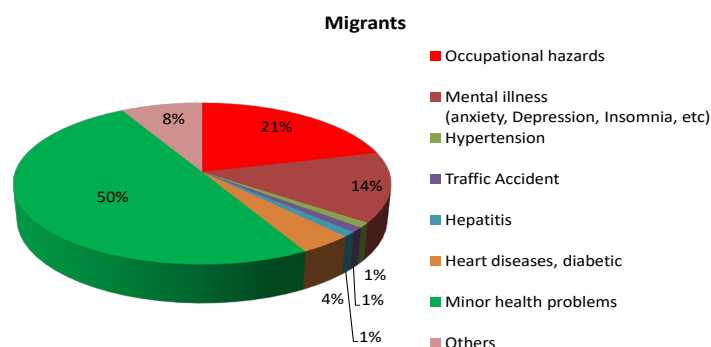
The Government of Bangladesh is committed to improving the health status of all categories of migrants including inbound, outbound and internal migrants throughout the migration process, and for the progressive realization and protection of their fundamental right to health.

- To establish a strategic Monitoring and information system (MIS) for Migrants' Health
- To enhance the policy and legal framework of migration health to provide support for the Health and social protection of migrants
- To enhance the policy and legal framework of migration health to provide support for the health and social protection of migrants
- To enhance policy and legal framework of migration health to provide support for the Health and social protection of migrants
- To enable existing health and social welfare system to be more migrant sensitive



Source: Health Vulnerabilities of Migrants from Bangladesh, Baseline assessment IOM, Dhaka, August 2015

Figure 4.6.3 Types of employment by departing & returnee migrants



Source: Health Vulnerabilities of Migrants from Bangladesh, Baseline assessment IOM, Dhaka, August 2015

Figure 4.6.4 Illness experienced abroad, as percentage of returnee migrants who fell ill in the destination country



Summary Fact Sheet

Table 4.6.1 Trend of health service coverage fact sheet 2016-2018 progress in IHR (2005) core capacities

Sl No	Programme indicators	2016 (Jee) %	2017 %	2018 %
1	Legislation And Financing	60	80	60
2	IHR coordination and national ihr focal point functions	60	89	80
3	Zoonotic events and the human-animal interface	50	92	80
4	Food safety	60	95	40
5	Laboratory	60	91	73
6	Surveillance	75	96	80
7	Human resources	70	82	40
8	National health emergency framework	25	60	47
9	Health service provision	-----	----	60
10	Risk communication	56	90	60
11	Poins of entry	40	73	60
12	Chemical events	20	27	40
13	Radiation emergencies	40	42	40

Source: Joint External Evaluation 2016-17 & State Party self assessment annual reporting Tools 2017,0218

Health Education, Promotion and Lifestyle Modification

Health education is an important part of public health. Through effective education and lifestyle modification, a country can reduce both mortality and morbidity rate and health expenditure. The 4th HPNSP have an operation plan on 'Lifestyle, and Health Education & Promotion (L& HEP)' under the DGHS. Main goal of this Operational Plan is to provide support to influence and promote healthy lifestyle and to identify different target audience by arranging workshop, seminar, and campaign.

OBJECTIVES

General objective

To influence the healthy behavior of individuals and community and living conditions that influence health by improving their knowledge, attitude, practices and skills by creating a 'health literate society'.

Specific objectives

- To establish legislative framework, communication strategy, implementation strategy, and inter sectoral collaboration for healthy lifestyle and healthy environment.
- To establish linkage with other OPs/units of the DGHS for implementation of respective SBCC, to improve the activities for implementation of lifestyle, health education & promotion at individual and community level.
- To identify different target audience and address communicable, non-communicable diseases, MNCAH issues and emerging and re-emerging health problems.
- To use and establish linkage with print, electronic, and social media and effective community engagement for SBCC.

ACTIVITIES IN 2017-2018

January-June 2017

- Communication campaign through different media (observance of World Health Day, Safe Motherhood Day & World No Tobacco Day)
- HPN coordination meeting, and Behavior Change Communication (BCC) Working group meeting has been done, SBCC materials developed according to the guideline specified in the strategy.
- Community engagement and mobilization, advocacy and coordination and ICT and innovation have not been done due to lack of fund
- Capacity development and logistic support (regular maintenance of motor vehicle and printing press).

- Production, distribution and dissemination of SBCC materials (80000 leaflets) on World Health Day.
- No behavioral research on health promotion due to lack of fund.

July 2017-June 2018

- Implementation of the comprehensive social and behavior change communication strategy (8 district level workshops, 10 awareness building advocacy workshops, 64 public education campaigns).
- Establishment of inter-sectoral collaboration on lifestyle and healthy environments including private sector engagement (16 workshops on promotion of safe and healthy lifestyle, 16 awareness building seminars).
- Communication campaign through different media for behavior change (airing of 5 TV spots telecast 450 times, TV skits, short drama, 10 messages for TV scroll telecast 2000 times, publishing different issues on newspaper, promotion by folk dramas & songs, 50 event, 20 billboards, 3 workshop on promotion of healthy lifestyle for non-communicable disease control, installation of LED screen and digital LED signage, 3 advocacy workshop on prevention of tobacco and cancer and other non-curable diseases, 16 workshop on prevention and control of malaria and dengue, Celebrating 5 national days, printed 40 thousand leaflet).
- Community engagement and mobilization (Workshops, 640 orientation programs on practical health education, 640 orientation programs school health program, 300 sessions for elderly health care, 16 workshops for counseling on Tobacco and drug abuse, 640 sessions on promotion of safe delivery, 640 sessions on personal hygiene practice, print media campaign through branding local buses, TV documentary telecast on 5 TV channels).
- Advocacy and coordination (workshop on reduction of NCD high risk behaviors, 32 coordination meetings).
- Capacity development and logistic support for SBCC (all target batches' training are completed-6 batches done in NIPSOM, 28 batches in Civil surgeon office and 2 batches in Thailand. Different types of logistics are supplied).
- ICT and Innovation (Now BHE & Digital Archive of BHE is up to date, all 64 batches completed e-learning courses).
- Production, distribution and dissemination of SBCC materials (1000000 poster, leaflet etc was printed and distributed on different health promotion issues, regular maintenance and upgradation of BHE printing press).

- HPN Co-ordination Meeting and BCC Working on group meeting have been done. SBCC materials developed according to the guideline specified in the strategy.
- Research and knowledge management (1 behavioural research on health promotion completed by NIPSOM).

TRAINING/ORIENTATION/WORKSHOP/ SEMINAR/ADVOCACY

January-June 2017

No training has been given because of lack of fund.

July 2017-June 2018

- Local training has been given to 510 men and 340 women on district level.
- Foreign training has been given to 8 men on central level and 4 men on district level.
- Workshop on healthy lifestyle, seminar on drowning and injury and Nipah virus, advocacy on tobacco, cancer, dengue and chikungunya has been done and total number of participants were 12 men and 6 women on central level and 5600 men and 4190 women on district level in MOHFW and 3560 men and 2630 women on district level in Non-MOHFW.

CHALLENGES FACED

January-June 2017

Challenge faced in implementing the OP activities were inadequate space of the BHE head quarter, lack of fund for procurement, delay of fund release and shortage of manpower.

July 2017-June 2018

Challenge faced in implementing the OP activities were: fund utilization was very difficult due to IBAS++, shortage of manpower in field level.

Suggestions /recommendations to overcome the above challenges:

- a) Need more training for the management of IBAS++ software
- b) Need to create the post of health education officer at upazilla level
- c) Need infrastructure

HEALTH INTERVENTIONS FOR FDMNS BY DGHS COORDINATION CENTER, COX'S BAZAR

INTRODUCTION

The Forcibly Displaced Myanmar Nationals (FDMNs) settlement in Cox's Bazar on the border with Myanmar is now the world's largest refugee camp having a total of 905000 FDMNs. During August 2017 - December 2018, Bangladesh received 722042 FDMNs since the conflict started in Rakhine State, Myanmar (Source: Population Factsheet, UNHCR, 30 June, 2018). It has been the fastest and largest movement of refugees in decades.

Since the arrival of FDMNs in Bangladesh, the Government of Bangladesh (GoB) through DGHS has been in the front line to deliver emergency response and health support. The FDMNs have been placed in 68 settlements across Ukhia upazila and Teknaf upazila in Cox's Bazar district as well as Naikkhongchari upazila in Bandarban district. Yet, in order to accelerate the emergency response, the GoB is bringing most of the FDMNs population together to a 3,000 acre area located in Ukhia.

Since the influx, the collaborative partnership between the government, private sector, civil societies and international development agencies have been able to provide static health centers in 199 fixed sites with 131 doctors, 681 additional staff members, 21 ambulances and 231 vaccination teams (Source: DGHS Dashboard). In terms of health information, DGHS has created a public dashboard namely "Forcibly Displaced Myanmar National (FDMN) Health Situation and Intervention in Bangladesh", which is being updated on a daily basis.

Given the fact that the FDMNs camps are located in the remote areas, and considering the speed and the size of population influx and the availability of amenities to provide healthcare services, the achievements have been quite significant and rightly praiseworthy nationally, regionally and internationally.

HEALTH SECTOR COORDINATION

The DGHS Coordination Center participates regularly at the Health Sector Committee, Strategic Advisory Group and Emergency Preparedness meetings and provides necessary directions and recommendations. Moreover, other coordinators participate in nine other sub-committee meetings on a regular basis that are usually held weekly or fortnightly and provide their valuable suggestions for providing health interventions toward the FDMNs. Beside these, DGHS Coordination Center regularly arrange coordination meetings with partners at the office of Coordination center and/or Civil Surgeon Office.

COORDINATION FOR ESTABLISHING PHCS/MPHCS

DGHS Coordination Center, Cox's Bazar has been relentlessly working for coordinating the work of different service providers and to improve the service quality. Initially, DGHS started delivering health services from makeshift camps which are not at all suitable for delivering quality services. In view of the lingering nature of the crisis and the monsoon season, DGHS Coordination Center has decided to set up 20 Primary Health Centers (PHCs) and 03 Multipurpose Primary Health Centers (MPHCs) in and around the camps both in Ukhia and Teknaf upazila including upgrading the existing makeshift camps. These centers will be semi-permanent, prefabricated, relocatable and of durable structures. Among the partners UNHCR is building 5 PHCs, IOM has already constructed 1 and will build 3 more PHCs, AFAD through Turkish Red Crescent will build 6 PHCs and 3 MPHCs, Swiss Red Cross built 3 PHCs, and HAEFA built 2 PHCs.

REFERRAL SOP

Referral SOP has been drafted with a view to providing possible health services inside the FDMNs camps, however whenever there is a need for referral services, it should follow referral pathway. A referral pathway and referral form also have been developed. These documents have been finalized and approved by Civil Surgeon, Cox's Bazar and other relevant authorities.

HEALTH INTERVENTIONS

Vaccination is ongoing through special immunization activities along with routine immunization. Routine EPI has also been initiated, targeting children less than 2 years

of age with BCG, Pentavalent, OPV, PCV and MR. Routine EPI has started from 3 June 2018 for the FDMNs in their camps situated at Ukhia Upazila. In total 56 teams has been working to operate 56 outreach centers in Ukhia, among which DGHS Coordination Center contributes 8 teams. Each team for outreach center consists of 2 vaccinators. They register eligible children and pregnant women and perform IPC in one day and next day they vaccinate registered children and pregnant women in each catchment area. By this way, each team covers 03 catchment areas in a week and 12 catchment areas in a month.

Besides these, 28 static centers are also functioning regularly, however it has been decided that these static centers will be operated for twice a week. Among the static centers NGOs/INGOs are operating 20 centers and DGHS Coordination Center is operating 2 static centers at Jamtoli and Moynarghona PHC. Meanwhile, in Teknaf, the DGHS Coordination Centre runs 4 outreach centers in camp no-21 (Chakmarkul), camp no-22 (Unchiprang), camp no-23 (Shamlapur), camp no-24 (Noyapara).

DIPHTHERIA OUTBREAK AMONG FDMN

In total, 5356 suspected cases of diphtheria and 44 deaths have been reported in Cox's Bazar since the outbreak started in November 2017. 199 of the suspected cases were from host communities (no death). To battle the situation, WHO has been coordinating and supporting the overall diphtheria response with the Ministry of Health and Family Welfare. WHO has provided guidelines and training to partners and health workers on the detection and clinical management of suspected cases, and the tracing of contacts for providing them preventive treatment Antibiotics and over 2,500 doses of diphtheria antitoxin have been provided to treat patients.

Table 4.8.1 Immunization activities for FDMNs from August 2017 to December 2018

Special Immunization Activities	1st round	2nd round	3rd round	4th Round
MR vaccine	135,519	354,982		
Oral Polio vaccine	72,334	236,696		
Vitamin A	68,941	156,505		
Oral Cholera vaccine	700,487	199,472	879,273	328,556
Td	165,927	223,065	259,016	
Penta, PCV, bOPV	149,962	169,617	172,432	

HEALTHCARE SERVICE DELIVERY

To ensure the availability of adequate and sufficient medical services, the Coordination Centers recruited a total of 117 people to provide humanitarian health response in Cox's Bazar. They were deployed in Cox's Bazar coordination centre, Cox's Bazar Sadar District Hospital, Civil Surgeon Office, Ukhia UHC, Teknaf UHC as well as Naikhangchari UHC.

Moreover, the coordination center deployed 12 medical teams to locations within Ukhia and Teknaf Upazilas. Each team consists of five staff including one doctor, one nurse, one field supervisor, and two MHC workers. The medical teams are equipped with devices to record patient data for the integrated health information system. Those data include the profile of the patient, disease trend and the distribution of medication. To date, the coordination center has been able to provide medical services to 203,082 patients. A total of 45,221 patients were served by DGHS Medical Camps/ Centers from 15

Table 4.8.2 Number of patient consultations provided by location

Team No	Camp Name	Total no. of patients			
		Period: 15November-31December 2017	Period: 01 March - 30 June 2018	Period: 16 July- 31 August 2018	Period: 01 September- 31 November
1	Jamtoli, Ukhia	4,027	13762	4946	11535
2	Lambashia, Ukhia	5,317	10605	5572	5327
3	Taznimarkhola, Ukhia	3,980	11250	5633	13113
4	Hakimpara, Ukhia	6,078	8812	3950	5022
5	Kutupalong 1, Ukhia	5,821	9323	3632	5916
6	Balukhali 1, Ukhia	4,875	9041	3328	5784
7	Modhurchora, Ukhia	2,892	7996	4062	5138
8	Balukhali 2, Ukhia	1,825	9561	5103	8498
9	Nayapara, Teknaf	3,825	7630	3525	5119
10	Uchingpran, Teknaf	3,825	5816	3268	7350
11	Chakmarkul, Teknaf	1,078	5294	3341	6469
12	Shaplapur, Teknaf	5,138	5347	2490	4281
13	Sadar Hospital	0	4572	0	0
Total		45,221	1,09,009	48,852	83,552
Grand Total		286,634			

November to 31 December, 2017, 1,09,009 patients from 1 March to 30 June, 2018 and 48,852 patients from 16 July to 31 August, 2018.

ON JOB TRAINING FOR CAPACITY BUILDING

- Training on EPI
- Training on Helping Babies Breath (HBB)
- ToT on Acute Watery Diarrhea(AWD)
- Training on Emergency Preparedness and Response (EPR)

- Training on major Non Communicable Diseases (NCDs)
- Orientation on strengthening field activities for Multipurpose Health Workers
- Training on nutrition
- Training on Health Center Management
- Training on Kobo software
- Training on EWARS
- Basic training on report writing and presentation
- Training on Geriatric Health Management

HEALTH SESSIONS BY FIELD STAFFS

Health sessions for different groups (Male group, Female

groups (Adolescent girls group, Pregnant and lactating mothers group, mixed group, Children group, School students) of FDMNs were performed on a regular basis by the field staffs of the DGHS Coordination center.

MAJOR ACHIEVEMENTS

Host communities were being deprived of usual health services after sudden influx of Myanmar Nationals (FDMN) in Cox's Bazar. Establishment of coordination center reduced the extra burden of Civil Surgeon and UH&FPOs so that they can focus on the host community.

1. DGHS Coordination Center, Cox's Bazar coordinates the whole health issues of FDMN community.
2. DGHS Coordination Center, Cox's Bazar has collaborated with all the health care providers, including NGOs, which in turn helps to avoid the duplication of health services, thus ensure equitable distribution and quality of the health services to the FDMN community.
3. 117 personnel have been recruited under DGHS Coordination Center, Cox's Bazar.
4. Different partners have committed to support DGHS Coordination Center, Cox's Bazar for construction of 19 PHCs and 3 MHCs. Several MoU have already signed to support these facilities.
5. Six Primary Health Center (PHC) at Jamtoli, Moynarghona, Tasnimarkhola, Balukhali, Balukhali 1 and Lambashia have been inaugurated and these centers are now providing limited curative care services with preventive and promotive care services by the DGHS Coordination Center, while development partners are providing nutrition, psycho social supports and maternity services.
6. 431,448 children (102.5% of the estimated target) have been immunized in the third round vaccination campaign, while 172432 children (aged 6 months to 7 years) have been vaccinated with penta and 259016 children (7 years to 15 years) have been vaccinated with Td.
7. 891,471 people (90.5% of estimated target) have been immunized in third round cholera vaccination campaign 2018, against the total target of 984,906 people.
8. Medical teams of DGHS Coordination Center, Cox's Bazar have provided services to 286,634 patients in FDMN camps through 12 medical teams.
9. A web based IT application for patient registration and treatment has been developed and now being reviewed by DGHS Coordination and support Center, Dhaka.
10. Disease data are reviewed daily to identify likelihood of developing outbreak/epidemic and provide early warning and alerts.
11. MSR stock positions of CS office Cox's Bazar, District Sadar hospital, Cox's Bazar; Teknaf UHC and Ukhia UHC were reviewed regularly to ensure uninterrupted supply of MSR.
12. Staff attendances are monitored regularly to ensure timely attendance and prevent unauthorized absence.
13. Relevant data analysis experts (data managers) from different development agencies were identified and workshops have been organized for those to share and collect experience and vice versa.
14. Diphtheria outbreak was successfully combated.

CONCLUSION

FDMN may stay longer than it was expected. Therefore, the initial response from the world community was sufficient to help them in short and intermediate term. However, for the long term settlement of FDMN, we need to rethink about the whole situation.

Nutrition is one of the basic human rights, which should be equally obtained by everyone. Nutrition is an essential component of the physical growth and mental development and every person needs this element. In the absence of nutrition, the fetus does not increase properly and is often born with low birth weight. Government of Bangladesh is committed to improve the nutritional status of the people of the country. The Constitution of Bangladesh cites nutrition in Article 18 (1) describing the principles of State governance: “...the state shall regard raising the level of nutrition and improvement of public health as among its primary duties..”. Moreover, the Bangladesh National Nutrition Policy 2015 and the National Plan of Action for Nutrition (NPAN-2) were developed to address the challenges relating to nutrition. The policy takes into consideration all relevant global policies such as Second International Conference of Nutrition (ICN2) and relevant national policies and strategies in areas such as health, food, agriculture, environment and education reflecting its multi-sectoral approach to ensure the improvement of nutritional status.

CURRENT NUTRITIONAL STATUS

Along with undernutrition, obesity and nutrition-related non-communicable diseases are major public health problems in Bangladesh. Despite not reaching an expected level, different interventions and programs related to the field of nutrition has contributed to the significant improvement in Bangladesh’s nutrition situation over the last few decades . Childhood malnutrition has been decreasing slowly in Bangladesh and the result is quite noticeable. 31% under five years aged children are now stunted while only 8% are wasted. About 34% of children 6-23 months aged are fed with appropriate feeding practices (BDHS 2017-18).

Detailed nutritional status indicators including maternal health are listed in the following table.

Table 4.9.1 Nutritional status indicators from BDHS 2017-18

Indicator	2007	2011	2014	2017-18	Remarks
Maternal and newborn care					
ANC from medically trained provider (percentage of women receiving at least one ANC from a medically trained provider)	53%	55%	64%	82%	
ANC-4+ (percentage of women receiving 4 or more ANC)	22%	26%	31%	47%	
Medically trained provider, by place of delivery (percentage of deliveries in last three years)	21%	32%	42%	53%	Home: 50% Public facility: 14% Private facility: 32% NGO: 4%
Facility deliveries by poorest wealth quintiles (percentage of facility deliveries in last three years)	6%	10%	15%	26%	
Facility deliveries by richest wealth quintiles (percentage of facility deliveries in last three years)	49%	60%	70%	78%	
Cesarean Section (percentage of deliveries in last three years)	9%	17%	23%	33%	Facility deliveries by CS: National facility : 67% Public facility : 36% Private facility: 84% NGO: 40%
Postnatal care for mothers (percentage of mothers receiving care from medically trained providers within 2 days of delivery)	20%	27%	34%	52%	Place of delivery (percentage of mothers receiving PNC from medically trained provider within 2 days of delivery) Facility deliveries: 97% Home deliveries: 7%
Trends in Child Nutritional Status					
Stunting	43%	41%	36%	31%	
Underweight	41%	36%	33%	22%	
Wasting	17%	16%	14%	8%	
Trends in IYCF (percentage of 6 – 23 month old children fed with 3 IYCF practices)		21%	23%	34%	11% points increase in last three years.
Childhood Mortality					
Under five (<5) children mortality per 1000 live births	65	53	46	45	
Infant mortality per 1000 live births	52	43	38	38	
Neonatal mortality per 1000 live births	37	32	28	30	Increased and contributing 67% of the total under-5 aged children deaths.
Childhood Vaccination					
Percentage of 12 – 23 month old children vaccinated by 12 months of age (All vaccines)	76%	83%	78%	86%	
Percentage of 12 – 23 month old children vaccinated by 12 months of age (Measles)	77%	84%	80%	88%	

Indicator	2007	2011	2014	2017-18	Remarks
Treatment of Diarrhea (5% under five children had diarrhea in last two weeks before the survey)					
ORT (ORS or homemade solutions)	81%	81%	84%	85%	
ORT & Zinc	20%	34%	38%	44%	Probably highest in the globe
Vitamin A supplementation (percentage of 6 – 59 month old children receiving supplement in the last 6 months)	84%	60%	62%	79%	

*ANC : Antenatal Care, CS : Civil Surgeon, PNC : Postnatal Care, ORT : Oral Rehydration Therapy, ORS: Oral Rehydration Salt

SUMMARY

Maternal Health

- A medically trained provider attends 53% of deliveries. In a decade, deliveries by Skilled Birth Attendant (SBA) increased from 21% to 53%, due to sharp increase in facility delivery.
- Half of the deliveries occur in a facility, mostly in private facilities (32%). The public and NGO health sectors account for 14% and 4% of deliveries, respectively.
- In equity in use of facility for delivery among the poor and rich is still high but declining over time.
- Eight out of 10 pregnant women are getting at least one ANC from a medically trained provider. There is a sharp increase in ANC visits from 64% to 84% between BDHS 2011 – 2017.
- Receiving ANC4+ increased from 31% to 47% in last three years.
- However, less than one in five (18%) pregnant women received quality ANC4+ at least one from MTP + all basic components.
- One third of deliveries are performed by caesarean section. This implies that one million C – section deliveries are taking place every year.
- 84% of deliveries in the private sector are performed by C – section.
- 52% mothers received PNC from a MTP within 2 days of delivery. But among non-institutional deliveries, only 7% of mothers received PNC within 2 days.

Childhood mortality and health care - seeking

- Under 5 Mortality Rate is 45 per 1000 live births.
- Neonatal mortality Rate is 30 per 1000 live births and it accounts for 67% of all under 5 deaths.
- Under five mortality rate has been declining gradually over the last two decades. However, between the 2014 – 2017 BDHSs, the decline in under 5 mortality rate has

slowed down noticeably.

- Infant and Neonatal mortality have not declined between 2014 BDHS and 2017 – 18 BDHS.
- 86% of 12 – 23 month old children had received all the recommended vaccinations before their first birthday
- 85% of children with diarrhea were given Oral Rehydration Therapy (ORT) and 44% received both ORT and Zinc.

Child Nutrition and Feeding

- There has been steady improvement in child nutritional status over the past decade.
- Stunting among children declined from 43% in 2007 to 31% in 2017.
- Wasting came down to 8% after remaining critically high for years.
- 65% of infant under age of 6 months are exclusively breastfed in 2017, which is markedly higher than the 2014 level (55%).
- IYCF practices among 6 – 23 month old children increased from 23% in 2014 to 34% in 2017.
- 79% of children received Vitamin – A supplementation, an increase from 62% in 2014.
- Hygiene and sanitation standards remain poor, only 3% of the caregivers wash their hands before feeding a child (FSNSP 2014).

MAJOR ACTIVITIES IN 2017

Bangladesh has achieved the Millennium Development Goal (MDG) targets for nutrition and is committed to achieve the targets of Sustainable Development Goals (SDGs) through Universal Health Coverage. The NNS OP 2017-22 focused mainstreaming of nutrition services through DGHS and DGFP during the implementing period 2017-22. This new OP will focus more on system strengthening and multi-sectoral coordination to ensure universal and

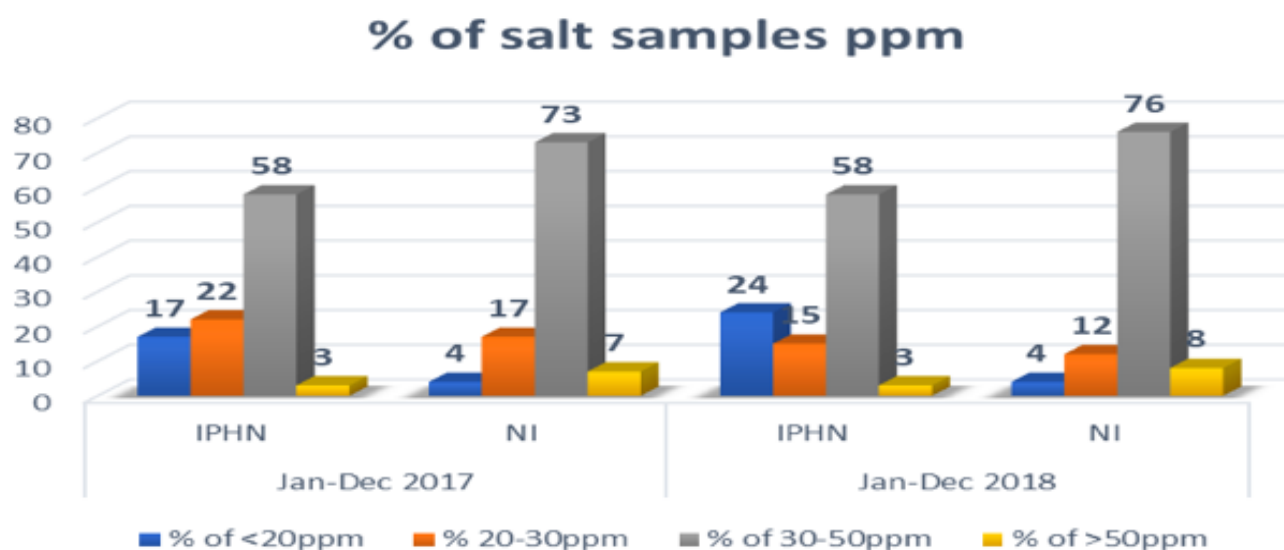


Figure 4.9.1. Percentage of salt in BSCIC salt samples

equitable coverage of both specific and sensitive nutrition interventions.

- The Institute of Public Health Nutrition (IPHN) under the Ministry of Health and Family Welfare (MoHFW) conducted validation test of BSCIC salt samples 888 no. from Jan-Dec 2017 and 1072 no. of salt samples from Jan-Dec 2018, which was supported by the Nutrition International (NI). This institute also tested 697 no. and 1022 no. of salt samples that were collected by sanitary inspectors of different sub-districts. Among the samples collected by NI >30PPM slightly increased from 2017 (80%) to 2018 (84%) where IPHN collected samples contains >30PPM is 61% to 66% in 2017 and 2018 respectively.
- Vitamin A plus campaign activities are discussed later in detail.
- Real time Monitoring and Response (RTMR) initiated in

last Vitamin A Plus campaign (NVAC+) in July 2018.

- The ministry approved national Advocacy and communication strategy.
- National Dietary Guideline has been developed and sent to the ministry for approval.
- Established Digital Archive at IPHN.
- Developed BCC eToolkit for field workers (link: www.dghs.gov.bd and <http://bdsbcc.org/>)
- Established 395 Integrated Management of Childhood Illness (IMCI) & Nutrition Corner in the district hospitals and UHC .
- Established 398 SAM Unit at District hospital and UHC.
- Established 592 BFHI.
- Creation of Nutritionist position in 64 districts.

NATIONAL VITAMIN A PLUS CAMPAIGN (NVAC+) 1ST ROUND, 14 JULY 2018

- Out of 21055529 (6-59 months age) children, total 20805290 childre (98.8%) received Vitamin A Capsule (including child to child searching).
- No significant differences were observed among the division's coverage.
- A certain number of district's coverage are 100%, while the remaining ones are between 95-99%.

Table 4.9.2 NVAC+ (1st round, 2018) coverage summary

Area	Target	Coverage	Percentage (%)
City Corporation(CC)	2059688	2038448	99.0
Pourashava (Municipality)	1006925	981388	97.5
Rural (Districts)	18995841	18766842	98.8
National (CC+Rural)	21055529	20805290	98.8

- The coverage of Gazipur City Corporation (CC) was 102% (6-11months age) and 101% (12-59 months age) which indicates that the target needs to be updated. The remaining of the CC's coverage are similar to the districts/divisions; (coverage was 95-100%).
- Among the 240 HTR unions, 49985 (6-59 months age) unreached children received Vitamin A capsule through Child to Child (CtC) searching. Over 2.8% unreached children of the 240 unions were covered through CtC (percentage (%) based on respective HTR Sub-division's target)

Table 4.9.3 NVAC+ (1st round,2018) Coverage of 46 Sub-district (240 HTR Unions)

HTR areas	6-11 month old child	12-59 month old child	6-59 month old child
NVAC + day in 46 Sub-Districts	96.3%	97.1%	97.1%
CtC subsequent 4 days	3.1%	2.7%	2.8%
Total	99.4	99.9	99.8%

COVERAGE BY DIVISION

The NVAC+ July 2018 coverage by division is given below (Figure 4.9.3). It shows that coverage is equal with poor variance. Therefore, in case of 6-11 month old children, Dhaka (97%) division's position is minor (for details please see districts below).

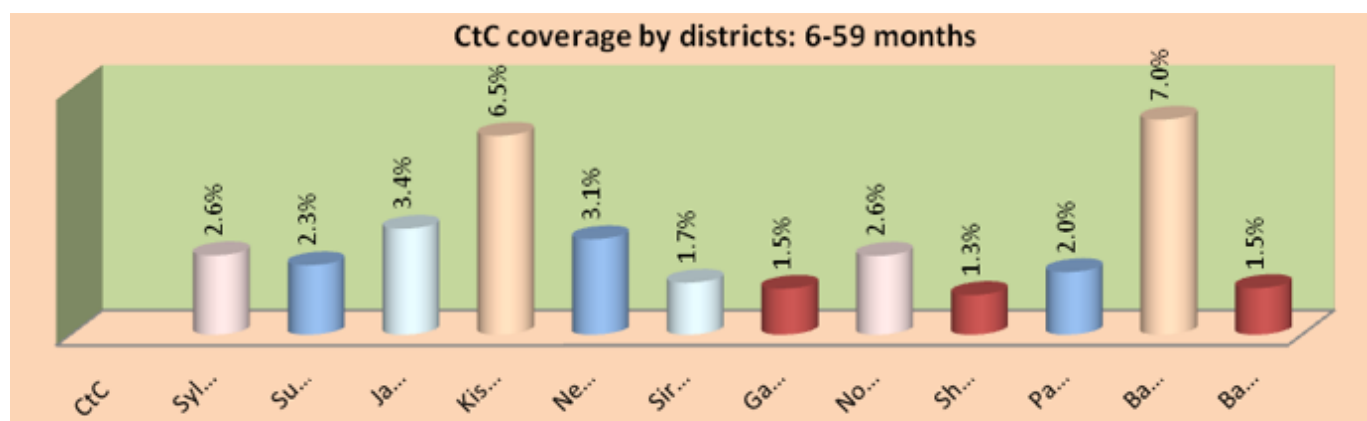


Figure 4.9.2 CtC Coverage by District (children aged 6-59 months) (source: IPHN)

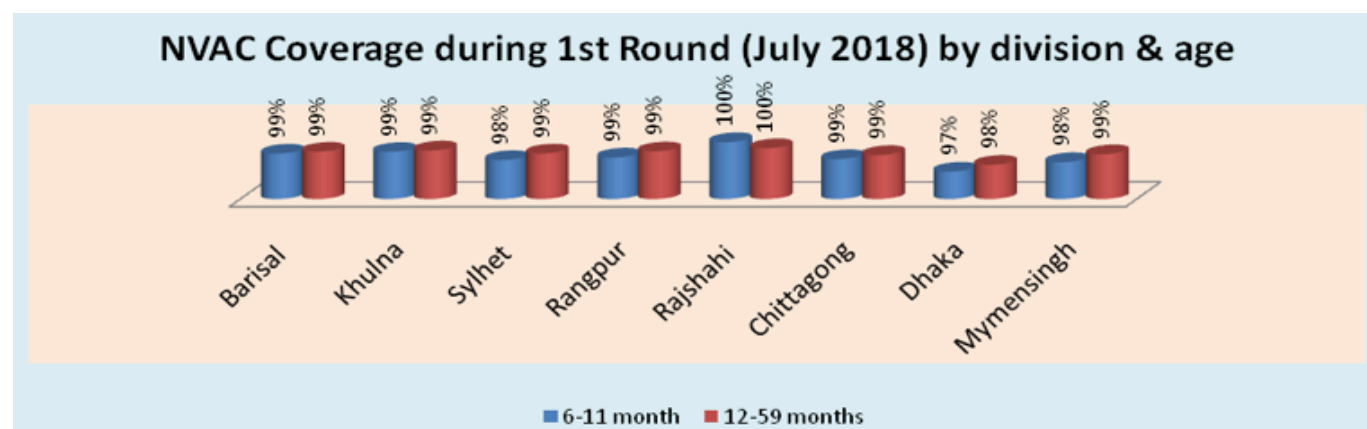


Figure 4.9.3 NVAC+ Coverage by Division (source: IPHN)

COVERAGE BY CITY CORPORATION:

More than 7 CCs coverage exceeded 100% (in both age groups) which indicates that uncounted floating people may have

been covered, although average coverage is 97.2% and 99.5% for 6-11 months and 12-59 months old children respectively.

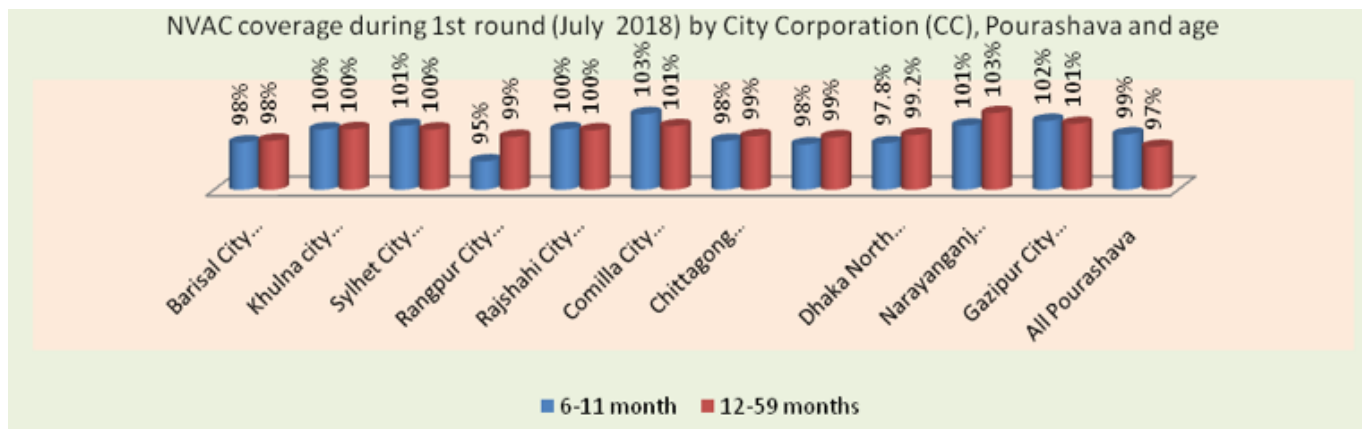


Figure 4.9.4 NVAC+ Coverage by City Corporation (source: IPHN)

FOOD SAFETY AND QUALITY CONTROL

Unsafe food has been a long-standing challenge in the country and poses a serious public health risk leading to increased prevalence of food borne diseases. The Institute of Public Health (IPH) is playing an important role in assuring safe and quality food providing laboratory analysis of food and water, and technical expertise and coordination support through its National Food Safety Laboratory (NFSL), Public Health Laboratory (PHL) and Food Safety Unit (FSU).

a. Laboratory analysis of food and water samples

The NFSL and PHL conduct tests for food and water samples according to the country's laws and regulations and standards as well as international standards. The Sanitary Inspectors working under the DGHS, municipalities, and city corporations send the food and water samples. Food and water samples are also sent from other sources, including private organizations, international organizations and UN bodies across the country. With the test results, the local authorities take legal actions against the offenders. It may be mentioned that some six hundred well-trained and equipped Sanitary Inspectors under the DGHS are rendering risk based food inspection services across the country.

The laboratories also conduct routine and regular tests for water samples sent from the Bangabhaban, Ganabhaban, Office of the Hon'ble Prime Minister, the Parliament and other important offices.

National food safety laboratory (nfsl)

With financial support from the European Commission and technical support from the FAO of UN, the National Food Safety Laboratory was established in 2009, and started functioning in 2012.

This service, research, training and reference Laboratory was inaugurated in October 2012 by Hon'ble Minister of the Ministry of Health and Family Welfare. Further development of the NFSL was done with the financial support of the Government of the Netherlands and technical support from the FAO of UN, during the period of 2012-2018. A group of scientists, chemists and technologists, trained in home and abroad, man the lab. The team has validated more than 80 methods for different food items. The Laboratory has been accredited by the Bangladesh Accreditation Board (BAB) under ISO/IEC 17025:2005.

The NFSL is playing a key role in the assessment of pesticide residues; heavy metals; antibiotics; formaldehyde, different additives including coloring agents and preservatives; myco-toxins and food borne pathogens in different food items. The samples are being

received from inspection activities, private sectors, UN and international, surveillance activities including food contaminants and food-borne disease surveillance as well food safety emergency response. The Lab is regularly participating in proficiency tests in home and abroad.

During the reporting period, this laboratory of international standard tested a wide range of parameters in different food items as shown in Table 4.9.4.

Table 4.9.4 Summary of the food samples tested by the National Food Safety Laboratory during 2013-2018

Year	No. of total samples	No. of Parameters tested
2013	214	1284
2014	363	2541
2015	1814	33875
2016	1134	13706
2017	835	9726
2018 (till Oct)	539	1390

PUBLIC HEALTH LABORATORY (PHL)

As mentioned earlier, the IPH started its journey with the Combined Public Health Laboratory in 1952. Recently, with the financial support from the Dutch Government and technical

support of the FAO of UN, the PHL has been refurbished, and upgraded with instruments and equipment, which was inaugurated by the Hon'ble Minister of the Ministry of Health and Family Welfare on 29 August 2016.

The number of tested food items and water samples over the last seven years (2012-2018), with satisfactory and unsatisfactory test results are shown below [in Figure 4.9.5 and Table 4.9.5 and 4.9.6]. It may be mentioned that the suspected samples being collected by the Sanitary Inspectors might be adulterated or contaminated.

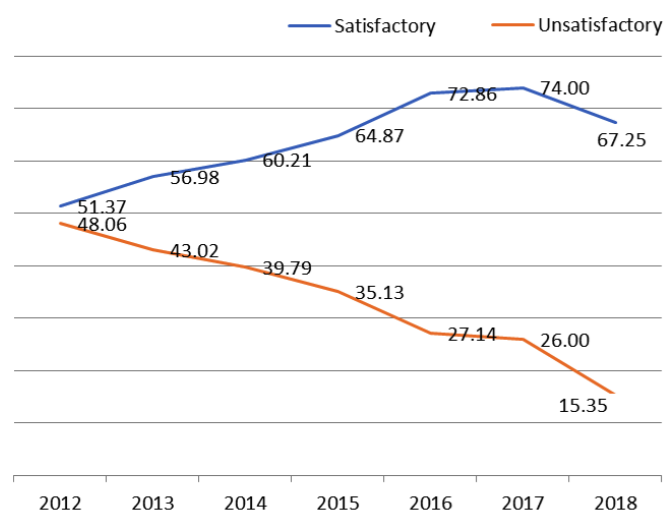


Figure 4.9.5 Food samples tested by Public Health Laboratory over the last 7 years (2012 -2018)

Table 4.9.5 Bacteriological test results of water samples over the last 7 years (2012- 2018)

Year	Total samples	Satisfactory		Unsatisfactory	
		No.	Percentage	No.	Percentage
2012	600	563	94.0	37	6.0
2013	706	666	94.0	40	6.0
2014	595	545	92.0	50	8.0
2015	502	434	86.0	68	14.0
2016	787	693	88.0	94	12.0
2017	320	308	96.3	12	3.7
2018 (till Oct)	231	170	73.6	61	26.4

Table 4.9.6 Chemical test results for water samples during 2012-2018.

Year	Total no. of samples	Satisfactory		Unsatisfactory	
		No.	Percentage	No.	Percentage
2012	350	347	99.20	03	0.80
2013	500	481	96.19	19	3.81
2014	482	453	94.52	29	5.48
2015	438	425	97.90	09	2.10
2016	644	602	95.48	42	6.52
2017	593	559	94.27	34	5.73
2018 (till Oct)	406	398	98.03	08	1.97

The NFSL and PHL also provided laboratory support to the IEDCR for response to the food safety emergencies and outbreak investigations during the reporting period. Furthermore, the NFSL also provided support for the research fellows and students of different institutions and universities for their theses and dissertations.

Initially, the PHL was the Public Health and Drug Testing Laboratory (PH&DTL). Until recently, the Drug Testing Laboratory wing was under the administrative and technical control of the IPH. However, as per recommendations of the WHO, the Ministry of Health and Family Welfare has placed the DTL under the administrative control of the Directorate General of Drug Administration. Nevertheless, staff salaries and other recurring expenditure are still met from the existing budget of the IPH.

b. Technical and coordination support for food safety activities

Aimed at providing technical and coordination support for food safety activities under the DGHS as well as for liaison with other agencies, the Ministry of Health and Family Welfare has established the Food Safety Unit in 2012.

Food Safety Unit

A Food Safety Programme, technically supported by the FAO of UN and financially supported by the Dutch Government and earlier by the European Commission, is housed within the administrative set-up of the Food Safety Unit of the IPH. The Programme is implementing its activities under the Operational Plan of the National Nutrition Services of the Health, Population and Nutrition Sector Programme. A group of experts in food safety and allied fields runs the Unit. A number of technical experts of the FAO also support the Unit. The Unit has also been contributing in the development of manuals, guidelines, strategies, policies and plans in the areas of food safety and quality control in the country. The IPH is taking necessary steps to make it a full-fledged unit to accomplish its objectives.

CURATIVE SERVICE

Ensuring health service for all

PRIMARY HEALTHCARE

- Primary health care addresses comprehensive and interrelated physical, mental and social health and wellbeing. It is an attitude to health and wellbeing centered on the essentials and circumstances of individuals, families and communities through their active participation and at an affordable cost.
- PHC is an integral part of a country's health system and, ideally, its main focus. It is the first level of contact for individuals, families and communities and enables health care to be delivered as close as possible to where people live and work. PHC is therefore the first element of the care continuum.
- Primary health care is rooted in a commitment to social justice and equity and in the recognition of the fundamental right to the highest attainable standard of health, as echoed in article 25 in the Universal Declaration on Human Rights: "Everyone has the right to a standard of living adequate for the health and wellbeing of himself and of his family, including food, clothing, housing and medical care and necessary social services [...]".
- Primary health care is well-positioned to be able to respond to rapid economic, technological, and demographic changes, all of which impact health and well-being. A recent analysis found that approximately half of the gains in reducing child mortality from 1990 to 2010 were due to factors outside the health sector (water and sanitation, education, economic growth).
- A primary health care approach draws in a wide range of stakeholders to examine and change policies to address the social, economic, environmental and commercial determinants of health and well-being. Treating people and communities as key actors in the production of their own health and wellbeing is critical for understanding and responding to the complexities of our changing world.

- Primary health care has been proven to be a highly effective and efficient way to address the main causes and risks of poor health and wellbeing today, as well as handling the emerging challenges that threaten health and wellbeing tomorrow. It has also been shown to be a good value investment, as there is evidence that quality primary health care reduces total healthcare costs and improves efficiency by reducing hospital admissions.
- Stronger primary health care is essential to achieving the health-related Sustainable Development Goals (SDGs) and universal health coverage. It will contribute to the attainment of other goals beyond the health goal (SDG3), including those on poverty, hunger, education, gender equality, clean water and sanitation, work and economic growth, reducing inequality and climate action.

Table 5.1: Primary healthcare facilities run by the DGHS, 2018 (December)

Type of facility	Type of service	Total no. of facilities	Total no. beds
Upazila health complex (50-bed)	Hospital	320	16000
Upazila health complex (31-bed)	Hospital	93	2883
Upazila health complex (10-bed)	Hospital	11	110
Subtotal of upazila health complexes		424	18993
50-bed hospital	Hospital	2	100
31-bed hospital	Hospital	6	186
30-bed hospital	Hospital	2	60
25-bed hospital	Hospital	1	25
20-bed hospital	Hospital	32	640
10-bed hospital	Hospital	19	190
Upazila health office	OPD	60	0
Union Sub Center	OPD	1312	0
Union health and family welfare center (UH & FWC, Netherland)	OPD	87	0
Urban dispensary	OPD	35	0
School Health Clinic	OPD	23	0
Grand total of total primary level facilities (except community clinic)		2003	20194
Community clinic (functional at present)	OPD	13,779	0
Grand total of total primary level facilities (including community clinic)		15782	20194

COMMUNITY CLINICS

- Community Clinic (CC) is the brainchild of Hon'ble Prime Minister Sheikh Hasina. It is being branded in her name. It is the lowest-tier public health facility.
- It is a one-stop service outlet on health, family planning and nutrition, each covering around 7000-10000 rural population. It is a unique example of public-private partnership as all the CCs have been constructed on land donated by the community people.
- Costs incurred for construction, medicines, service providers, logistics, and all other inputs are borne by the Government but these are managed by both community and the Government through Community Groups (CGs).
- The CG (CC management body) for each CC comprises 17 members and is headed by Elected Member of the Union Parishad; other members in the CG are from different walks of life. To support a CG in the management and community engagement, there are 3 Community Support Groups (CSGs) for each CC. In both CG and CSG, at least one-third of the members are female, including adolescents. During 1998-2001, more than 10,000 CCs were constructed, of which about 8,000 started functioning but, in 2001, shortly after commissioning, the CC activities were closed due to political change in the Government and continued as such till 2008.
- In 2009, the Government took the initiative for revitalization of CCs as a topmost priority under a project titled "Revitalization of Community Healthcare Initiatives in Bangladesh" (RCHCIB), setting the implementation period from 1 July 2009 to 30 June 2015.
- Under this project, the closed CCs were made functional in phases after necessary repair, recruitment, and capacity-building by creating the position of a new category of service provider (Community Healthcare Provider or CHCP) for each CC. Medicines, with all necessary logistics, were provided.
- At present (as of November 2018), 13,779 CCs are operational. From July 2015, all the activities of CCs were being carried out under an Operational Plan of the DGHS, namely Community-based Healthcare (CBHC) under the Health, Population and Nutrition Sector Development Program (HPNSDP) 2011-2016.
- Thousands of villagers, particularly the poor and the underprivileged mothers and children, are getting services from the nearby CCs. On an average, 9.5-10.00 million service seeker-visits take place nationwide each month in the CCs.
- It is to be mentioned that normal child delivery is being conducted in a good number of CCs (about 20% of the functional CCs), at the demand of the community, decision of the CG based on the availability of skilled

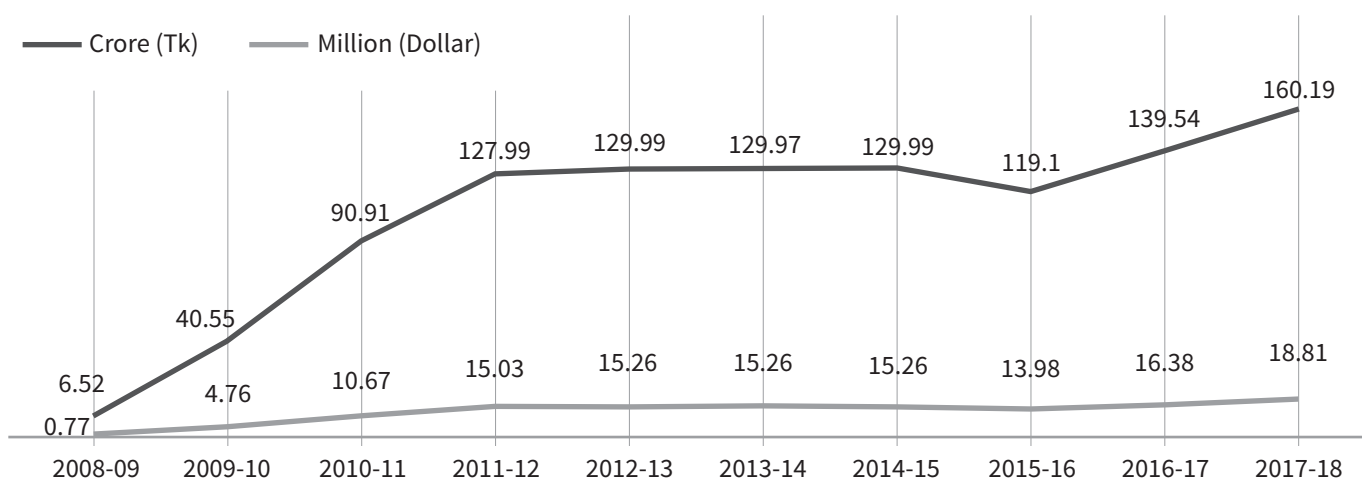


Figure 5.1: Total expenditure of Government for supply of medicines to community clinics

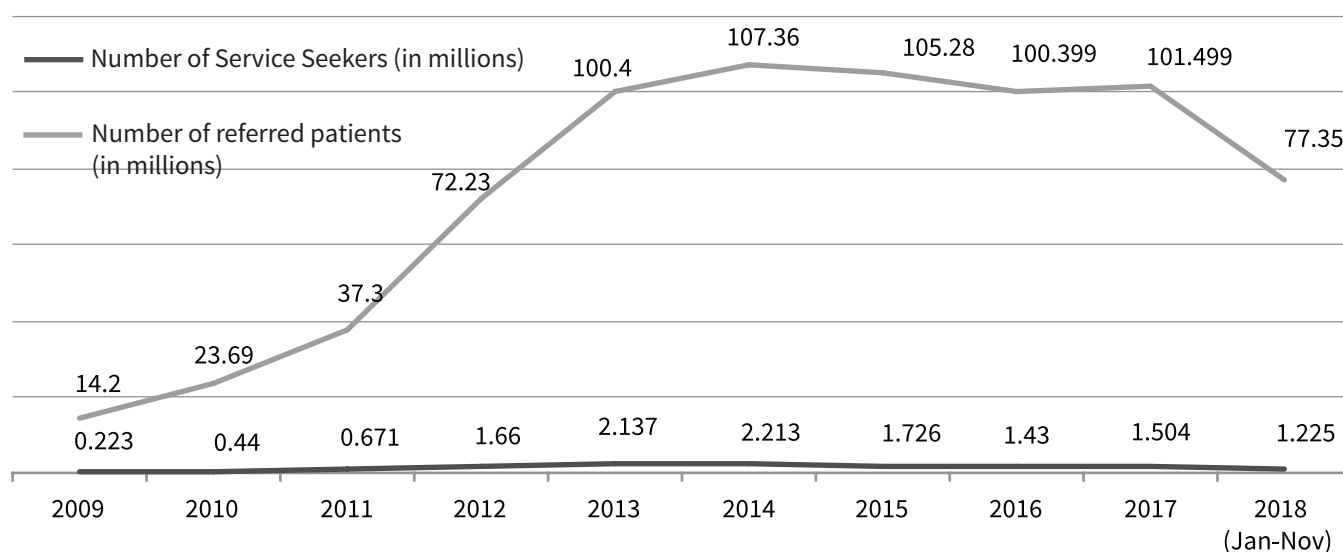


Figure 5.2: Total number (in millions) of service-seekers at CCs and cases referred from community clinics to higher facilities for proper management.

manpower. In total, 1,935 CHCPs have been trained on child delivery and, based on the decision of the committed local management; cases can be referred easily to the nearby UHC, if necessary. From 2009 till 2018 (November) 66,529 normal deliveries have been conducted in CCs without notable casualties to the mother and their babies.

- The use of ICT by CCs for data management and service provision is quite impressive. The Management Information System (MIS) of the DGHS is proving all-out support to develop and maintain the ICT backbone from April 2014. In some upazilas, monitoring of CCs is being done through Skype; telemedicine services are also being organized between CCs and UHCs, with a patient at the CC and a doctor at the other end.
- From 2011, with the deployment of CHCPs after basic training, the number of service-seekers has been increasing. From April 2009 to November 2018, there were 739.70 million visits by rural people to CCs all over the country and 13.229 million emergency and complicated cases were referred to higher facilities for better management.
- Community clinic is an unprecedented instance of community participation and public-private

partnership. Being inspired by community participation, some UN agencies and NGOs have started working for the community clinics. In Bangladesh the 4th Health, Population and Nutrition Sector Program (HPNSP) is under implementation for a period from January 2017 to June 2022. Operational Plan of the Community-based Healthcare (CBHC) will contribute substantially in achieving the targets of SDGs, like the MDGs and will continue till the achievement of the SDGs by 2030.

- Due to the flagship nature of the program, community clinics deserve special mention. In addition to the community clinics, important components of primary healthcare, among others, include domiciliary healthcare, essential service delivery, along with urban primary healthcare, maternal healthcare (inclusive of some screening programs for women's health), child healthcare, school health program, and adolescent health program.
- The MIS-DGHS is playing the leading role in providing computers with Internet connections, which now extends down to the grassroots-level health facilities as well as the health workers. The union health centers and community clinics have laptop computers and wireless modems, and the community health workers have Android tablets.

DOMICILIARY HEALTH SERVICES IN RURAL BANGLADESH

- There are domiciliary workers—one for every 5 to 6 thousand people at the ward or village level. Under the DGHS, there are 26,538 sanctioned posts of domiciliary workers, of which 20,908 are for health assistants (HA), 4,220 for assistant health inspectors (AHI), and 1,410 for health inspectors (HI). As of now, 78.54% posts were filled up.
- Like the DGHS, the DGFP also has domiciliary workers to work at the ward or village level. These staff members are called Family Welfare Assistants (FWA) and Family Planning Inspectors (FPI).

ESSENTIAL SERVICE DELIVERY AND URBAN PRIMARY HEALTHCARE

- There was an Operational Plan, namely “Essential Service Delivery” (ESD) mainstreamed under the DGHS

(HPNSDP 2011-2016) to improve service, particularly at the upazila level and below and complement urban primary healthcare. ESD will continue under the current HPNSP also.

- The areas of services include: limited curative care, support services and coordination, medical waste management, urban health, mental health, and tribal health.
- The urban primary healthcare in Bangladesh is principally the responsibility of the Ministry of Local Government, Rural Development and Cooperatives (MOLGRD), carried out through the city corporations and municipalities.
- Besides, large-scale primary healthcare activities under Urban Primary Healthcare Project (UPHCP) and Smiling Sun Franchise Program are run by NGOs in collaboration with the city corporations, Municipality and with financial assistance from donors.

Table 5.2: Secondary and tertiary hospitals/health centers under the DGHS, 2018 (December)

Type	Level of facilities	Name of facility	No. of facilities	No. of functional beds
Airport Health Office	Secondary	Dhaka Airport Health Office	1	0
		Chittagong Airport Health Office	1	0
Chest disease clinic	Secondary		42	0
Chest hospital	Secondary		14	866
District level office (Civil Surgeon Office)	Secondary		65	0
District level hospital (District/General hospital)	Secondary		59	9,600
Divisional level office (Divisional health Office)	Tertiary		8	0
Dental college hospital	Tertiary		1	200
Hospital of alternative medicine	Tertiary		3	200
Infectious disease hospital	Secondary		5	180
Leprosy hospital	Secondary		3	130
Medical college hospital	Tertiary		18	15,013

Type	Level of facilities	Name of facility	No. of facilities	No. of functional beds
Port Health Office	Secondary	Chittagong Port Health Office	1	0
		Khulna Port Health Office	1	0
Specialized Health Center	Tertiary	Central Skin & Social Hygiene Center, Chittagong	1	0
		National Center For Control Of Rheumatic Fever & Heart Disease	1	0
		National Asthma Center	1	70
Specialized hospital	Tertiary		5	1,450
Specialty postgraduate institute and hospital	Tertiary		11	3,284
Trauma center	Secondary		6	120
Other hospitals	Tertiary	Kurmitola 500 Bed General Hospital, Dhaka	1	500
	Tertiary	Mugda 500 Bed General Hospital, Dhaka	1	500
	Secondary	Narayanganj 300 Bed Hospital	1	300
		Narsingdi 100 Bed Zilla Hospital	1	100
		Bangladesh Korea Moitree Hospital, Savar, Dhaka	1	0
		Kuwait Bangladesh Friendship Govt. Hospital, Uttara, Dhaka	1	0
		Saidpur 100 Bed Hospital, Nilphamari	1	100
Total secondary & tertiary level facilities		254	32,613	

- Available number of beds is one of the good proxies for measuring the strength of healthcare infrastructure in different geographic areas. These are run under the administrative control of the DGHS.

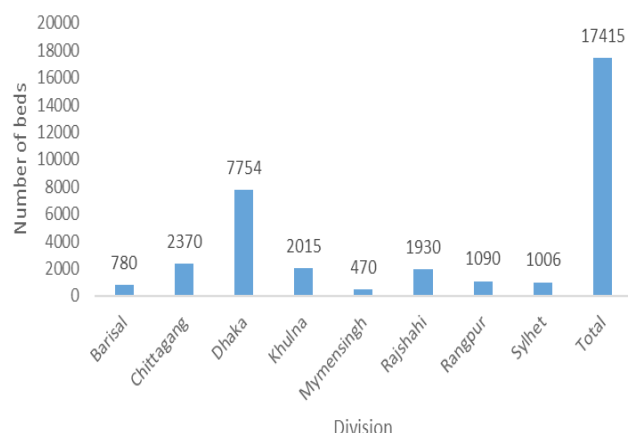


Figure 5.3: Number of beds in all secondary and tertiary-care hospitals other than medical college hospitals by administrative division, 2018

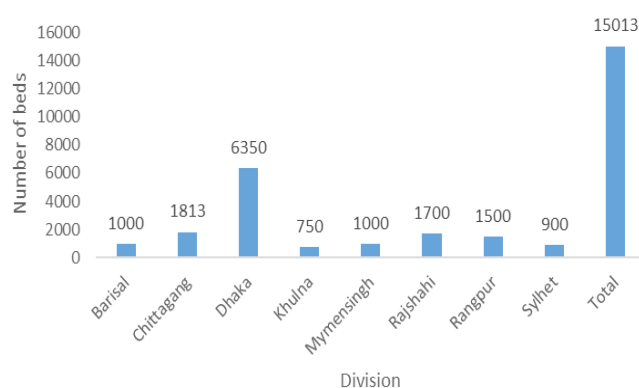


Figure 5.4: Number of beds in all medical college hospitals by administrative division of Bangladesh, 2018

PRIVATE HOSPITALS, CLINICS, AND DIAGNOSTIC CENTERS

- As of June 2016, the DGHS provided registration to 14,583 private hospitals, clinics, and diagnostic centers in Bangladesh.
- The number of registered private hospitals and clinics is 5,054, and that of registered private diagnostic centers is 9,529. The total number of beds in these registered private hospitals and clinics is 90,587.

UTILIZATION OF PUBLIC HEALTH FACILITIES

The information is based on hospital statistics available from the DHIS2 of 2017 (upto December 2017) data, which should be interpreted with caution due to incomplete progress reporting.

1. Outpatient attendance

Outpatient attendance at different hospitals reporting through DHIS2 in 2017 was as follows:

- at specialized hospital ranged from 34101 at DBangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat to 930745 at Kurmitola General Hospital (500 Bed). Dental college and hospital has less number may be due to less reporting. (table 5.4)
- at medical college hospital ranged from 199437 at M. Abdur Rahim MCH, Dinajpur to 935876 Dhaka MCH. (table 5.3)
- at district/general hospital ranged from 54205 at Rangamati district hospital to 436973 at Jessore district hospital. (figure 5.5)
- 423 upazila health complexes reported OPD attendance data which is not presented in the annex.

Table 5.3: Outpatient attendances at Gov. Medical college hospitals, 2017

OPD visit in MCH	No. of visit
M Abdur Rahim MCH, Dinajpur	199437
Faridpur MCH, Faridpur	205758
Shaheed Tajuddin Ahmad MCH, Gazipur	236035
Khulna MCH, Khulna	261486
Comilla MCH, Comilla	264704
Rangpur MCH, Rangpur	324662
Shahid Ziaur Rahman MCH, Bogra	432042
Chittagong MCH, Chittagong	620667
Sir Salimullah MCH, Mitford, Dhaka	656513
Rajshahi MCH, Rajshahi	720558
Dhaka MCH, Dhaka	935876

Table 5.4: Outpatient attendances at Gov. Specialized hospital, 2017

OPD attendance at Specialized hospital	no. of visit
Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat	34101
Pabna Mental Hospital, Pabna	38063
Sheikh Fazilatunnessa Mujib Eye Hospital And Training Institute	47911
National Institute of Mental Health and Research, Dhaka	50114
National Institute Of Ent In Dhaka	85884
Institute of Child & Mother Health (ICMH) Matuail	105630
National Institute of Kidney Diseases & Urology, Dhaka	113377
National Institute of Traumatology & Orthopaedic Rehabilitation, Dhaka(NITOR)	114371
National Institute of Cardiovascular Diseases, Dhaka	118261
National Institute of Cancer Research and Hospital, Dhaka	132084
National Institute of Diseases of the Chest and Hospital, Dhaka	136205
National Institute Of Neuro Science (NINS)	240764
National Institute of Ophthalmology, Dhaka(NIO)	266022
Mugda 500 beded General Hospital	361696
Kurmitola General Hospital (500 Bed)	930745

2. Hospital Emergency ward attendance

- at specialized hospital ranged from 278 at Sheikh Fazilatunnessa Mujib Eye Hospital and Training Institute to 101144 at National Institute of Cardiovascular Diseases, Dhaka. (table 5.7)
- at medical college hospital ranged from 13838 at MAG Osmani MCH, Sylhet to 462438 at Dhaka medical college hospital, Dhaka. (figure 5.5)

Table 5.5: Emergency ward attendances at Gov. Medical college hospitals, 2017

Emergency ward attendance 2017	no. of visit
MAG Osmani MCH, Sylhet	13838
Mymensingh MCH, Mymensingh	15282
Rajshahi MCH, Rajshahi	19486
M Abdur Rahim MCH, Dinajpur	51804
Comilla MCH, Comilla	53229
Faridpur MCH, Faridpur	54156
Shahid Ziaur Rahman MCH, Bogra	72270
Shaheed Tajuddin Ahmad MCH, Gazipur	102133
Rangpur MCH, Rangpur	105036
Sir Salimullah MCH, Mitford, Dhaka	168572
Shaheed Suhrawardy MCH, Dhaka	187838
Chittagong MCH, Chittagong	237222
Dhaka MCH, Dhaka	462438

Table 5.6: Inpatient attendances at Gov. Medical college hospitals, 2017

Inpatient attendance	no. of visit
Shaheed Tajuddin Ahmad MCH, Gazipur	22451
M Abdur Rahim MCH, Dinajpur	48572
Comilla MCH, Comilla	56324
Faridpur MCH, Faridpur	56363
Khulna MCH, Khulna	57907
Sir Salimullah MCH, Mitford, Dhaka	65797
Shaheed Suhrawardy MCH, Dhaka	75006
Shahid Ziaur Rahman MCH, Bogra	76251
Rangpur MCH, Rangpur	110582
MAG Osmani MCH, Sylhet	146815
Rajshahi MCH, Rajshahi	170596
Dhaka MCH, Dhaka	172343
Chittagong MCH, Chittagong	203521
Mymensingh MCH, Mymensingh	205729

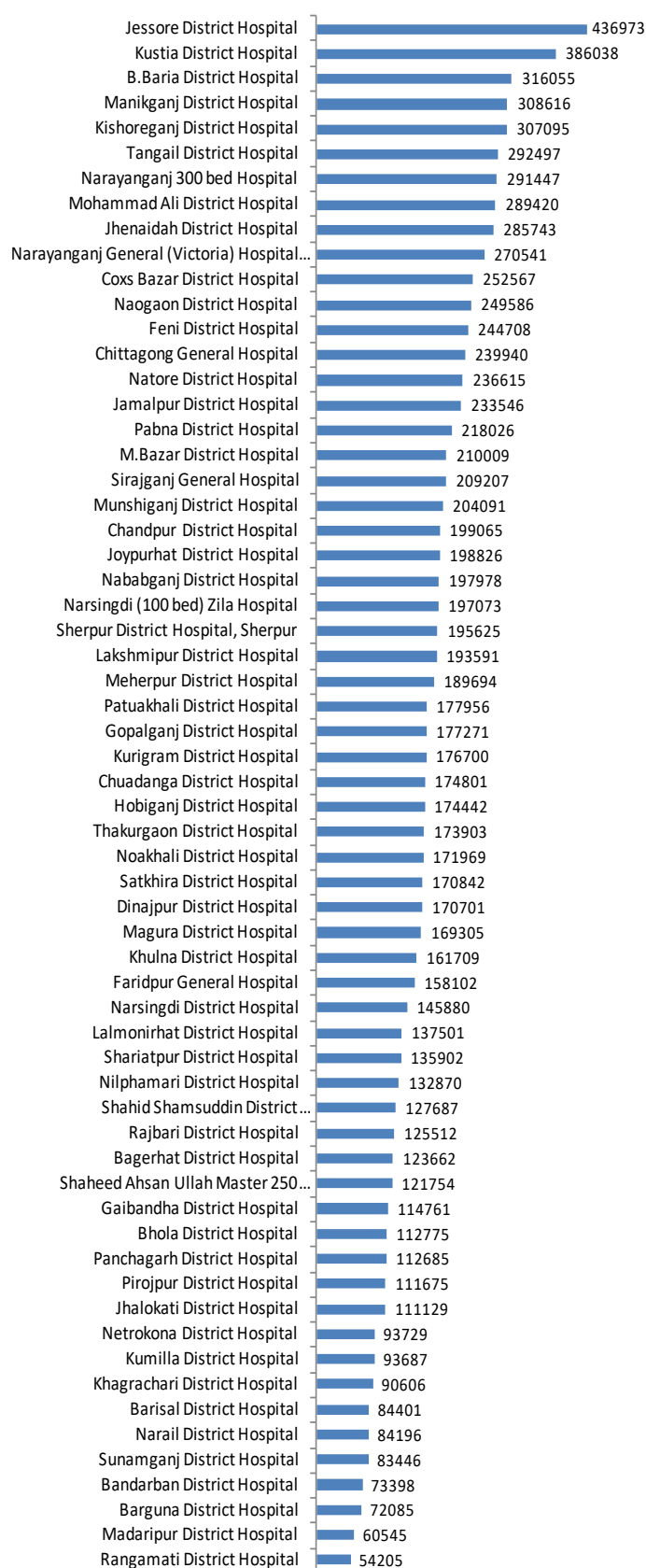


Figure 5.5: Outpatient attendances at district/general hospital, 2017

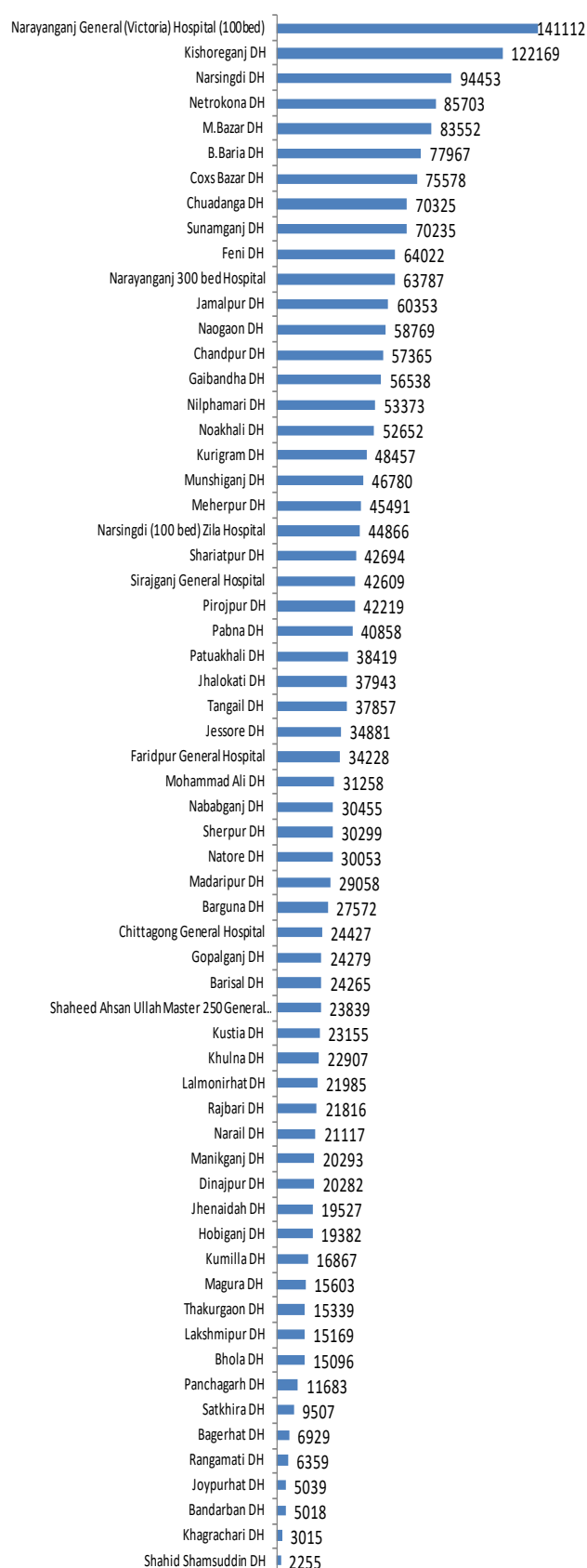


Figure 5.6: Emergency ward attendances at district/general hospital, 2017

Table 5.7: Emmergency attendances at Gov. Specialized hospital, 2017

Emmergency attendances	no. of visit
National Institute of Cardiovascular Diseases, Dhaka	101144
Kurmitola General Hospital (500 Bed)	58891
National Institute of Traumatology & Orthopaedic Rehabilitation, Dhaka(NITOR)	50555
Institute of Child & Mother Health (ICMH) Matuail	45156
Mugda 500 beded General Hospital	34105
National Institute Of Neuro Science (NINS)	26950
National Institute of Diseases of the Chest and Hospital, Dhaka	9963
National Institute of Kidney Diseases & Urology, Dhaka	7980
National Institute Of Ent In Dhaka	7104
National Institute of Cancer Research and Hospital, Dhaka	5688
National Institute of Mental Health and Research, Dhaka	3542
National Institute of Ophthalmology, Dhaka(NIO)	3472
Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat	3304
Sheikh Fazilatunnessa Mujib Eye Hospital And Training Institute	278

- at district/general hospital ranged from 2255 at Shahid Shamsuddin district hospital, Sylhet to 141112 at Narayanganj General (Victoria) 100bed Hospital . (figure 5.6)
- 423 upazila health complexes reported Emergency attendance data which is not possible to show in graph for this please refers to annex .

3. Hospital Inpatient attendance

- at specialized hospital ranged from 1676 at Pabna Mental Hospital to 69879 at National Institute of

Table 5.8: Inpatient attendances at Gov. Specialized hospital, 2017

Inpatient attendance at Specialized hospital	no. of visit
Pabna Mental Hospital, Pabna	1676
Sheikh Fazilatunnessa Mujib Eye Hospital And Training Institute	2308
National Institute Of Ent In Dhaka	2942
Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat	3369
National Institute of Mental Health and Research, Dhaka	4237
National Institute of Cancer Research and Hospital, Dhaka	5455
National Institute of Kidney Diseases & Urology, Dhaka	5636
Institute of Child & Mother Health (ICMH) Matuail	8332
National Institute Of Neuro Science (NINS)	10471
National Institute of Ophthalmology, Dhaka(NIO)	12784
National Institute of Diseases of the Chest and Hospital, Dhaka	16342
Mugda 500 beded General Hospital	22426
National Institute of Traumatology & Orthopaedic Rehabilitation, Dhaka(NITOR)	23049
Kurmitola General Hospital (500 Bed)	32574
National Institute of Cardiovascular Diseases, Dhaka	69879

Cardiovascular Diseases. (figure 5.8)

- at medical college hospital ranged from 22451 at Shaheed Tajuddin Ahmad MCH, Gazipur to 205729 at Mymensingh MCH, Mymensingh (figure 5.6)
- at district/general hospital ranged from 2861 at Shahid Shamsuddin district hospital, Sylhet to Pabna district hospital. (figure 5.7)
- 423 upazila health complexes reported Inpatient attendance data which is not possible to show in

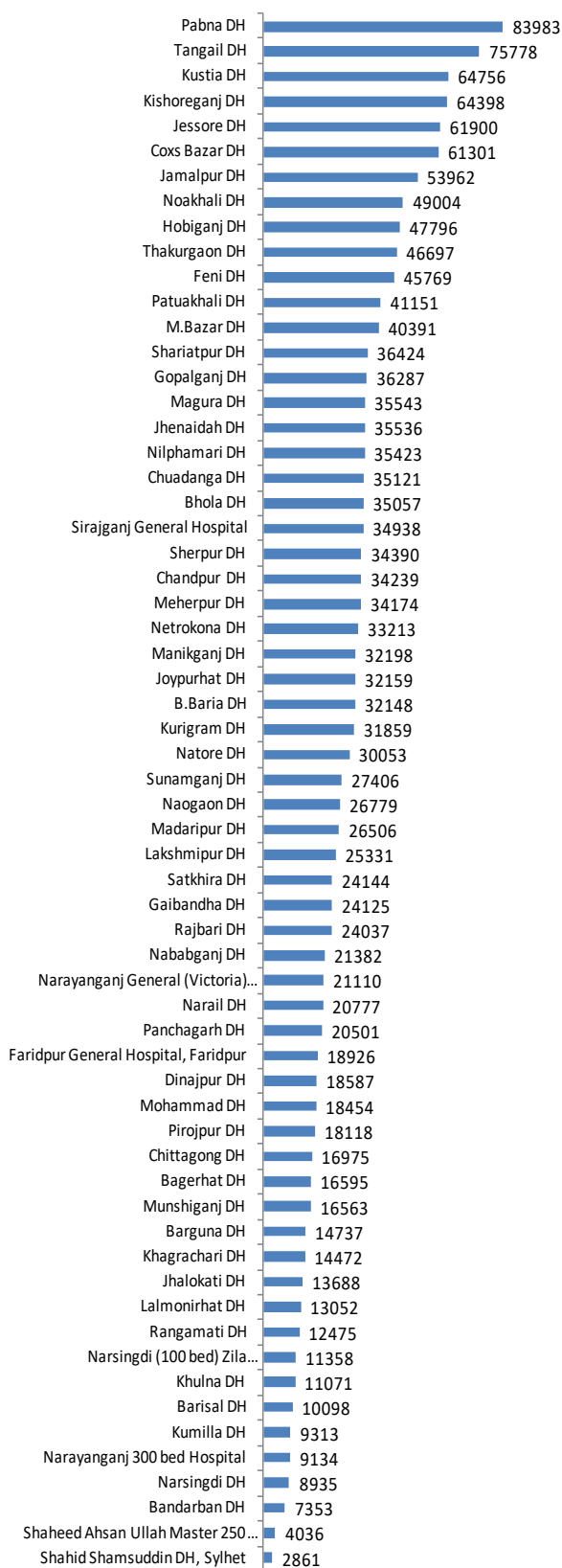


Figure 5.7: Inpatient attendances at district/general hospital, 2017

graph for this please refers to annex .

4. Hospital reported death

The total number of deaths reported from different health facilities under DGHS were 109557. Table 5.9 shows the number of institution reported death according to age and sex. Figure 5.8 shows total number of death according to age and sex in 2017.

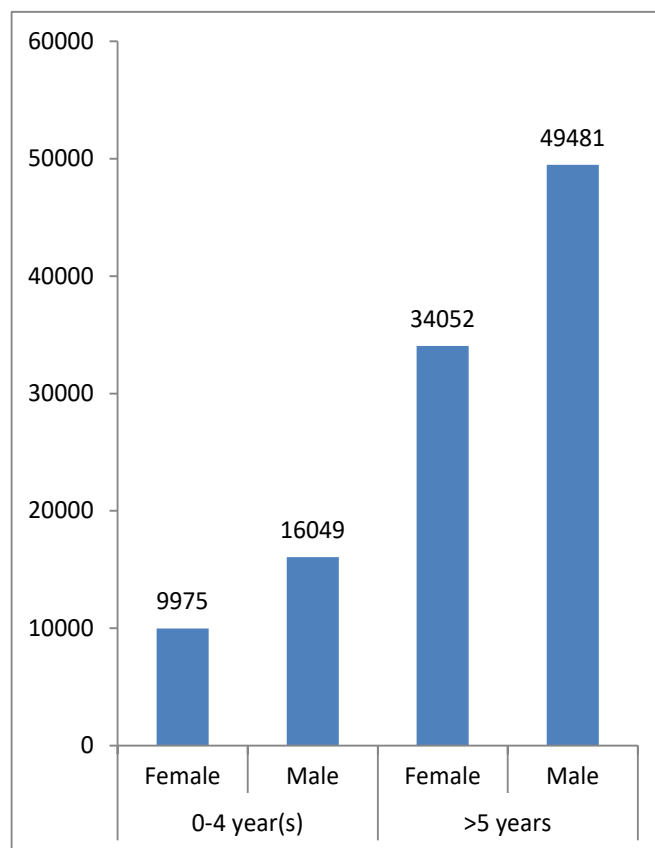


Figure 5.8: Number of total death according to age and sex reported from all Gov. health facilities, 2017

5. Cause of Morbidity

Major causes for admission in healthcare facilities were different in under 5 year olds and in persons aged 5 or above. Diarrhoea was a major cause in both age group. The morbidity profile of both the age group are visualised in the following graphs.

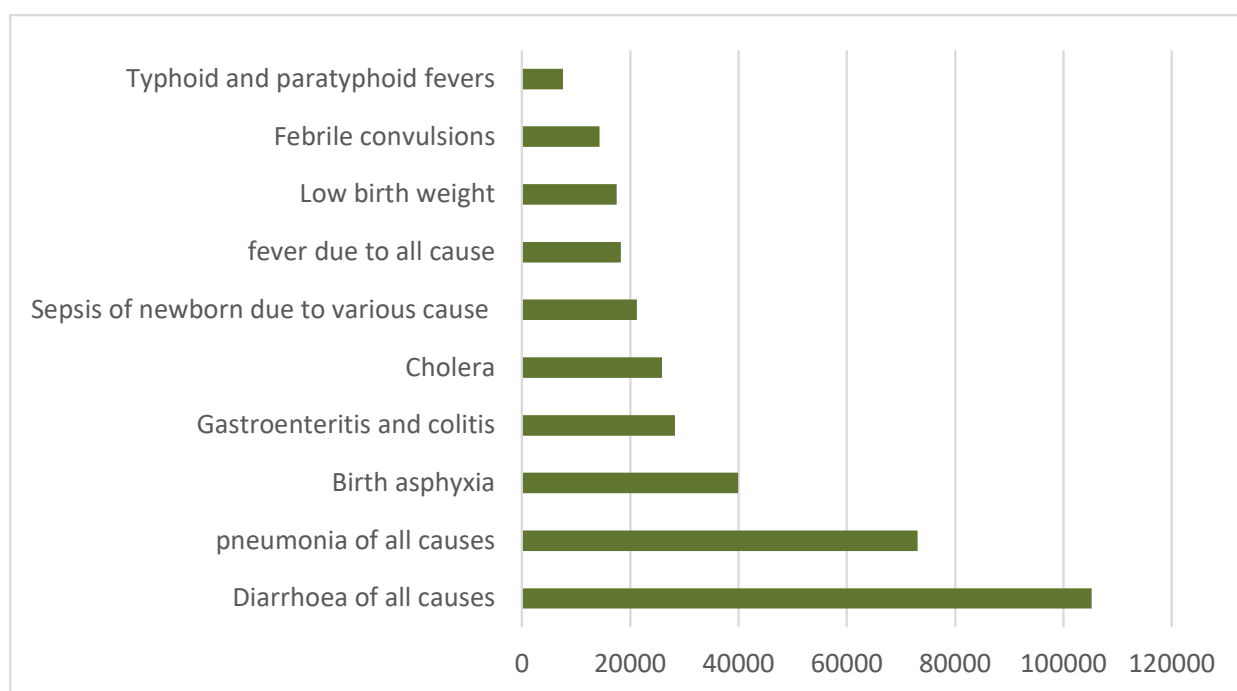


Figure 5.9: Top 10 diseases in under 5 children which required admission in hospital, 2017

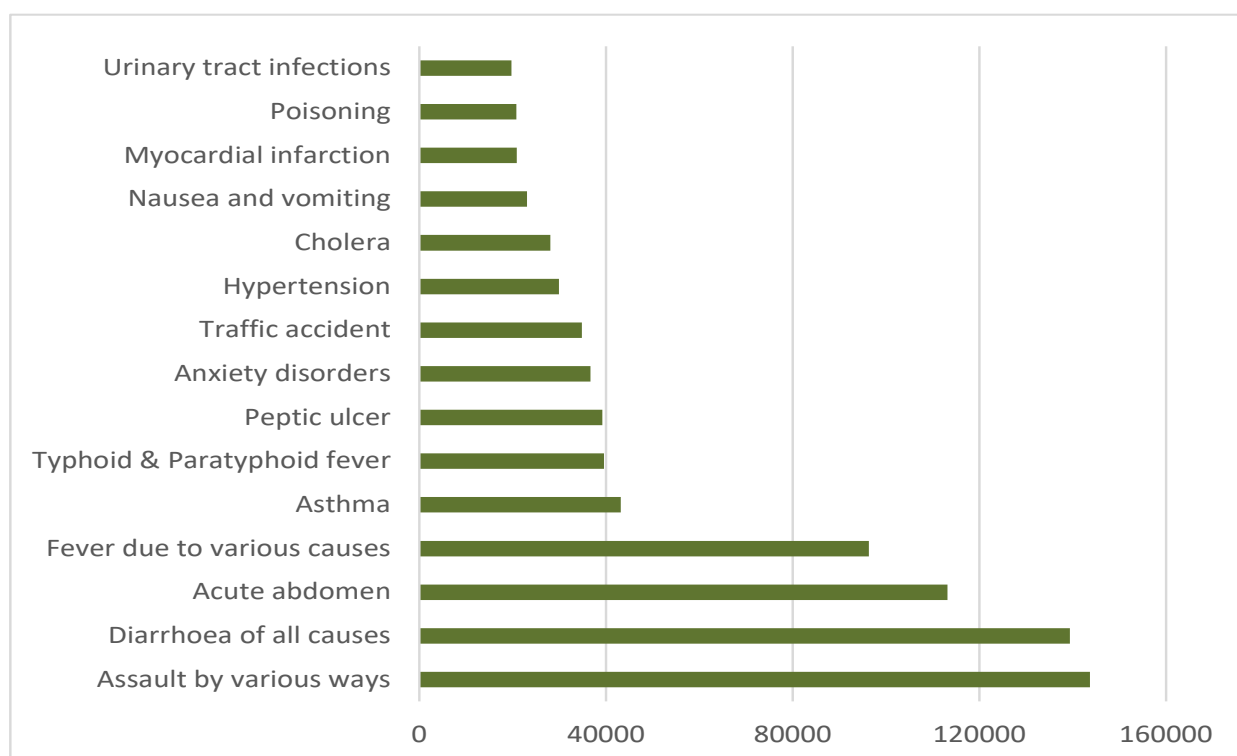


Figure 5.10: Top 15 diseases in person aged 5 or above which required admission in hospital, 2017

6. Cause of Death

Following table and graphs shows the inpatient cause of death reported from medical college hospital,

district/general hospital and upazila health complex in 2017 through through DHIS2. The cause of death has disaggregated according to age i.e. under five causes of death and above five causes of death.

Table 5.9: Number of institutional deaths reported by different health facilities in 2017

Level of care	Type of facilities	Number of facilities reported death	Total number of death	0-4 year(s)		>5 years	
				Female	Male	Female	Male
Primary	10 bed hospital	0	0	0	0	0	0
	20 bed hospital	2	27	19	3	5	0
	31 bed hospital	1	2	0	0	1	1
	100 bed hospital, Saidpur, Nilphamari	1	80	6	12	26	36
	Upazila health complex	387	6287	445	640	2152	3050
Secondary	Distric hospital/general hospital	62	25162	3118	4253	7615	10176
Tertiary	500 bed hospital	2	124	5	6	9	104
	Medical college hospital	14	69260	6274	9326	22057	31603
	Other tertiary hospital (national instuites)	14	8615	108	1809	2187	4511
Total		483	109557	9975	16049	34052	49481

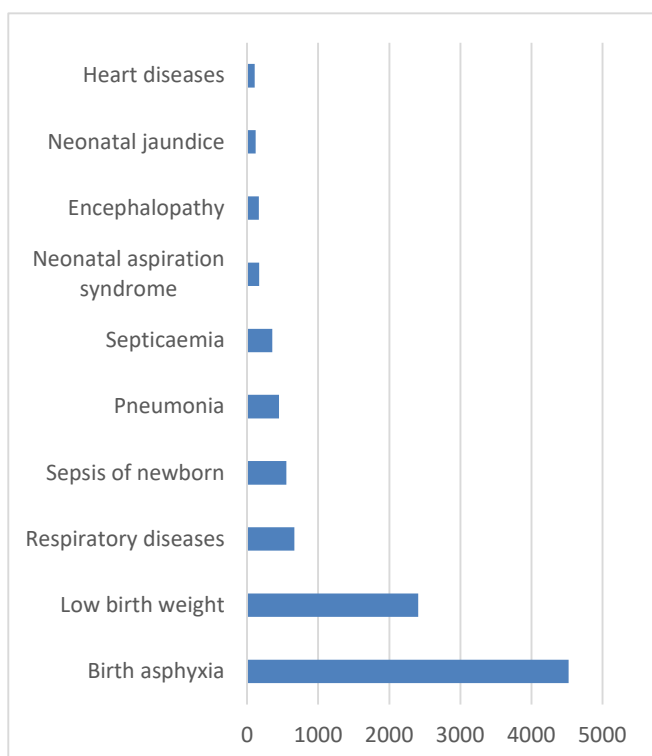


Figure 5.11: Top 10 causes of death among under five year of age, 2017

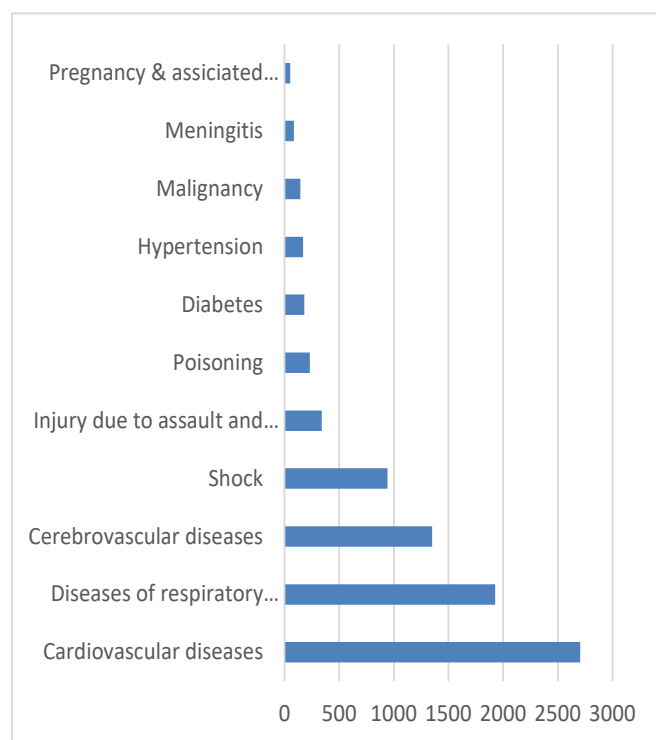


Figure 5.12: Top 10 causes of death among five year olds and above, 2017

7. Average length of stay

- at specialized hospital ranged from 2.7 days in Bangladesh Institute of Tropical and Infectious Diseases to 15 days at National Institute of Disease of the Chest and Hospital (NIDCH). (figure 5.17)
- at medical college hospital ranged from 3.2 at Osmani Medical College Hospital to 8 days in Dhaka Medical College Hospital, Dhaka. (figure 5.18)
- at district/general hospital ranged from 1.7 days at

Hobiganj district hospital and 1.8 days Naogaon district hospital to 11.3 days in Narayanganj 300 bed hospital. (figure 5.19)

- at upazila health complexes range within highest 20 UHC 4.1 days at Sariakandi and Wazipur UHC to 29.3 days at Haziganj UHC (figure 5.20); UHC ranges within lowest 20 shown in figure 5.21)

Table 5.10: Average length of stay (ALS) at Gov. Medical College Hospital, 2017

organisation name	ALS
Dhaka Medical College Hospital, Dhaka	8.1
Shaheed Suhrawardy Medical College Hospital, Dhaka	6.7
Chittagong Medical College Hospital, Chittagong	5
Shahid Ziaur Rahman Medical College Hospital, Bogra	5
Khulna Medical College Hospital, Khulna	4.9
Sir Salimullah Medical College Hospital, Mitford, Dhaka	4.9
Rangpur Medical College Hospital, Rangpur	4.7
Comilla Medical College Hospital, Comilla	4.5
Sher-e-Bangla Medical College Hospital, Barisal	4.5
M Abdur Rahim Medical College Hospital, Dinajpur	4.2
Mymensingh Medical College Hospital, Mymensingh	4
Faridpur Medical College Hospital, Faridpur	3.9
Rajshahi Medical College Hospital, Rajshahi	3.4
Shaheed Tajuddin Ahmad Medical College Hospital, Gazipur	3.3
MAG Osmani Medical College Hospital, Sylhet	3.2

Table 5.11: Average length of stay (ALS) at Gov. Specialized hospital, 2017

Organisation name	ALS
National Institute of Diseases of the Chest and Hospital, Dhaka	14.5
National Institute of Mental Health and Research, Dhaka	12.5
National Institute Of Neuro Science (NINS)	10.5
National Institute of Traumatology & Orthopaedic Rehabilitation, Dhaka(NITOR)	9.4
National Institute of Kidney Diseases & Urology, Dhaka	8.8
National Institute Of Ent In Dhaka	7.6
Kurmitola General Hospital (500 Bed)	6.8
Mugda 500 beded General Hospital	4.8
National Institute of Cardiovascular Diseases, Dhaka	4.2
National Institute of Ophthalmology, Dhaka(NIO)	4.1
Institute of Child & Mother Health (ICMH) Matuail	3.6
Dhaka Shishu Hospital	3
Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat	2.7

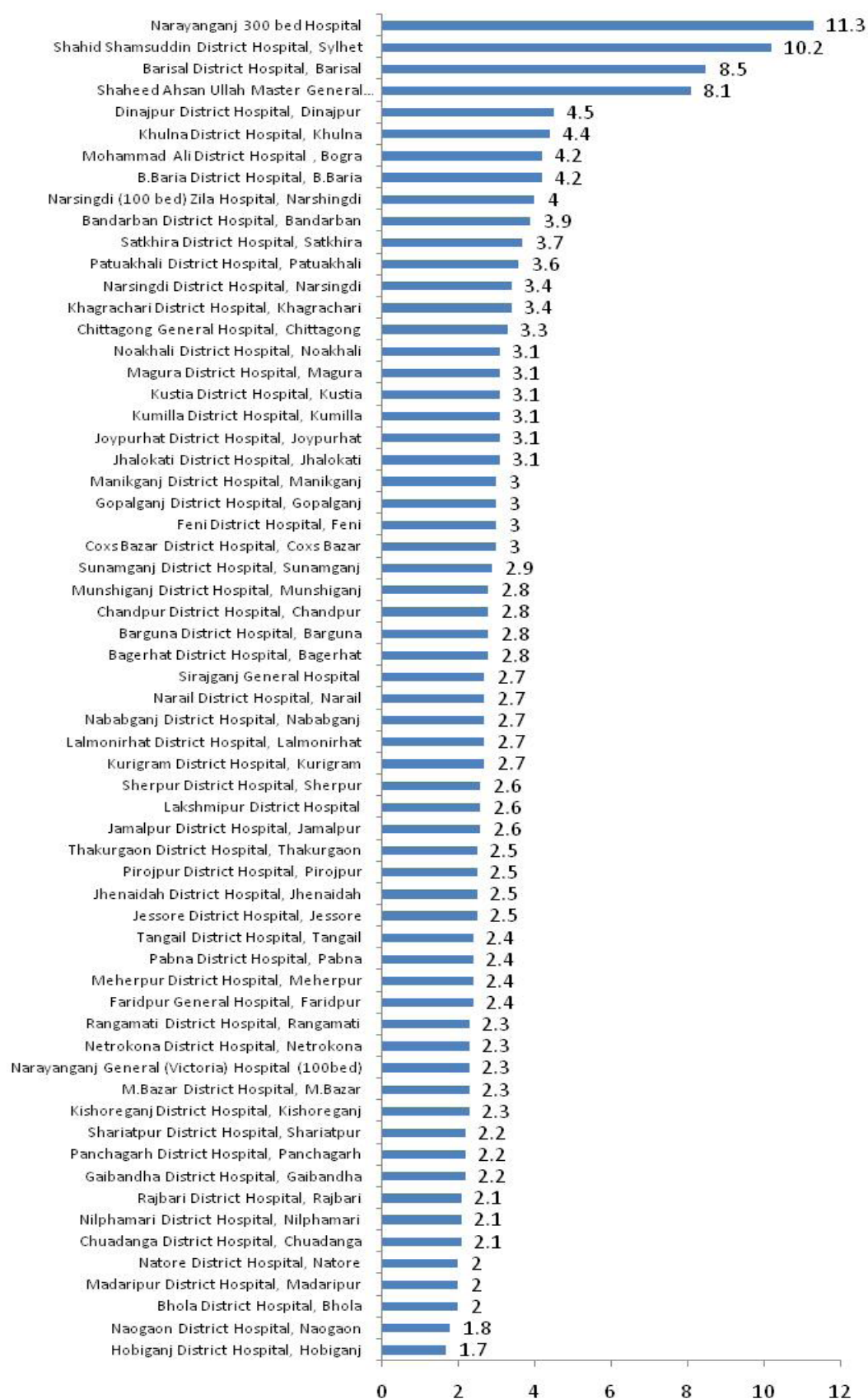


Figure 5.13: Average length of stay by inpatient in District/general hospital, 2017

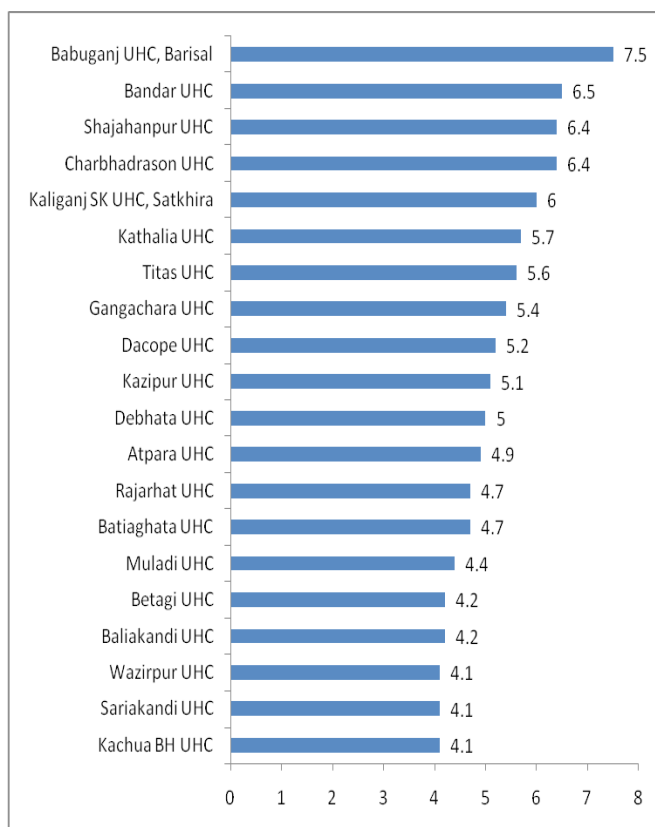


Figure 5.14: Average length of stay by inpatient in highest 20 Upazila health complex, 2017

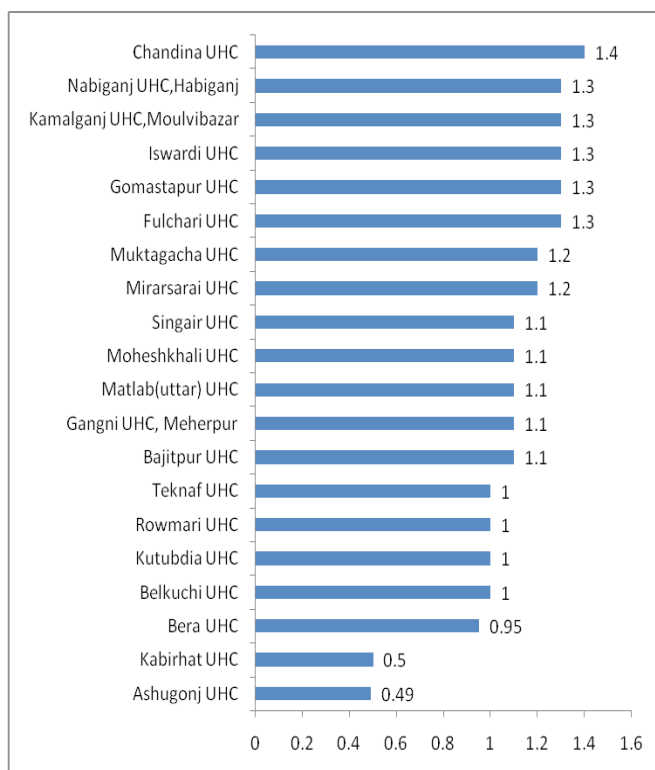


Figure 5.15: Average length of stay by inpatient in lowest 20 Upazila health complexes, 2017

8. Bed-occupancy ratio, 2017

- At specialized hospitals that report through DHSI2 on hospital bed statement ranged from 20.7 percent in Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat to 191.5 percent in National Institute of Cardiovascular Diseases, Dhaka, with an average BOR of 80.54% (table 5.13)
- At Gov. medical college hospitals ranged from 105.8 % in Shaheed Tajuddin Ahmed Medical College Hospital, Gazipur to 238.7 % in Mymensingh Medical College Hospital, with an average bed occupancy ratio of 153.57 %. (table 5.12)
- At district/general hospital ranged from 66.6% at Chittagong general hospital to 317.9 % at Thakurgaon district hospital, with an average bed occupancy ratio of 144.82 (figure 5.16).
- At upazila health complex with 20 highest and lowest BOR with an average bed occupancy ratio of 81.02 %, figure 5.17 and figure 5.18 respectively.
- All the data are from DHIS2. Data of some hospitals could not be gathered from DHIS2 and thus not mentioned here.

Table 5.12: Bed occupancy ratio (%) of Gov. Medical College Hospital, 2017

organisation name	Bed occupancy rate	No. of sanctioned beds
Shaheed Tajuddin Ahmad Medical College Hospital, Gazipur	105.8	214
M Abdur Rahim Medical College Hospital, Dinajpur	112.5	500
Faridpur Medical College Hospital, Faridpur	114.5	500
Shaheed Suhrawardy Medical College Hospital, Dhaka	132.7	850
Dhaka Medical College Hospital, Dhaka	133.3	2600
Rajshahi Medical College Hospital, Rajshahi	145.8	1200
Rangpur Medical College Hospital, Rangpur	149.4	1000
Sir Salimullah Medical College Hospital, Mitford, Dhaka	151	600
MAG Osmani Medical College Hospital, Sylhet	156.4	
Comilla Medical College Hospital, Comilla	156.9	500
Sher-e-Bangla Medical College Hospital, Barisal	159.2	1000
Khulna Medical College Hospital, Khulna	162.8	500
Chittagong Medical College Hospital, Chittagong	190.3	1313
Shahid Ziaur Rahman Medical College Hospital, Bogra	194.2	500
Mymensingh Medical College Hospital, Mymensingh	238.7	1000

Table 5.13: Bed occupancy ratio (%) of specialized hospitals, 2017

organisation name	Bed occupancy rate	No. of sanctioned beds
National Institute of Cancer Research and Hospital, Dhaka	92.0	300
Bangladesh Institute of Tropical and Infectious Diseases, Fouzdarhat	20.7	100
National Institute Of Ent In Dhaka	56.7	120
Mugda 500 beded General Hospital	66.7	500
National Institute of Mental Health and Research, Dhaka	76	200
National Institute of Ophthalmology, Dhaka(NIO)	77.5	250
Institute of Child & Mother Health (ICMH) Matuail	80.1	200
National Institute of Diseases of the Chest and Hospital, Dhaka	90.6	670
National Institute of Kidney Diseases & Urology, Dhaka	94.1	150
National Institute Of Neuro Science (NINS)	100	300
Kurmitola General Hospital (500 Bed)	102.7	500
National Institute of Traumatology & Orthopaedic Rehabilitation, Dhaka(NITOR)	105.8	500
National Institute of Cardiovascular Diseases, Dhaka	191.5	414

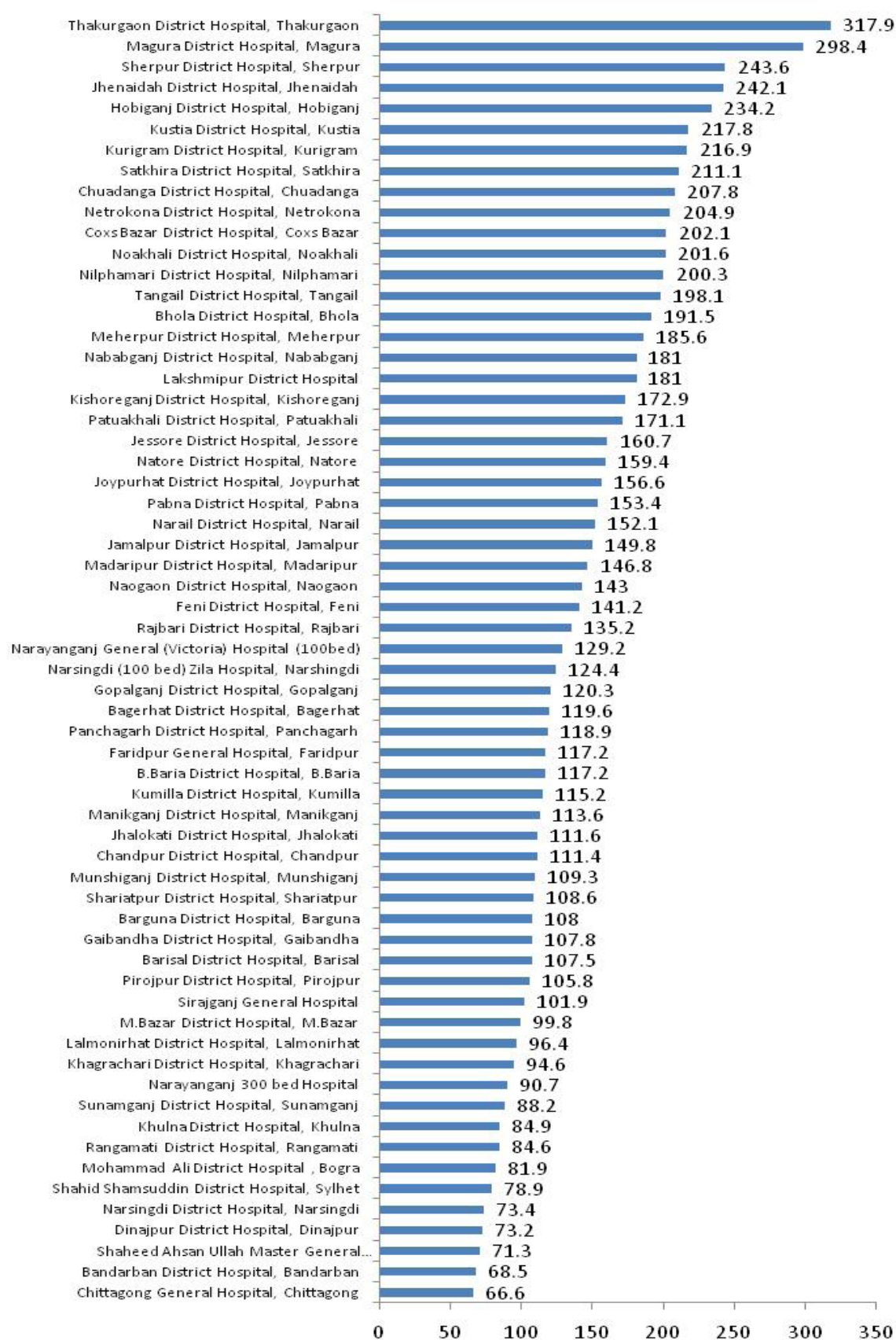


Figure 5.16: Bed occupancy ratio (%) of Gov. district/general hospitals, 2017

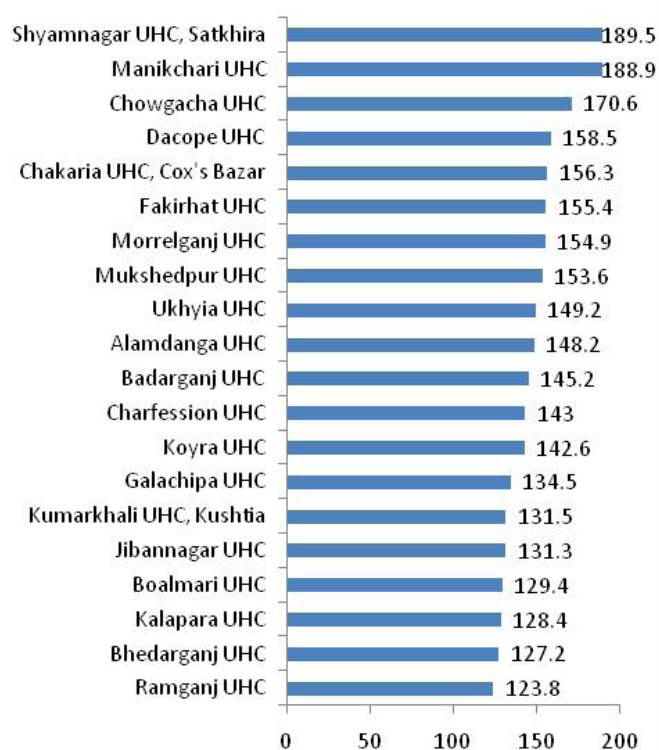


Figure 5.17: BOR in highest 20 Upazila health complexes, 2017

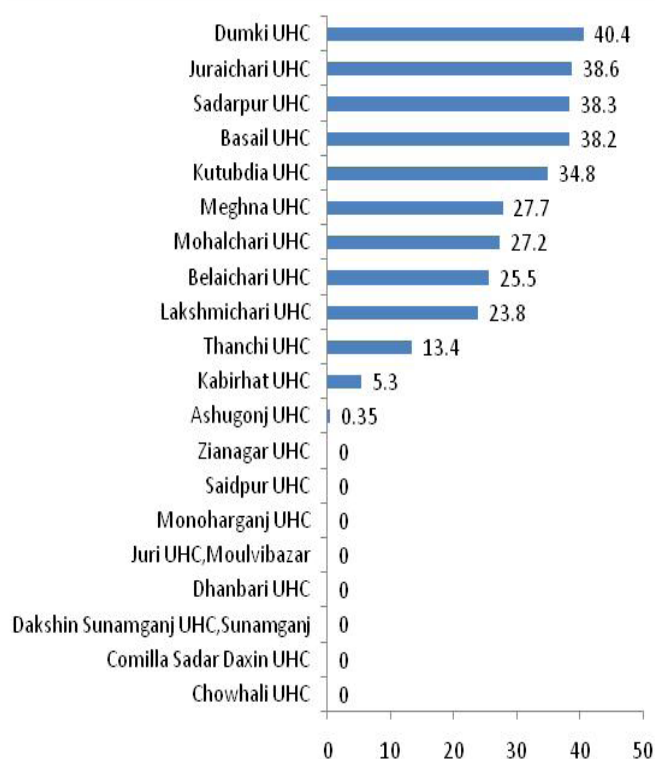


Figure 5.18: BOR in lowest 20 Upazila health complexes, 2017

9. Surgeries

Total major surgeries performed were 296853 and total minor surgeries were 1197646 including all government health facilities.

Table 5.14: Number of major and minor surgeries performed at Gov. specialized hospitals, 2017

Hospital name	Major Surgery	Minor Surgery
National Institute of Diseases of the Chest and Hospital, Dhaka	853	2533
National Institute of Kidney Diseases & Urology, Dhaka	1058	1696
National Institute Of Neuro Science	1489	86
National Institute Of Ent In Dhaka	1749	556
Mugda 500 beded General Hospital	2897	1360
Institute of Child & Mother Health (ICMH) Matuail	3887	1044
National Institute of Ophthalmology, Dhaka(NIO)	14902	1967

Table 5.15: Number of major and minor surgeries performed at Gov. medical college hospital, 2017

Hospital name	Major Surgery	Minor Surgery
Shaheed Tajuddin Ahmad MCH	1486	3008
M Abdur Rahim MCH, Dinajpur	2845	13989
Faridpur MCH, Faridpur	4299	1802
Khulna MCH, Khulna	4346	1946
Shahid Ziaur Rahman MCH, Bogra	6514	14005
Rangpur MCH, Rangpur	8598	16818
Sher-e-Bangla MCH, Barisal	10383	11020
Comilla MCH, Comilla	10711	8636
Sir Salimullah MCH, Mitford, Dhaka	11498	8981
Rajshahi MCH, Rajshahi	15134	22167
Chittagong MCH, Chittagong	15382	12773
Mymensingh MCH, Mymensingh	16101	34009
Shaheed Suhrawardy MCH, Dhaka	16925	9220
MAG Osmani MCH, Sylhet	19272	21729
Dhaka MCH, Dhaka	29054	39819

Table 5.16: Number of major and minor surgeries performed at district/general hospitals, 2017

Hospital Name	Minor Surgery	Major Surgery
Barisal DH	277	430
Panchagarh DH	347	899
Rajbari DH	414	363
Rangamati DH	635	422
Shahid Shamsuddin DH, Sylhet	637	1460
Gopalganj DH	707	820
Chittagong General Hospital	920	2040
Khagrachari DH	975	43
Shariatpur DH	1182	748
Natore DH	1347	826
Kumilla DH	1408	1342
Manikganj DH	1458	2772
Bandarban DH	1465	79
Hobiganj DH	1508	657
Jhalokati DH	1657	413
Coxs Bazar DH	1829	1141
Noakhali DH	1953	2986
Bhola DH	1962	434
Dinajpur DH	2013	1302
Chandpur DH	2133	1098
Thakurgaon DH	2135	326
Pirojpur DH	2311	143
Kurigram DH	2323	1085
Munshiganj DH	2459	813
Nababganj DH	2635	184
Jamalpur DH	2913	3119
Gaibandha DH	3204	341
Narsingdi DH	3806	502
Narsingdi (100 bed) Zila Hospital	3855	641
Joypurhat DH	3860	1775
Satkhira DH	3918	1093

Hospital Name	Minor Surgery	Major Surgery
Mohammad Ali DH , Bogra	4166	2244
Chuadanga DH	4333	1579
Barguna DH	4554	38
Nilphamari DH	4865	732
Lalmonirhat DH	4941	394
Narayanganj General Hospital	5197	1526
Narayanganj 300 bed Hospital	5395	1543
Feni DH	5686	945
Jhenaidah DH	6110	1596
B.Baria DH	6113	2512
Lakshmipur DH	6301	1007
Sherpur DH	6337	1728
Pabna DH	6471	2969
Sirajganj General Hospital	6482	953
M.Bazar DH	6788	1750
Magura DH	6944	1342
Khulna DH	7165	830
Kustia DH	7477	3822
Madaripur DH	8377	1143
Faridpur General Hospital	8397	1381
Bagerhat DH	8784	208
Narail DH	9739	738
Patuakhali DH	10188	2075
Meherpur DH	11336	555
Sunamganj DH	13561	159
Naogaon DH	13990	1112
Tangail DH	14945	5890
Jessore DH	18240	4628
Kishoreganj DH	23676	2938
Netrokona DH	53999	62

ALTERNATIVE MEDICAL CARE

Table 5.17: Total Patient of Govt. Homoeopathic Medical College Hospital 2017

No	Division	Male Patient	Female Patient	Child Patient	Total Patient
01.	Outdoor	13308	18502	16604	48414
02.	Pathology	529	655	77	1262
03.	Emergency	793	1584	1792	4169
04.	Indoor admission patient	168	205	83	456
05.	Antenatal care	-	-	-	65

Table 5.18: Total Patient of Govt. Unani-Ayurvedic Medical College Hospital for Unani treatment, 2017

Unani					
No	Division	Male Patient	Female Patient	Child Patient	Total Patient
01.	Outdoor	5554	8874	1061	15489
02.	Indoor admission patient	4566	5270	60	9896

Table 5.19: Total Patient of Govt. Unani-Ayurvedic Medical College Hospital for Ayurvedic treatment, 2017

Ayurvedic					
No	Division	Male Patient	Female Patient	Child Patient	Total Patient
01.	Outdoor	5031	4034	596	9661
02.	Indoor admission patient	3508	4723	47	8278

Table 5.20: Total Patient of Govt. Homoeopathic Medical College Hospital up to November, 2018

No	Division	Male Patient	Female Patient	Child Patient	Total Patient
01.	Outdoor	15937	23673	12354	51964
02.	Pathology	-	-	-	-
03.	Emergency	1395	2006	961	4362
04.	Indoor admission patient	180	215	88	483
05.	Antenatal care	-	-	-	329

Table 5.21: Total Patient of Govt. Unani-Ayurvedic Medical College Hospital for Unani treatment up to November, 2018

Unani					
No	Division	Male Patient	Female Patient	Child Patient	Total Patient
01.	Outdoor	5056	8007	846	13909
02.	Indoor admission patient	3000	3420	92	6512

Table 5.22: Total Patient fof Govt. Unani-Ayurvedic Medical College Hospital for Ayurvedic treatment up to November, 2018

Ayurvedic					
No	Division	Male Patient	Female Patient	Child Patient	Total Patient
01.	Outdoor	6676	6321	776	13773
02.	Indoor admission patient	3289	3784	156	7229

Table 5.23: Top ten diseases of the admitted patients under Govt. Homoeopathic Medical College Hospital, 2017

SL	Name of Disease	Number of patient	Percent
01	Undiagnosed	95	20.83%
02	Joint pain/ Rheumatic pain	38	8.33%
03	Low back pain	27	5.92%
04	Fever	24	5.26%
05	Abdominal pain	17	3.73%
06	Asthma	17	3.73%
07	Paralysis	17	3.73%
08	Vertigo	12	2.63%
09	Hemorrhoids/piles	12	2.63%
10	Lower Abdominal Pain	10	2.19%
	Total admitted patient	456	

Table 5.24: Top ten diseases of the admitted patients under Govt. Homoeopathic Medical College Hospital up to November, 2018

SL	Name of Disease	Number of patient	Percent
01	General Weakness	100	39.2%
02	Arthritis	32	12.54%
03	Peptic ulcer Disease	26	10.19%
01	General Weakness	100	39.2%
02	Arthritis	32	12.54%
03	Peptic ulcer Disease	26	10.19%
04	Asthma	21	8.23%
05	Stroke & Paralysis	20	7.8%
06	Cancer	18	7.05%
07	Skin Disease	12	2.63%
08	Back pain	11	4.31%
09	Diabetes Mellitus	8	3.13%
10	Eczema	7	2.74%
Total		255	

Table 5.25: Top ten diseases of the admitted Unani patients under Govt. Unani-Ayurvedic Medical College Hospital 2017-2018

Unani	
SL	Name of Disease
01	Joint pain
02	Skin Diseases
03	Leucorrhoea
04	Dysentry
05	Common Cold
06	Diabetes
07	Piles
08	Scabies
09	Menorrhagia
10	Asthma

Table 5.26: Top ten diseases of the admitted Ayurvedic patients under Govt. Unani-Ayurvedic Medical College Hospital 2017-2018

Ayurvedic	
SL	Name of Disease
01	Neurological Disorders, Rheumatic Arthritis, Joint pain
02	Hyper acidity
03	Skin Diseases
04	Dyspepsia
05	IBS
06	Asthma
07	Anemia
08	Leucorrhoea
09	Piles
10	Retension of urine

HEALTH WORKFORCE SITUATION IN BANGLADESH

Motivation through promotion and training

INTRODUCTION

The health workforce situation, particularly in the DGHS, are summarized in this chapter. An overview of the number of sanctioned and filled-up posts is presented along with vacancies. Health workforce deployment and redeployment are ongoing processes; attrition, due to death, retirement, resignation, termination, migration, transfer, replacement, and filling-in are constantly occurring. Therefore, the status of health workforce, as shown in this report, may not remain the same by the time this bulletin is published.

An overall health workforce situation of DGHS is demonstrated in Table 6.1. A division-wise distribution is shown in the Annex to this chapter. Some key points are highlighted below:

- Out of 103743 sanctioned posts under the DGHS, about half of them are of Class III category. Physicians (Class I) comprises 25.04 %, and Class IV employees comprises 22.83% of the total sanctioned posts.
- For class-II staff, nursing posts are not provided here.
- The Class I (non-doctor) comprises 0.18% of the sanctioned posts and 0.21% of the available staff posts.
- 28758 sanctioned posts remained vacant in November 2018, which constituted 27.72% of the total sanctioned posts. Vacancy rate was 19.50% (5066 posts) for doctors, 28.67% (14693 posts) for Class III staff and 23.02% (8365posts) for Class IV staff.
- Rate of vacancy this year increased from the rate of vacancy in 2017. However, the number of sanctioned posts have also increased from 2017. This might explain the increase in the rate of vacant posts.

Out of
103,743
sanctioned posts under the
DGHS, **27.72%** are still
vacant. Vacancy rate for doctors
is now **19.50%**

Table 6.1. Number of sanctioned, filled up and vacant posts under DGHS (November 2018)

Category		Sanctioned post	Filled up Post	Vacant Post	Rate of Vacancy
Class I	Doctors	25980	20914	5066	19.50%
	Non-Doctor	526	227	299	56.84%
Class II*		953	618	335	35.15%
Class III		51246	36553	14693	28.67%
Class IV		25038	16673	8365	33.41%
Total		103743	74985	28758	27.72%

*excluding nursing posts

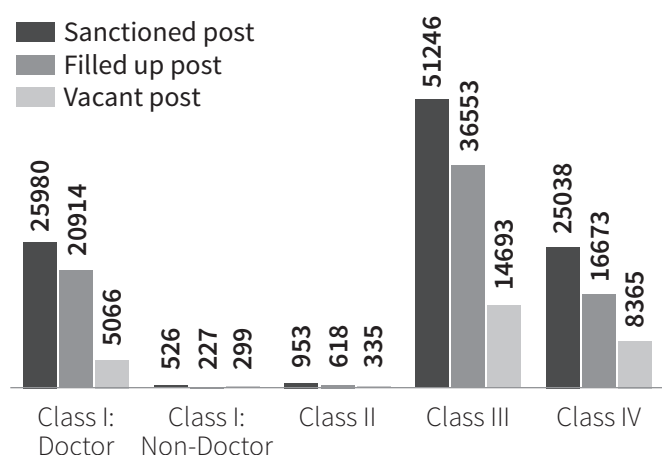


Figure 1: Number of sanctioned, filled up and vacant posts under DGHS (not including nursing posts)

ADMINISTRATIVE, MANAGERIAL, ACADEMIC AND CLINICAL POSITIONS

Table 6.2 shows the number of sanctioned, filled-up and vacant posts at administrative, managerial, academic and clinical positions under the DGHS in November 2018.

Table 6.2. Number of sanctioned, filled up and vacant posts at managerial, teaching and certain clinical positions (November 2018)

Name of Post	Sanctioned	Filled up	Vacant
Director General	1	1	0

Name of Post	Sanctioned	Filled up	Vacant
Addl. Director General/ equivalent	5	5	0
Director/ Principal/ Vice Principal/OSD/equivalent	125	230	-105
Deputy Director/OSD/ equivalent	133	698	-565
Assistant Director/Civil surgeon/OSD/equivalent	251	715	-464
Deputy civil surgeon/ UHFPO/OSD	976	1834	-858
Professor	640	319	321
Associate professor	1063	585	478
Assistant professor	1794	779	1015
Senior consultant	597	274	323
Senior Lecturer	8	6	2
Junior Lecturer	32	32	0
Junior consultant	3850	1766	2084
Assistant surgeon/OSD/ equivalent	16064	13240	2824
Other post (Pay Scale: 7+8)	441	430	11
Total	25980	20914	5066

Table 6.3. Number of sanctioned, filled up and vacant posts of medical technologists by discipline (November 2018)

Category	Sanctioned	Filled up	Vacant
Pharmacy	2905	1546	1359
Laboratory	2237	1488	749
Radiography	778	573	205
Radiotherapy	83	41	42
Physiotherapy	296	109	187
Dental	624	469	155
MT (S.I.)	498	475	23
MT (EPI)	499	483	16
Total	7920	5184	2736

MEDICAL TECHNOLOGISTS

Figure 6.2 shows the number of sanctioned, filled-up and vacant posts for medical technologists.

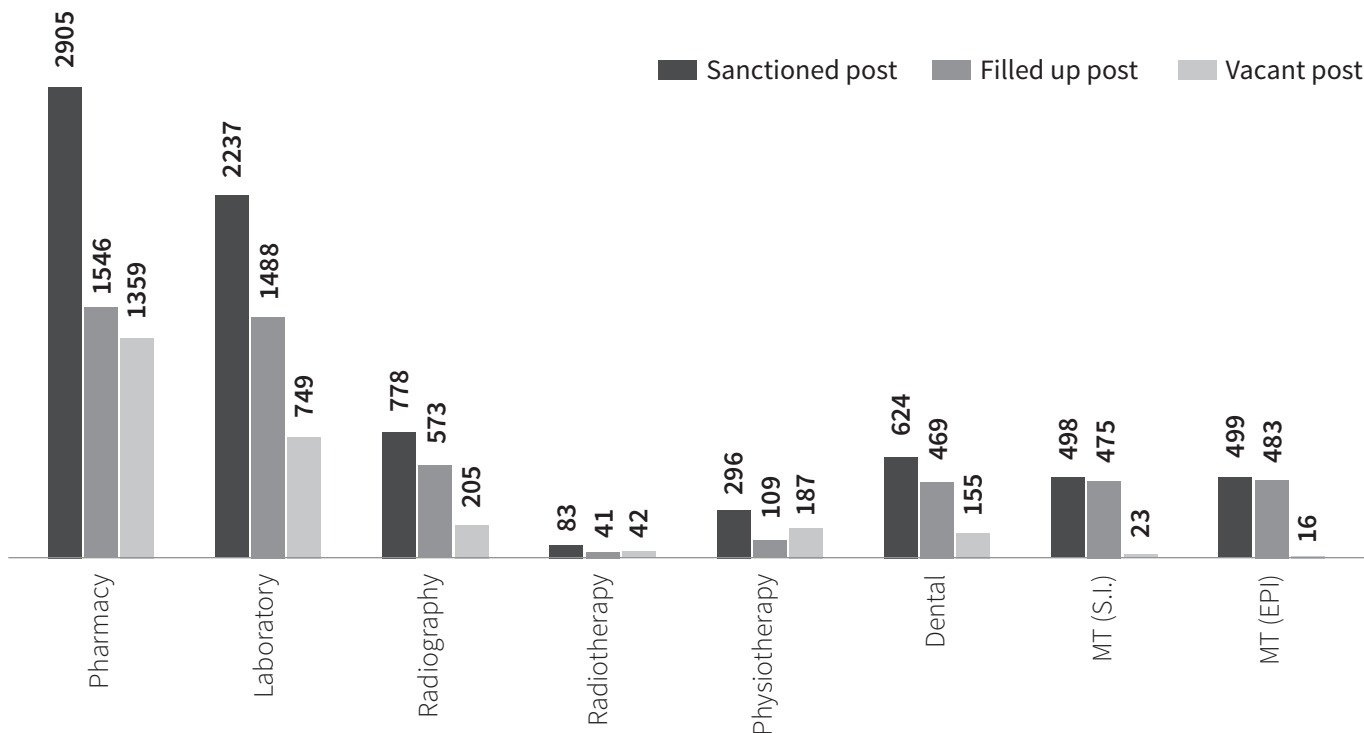


Figure 6.2. Number of sanctioned, filled up and vacant posts of medical technologists by discipline (November 2018)

SUB-ASSISTANT COMMUNITY MEDICAL OFFICERS AND DOMICILIARY STAFFS

Table 6.4. Number of sanctioned, filled-up and vacant posts of Sub-Assistant Community Medical Officers (SACMO) and domiciliary staffs under the DGHS.

Category	Sanctioned	Filled up	Vacant
Sub Asstt. Community Medical Officer (SACMO)	5368	3801	1567
Domiciliary staff			
Health Inspector	1410	1047	363
Assistant Health Inspector	4220	3636	584
Health Assistant	20908	15420	5488
Total	26538	20103	6435

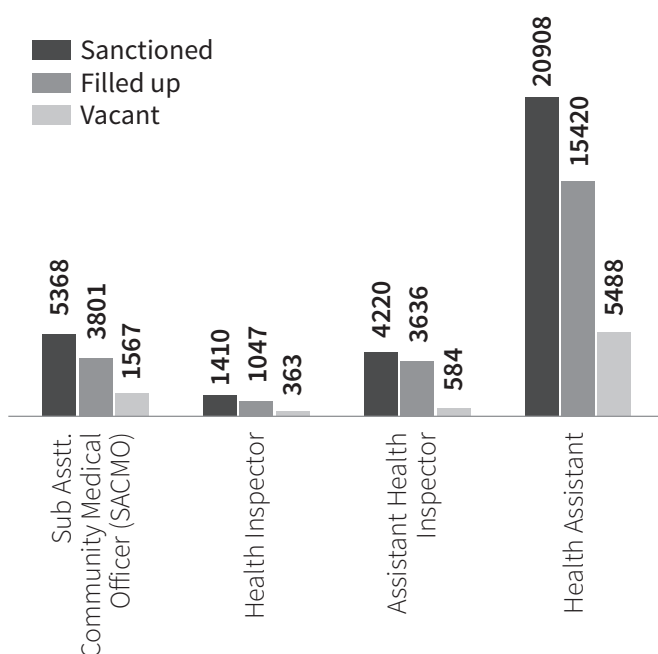


Figure 6.3. Number of sanctioned, filled up and vacant posts of SACMO, Health Inspectors, Assistant Health Inspectors and Health Assistants

Table 6.5 shows the number of sanctioned, filled-up and vacant posts of various officers and staff in alternative medicines under the DGHS as of December 2018.

NEW RECRUITMENT

Bangladesh government is regularly recruiting health manpower through various recruitment procedure. In last 10 years, Bangladesh government have recruited over 15 thousand doctors. **In 2017 and 2018, 624 newly recruited doctors (grade 9) joined our health workforce.** These recruitment was done under BCS 35 and BCS 36. No adhoc recruitment was done in the period 2017-18.

Table 6.5. Number of sanctioned, filled-up and vacant posts of various officers and staff in alternative medicines under the DGHS as of December 2018 (HRIS,DGHS)

Name of Post	Sanctioned		Filled up		Vacant	
	Rev.	Dev.	Rev.	Dev.	Rev.	Dev.
Director	1	0	1	0	0	0
Line Director	0	1	0	1	0	0
Principal-cum-Superintendent	3	0	3	0	0	0
Professor	2	0	0	0	2	0
Deputy Director	1	0	1	0	0	0
Program Manager	0	1	0	1	0	0
Assistant Director	1	2	1	0	0	2
Assistant Professor	16	7	13	2	3	5
Deputy Program Manager (U/A/H)	0	3	0	3	0	0
RP(U/A/H)	0	3	0	2	0	1
Register (U/A/H)	0	3	0	2	0	1
Asst. Register (U/A/H)	0	3	0	0	0	3

Name of Post	Sanctioned		Filled up		Vacant	
	Rev.	Dev.	Rev.	Dev.	Rev.	Dev.
RMO	0	1	0	1	0	0
Lecturer	21	46	15	31	6	15
Medical Officer (U/A/H), DGHS	0	3	0	3	0	0
Medical Officer (Unani Medicine)	76	99	25	53	51	46
Medical Officer (Ayurvedic Medicine)	75	96	24	49	51	47
Medical Officer (Homeopathic Medicine)	65	96	14	60	51	36
IMO (U/A/H),	6	18	0	15	6	3
Drug Superintendent	0	3	0	0	0	3
Other posts equivalent of medical officer	14	2	14	2	0	0
Research Officer (U/A/H),	0	4	0	3	0	1
Deputy Superintendent	1	0	1	0	0	0
Nurse/Staff nurse	12	0	12	0	0	0
Secretary	1	0	1	0	0	0
Pharmacist	0	3	0	0	0	3
Inspector	0	9	0	0	0	9
Accountant Officer	0	1	0	0	0	1
Accountant	0	1	0	0	0	1
Production Assistant	0	4	0	0	0	4
Support Personnel (Compounder)	220	214	65	0	155	214
Herbal Assistant (Gardener)	0	477	0	430	0	47
Diver	2	6	2	0	0	6
Others Class III personnel	100	2	82	0	18	2
Others Class IV personnel	184	2	124	2	60	2
Total	801	1110	398	660	403	452

PROMOTION AND RECOGNITION OF SERVICE

The promotion procedure of the health system in the past was complex and slow. The promotion process was managed by Public Service Commission. Since 2012, the promotion process for most posts (excluding professors and directors) is now being finalized by Departmental Promotion Committee (DPC) in the MoHFW. Thus the promotion rate has increased and delay in getting promotion has been decreased. This has motivated the health workforce in positive way. Number of promotions is provided in the table 6.6.

MoHFW is now recognising high performing health facilities (Upazila health complex, civil surgeon office, district hospital, medical colleges and specialized hospitals). MoHFW selects promising health personnel for different inservice training both in country and abroad. This is another way of recognising good service.

IN SERVICE TRAINING

Number of person receiving inservice trainings is presented in table 6.7.

BIOMETRIC ATTENDANCE

DGHS introduced biometric attendance system in different health centers. Despite improvements, the biometric attendance system is still facing some challenges, including electricity blackouts and technical issues. Furthermore, the number of doctors/staffs who are on leave or in an assigned task or in a training are not used in the calculation of attendance rate. Rate of attendance is thus under-representation

Table 6.6. Number of promotions in various health posts

Year	Promotion in July 2017- June 2018
Professor	127
Associate Professor	543
Assistant Professor	0
Senior Consultant	0
Junior Consultant	736
Director	32
Deputy Director	102
Assistant Director	268
Total	1808

of the true attendance rate. Information regarding biometric attendance is provided below in figures and charts.

Number of active machine has increased significantly in 2017-18. Also, the number of organization sending data have risen. Biometric attendance is now being used by the health managers in monitoring.

A dashboard has been created to monitor the attendance system in DGHS website. The screenshot of the dashboard is provided below.

Table 6.7. In service training number in last 10 years and in 2017-18

Training	Number of Batches (2009-2018)	Total Participants (2009-2018)	Training in 2017 -2018	
			Batch	Participants
Local Training	8003	2,00,295	1299	32405
Foreign Training	155	1056	31	325

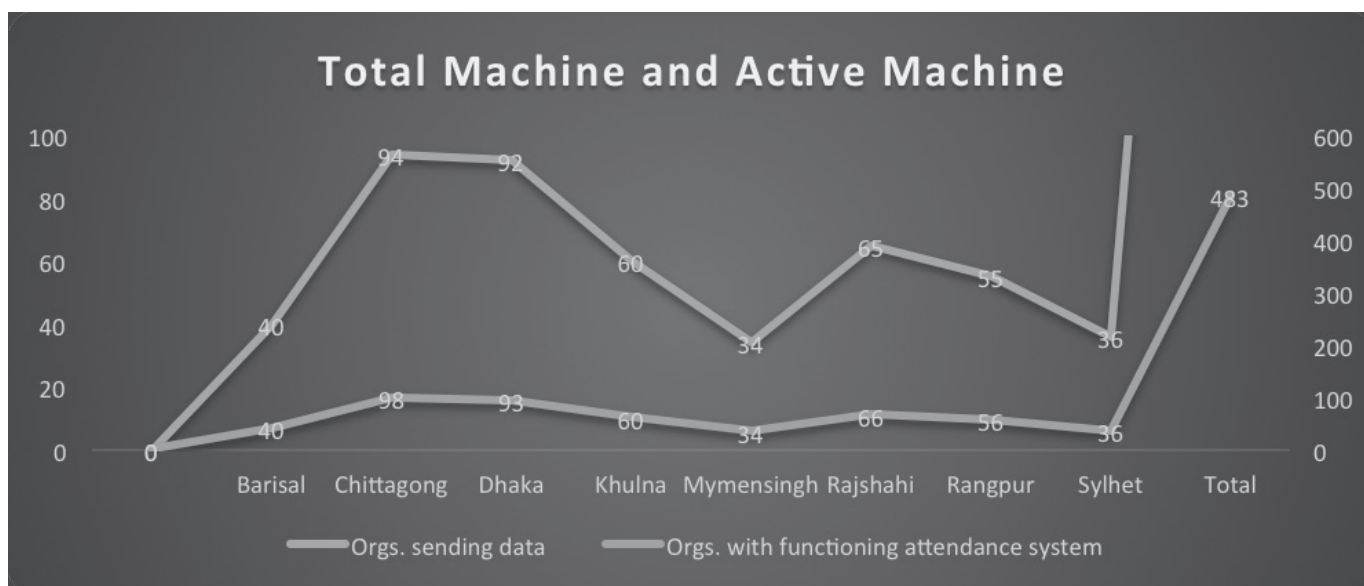


Figure 6.4. Number of active machine in the biometric attendance system



Figure 6.5. Biometric attendance system website homepage



Figure 6.6. Health workforce management dashboard at DGHS website with biometric attendance system

MEDICAL EDUCATION

PROGRESS IN MEDICAL EDUCATION

INTRODUCTION

Medical education system in Bangladesh has seen great development in the last decade. Numbers of doctors, nurses, medical assistants and technologists have increased multiple folds. To meet the demand of the large population of Bangladesh and achieve the WHO standard of doctor-patient ratio, many education institutes have been established. Current situation of medical education system is described in the following segments.

RESPECTIVE/IMPLEMENTING ORGANIZATIONS

- Medical Education and Health Manpower Development (ME&HMD) Unit of DGHS
- Nursing Education Unit of DGNM
- Bangladesh Medical and Dental Council
- Bangladesh Nursing and Midwifery Council

MEDICAL EDUCATION AND HEALTH MANPOWER DEVELOPMENT (ME&HMD) UNIT OF DGHS

Overview

- Medical Education and Health Manpower Development (ME&HMD) Unit is an indispensable part of the DGHS since the beginning of its journey;
- To achieve the Vision-2021 and 2041 including Sustainable Development Goals (SDGs), ME&HMD, DGHS is engaged with the delivery of quality education for the production of a competent and motivated health workforce;
- It also looks after in-service training of the existing health workforce under DGHS.
- ME&HMD leads the implementation of the medical education operational plan under the 4th health sector program (HPNSP 2017-2022) for institutional as well as individual capacity development of health workforces.



New medical colleges are being established with state of the art design (design of President Abdul Hamid Medical College and Hospital, Kishorganj)

Major Activities

- Streamlining post-graduate medical education.
- Ensuring qualitative graduate medical education.
- Ensuring qualitative para-professionals education.
- Ensuring need based proper health manpower development.
- Conducting Admission test for MBBS, BDS, IHT, MATS, AMC courses
- Procurement of furniture, machineries & equipment for all government medical & dental colleges.
- Strengthening libraries with journals and books and establishing a unique e-library.

MEDICAL EDUCATION OFFERED IN BANGLADESH

Bachelor or lower level courses

- Bachelor of Medicine & Surgery (MBBS)
- Bachelor of Dental Surgery (BDS)
- BSC in Nursing and Diploma in Nursing
- BSC in Health Technology/Physiotherapy and Diploma in Health Technology/Physiotherapy
- Medical Assistant Training Course

Post-graduate courses

- MD or MS courses in various disciplines including clinical, basic and public health subjects
- Fellowship course (FCPS, MCPS)



36 public &
69 private



1 Armed Forces Medical colleges &
5 Army Medical colleges.



39 post graduate medical teaching institutes including **04** medical universities



9 Public &
200 Private MATS

11 Public & **97** Private IHT

Figure 7.1. Current status on enrolment capacity of different tier of medical education

MBBS DEGREE STATUS

- 36 government medical colleges with 4068 seats for MBBS
- 1 Armed Forces Medical college & 5 Army Medical colleges with 375 seats for MBBS
- 69 private medical colleges with 6231 seats for MBBS
- Total number of seats for MBBS increased by 3% from last year and by 192% in the last 10 years

Table 7.1. Medical colleges and MBBS enrolment capacity in 2017

Type	Number of Colleges	Enrolment capacity
Government medical colleges	36	4068
Armed forces and Army medical colleges	6	375
Private medical college	69	6231
Total	111	19674

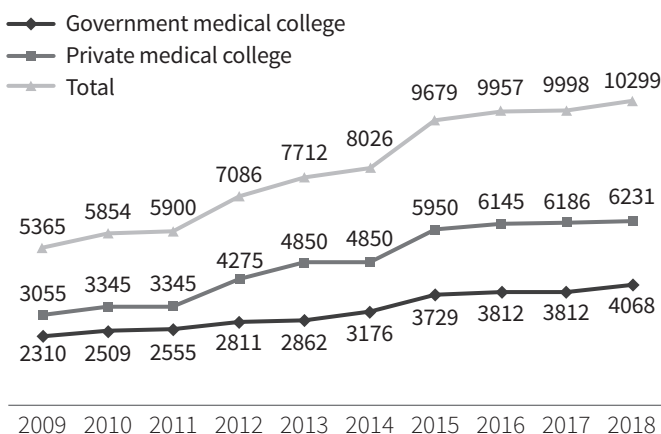


Figure 7.2. Total MBBS seat capacity by year (excluding Armed forces and Army medical colleges)

BDS DEGREE STATUS

- 9 government medical colleges with 532 seats for BDS
- 26 private Medical Colleges with 1405 seats for BDS
- Total number of seats for BDS increased by 198% in last 8 years, however only 5 seats for BDS was added in last 1 year.

Table 7.2. Dental colleges and BDS enrolment capacity

	Number of Colleges	Enrolment capacity
Government dental colleges	9	532
Private dental colleges	26	1405
Total	35	1937

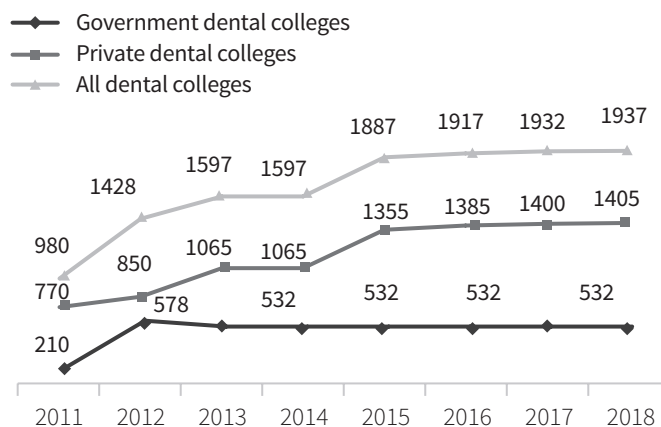


Figure 7.3. BDS enrolment capacity by year

FOREIGN STUDENT ENROLMENT IN MBBS AND BDS COURSES

- There are 1896 foreign students in MBBS and BDS course in different institutes of Bangladesh
- Total number of foreign student in MBBS and BDS courses in 2018 across government and private institutes increased by 235% in the last 10 year
- Foreign students number in private medical colleges saw the biggest gain

Table 7.3. Foreign student enrollment during period 2009-2018

Name of the Institute	Number of Colleges
Government Medical College	81
Private Medical College	1702
Government Dental College and Unit	15
Private Dental College & Unit	98
Total	1896

Data source: Administrative data (ME&HMD, DGHS)

Table 7.4. Number of nursing institutions, along with numbers of seats, offering different types of nursing degrees (June 2017)

Course	Ownership	No.	No. of seats
Basic BSc	Government	15	1100
	Private	45	2185
	Sub-total	60	3320
Post-BSc/Public Health Nursing	Government	4	375
	Private	37	1475
	Sub-total	41	1875
Specialized Diploma	Government	1	20
	Private	3	60
	Sub-total	4	80
Diploma in Nursing & Midwifery	Government	43	2,580
	Private	140	50
	Sub-Total	183	2,630
	Total	288	7905

NURSING DEGREES STATUS

The number of institutions, along with the number of seats in both government and private sectors, offering different types of nursing or midwifery degrees are shown in Table 7.4 and 7.5.

Table 7.5. Number of institutions producing midwives, along with numbers of seats, offering different types of midwifery degrees (June 2017)

Course	Number of Institute	No. of seats
diploma in midwifery	54	1545
Junior Midwives	13	360
Total	67	1905

Detailed list of institutes providing diploma in midwifery course is provided in the Annex

COURSES FOR FAMILY WELFARE VISITORS (FWV), COMMUNITY PARAMEDICS AND CSBA

Table 7.6. Number of institutes with available seats providing courses to FWV, community paramedics, and CSBA

S.I	Course	Duration	Government		Private		Total	
			No. of institutes	No. of seats	No. of institutes	No. of seats	No. of institutes	No. of seats
1	Family Welfare Visitor (FWV)	18 months	12	1000	0	0	12	1000
2	Community Paramedic	2 years	0	0	27	1270	27	1270
3	Community Based Skill Birth Attendants (CSBA)	6 months	48	960	4	80	52	1040

Table 7.7. Number of institutes with available seats providing courses to health technologists

	Number of Colleges	Enrolment capacity
Government Institute of Health Technology	11	2685
Private Institute of Health Technology	97	8940

HEALTH TECHNOLOGY DEGREE STATUS

- There are 11 government institute of technology with an enrolment capacity of 2685 seats
- There are 97 private Institute of Technology with an enrolment capacity of 8940 seats
- These institutes offer both BSC and diploma courses in health technology and physiotherapy

MEDICAL ASSISTANT TRAINING STATUS

- There are 9 government Medical Assistant Training Schools (MATs) with an enrolment capacity of 818 seats
- There are 200 private MATs with an enrolment capacity of 13540 seats

Table 7.8. Number of institutes with available seats providing courses to medical assistants

	Number of Colleges	Enrolment capacity
Government Medical Assistant Training School	9	818
Private Medical Assistant Training School	200	13540

POST GRADUATE EDUCATION

- There are 39 institutions providing post graduate degree courses (Masters and Diploma) for MBBS and BDS graduates in various disciplines
- 609 seats increased in year 2017 from year 2016, which indicates an increase of 27%
- In addition BCPS provides postgraduate fellowship courses in various disciplines

Table 7.9. Number of post graduate medical institutes in Bangladesh

Post Graduate Institutes	
Government	22
Autonomous	7
Non -Government	10
Total	39

Table 7.10. Number of seats in different Post Graduate Medical Institutes in Bangladesh

Total Number of Seats	Provide Degree
Government	1048
Non -Government	1112
Foreign	112
Total	2272

Data Source: CME Survey 2017

FCPS, MCPS DEGREES FROM BCPS

- In January 2017, 204 doctor earned their FCPS degree and 41 doctors earned their MCPS degree
- In July 2017, 199 doctors earned their FCPS degree and 42 doctors earned their MCPS degree

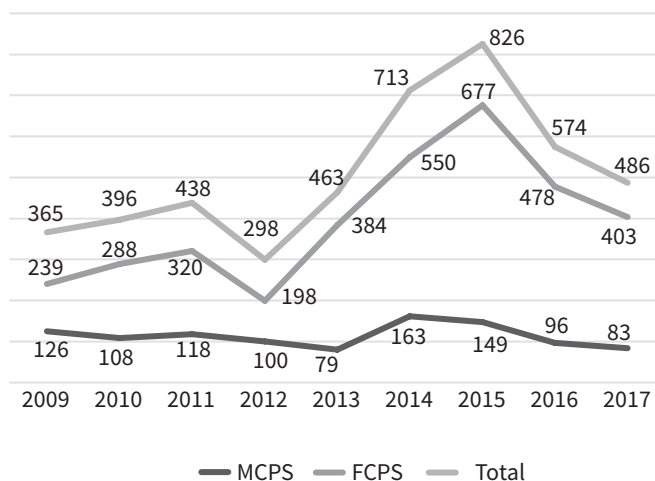


Figure 7.4. Trend of completion of FCPS and MCPS fellowship from BCPS over the last decade

PROGRESS IN MEDICAL EDUCATION AND INNOVATIONS

- Number of medical education courses, enrolment capacity both increased multiple folds in last decade
- National level admission test for MBBS, BDS, IHT, MATs and alternative medical care courses are being conducted digitally from online application submission up to result declaration. Global positioning system (GPS) is used during transportation of question papers and dispense of examination materials. After evaluation examination results are informed to the examinee through SMS by teletalk telecom network.

Table 7.11. Institution-based progress in medical education scenario in the last 10 years

Institute	2009	2018	Increased in Number	Increased in Per-centage (%)
Medical University	1	4	3	300
Government Medical College	17	36	19	112
Private Medical College	40	69	29	73
Army And Armed Forces Medical College	1	6	5	500
Government Dental College and Unit	3	9	6	200
Private Dental College and Unit	11	26	15	136
Government Institute of Health Technology (IHT)	3	11	8	267
Private Institute of Health Technology (IHT)	38	97	59	155
Government Medical Assistant Training School (MATS)	7	9	2	29
Private Medical Assistant Training School (MATS)	23	200	177	770
Total Institute	144	467	323	224

Data source: Administrative data (ME&HMD, DGHS)

CENTRE FOR MEDICAL EDUCATION (CME)

Achievements of in 2017-18 in the sector of Medical Education (A regional centre of excellence of WHO)

Curriculum development

- Updating the BDS curriculum as “BDS curriculum -2017”

Hand book development

- “Handbook on Tuberculosis for Medical Students and Physicians” Targeting SDG was developed

Activities on quality assurance of medical colleges of Bangladesh

- Updating the “Guidelines and tools for Quality Assurance Scheme for undergraduate medical education in Bangladesh”
- Acting as secretariat on the Quality Assurance Scheme
- Field visit: Total 15 medical colleges were visited
- Consultative Meeting on “Assessment of the Implementation of National Quality Assurance Scheme at Different Medical Colleges in Bangladesh” 18th October 2017, IPH, Conference Room

Accreditation of Medical institutes of Bangladesh

- Draft proposal developed on “Regulation of Medical Accreditation of Bangladesh-2017” (Submitted to MOH&FW)

Research

- Post Graduate Medical Education in Bangladesh: A Situation Analysis: It has found out total number of different categories postgraduate doctors of

Bangladesh. (Submitted to MOH&FW)

Training

- **“Training on Teaching Methodology and Assessment”** - In 2017 total 183 teachers and in 2018 total 84 teachers were trained from different government and nongovernment medical and dental colleges and institutes

Conduction of courses

- **MMEd** (Masters in Medical Education) – Two years duration
 - In 2017 total 12 students were admitted and 10 have passed (2 are enrolling thesis part)
 - In 2018 total 7 students were admitted and they are enrolling in part 1
 - Till today total 92 have passed the MMED course
- **DHPed** (Diploma in Health Professional Education)- One year duration: The course has been started

Conference

- A conference on “National Conference on Health Professional Education -2018” was held, duration: 2 days

e-learning studio

- e-learning studio for web based education-2018 was developed

Infrastructure development

- Vertical extension of the CME building – 4th and 5th floors completed

HEALTH INFORMATION SYSTEM AND e-HEALTH

Strengthening, supervision and monitoring of healthcare through ICT

BACKGROUND

Management Information System (MIS) is a department of Directorate General of Health Services (DGHS) under the Ministry of Health and Family Welfare (MoHFW). The main objective of MIS is to establish Health Information System (HIS) and e-health in Bangladesh. As a developing country in the South-East Asia, Bangladesh has harnessed impressive rollout of e-health in its public health sector. As a first nation in the world to announce “Digital Bangladesh 2021” vision in 2009, Bangladesh is arguably among one of the few countries of the world, where the public health sector has made remarkable and astounding progress in all areas of e-Health and Health Information System (HIS). The entire health sector, national, sub-national and grassroots community health workforce, is digitally connected to the robust national databases. The country follows a model of shared e-Health resource, hosted by the Ministry of Health; these resources are used by development partners, NGOs, private sectors, and other government and non-government organizations. MIS always follow e-health data standardization and interoperability. In 2011, the United Nations honored Bangladesh Prime Minister Sheikh Hasina, with a “Digital Health for Digital Development” award. In 2014, Bangladesh’s e-Health progress and model was recognized as the World’s best practice by the German government; it was exemplified through the publication of a book titled A Quiet Revolution: Strengthening the Routine Health Information System in Bangladesh. Improvement of e-Health in Bangladesh is now consistent and rapid, and the country is now focusing more on innovations and quality control. An inventive program is now in progress under the HIS and e-health Operational Plan, under the 4th sector-wide program (HPNSP 2017-2022).

In 2011, the United Nations honored Bangladesh Prime Minister Sheikh Hasina, with a
“Digital Health for Digital Development”
award.

OBJECTIVES AND STRATEGIES OF MIS

GENERAL OBJECTIVE:

To improve health information system, e-health and medical biotechnology.

SPECIFIC OBJECTIVES

i. To improve the existing health information system and e-health through:

- Strengthening population based and institution-based human resources and program based HIS.
- Strengthening and improving of electronic health records for all citizens.
- Developing and strengthening logistic tracking and inventory management and procurement system.
- Expanding GIS in health services.
- Improving the infrastructure and human resource capacity necessary for HIS and e-health.
- Developing more health call center, mobile phone health service and other m-Health and maintaining their progress.
- Strengthening and expanding video conferencing and telemedicine service.
- Introducing more e-health services and programs.
- Sustaining the e-health initiatives and encouraging public-private partnership.

ii. To improve Medical Biotechnology through:

- Achievement of short and medium term deliverables, mentioned in the National Guidelines on Medical Biotechnology.
- Creation of conditions for achieving the long term deliverables of the National Guidelines on Medical Biotechnology.

STRATEGIES

- To fully digitize the entire routine health information system in Bangladesh, by 2021, to build a strong and powerful evidencebase for decision making.
- To strengthen the monitoring and evaluation (M&E) processes for the sector program, through effective utilization of evidences that are available from all possible sources (routine data, surveillance, CRVS, survey, research etc.)
- To establish partnerships with relevant institutions for facilitating data demand; also, to update and utilize them in MoHFW and its agencies and departments.
- To develop and implement multi-year, comprehensive HIS, e-health and M&E training plans for capacity building of managers and staffs.
- To increase budgetary allocation for HIS, e-health and M&E activities to sustain and further build on the progress.

COMPONENT OF MIS



Health Information System (HIS)



e-Health



Medical Biotechnology Program

MAJOR ONGOING ACTIVITIES OF HIS AND e-HEALTH PROGRAM UNDER MIS, DGHS

The following innovative activities were conducted on a regular or ad-hoc basis in 2017:

a. Online local health bulletin

MIS-DGHS has been yearly publishing facility wise online local health bulletin since fiscal year 2011-12. In 2016, automation of LHB was completed by the involved re-engineering and automation of the online Local Health Bulletins (Figure 8.1). From community clinic to specialized hospital, almost all the health facilities under DGHS produced it. The bulletin contains indicator wise aggregated information such as, the number of patient-visits in outpatient and emergency departments, number of hospital admissions, bed-occupancy and turnover ratios, average length of stay, EmONC, etc. as well as other major public-health indicators. The bulletins are available for downloading and can also be printed and distributed locally.

Please Visit [>>Dashboard >> Local Health Bulletin](http://www.dghs.gov.bd) for further inquiries.

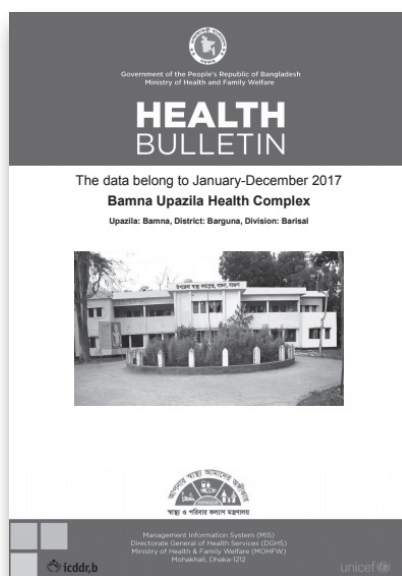


Figure 8.1. Online local health bulletins in dashboard

b. DHIS2

Ministry of Health and Family Welfare (MoHFW) of Bangladesh has been using District Health Information System 2 (DHIS2) since 2010. Data from community clinics to tertiary level health facilities and programs, like MNCAH, IMCI, EPI, TB, NCDs, communicable diseases, HIV/STD,

nutrition, cervical cancer and breast cancer screening, obstetric fistula screening and care program, are channeled to the national HMIS through this software. Figure 8.2 shows the dashboard of DHIS 2.29 version.

Using DHIS2, the MoHFW developed an electronic central repository for national health data called the National Data Warehouse. It aims to bridge the gap between the fragmented systems while bringing together information from various databases. In addition, the Data Warehouse provides rich data-mining functions to generate reliable and accurate data. These are such data that decision-makers need for planning and monitoring health interventions across all levels of the health system. However, this HMIS innovations are empowering Bangladesh to collect data from a wide range of sources to generate real-time information on health. It also knows that improving national health data is the first step in tracking progress towards the Sustainable Development Goals (SDG). The country is progressively

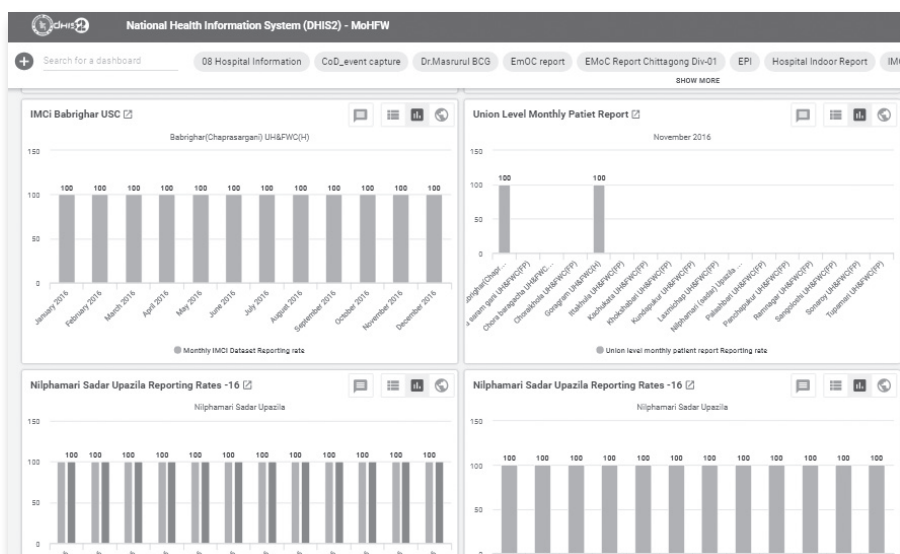


Figure 8.2. DHIS 2.29 screenshot

moving towards its aims.

c. Dashboard

An innovative system called Health Dashboard has been developed by the Management Information System (MIS) of the DGHS to connect modern digital technology to health services. We are using renowned Business Intelligence (BI) tools named Tableau to visualize this dashboard. Through this dashboard, information on routine health service data and survey data can be viewed. Also, almost all activities' data can be shown through this dashboard. Along with this various important health indicators, human resources, and activities in biometric mode, etc. have been displayed for public view. The features are displayed in left side, and they are very rich in information. This dashboard is placed inside the DGHS website. Figure 8.3. shows the screenshot of the DGHS dashboard.

d. Human Resource Information System (HRIS)

Central Human Resource Information system (HRIS) has been designed in such a way so that it can electronically capture and maintain

human resource information for the entire health sector. It is based on 4 registries; Geo-location Registry, Facility Registry, Sanctioned Post Registry, and Provider Registry. All human resource-related activities, including transfer, promotion, deputation, leave, ACR, and others, are automated through this system. The DGHS is the main stakeholder of this system. Initially, it was named as Human Resource Management System (HRMS) and was developed by MIS-DGHS. Currently, the MIS-DGHS plays a vital role in updating the HRIS system. This system has recently received three awards in different award functions. Figure 8.4 shows the screenshot of HRIS.

e. Health facility registry and GIS

Almost all the public health facilities

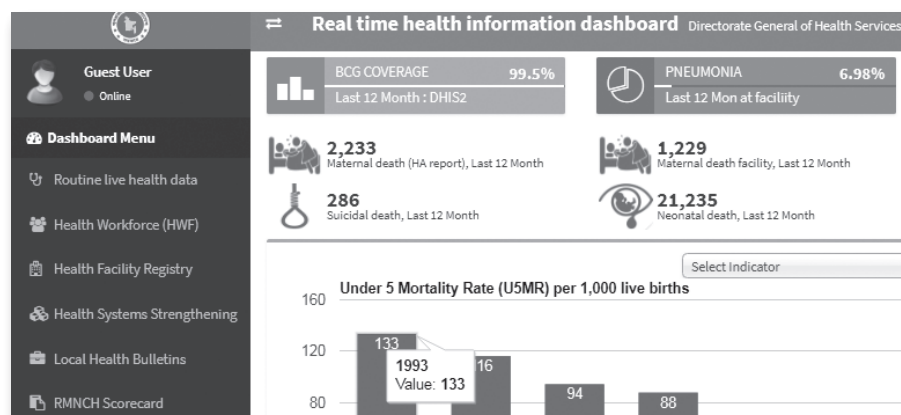


Figure 8.3. Screenshot of DGHS dashboard



Figure 8.4. Screenshot of the Human Resource Information System (HRIS)

under DGHS are registered in the health facility registry and it has a unique code number for each facility. It is very important for interoperability and e-health strategy. Geo-location Registry is also integrated within this facility registry. All the necessary information about health facilities under MoHFW such as Facility head name, contact no., address, email id etc. can be found here. Some of the private health facilities are also registered in this system. A total of 23,603 health facilities have been registered here till now. Health facility registry can be access through this link <http://facilityregistry.dghs.gov.bd/>. Figure 8.5. shows the screenshot of this health facility registry.

f. Health System Strengthening (HSS)

The Management Information System (MIS) unit of the Directorate General of Health Services has launched a revamped performance management platform. WHO provided technical assistance in the conceptualization and development of this revamping process, as part of its efforts to support the strengthening of health systems in Bangladesh. Other partners, such as icddr, b and UNICEF, have also been involved since the inception of the initiative. The Health Systems Strengthening (HSS) Project, as it is more commonly known, aims to promote best practices, transparency and accountability in health services management by incentivizing better use of the existing resources and generation of better results. Best-performing ones in each tier receive the Health Minister's National Award.

To guide the performance measurement approach, the building blocks of the WHO health system

Organization Name	Organization Code	Division	District	Upazila	Paurasava	Union	Ward	Agency	Org Type	Photo
Bandarban District Hospital	10000655	Chittagong	Bandarban	Bandarban Sadar	Bandarban Paurashava	Urban Ward No-03		DGHS	District Hospital	

Figure 8.5. Screenshot of the Health facility registry

and the result chain analytical frameworks were adopted in line with the local context. Performance indicators were defined at input, output and outcome levels, and to some extent, at the impact level for facilities and community health services. In total, four different tools are being used to assess their performance. The first is the online measurement tool where facilities report on selected indicators through the existing systems used in MIS. The second tool is the onsite monitoring which is based on a form completed by line managers responsible for oversight of the different facilities. Figure 8.6. shows the screenshot of a live dashboard; it was created (by using a score board) for performance evaluation. Also, it will automatically measure the performance of health facilities by pulling the routine data inputted into the District Health Information System (DHIS2), Human Resource Management (HRM), biometric attendance system, SMS complaint/suggestion system, etc. as well as the results of the onsite monitoring system. The third tool is

used for physical verification by an independent team of assessors of the short-listed facilities, following the results of the score board. Finally, the fourth tool involves a patient satisfaction survey to measure the extent to which the demand side is receiving the care they expect. A comprehensive physical assessment, along with patient satisfaction survey tools, is being used for the assessment, in short-listed health facilities. Also, it is used to nominate the best-performing health facilities, awarded by the Honorable Health Minister every year.

Through the initiative, a model of systematic mechanism was developed for measuring performance and evaluation of the management of health services. The endeavor of independent assessment has created an opportunity for the assessor team to understand and observe the various local-level positive initiatives by the healthcare providers. It also identifies the gaps for better health outcome in the public health sector. For more information, please visit:

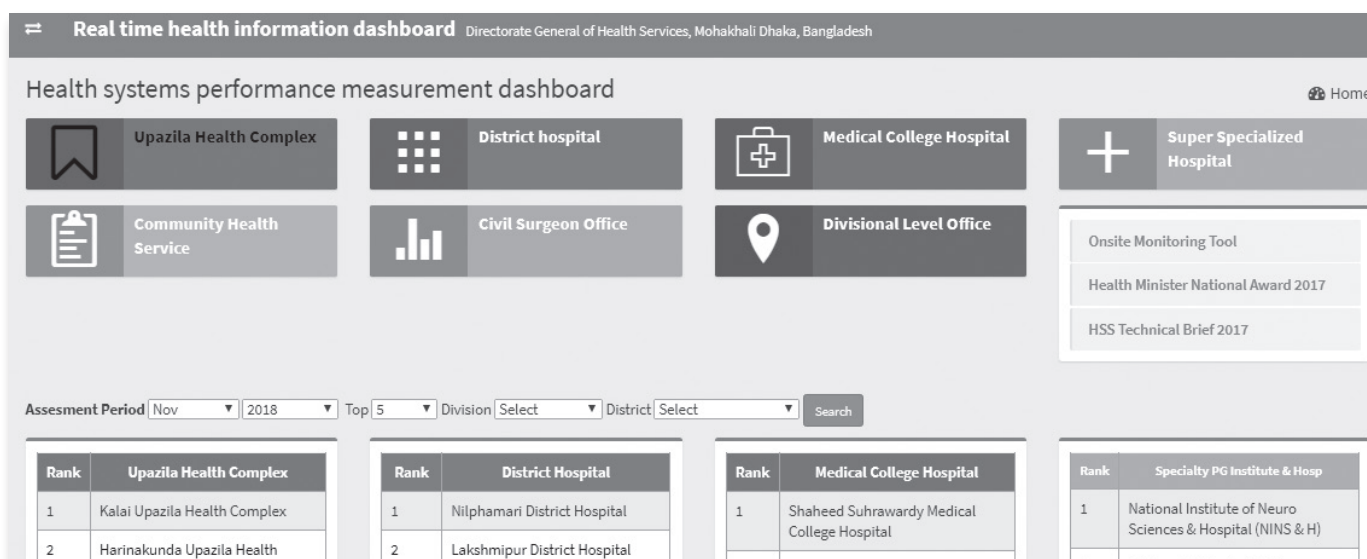


Figure 8.6. Screenshot of Health System Strengthening (HSS) scoreboard

http://103.247.238.81/webportal/pages/hss_menu.php.

g. Shared Health Record and OpenMRS

Shared Health Record (SHR) is a system that stores the lifetime medical history of citizens in electronic format and Open Health Information Exchange is a gateway to different systems and data. The electronic medical records (EMR), stored in SHR, are shared through OpenHIE, across different medical or health facility management systems (like OpenMRS+ combination of OpenMRS, OpenERP and OpenELIS, OpenSRP, etc.), with proper access credentials and privileges. Here, sensitive information is protected with special security. In this case, patients do not need to carry their medical files when they visit any medical center or a physician.

Currently, SHR has been implemented in Kaliganj Upazila Health Complex, Gazipur, and Gazipur General Hospital. Apart from SHR, the hospital automation

system in Dhaka Medical College Hospital (DMCH) and the National Institute of Neurosciences & Hospital (NINS) has been almost done in 2016-2017. LAN and infrastructure have been set up, and they will fully be in operation by the end of this year. These are considered as new additions to hospitals that already have started the automation system. These hospitals include, National Institute of Kidney Diseases & Urology (NIKDU), Government Employees' Hospital, Azimpur

Maternity Hospital, Bangladesh Secretariat Clinic, National Institute of Traumatology, Orthopedics and Rehabilitation (NITOR), and National Institute of Cardiovascular Diseases (NICVD). However, the real expansion of automation for a reasonable number of hospitals is illustrated through the scaling-up of the Shared Health Record. Figure 8.7. shows the architecture of SHR and OpenHIE.

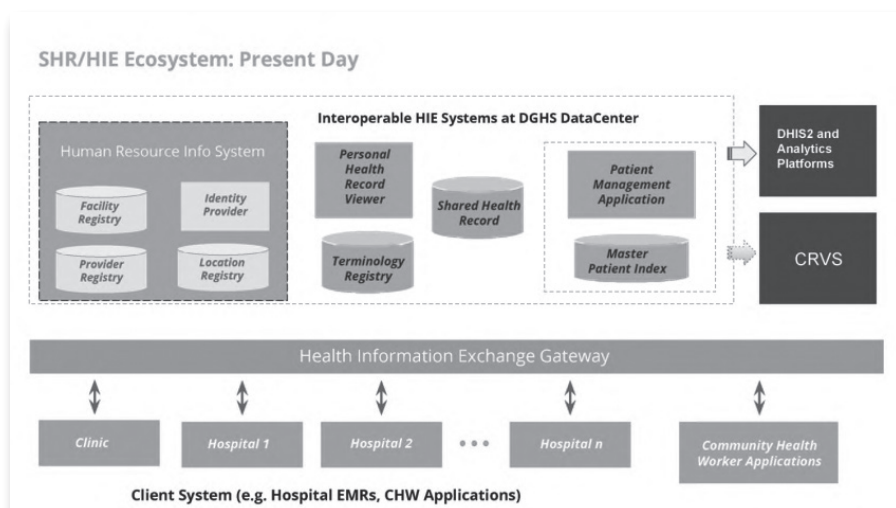


Figure 8.7. Architecture of SHR and OpenHIE



Figure 8.8. Video-conferencing with DHO, CSO from DG's office

h. Video- conferencing

Video-conferencing has been started to ease the administrative work. It can directly convey messages from the Ministry and DG office to all divisional and civil surgeon's offices, as well as Medical Colleges and Medical College Hospitals on virtual settings. The system is providing an opportunity for arranging remote meetings, conferences, trainings, and emergency orders (during natural disasters, like flood, landslide, Rohingya influx etc.). Thus, it is minimizing communication gaps among the Ministry, DG office, and peripheral health offices. This technology has made significant improvement in administration. The system has been installed initially in 77 strategic locations : MoHFW, DGHS, DGFP, offices of 7 division directors, and 64 civil surgeons; 64 UHCs having telemedicine centers are using video-conferencing since 2016-17. In the fiscal year 2017-18, additional 38 video-conferencing have been installed in different strategic locations. Now, there are altogether 115 video conferencing centers. The MIS-DGHS has taken the initiative to extend this video-conferencing

system in all tertiary-level hospitals/ institutes, medical colleges, medical college hospitals as well as in all district hospitals. Furthermore, this service will gradually cover all the public health facilities/hospitals in upcoming years. Every week, video conference meetings are held at the DG office. Through these meetings, the DG office communicates with DHO, CSO, MCH, DH, UHC offices and different specialized hospitals. Figure 8.8. shows such video-conferencing with DHO, CSO from DG's office.

i. Telemedicine service

It has started from 8 telemedicine centers in the fiscal year 2009-2010. The Honorable Prime Minister Sheikh

Hasina formally inaugurated the Telemedicine Services on 6 July 2011 from the National Digital Innovation Fair held in Bangabandhu Novo Theater. Following that event, about 10 additional centers have been added each year. Until now, there are 94 telemedicine centers, including the Telemedicine Monitoring Unit at the MIS-DGHS.

There have been 94 telemedicine centers set up in 2017-18 fiscal year. Among them, nine are situated at the specialized hospitals, eight at MCHs, 11 at DHs, 65 at UHCs, and one at the MIS monitoring cell. Sometimes, they even provide consultation services as well (figure 8.9). Telemedicine

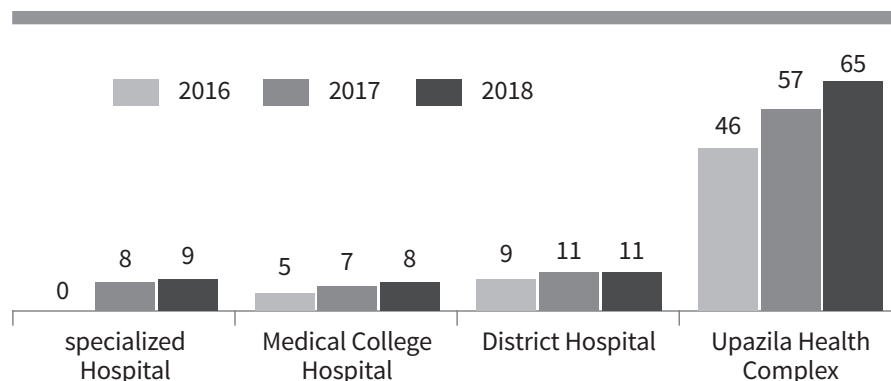


Figure 8.9. Number of telemedicine centers, institutions wise from 2016-18

Monitoring Unit of the MIS-DGHS especially received complaints and problems, faced by other telemedicine centers, regarding hardware, software, unavailability of physicians at the sites, etc. The number of patient consultation through this service has been on the increasing trend (figure 8.10). Telemedicine centers are

equipped with tele-gadgets, like tele-stethoscope, tele-ECG, tele-microscope, tele-glucometer, etc. and they also have high-quality cameras, big screen monitors and computers. Figure 8.11. shows that mostly people, suffering from skin diseases, have received the telemedicine service, due to a shortage of skin specialists in remote areas.

j. Citizen's Grievances handling system (Complaint and suggestion system)

Since 2012, an exciting and effective innovation 'complaint-suggestion box' has been added to the existing service line of the MIS-DGHS. Citizens are now actively participating to ensure the accountability of the service providers in health service delivery system. The SMS-based complaint-suggestion box continues to remain as the innovative and effective mechanism to know citizens' feedback on the quality of service in public hospitals. This system is frequently recommended for further promotion to improve accountability and transparency of public hospitals. It is one of the components to measure DLI (Disbursement Linked Indicator). This system is working in 739 public hospitals and health organizations. In each of these facilities, a display

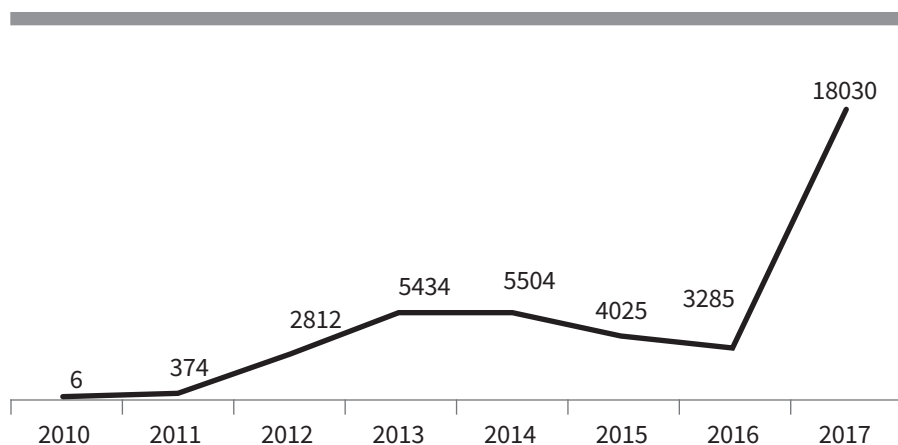


Figure 8.10. Year wise number of patient consultation through Telemedicine service from 2010 to 2017



Figure 8.11. Disease profile obtained from telemedicine service in 2017

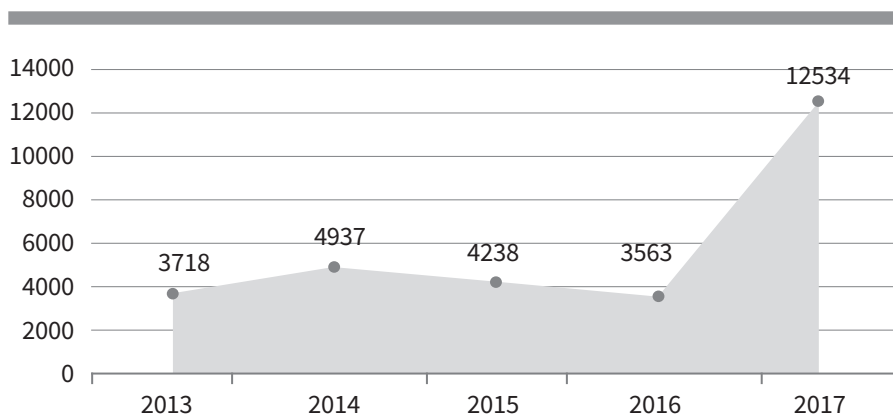


Figure 8.12. Number of SMS received from Complaint and suggestion system from 2013 -2017

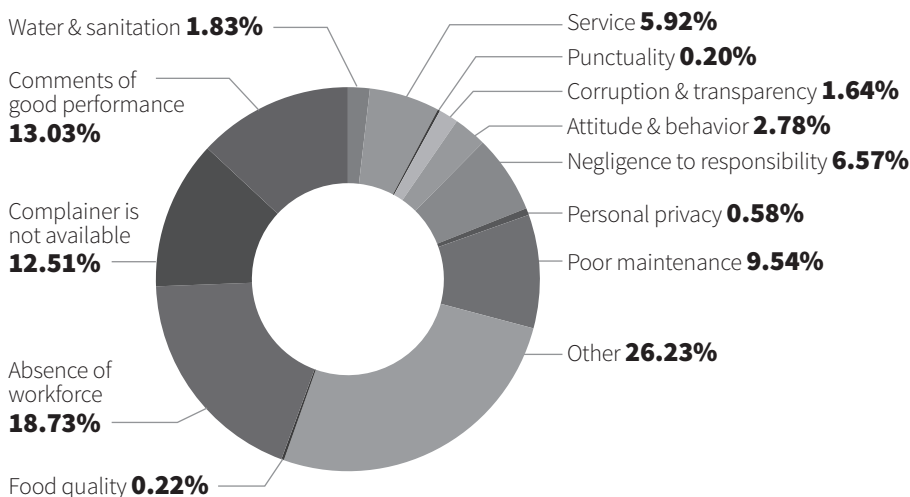


Figure 8.13. Type of categories in complaint and suggestion system received in 2017

board is mounted on the wall of a prominent place of the facility. The display board describes how to send complaints about the quality of services or suggestions for improvement of services. Presently, on an average, the system receives 30 SMS per day. A web-server located at the MIS-DGHS instantly receives the complaints and suggestions displayed on the web portal; some details are also provided for public viewing. One dedicated responsible person addresses every message. Nowadays, lodging messages have significantly been increased; this indicates more participation of citizens who

visit hospitals (figure 8.12) due to continuous follow up.

The number of SMS in 2017 has been notably increased. Figure 8.13 shows various type of categories in complaint and suggestion system received in 2017. Among the following categories “absence of workforce” received the highest number of complaints. Concerned authority should take necessary action to address such problems. The public view of the complaint-suggestion box is available at www.dghs.gov.bd>>Database Login >>Complaint & Suggestion Box.

k. Monitoring staff attendance through Biometric attendance

Particularly in remote areas of Bangladesh many service providers, working at hospitals and health centers, remain absent. This is a common complaint not only in Bangladesh but also in many other countries. To track the office attendance of government health staff in workplaces, the MIS-DGHS installed remote biometric time-attendance machines in all upazila and district hospitals and in some tertiary hospitals. These machines have been installed since 2012; by phases, all UHCs and DHs have been covered by now. These are low-cost fingerprint biometric machines, and the recorded touch-encounter scan can be tracked from the central office of the MIS-DGHS. During installation, the staff member's fingerprints are recorded in the database. Every day, the staff members need to touch the sensor of the machine during their check-in and check-out. The machine itself can store a memory of 30,000 encounters. Figure 8.14. shows the number of

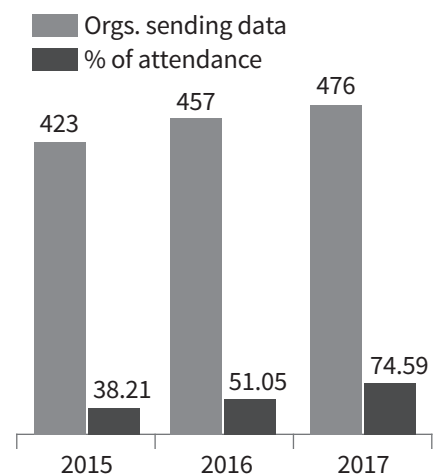


Figure 8.14. Number of organization actively sending data along with the percentage of attendance by Biometric system.

organizations actively sending data. It also shows how, after adopting this method, the percentage of attendance increased by 36.38% from 2013 to 2017.

At the MIS-DGHS, a web-server captures the attendance data whenever the server finds the local computers switched on and connected to the Internet for any purpose. Pre-defined web-based reports can be generated on the server-side, which can be accessed through web-browser from any place. Furthermore, the GIS map of office attendance also can be viewed from <http://www.dghs.gov.bd>>Data>>Office Attendance Monitoring>>Office Attendance GIS Map>

1. Health call center 16263 (Shastho Batayon)

Ministry of Health and Family Welfare has established the largest and the first ever One-Stop and Integrated National Health Call Center, known as “Shastho Batayon 16263” since September, 2015.

This service is also called as the National Health Helpline where people from all over the country can get health service from any part of Bangladesh 24/7 by dialing the short code 16263 of Shastho Batayon from any mobile or land line number. Under the supervision of MIS division of DGHS, call center service is being implemented and operated by Synesis IT Limited.

Key services of Shastho Batayon 16263:

1. Doctors’ Advice and treatment:

A team of trained medical doctors provide these services. People call 16263 and directly communicate with doctors about their health-



related problems and queries, any time they want.

2. Health Information: Sometimes it is hard to find a doctor nearby or to get information about any hospital. Shastho Batayon 16263 provides public as well as private hospitals, clinics and doctors’ information and their phone numbers.

3. Ambulance Information: Shastho Batayon 16263 has developed a database of ambulance contact numbers with location information. Anyone can get emergency ambulance information for any specific location by just dialing 16263.

4. Health Complaints Management:

Before, people faced different types of problems while taking any health-related services because of the mismanagement and other malpractices in this sector. However, there were no one to hear to them; there was a little scope to submit any complain. To mitigate such situation and improve the quality of overall health sector of Bangladesh, Shastho Batayon 16263 are now taking and solving complaints from any Public-Private health service. Not only this, Shastho Batayon also informs the complainant about the measures taken based on their complaints.

5. Emergency Accident Information:

Shastho Batayon 16263 also receives emergency accidental calls. As soon as they get such calls, they quickly inform the concerned authorities.

The total numbers of calls received in 2016 and 2017 were respectively 1205898 and 1016317 (shown in Figure 8.15). Mostly people suffering from viral

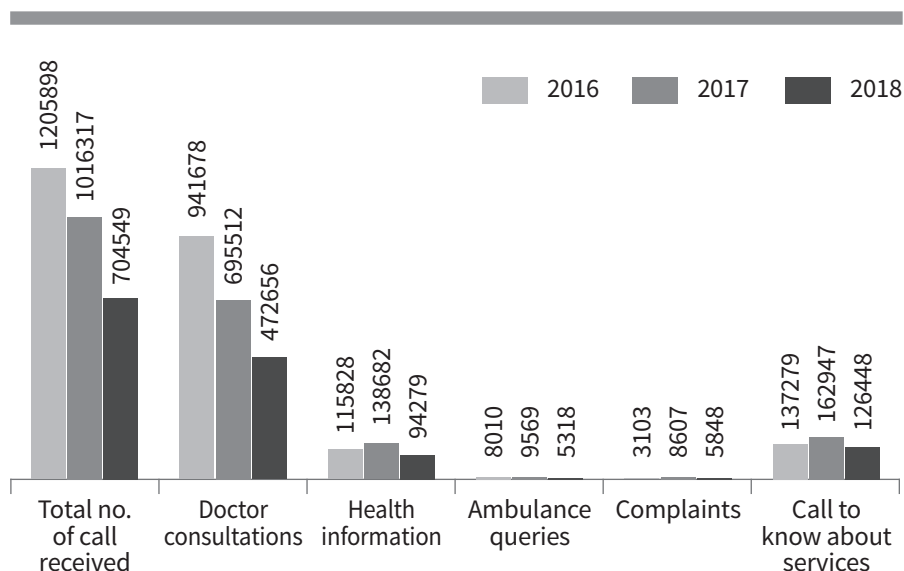


Figure 8.15. Number of call received through Shaysho Batayon in the last three years

fever and diarrhea received service from shasthaya batayan shown in figure 8.15. This is becoming popular, and it has transformed into an easy way to seek medical consultation for acute health problems. More details on month-wise data can be viewed from the public dashboard of the DGHS web portal, i.e. http://103.247.238.81/webportal/pages/shastho_batayon.php.

m. Bulk SMS and quick email service and update of social media portals (Facebook, Twitter, Google +, Youtube)

MIS, DGHS operate and update social media portals such as facebook, twitter, google + youtube etc for sharing information publically. Bulk SMS is active for sending short messages to respective group and every UHC, CSO, DH has official email id which are very functional. Every office/health facilities has own email id. (e.g. alluhfpo@dghs.gov.bd, allld@

ld.dghs.gov.bd, allcs@cs.dghs.gov.bd, allchcp@cc.dghs.gov.bd, alldivdir@mis.dghs.gov.bd, allmis@mis.dghs.gov.bd, allhospi@hospi.dghs.gov.bd).

n. Civil Registration and Vital Statistic (CRVS)

In 2014, with the assistance from WHO, the MoHFW conducted a comprehensive assessment of the CRVS system, using the standard assessment tool developed by WHO and the University of Queensland. Following that sample, Bangladesh developed a CRVS Strategic Action Plan (SAP), including key recommendations, identification of responsible agencies as well as estimated budgets and timelines. Recently, Enterprise Architecture (EA) of the CRVS has been developed, and review of the legal framework has been done. Therefore, the SAP is now under the process of being revised.

DGHS and DGFP under MoHFW

support CRVS through notifications of all the birth and death events in a community and in facilities to the local registrar office. Hospitals use Form 804 as well as their perceived cause of death forms for death declarations. Birth registration is mandatory for the death registration. Bangladesh has recently started practicing the international standard medical certificate of cause of death in 4 hospitals and introduced MCCoD in another 19 hospitals on a piloting basis. At selected large hospitals, where many deaths occur, physicians have been trained to use the ICD certificate for cause of death, and statistical coding is done according to the Startup Mortality List (ICD-10-SMoL). Verbal Autopsy (VA) also has been introduced in selected areas of the country for community deaths for which there is no medical certificate on cause of death. Death registration is mandatory for conducting VA. Since there are no electronic linkages between the ORG's BRIS and DGHS database, these deaths, along with causes of deaths, are not being notified directly.

KEY ACHIEVEMENTS (CRVS) DURING 2016-2017

The MoHFW, with coordinating support from the CRVS Secretariat, also made progress on improving the links between the health sector and registration services, including piloting programs for training in the use of MCCoD, VA and ICD-10 (including SMoL) as well as improving the quality and collection of VS data by incorporating key variables into the DHIS2 to enable real-time data tracking.

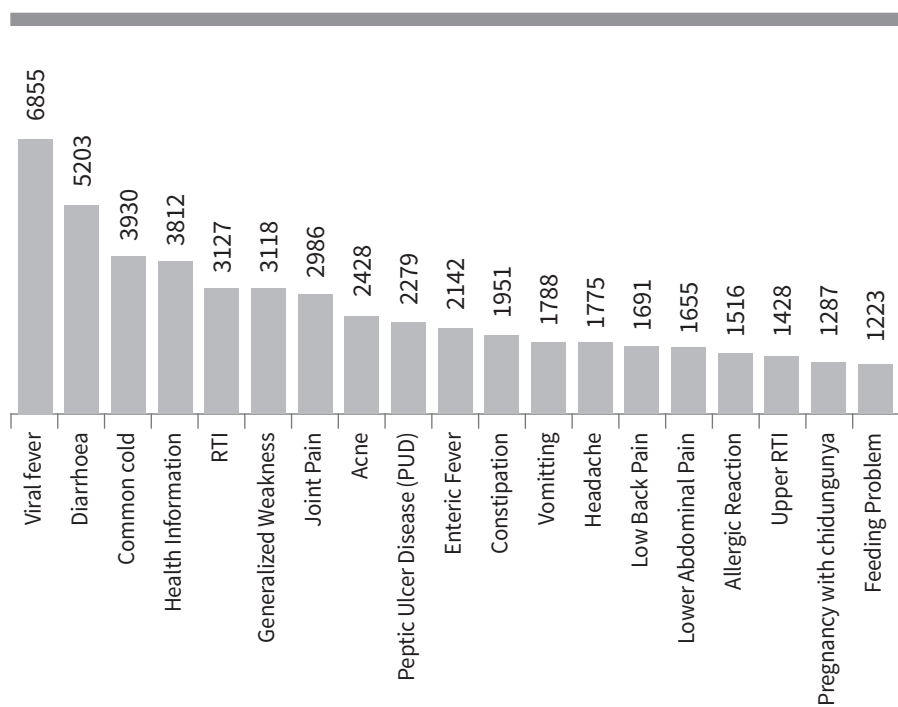


Figure 8.16: Number of people giving calls to Shastho batayon according to types of health problems in 2017

PARTNER-SUPPORTED ACTIVITIES OF CRVS (2017–2018)

Nationally focused activities

- Under the leadership and management of the CRVS Secretariat, WHO and D4H are working with the MoHFW to increase the use of medical certification of cause of death, using ICD coding and verbal autopsies. So far, 17,498 VAs are done, 26,482 MCCoDs are issued and 20470 MCCoD forms have been transmitted to SMoL module in DHIS-2 of the DGHS
- D4H is working with the DGHS to improve the data quality and collection for vital statistics reporting
- UNHCR is working to end statelessness for migrants
- Recently, the MoHFW has decided to incorporate detailed cause of death in the undergraduate medical curriculum and has also created permanent post of coders
- UNICEF is helping the DGHS in streamlining service delivery by tracking health information
- UNFPA is working to strengthen the national statistical system for data collection and analysis so that data for development (D4D) policies can be identified and made available
- Under the leadership and management of the CRVS Secretariat, the D4H is working with the MoHFW and ORG at 13 upazilas, to facilitate birth and death registration by using frontline health and to engage family planning workers to assist families with the registration process
- Birth and death registration within 45 days has been enhanced remarkably in Kaliganj. The Government has decided to scale up the Kaliganj

model of birth and death notification and registration using Verbal Autopsy in 12 subdistricts and MCCoD in 69 district-level hospitals, including the Medical College Hospitals as well as in private facilities.

MAJOR ACTIVITIES CARRIED OUT BY MIS IN FISCAL YEAR 2017-18

The following are the major activities conducted by MIS in 2017-18:

- It has conducted annual divisional HMIS workshop in all 8 Divisional health office.
- It has conducted digitally managed medical and dental admission tests.
- It has successfully conducted yearly National Health Award 2017 as part of the health service strengthening program with technical support from WHO.
- It has prepared, printed and distributed the National Health Bulletin 2017.
- It has strengthened and continued the online reporting of routine data through DHIS2 from community clinic to central and specialized hospitals/institutions.
- It has conducted several HMIS training and workshop for health managers at all level, doctors, statisticians, nurses and field workers throughout the country in 2017. In local training, 23,838 officers and staff members participated in the training courses, workshops, and seminars in FY 2017-2018. About 1120 officers participated in trainings/workshops supported by development partners (UNICEF) (see Annex).
- Number of desktop computers (57), printers (20), monitors (15), and tablets (33) were maintained and repaired at MIS, DGHS received from upazila to central level. However, due to decentralization of repair and maintenance to divisional and district level from this year, these numbers are now reduced, in comparison to previous years.
- It has procured hardware and machinery to strengthen and sustain HMIS service (table 8.1)
- It has maintained and updated the Data center (figure 8.17)



Figure 8.17: Picture of Data Center, MIS-DGHS

TECHNICAL PARTNERS

Along with the MOHFW, other technical partners assist the MIS-DGHS to make technology-related solutions, training, and capacity-building; these include a2i Project, World Bank, WHO, UNICEF, DFID, UNFPA, Rockefeller Foundation, JICA, USAID, icddr, Measure Evaluation, CIDA, UNESCAP, JPGSPH-BRAC University, BRAC, JHU, MSH (SIAP), Save the Children, D.Net, CIRPB, CIDA, etc.

Supply of ICT equipment and computer stationeries

Table 8.1. shows different types of hardware and machinery procured and distributed from FY 2015-16 through 2017-2018. However, no procurement was done in FY 2016-2017.

Medical Biotechnology: the booming future

In comparison to the advancements of our neighboring countries in biotechnology, the situation in Bangladesh is just in the sprouting stage. A situation analysis on

biotechnology, funded by FAO and UNDP (that ended in 2003), identified that most of the work done in this sector is related to agriculture. Tissue cultures of banana, orchid, jute, wheat, etc. are some examples of regular work. In 2013, the complete genome sequence of a local variety of jute was decoded by the famous scientist Dr. Maqsoodul Alam.

With the recent scientific advancement, medical biotechnology is playing a vital role in patient management, starting from diagnosis and prognosis to treatment. Currently, there are private companies in Bangladesh that use medical biotechnology techniques for diagnostic tests and send samples to neighboring countries for around 1,700 tests costing more than 1.9 million US dollar per year (Source: CMBT Survey 2016). This amount is being transferred out to neighboring countries only because medical biotechnology is not developed in our country.

The Government of Bangladesh has shown interest in the application of medical biotechnology for the

development of health and nutrition sectors and has developed the first National Guideline on Medical Biotechnology in 2006. Later, the guideline was revised and republished as gazette on 21 December 2010. Six successive meetings of the National Technical Committee on Medical Biotechnology (NTCMB) were held between 2004 and 2012 on direct recommendations of the National Executive Committee on Biotechnology (NECMB) and supervision of the National Taskforce on Biotechnology of Bangladesh (NTBB), with the Honorable Prime Minister as the Chair. These ended up with the establishment of the Center for Medical Biotechnology at the Institute of Public Health. National Guidelines on Medical Biotechnology 2010 includes guidance to implement provisions in the National Biotechnology Policy and describes deliverables for short, medium and long term goals. To implement these goals, Medical Biotechnology Program was included in HIS and e-health Operational Plan of the HPNSDP 2011-2016, and it is now being continued in

Table 8.1. Hardware and machinery procurement from fiscal 2015-16 through 2017-2018

Sl. No.	Hardware	FY 2015-2016	FY 2016-2017	FY 2017-2018	Distribution
1	Desktop Computer	1725	-	-	Hospitals, health offices, academic and training institutions across the country
2	UPS (offline 600VA)	1500	-	-	accompanies one for each desktop computer
3	Tablet device/PDA	4025	-	4000	To community healthcare providers
4	Video conferencing equipment			60	
5	Data center equipment/Disaster Recovery center equipment	-	-	1 Lot	
6	Medical Biotechnology equipment	-	-	1 lot	
7	Equipment for LAN	43			In 32 CS office and 11 tertiary hospitals
8	Printer	3450			
9	Modem			100000	
10	Motorcycle			200	



Figure 8.18: Hands-on-training for teachers and scientists in CMBT-ideSHi Laboratory

the 4th sector-wide program (HPNSP 2017-2022) as a component of HIS and e-health.

Center for Medical Biotechnology

Center for Medical Biotechnology, a program under MIS, DGHS, was established in 2012 at the Institute of Public Health (IPH) building to plan, implement, and monitor activities related to medical biotechnology as per the National Guideline on Biotechnology 2010. The primary goals and objectives of the center were to create a proper environment for the expansion of medical biotechnology in Bangladesh, preliminarily in the government sector, by providing both technical and resource support in collaboration with other stakeholders.

From the very beginning of its establishment Center for Medical Biotechnology (CMBT) began an inclusive effort in cooperation with ideSHi (Institute for Developing Science and Health Initiatives) and relevant stakeholders for capacity development. Till today, over 1,700

stakeholders including medical professionals from various disciplines, journalists, entrepreneurs, scientists, and members of Medical Curriculum Committee were trained through various training sessions and workshops, including core committee meetings, consultative workshop for updating medical curriculum, training workshop for journalists, sensitization workshop, and a 14-day (currently 10 days) hands-on training for medical teachers and scientists. MBT-related books were distributed among the medical university, medical colleges, and other health science institutions, and necessary equipments for setting up biotechnology labs were distributed among different institutions.

According to the deliverables for short- and medium-term goals, mentioned in National MBT Guideline, CMBT is conducting its activities under the Operational Plan of HPNSP 2017-2022 from July 2017.

Activities conducted by CMBT in 2017-2018 sessions were as follows:

Table 8.2. Training and workshop

Type of training/workshop	Venue	Duration (day)	Batch (No.)	Participants (No.)
Hands-on training for medical teachers and scientists	Dhaka	10	4	40
Training/workshop for medical teachers	Dhaka	4	30	120
Consultative workshop for medical teachers	Dhaka	4	30	120
Consultative workshop for review of medical curriculum	Dhaka	1	1	10

1. An International workshop was arranged at CMBT in collaboration with ideSHi in January 2018.

2. A set of equipment comprising 17 instruments, related to medical biotechnology, has been supplied to six medical colleges: Dhaka Medical College, Sir Salimullah Medical College; Shahid Suhrawardy Medical College, Rajshahi Medical College; MAG Osmani Medical College, Sylhet; Chittagong Medical College; and two institutions: CMBT-ideSHi and National Institute of Preventive and Social Medicine (NIPSOM).

3. A draft Development Project Pro-forma has been completed and submitted for conversion of the Center for Medical Biotechnology to National Institute of Medical Biotechnology as a center of excellence. Here, academic and clinical services will be provided to develop our national capacity in this sector and to lift it up to international level for global competition.

4. A tripartite memorandum of understanding (MOU) among Center for Medical Biotechnology, Bangabandhu Sheikh Mujib Medical

University and Barisal Biotech, UK has been signed to promote and expand research activities in medical biotechnology in Bangladesh.

5. A list, comprising the required training for human resource development in Medical Biotechnology, has been sent to the Ministry of Science and Technology.

6. CMBT, in collaboration with ideSHi, provides facilities to Ph.D. students doing thesis.

Table 8.3. List of Doctoral students with their area of work

Name	Designation	Title of thesis
Dr. Nusrat Sultana	Assistant Professor, Department of Virology, Dhaka Medical College	Mechanistic basis of low blood pressure and association of red cell distribution width with beta-globin gene mutation patterns in Bangladeshi patients with E/beta-thalassemia major
Suprovath Kumar Sarker	Ph.D. Student, ideSHi	Profiling of cut off values of amino acids and acylcarnitines for universal screening of inborn errors of metabolism
Mst. Noorjahan Begum	Ph.D. Student, ideSHi	Hypothyroidism in pregnant women and children: genetic spectrum analysis of hypothyroidism in hospital-care settings in Bangladesh.
Dr. Rosy Sultana	Ph.D. Student, ideSHi	Influence of hepatitis E viral load and genotypes on pregnant urban dwellers of Bangladesh
Farjana Akther Noor	Ph.D. Student, ideSHi	Prevalence pattern of thalassemia in the Comilla district of Bangladesh and thalassemia-induced transcription factors to decipher the disease severity

Table 8.4. Diagnostic services provided by CMBT in collaboration with ideSHi

Name of service	No. of services provided
Basic metabolic screening (inborn errors of metabolism diagnosis)	450

Table 8.5. Others research works

Sl. No.	Title of research
1	Services agreement for training and technical support to BITID
2	Profiling of cut off values of amino acids, acylcarnitines and urine organic acids for universal screening of inborn errors of metabolism
3	Collaborative management platform for detection and analysis of re-emerging and foodborne disease outbreaks in Europe (COMPARE)
4	Innovative approaches for the development of diagnostic methods for detecting IEM and genetic disorders using high throughput metabolomics profiling as well as monoclonal antibody
5	Strategic Typhoid Alliance across Africa and Asia (STRAATAA)
6	Foodborne illness surveillance (FBIS)

Table 8.6. Scientific papers published in international journals from CMBT in collaboration with ideSHi

Title of article	Name of journal
High resolution melting curve analysis enables rapid and reliable detection of G6PD variants in heterozygous females	BMC Genetics
Impaired acylcarnitine profile in transfusion-dependent beta-thalassemia major patients in Bangladesh	Journal of Advanced Research
High resolution melting curve analysis targeting the HBB gene mutational hot-spot offers a reliable screening approach for all common as well as most of the rare beta-globin gene mutations in Bangladesh.	BMC Genetics

The center for Medical Biotechnology was established with a goal of exploring the possibilities of this booming field in the government sector through various activities. Achievement of the organization is still limited to human resource

development and research activities, primarily due to less interest of the physicians in research. Center for Medical Biotechnology is currently working to institutionalize biotechnology research at the medical college level by different

activities. If this endeavor is successful, involvement of medical doctors in the current program can be increased to the desired level. This will assist in bringing fruitful results.

HEALTH RESEARCH

Research initiatives under DGHS

BACKGROUND

Health has been accepted as a fundamental right of all people by the Constitution of the World Health Organization (WHO) and in the International Declaration of Human Rights. Several research organizations both in private and public sectors and departments are conducting regular health research in their focused area. Significant research works are carried out in public medical colleges, national institutes, BSMMU, autonomous research organizations/ institutes such as BIRDEM, Heart Foundation, general universities and other academic institutions and by a variety of private organizations, including the pharmaceutical industry. This chapter will only includes research done by various departments of DGHS, BMRC, IEDCR and CME in the period 2017-18.

RESEARCH BY PMR

Effective planning based on evidence and priorities can support in achieving the sector program's objectives. Timely monitoring is essential for proper implementation of the program. Research helps to generate evidences based on which appropriate management decisions can be taken. Therefore, Planning, Monitoring and Research (PMR) are one of the most important operational plans in 4th Health, Population and Nutrition sector programs. This OP is responsible for periodic monitoring, performance evaluation and evidence based re-planning of other 13 OPs. Researches are conducted in following research areas by PMR in 2017-2018:

- **Communicable disease:** Study had been occurred in Communicable disease area like chronic hepatitis B Virus infection, prevalence of cytomegalovirus among healthy blood donor etc.

Table 9.1. Research by PMR in 2017

Research Areas	Number of research	Percentage
Communicable diseases	2	7.69
Non-communicable diseases	12	46.15
Health Education	1	3.85
Urban Health Problem	1	3.85
Medical Education	2	7.69

Research Areas	Number of research	Percentage
Internal Medicine	2	7.69
Surgery	2	7.69
Maternal & Child Health	3	11.54
Bioethics	1	3.85
Total	26	100

- **Non-communicable diseases:** Researches had also been conducted in Non-communicable diseases like consequences of Childhood cancer, Autism spectrum and neuro developmental disorders, Nephrotic Syndrome, type 2 Diabetes Mellitus, Cirrhosis of liver, Stroke, Asthma, inborn errors of metabolism (IEM) etc.
- **Health Education:** Provided to mothers by adolescents for promoting infant feeding practice
- **Urban health problem:** among the rickshaw drivers in the Dhaka city
- **Medical Education:** Evaluation of the curriculum of 2012 and assessment of Quality Assurance scheme of under graduate medical education program in Bangladesh
- **Internal medicine:** Study had been occurred in Anaesthesia and its medical practices, peptic ulcer diseases
- **Surgery:** A histopathological study of thyroid swelling and outcome of tympanoplasty by inlay procedure had also been occurred.
- **Maternal & child health:** Nutrition status among reproductive women, consequences of DSF program, introducing mobile phone technology to pregnant mother for better MNCH outcome etc.
- **Bioethics:** Attitude of ethics committee (EC) members on bioethics and structure and function on EC in Bangladesh etc.

RESEARCH BY BMRC

- **Communicable disease:** Study had been occurred in Communicable disease area like chronic hepatitis C Virus infection, prevalence of Dengue virus, antimicrobial resistance, tubercular ascities etc.
- **Non-communicable diseases:** Researches had also been conducted in Non-communicable diseases like consequences of Childhood cancer, thalassaemia, Geriatrics, Road traffic accident, Gastroesophageal reflux disease, Cancer, type 2 Diabetes Mellitus, cirrhosis of liver, stroke, asthma, inborn errors of metabolism (IEM) etc.
- **Health Education:** Provided child to parents approach and health seeking behavior of adolescent girls.
- **Urban health problem:** Study on Obesity and its impact on child health, consumption of artificial color soft drinks, childhood overweight, prevalence of Diabetes mellitus among tuberculosis patients.
- **Health system:** Study on economic analysis of the existing services, assessment of delivery of quality care, common factors influencing the utilization of Union Health and family Welfare centers.

Table 9.2. Research by BMRC in 2017

Research Areas	Number of research	Percentage
Communicable diseases	8	12.70
Non-communicable diseases	35	55.56
Health Education	3	4.76
Urban Health Problem	4	6.35
Health System	4	6.35
Surgery	1	1.59
Maternal & Child Health	7	11.11
Bioethics	1	1.59
Total	63	

- **Surgery:** Research on randomized control trial comparing stapled versus open and semiclosed Haemorrhoidectomy.
- **Maternal & child health:** Assessment of quality management of delivery in different hospital, feeding practice and nutritional status of under two years children, reproductive health status of female garments workers etc.
- **Bioethics:** Status of knowledge on medical ethics in a sample of medical professional at DHAKA city.

RESEARCH BY IEDCR

IEDCR stands for Institute of Epidemiology, Disease Control and Research. The present Institute of Epidemiology, Disease Control and Research (IEDCR) was established in 1976 through a bill approved in the Parliament which called for the establishment of an institute for epidemiological and communicable disease research as well as functioning of disease control programs mainly in the form of parasitic and entomological containment of vector borne diseases through application of epidemiological principles. IEDCR is the focal institute for conducting disease surveillances and outbreak investigations in Bangladesh. IEDCR has been engaged in conducting research on events of public health importance and developing public health workforce.

Following research works are done by IEDCR in 2017-2018

- Assessment of Vaccination Coverage and Vaccine-Preventable Disease Serology among Forcibly Displaced Myanmar Nationals in Cox's Bazar, Bangladesh
- Behavioral Adaptation in Live Poultry Trading and Farming System and Zoonoses Control in Bangladesh
- Behavioral Risk Characterization at Animal-Human Interface for Zoonotic Diseases in Bangladesh
- Cambridge Program to Assist Bangladesh in Life Style and Environmental Risk Reduction
- Developing Early Warning System for Climate Sensitive and WASH Associated Diseases through

health surveillance

- Evaluation of Mechanism to Improve Mobile Phone Survey Metrics
- One Health Economic Analysis of Zoonotic Diseases in Bangladesh
- Oral Cholera Vaccine Coverage Evaluation Survey among Forcibly Displaced Myanmar Nationals and host community in Cox's Bazar, Bangladesh
- The Evaluation of mHealth Platform for Diarrheal Disease Decision Support in Hospitals: A Cluster Randomized Control Trial
- The Effect of Climate Variability, Seasonal Variations and Environmental Events on Drinking Water Quality, Diarrhea Prevalence and WASH Behavior in Bangladesh
- Understanding the Ecology of Antimicrobial Resistance in Wildlife
- PhD Research: Transmission of Avian Influenza from Animal to Human at the Human-Animal Interface in Bangladesh

M.Sc Researches

- Mental Health Status in Newly Arrived Adult Forcibly Displaced Myanmar Nationals (FDMNs) Living in Bangladesh
- Quality of Non-communicable diseases (NCD) in Two Primary Healthcare Facilities in Bangladesh
- Child Injury in a Rural Community of Narshingdi District
- Needs Assessment for Reducing Childhood Overweight and Obesity in a Selected School of Dhaka City

Details of these researchs and other activities are presented in the annex.

RESEARCH BY IPH

Institute of Public Health (IPH) is one of the oldest and largest institutions in the country. It is mandated to support preventive, promotive and curative healthcare services through laboratories, production of biochemicals, quality control, training, education, research, and publications in the field of public health.

In recent days, the IPH has completed the following research works:

- Monitoring and evaluation of horticulture products and

Table 9.3. Research by IPH in 2017

Sl. No.	Title of research
1	Services agreement for training and technical support to BITID
2	Profiling of cut off values of amino acids, acyl-carnitines and urine organic acids for universal screening of inborn errors of metabolism
3	Collaborative management platform for detection and analysis of re-emerging and foodborne disease outbreaks in Europe (COMPARE)
4	Innovative approaches for the development of diagnostic methods for detecting IEM and genetic disorders using high throughput metabolomics profiling as well as monoclonal antibody
5	Strategic Typhoid Alliance across Africa and Asia (STRAATAA)
6	Foodborne illness surveillance (FBIS)
7	Lower Acute Respiratory Infections (LARI) etiology research among displaced Rohingya populations
8	Screening and profiling of plasmid harboring mcr-1 and mcr-2 genes in pathogenic bacteria and investigating underlying mechanism of dissemination in Bangladesh

other food commodities for chemical contamination: Phase-I;

- Monitoring and evaluation of horticulture products and other food commodities for chemical and microbiological contamination: Phase-II;
- Monitoring and evaluation of horticulture products and other food commodities for chemical and microbiological contamination: Phase-III;
- Assessment of Food Safety Culture in Food Business Operators in Bangladesh;
- Study on microbiological quality of water in Dhaka metropolitan;
- Detection of heavy metals, drugs and pesticide residues, mycotoxins and microbial contaminants in raw and packed milk of cow, cow feed and curd

RESEARCH BY CENTER FOR MEDICAL BIOTECHNOLOGY

The publications list from the research works done by the center for medical biotechnology is provided in following table.

Table 9.4. Research by CMBT in 2017

Title of article	Name of journal
High resolution melting curve analysis enables rapid and reliable detection of G6PD variants in heterozygous females	BMC Genetics
Impaired acylcarnitine profile in transfusion-dependent beta-thalassemia major patients in Bangladesh	Journal of Advanced Research
High resolution melting curve analysis enables rapid and reliable detection of G6PD variants in heterozygous females	BMC Genetics
Impaired acylcarnitine profile in transfusion-dependent beta-thalassemia major patients in Bangladesh	Journal of Advanced Research
High resolution melting curve analysis targeting the HBB gene mutational hot-spot offers a reliable screening approach for all common as well as most of the rare beta-globin gene mutations in Bangladesh.	BMC Genetics
Molecular Analysis of Glucose-6-Phosphate Dehydrogenase Gene Mutations in Bangladeshi Individuals	PloS one
Bacterial and viral pathogen spectra of acute respiratory infections in under-5 children in hospital settings in Dhaka city	PloS one
Examination of Huntington's disease with atypical clinical features in a Bangladeshi family tree	Clinical case reports

RESEARCH BY CME

- Post Graduate Medical Education in Bangladesh: A Situation Analysis to find out total number of different categories postgraduate doctors of Bangladesh. (Submitted to MoHFW).

FINANCING HEALTHCARE

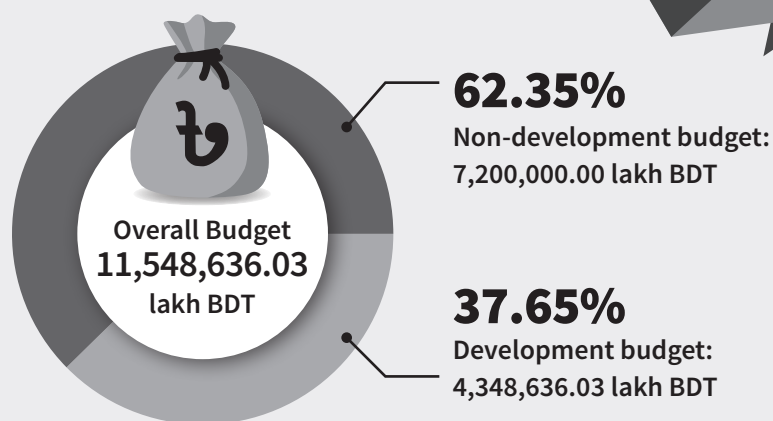
Around 90% utilization in 2017-18 financial year

BACKGROUND

Directorate General of Health Services (DGHS) performs its regular activities being funded by revenue budget. However, DGHS finances its development activities through Health, Population, and Nutrition Sector Program (HPNSP). Currently, 4th Health sector wide program is going on, which started in 2017. There are 14 Operational Plans in this program that fall under DGHS. An overview of the performance of these Operational Plans is provided in this chapter. In addition, a summary of revenue budget spending is shown in a later part of this chapter.

4th Health, Population, and Nutrition Sector Program (HPNSP)

Approved in ECNEC: 21 March, 2017



14
Operational Plans in DGHS

24
Development Projects

PERFORMANCE OF OPERATIONAL PLANS

The allocation, release, expenditure, and utilization in Fiscal Year (FY) 2017-2018 (revised ADP) of HPNSDP 2017-2022 fund against different Operational Plans of the DGHS are shown in Table 10.1. In this financial year, total allocated budget for 14 Operational Plans of DGHS was 386,137.00 lakh taka. Among that, 380,031.58 lakh taka was released and finally 357,270.18 lakh taka was utilized. This entire amount was significantly higher than last FY 2016-17. Utilization rate in 2016-17 was 68.28% of the allocated amount, where in 2017-18 the utilization rate was 92.52% of the allocated amount (figure 10.2).

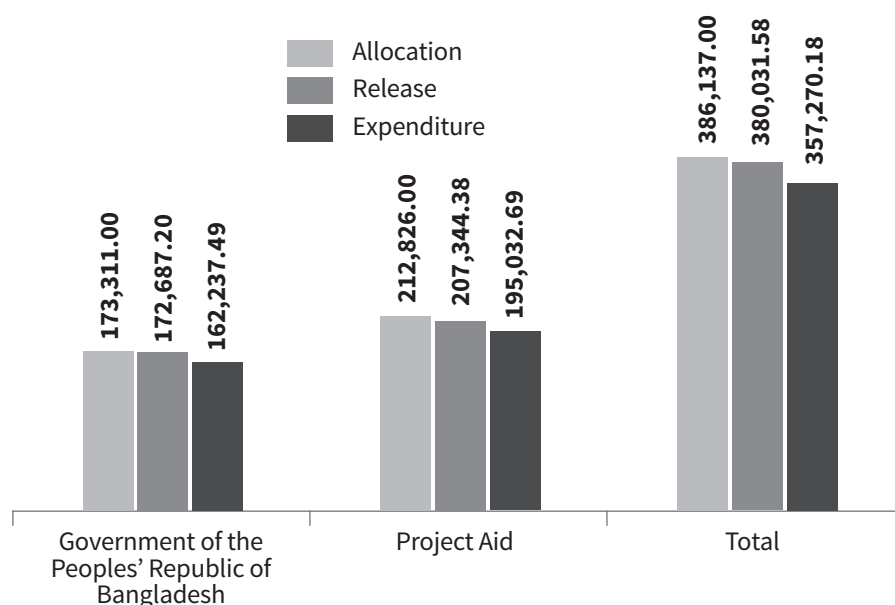


Figure 10.1: Financial Status (in lakh taka) for Operational Plans of DGHS in FY 2017-2018

Table 10.1: Allocation, release, expenditure, and utilization in FY 2017-2018 of 4th sector program fund against different Operational Plans(OP) of the DGHS (added percentage utilized in formatted document)

Sl No	OP/Project Name	Allocation			Release			Expenditure		
		GoB	PA	Total	GoB	PA	Total	GoB	PA	Total
1	Planning, Monitoring and Research	800.00	1,130.00	1,930.00	800.00	929.30	1,729.30	597.96	697.72	1,295.68
2	Health Information System and e-Health	6,850.00	1,400.00	8,250.00	6,850.00	1,400.00	8,250.00	6,807.64	1,400.00	8,207.64
3	Procurement, Storage and Supplies Management	16,657.00	500.00	17,157.00	16,657.00	375.00	17,032.00	15,486.93	73.23	15,560.16
4	Medical Education and Health Manpower Development	33,500.00	7,500.00	41,000.00	33,500.00	7,500.00	41,000.00	31,391.38	4,788.12	36,179.50
5	Maternal, Neonatal, Child and Adolescent Health	6,200.00	64,086.00	70,286.00	6,200.00	61,282.08	67,482.08	5,383.68	60,705.08	66,088.76
6	National Nutrition Services	1,000.00	11,000.00	12,000.00	936.00	11,000.00	11,936.00	394.54	9,630.95	10,025.49

Sl No	OP/Project Name	Allocation			Release			Expenditure		
		GoB	PA	Total	GoB	PA	Total	GoB	PA	Total
7	Communicable Diseases Control	7,621.00	12,300.00	19,921.00	7,621.00	15,019.89	22,640.89	7,089.70	14,123.22	21,212.92
8	TB-Leprosy & AIDS/STD Program	2,898.00	18,592.00	21,490.00	2,898.00	13,606.23	16,504.23	1,917.79	12,907.85	14,825.64
9	Non-Communicable Diseases	10,480.00	6,050.00	16,530.00	9,920.00	5,963.84	15,883.84	9,175.71	4,917.29	14,093.00
10	National Eye Care	400.00	1,000.00	1,400.00	400.00	1,000.00	1,400.00	362.93	949.74	1,312.67
11	Community Based Health Care	56,771.00	20,100.00	76,871.00	56,771.20	20,100.04	76,871.24	54,800.32	19,007.91	73,808.23
12	Hospital Services Management	23,800.00	67,500.00	91,300.00	23,800.00	67,500.00	91,300.00	22,655.87	64,292.34	86,948.21
13	Life Style and Health Education & Promotion	1,832.00	1,168.00	3,000.00	1,832.00	1,168.00	3,000.00	1,682.38	1,133.51	2,815.89
14	Alternative Medical Care	4,502.00	500.00	5,002.00	4,502.00	500.00	5,002.00	4,490.66	405.73	4,896.39
		173,311.00	212,826.00	386,137.00	172,687.20	207,344.38	380,031.58	162,237.49	195,032.69	357,270.18

Source: ADP fund monitoring system, MOHFW

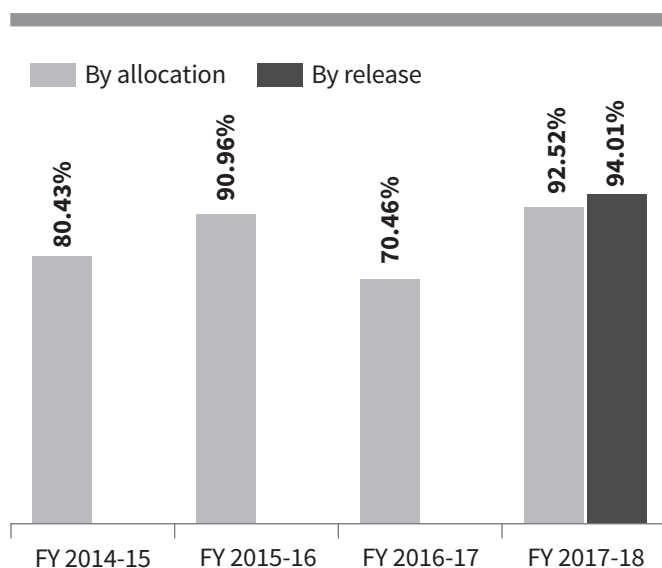


Figure 10.2: Fund utilization rate (in percentage) against allocation (GoB+PA) in different Fiscal Years

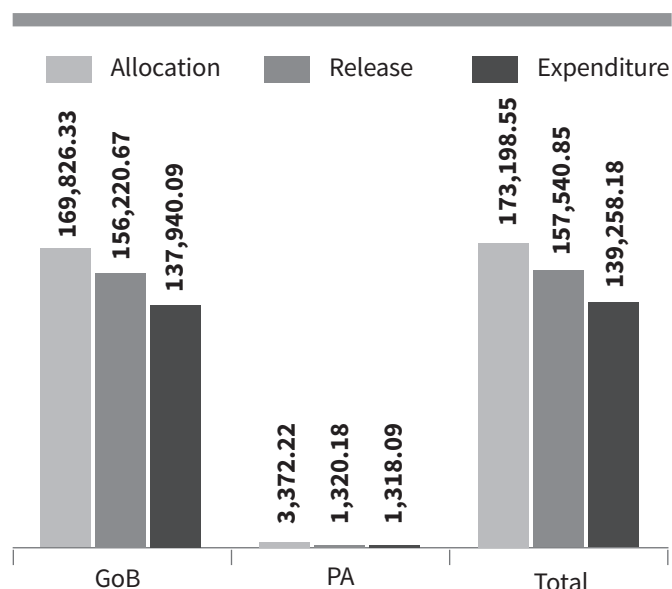


Figure 10.4: Financial status of development projects under DGHS in 2017-18

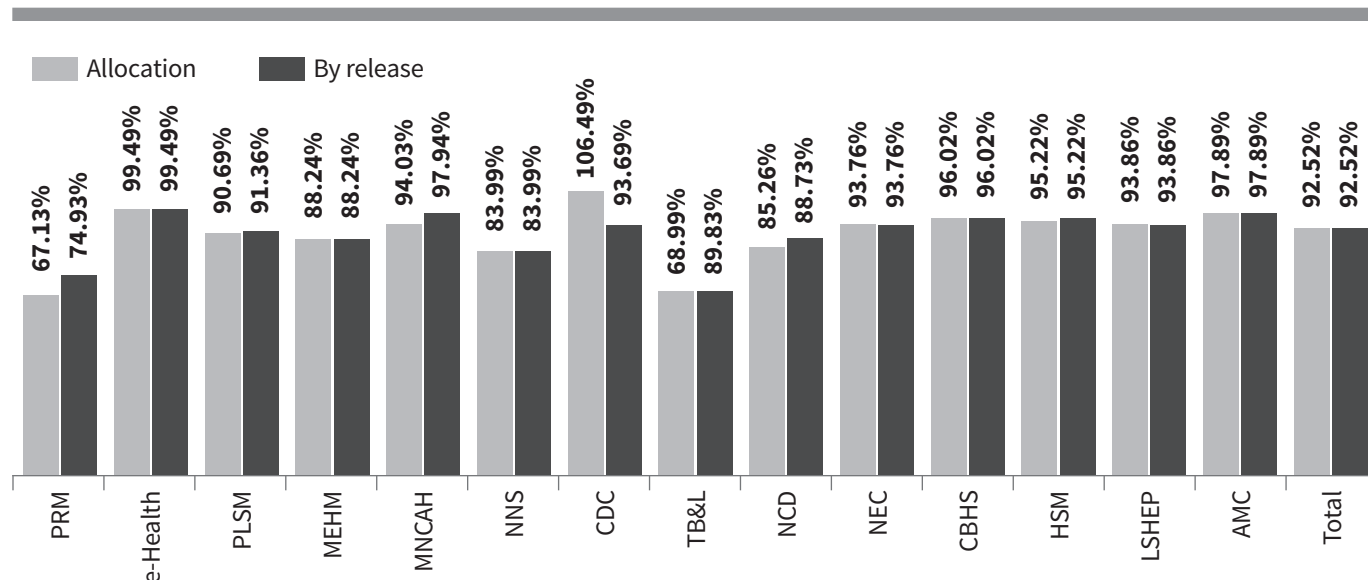


Figure 10.3: Fund utilization rate (in percentage) of the DGHS Operational Plans against allocated and released funds in FY 2017-2018

Progress status of 14 Operational Plans is presented in Figure 10.3. The entire Operational Plans performed well, fund utilization rate was highest for Communicable Disease Control, as well as HIS and e-Health. On the contrary, PMR Operational Plan had lowest utilization rate. Utilization rate increased for all Operational Plans compared to Fiscal Year 2016-17.

PERFORMANCE OF DEVELOPMENT PROJECTS

Currently, 25 development projects are ongoing under the supervision of DGHS. Total allocated budget, release and expenditure are provided in Table 10.2 and Figure 10.4. Utilization rate of allocated budget in 2017-18 was 88.39% compared to fund utilization rate of 62.37% in 2016-17.

Table 10.2: Development projects under DGHS in 2017-18 with allocation, release, expenditure, and utilization in lakh taka (percentage is also given)

Sl No	Investment project	Allocation	Release	Expenditure	Utilization rate of allocation (%)	Utilization rate of release (%)
1	Sheikh Hasina National Institute of Burn and Plastic Surgery, Dhaka	34,965.00	34,965.00	34,704.48	99.25	99.255
2	Establishment of 250- bed National Ophthalmology Institution and Hospital (1st Phase: 250 beds)	28.00	19.50	18.52	66.16	94.997
3	Advancement of National Institute of Cancer Research and Hospital from 50 bed to 300 beds	1,301.00	909.38	888.18	68.27	97.669
4	Establishment of National Institute of Laboratory Medicine and Referral Centre	5,437.00	4,835.75	1,494.08	27.48	30.897
5	Establishment of Sheikh Sayera Khatun Medical College and Hospital and Nursing Institute, Gopalganj.	6,185.50	5,938.67	5,841.15	94.43	98.358
6	Establishment of Shatkhira Medical College and Hospital	7,500.00	7,500.00	6,605.63	88.08	88.075

Sl No	Investment project	Allocation	Release	Expenditure	Utilization rate of allocation (%)	Utilization rate of release (%)
7	Establishment of Faridpur Medical College and 500-bed Hospital	1,800.00	1,800.00	1,800.00	100	100.000
8	National Institute of Digestive Diseases Research and Hospital	12,930.00	12,930.00	10,024.59	77.53	77.530
9	Establishment of Kushtia Medical College and Hospital Project	1,500.00	0.00	0.00	0	0.000
10	Establishment of Shaheed Sayed Nazrul Islam Medical college , Kishorganj	9,000.00	9,000.00	7,528.27	83.65	83.647
11	Extension of Shaheed Sheikh Abu Naser Specialized Hospital, Khulna	700.00	524.31	523.20	74.74	99.788
12	Establishment of Trauma Center at Gopalgong	445.00	443.78	366.67	82.4	82.624
13	Extension of National Institute of Traumatology and Orthopedic Hospital and Rehabilitation center (NITOR)	14,572.00	14,423.89	13,869.08	95.18	96.154
14	Establishment of Jamalpur Medical College and Hospital, as well as Jamalpur Nursing College	5,500.00	3,750.00	3,799.98	69.09	101.333
15	Establishment of Potuakhali Medical College and Hospital	10,000.00	6,843.27	6,397.68	63.98	93.489
16	Establishment of Sheikh Lutfar Rahman Dental College	2,500.00	2,500.00	526.13	21.05	21.045
17	MNCH and Health System Improvement Project Comp-2: Strengthening of Diagnostic Imaging System of 8 Divisional MCH	600.00	30.00	18.51	3.08	61.700
18	Col. Malek Medical College and 250- bed Hospital, Manikganj	18,104.00	18,104.00	17,409.42	96.16	96.163
19	Shaheed M.Monsur Ali Medical College and 500- bed Medical College, Sirajganj	19,018.83	12,623.50	12,936.82	68.02	102.482
20	Establishment of Shiekh Hasina Medical College and Hospital, Tangail	20,000.00	19,914.00	14,022.09	70.11	70.413
21	Strengthening public health actions for emerging infectious events in bangladesh	1,112.22	485.80	483.71	43.49	99.570
22	Expansion of National Institute of Neuroscience and Hospital	0.00	0.00	0.00	0	0.000
23	Shaheed Tajuddin Ahmed Medical College and Hospital, Gazipur	0.00	0.00	0.00	0	0.000
24	Eye Health Promotion and prevention of blindness in selected areas of Bangladesh	0.00	0.00	0.00	0	0.000
25	Establishment of 500-bed Hospital and Ancillary Building in Jessore, Cox's Bazar, Pabna and Noakhali	0.00	0.00	0.00	0	0.000
Total		173,198.55	157,540.85	139,258.18	90.96	88.39

Source: ADP fund monitoring system, MOHFW

PERFORMANCE OF REVENUE BUDGET

In the financial year 2017-2018, a total 41 billion BDT were allocated for DGHS and organizations under DGHS (excluding medical colleges). About 89% of this fund was used, summary of the expenditure is presented in the following Table 10.3.

FINANCIAL STATUS OF PROCUREMENT, STORAGE & SUPPLIES MANAGEMENT- HEALTH SERVICES (PSSM-HS)

After its establishment in 1947, under the 1st Health Sector Program (HPSP), Central Medical Stores Depot (CMSD) was given responsibility for centralized procurement of

Table 10.3. Revenue budget allocation and expenditure summary (in thousands taka) for DGHS and organizations under DGHS (excluding medical colleges) in the FY 2017-18

Institute name	Allocation	Expenditure	Utilization rate of allocation (%)
DGHS (central office)	6,164,475.00	5,999,459.00	97%
Divisional offices	161,966.00	134,085.00	83%
Civil surgeon offices	1,295,907.00	1,107,543.00	85%
Upazila health offices	10,358,600.00	9,401,899.00	91%
District hospitals	7,008,827.00	6,161,573.00	88%
Upazila health complexes	15,772,212.00	13,333,560.00	85%
TB hospitals	190,006.00	164,279.00	86%
TB centers	299,755.00	243,474.00	81%
TB control and training institutes	59,113.00	59,003.00	100%
Leprosy hospitals	63,861.00	51,493.00	81%
School health centers	73,005.00	51,245.00	70%
Model family planning clinics	73,859.00	57,480.00	78%
Para medical institutes	107,379.00	107,001.00	100%
Medical assistant training schools	205,884.00	196,830.00	96%
Total	41,834,849.00	37,068,924.00	89%

goods for all the Line Directors of DGHS and Ministry of Health Family & Welfare (MOHFW) in 1998.

- Ambulance, Jeep
- MSR Products

Locally, Upazila Health Complexes, District hospitals and Medical college hospitals also procure goods for the facility.

- CMSD procures –
 - Medical Equipment
 - Non-Medical Equipment
 - Pharmaceuticals
 - Vaccines (Meningitis, Influenza, Anti Rabies, Hepatitis-B Vaccine etc.)

- The source of fund for these procurements are:
 - RPA (GoB)
 - GoB (Development)
 - GoB (Revenue)
 - JICA Fund
 - Saudi Fund (SDF)
 - Nutrition International (NI)

● Major activities of CMSD are:

Sl No	Activity	Comments
1	Initiation of e-GP procurement for the first time in CMSD	12 packages executed through e-GP
2	Construction of new, modern warehouse in CMSD	Maximum construction work was completed. Electromechanical work is under process.
3	Restructuring of CMSD	The proposal of restructuring of CMSD has been sent to Ministry for approval.
4	Completion of huge procurement activity	Contract value in FY 2017-2018 was BDT 89074.26 Lakh

Table 10.3: Current Procurement Status 2017-2018

Nature of Package	Number of Packages	Estimated Value (lakh tk.)	Contract Value (lakh tk.)
GoB (Dev)	26	7884.99	6927.63
HSM, DGHS (Carried Over)	15	14998.06	11116.8
HSM(New)	21	26058.21	14280.4
JICA Fund	16	5443.7	4208.23
Nutrition International for NNS	2	240.57	136.04
Service Package for PSSM-HS	5	1093	96.84
SFD	4	1839.37	1702.94
GoB (Rev)	33	30005.24	21131.2
RPA(GOB)	46	26796.41	22156.5
GoB (Rev.) (Haji Vaccine & Non EDCL)	2	6183.55	5977.21
NILM&RC	10	4866.81	1340.43
Total	180	119226.3	89074.3

DEVELOPMENT STATUS

Extraordinary achievements in health sector

With limited resources Bangladesh is providing health care service to 160 million people. Health network is expanding and new structures as well as services are being included every year. These recent developments are listed below.

1. NEW FRONTIERS IN HEALTH SERVICES AND MANAGEMENT.

a. Structural development for health administration

- i. Completed during 2014-2018: District Family Planning Office, Chattogram
- ii. Ongoing projects since 2017 are described in table 11.1

Table 11.1. Ongoing projects since 2017

Structure	No.
Divisional Director and Deputy Director Office (Family Planning), Barishal	1
Deputy Director Office (Family Planning)	31
Health Engineering Building	5
Divisional Director Office (Health)	3
Upazila Family Planning Office and Storage Facility	104
Civil Surgeon Residence, Bogura	1
Total	145

b. Establishment of

- i. Medical Universities and hospitals.
 - Sylhet Medical University (2018)
- ii. Specialized postgraduate medical institutes and hospitals.
 - Three completed projects in 2018
 - Sheikh Hasina National Institute of Burn and Plastic Surgery and Hospital [World's largest facility of this kind]
 - Sheikh Rasel Gastro Liver Institute and Hospital
 - National Institute of Laboratory Medicine

- Three ongoing projects
 - Waste treatment facility of National Institute of Ophthalmology
 - National Institute of Physical and Rehabilitation Medicine, Dhaka.
 - Bangamata National Cellular and Molecular Research Center.

iii. Medical and Dental Colleges/Units and hospitals.

- Completed projects during 2014-2018 and ongoing projects are presented in following tables

Table 11.2. Completed projects during 2014-2018

	Medical College	Dental College/Unit	Total
Government	13	-	13
Armed Forces	5	-	5
Private	13	6	19
Total	31	6	37

Table 11.3. Ongoing projects of medical institutes

Name	No.
Burn and plastic surgery units in Sylhet, Barishal, Rangpur, Rajshahi and Faridpur Medical Colleges	5
Sheikh Hasina Medical College, Habiganj	1
Chandpur Medical College	1
Total	7

iv. Multidisciplinary hospitals

- One ongoing project
 - State-of-the-art 100 bedded Cancer Hospital

v. District hospitals

Table 11.4. Ongoing projects of district hospitals

Name	No.
Kurigram District Hospital	1
250-bedded Hospital, Sylhet	1
Development works at Manikganj 250-bedded Hospital	1

Name	No.
Development works at Dinajpur 250-bedded Hospital	1
Development works at Noakhali 250-bedded Hospital	1
Medical gas systems at district hospitals	20
Burn and plastic surgery unit at Mohammad Ali 250-bedded Hospital	1
Total	26

vi. Upazila hospitals

- Ongoing projects since 2017: Twelve (12) 50-bedded hospitals in new upazilas.

vii. 10, 20, 31 or 50-bedded hospitals

- Completed projects during 2014-2018: Eighteen (18)
- Ongoing projects since 2017: Sixty one (61)

viii. Other hospitals

Table 11.5. Ongoing projects of other hospitals

Name	No.
Children's Hospitals in Mymensingh, Chattogram, Kurigram, Khulna, Rajshahi and Barishal	6
Trauma center, Gopalganj	1
Total	7

ix. Health centers at union level

- Ongoing projects since 2017 are shown below

Table 11.6. Ongoing projects at union level

Name	No.
Union Health and Family Welfare Center	208
Construction of boundary walls	438
Mother and Child Care Center	97
Total	743

c. Community clinic: A mass-centered approach to healthcare

- Ongoing projects since 2017: To establish 1029 new clinics.

d. Increment of the number of hospitals and hospital beds during 2014-2018

Table 11.7. New hospitals and beds during 2014-2018

	New hospitals	New hospital beds
Government	11	3397
Private	2156	41015
Total	2167	44412

e. A great Number of patients received care at government healthcare facilities during January-December, 2017

Table 11.8. Number of patients receiving care at government healthcare facilities during 2017

Type of Facilities	Institution		No. of patients (outdoor)		No. of patients (emergency)		No. of patients (indoor)	
	No.	Reported (%)	Reported	Estimated	Reported	Estimated	Reported	Estimated
Postgraduate and specialized	12	59	1396813	2364674	264858	448380	162479	275062
Medical college	17	92	7192689	7860862	1451528	1586369	1445466	1579744
District	64	99	11430645	11518183	2497397	2516523	1862962	1877229
10, 20, 31, 50 and 100-bedded	48	65	509557	786353	61636	95117	35256	54407
Upazila health complex	432	99	23725957	23987420	3352944	3389894	2245262	2270005
Chest disease/TB	56	29	2551	8778	0	0	620	2134
Infectious disease	5	43	34101	80238	6890	16212	6579	15480
Community clinic	13105	98	88485294	90438771	-	-	-	-
Total			132777607	137045279	7635253	8052495	5758624	6074061

2. MODERN EQUIPMENT IN HEALTHCARE PROCURED AND SUPPLIED TO RELEVANT HOSPITALS AND/OR INSTITUTES DURING 2014-2018

Table 11.9. Modern equipment in healthcare procured and supplied to relevant hospitals and/or institutes during 2014-2018

Modern medical equipment	No.
MRI machine	15
CT scan machine	20
Auto analyzer machine	90
Echo color doppler machine	43
Bronchoscope machine	17

Modern medical equipment	No.
Brachytherapy machine	5
Cobalt machine	4
Ultrasound machine	127
X-ray machine	98
Total	419

3. MODERN HEALTHCARE UNITS AND SPECIALIZED CARE DURING 2014-2018

Table 11.10. New health care units during 2014-2018

Unit and/or care	No.
Operation theatre	85
ICU	15
CCU	10
HDU	8
NICU	4
SCANU	22
Dialysis	12
Cath lab	5
Burn unit	8
Medical waste management	5
Total	174

4. EXPANSION OF VEHICLE POOL FOR HEALTH MONITORING

- a. Field level officials are given 200 motor cycles in 2018.

5. ACHIEVEMENTS IN MEDICAL EDUCATION

- a. New establishments during 2014-2018

Table 11.12. New teaching institutes during 2014-2018

Type/Name	No.
Constructions in Sir Salimullah, Noakhali, Cox's Bazar, Jashore, Pabna, Barishal, Chattogram, Cumilla and Khulna Medical Colleges	12
Nursing College and/or Institute (Government)	38
Nursing College and/or Institute (Private)	132
Medical Assistant Training School (MATS) (Government)	1
Medical Assistant Training School (MATS) (Private)	60
Institute of Health Technology (IHT) (Government)	3
Institute of Health Technology (IHT) (Private)	12
Total	258

b. Ongoing projects since 2017

Table 11.13. Ongoing projects regarding teaching institutes

Type/Name	No.
Constructions in Rangpur and Pabna Medical College	2
Sheikh Hasina Medical College, Jamalpur	1
Sheikh Hasina Medical College, Tangail	1
Sheikh Lutfur Rahman Dental College, Gopalganj	1
Nursing College and/or Institute (Government)	12
Medical Assistant Training School (MATS) (Government)	9
Institute of Health Technology (IHT) (Government)	11
Total	37

c. Increment of seats in different institutes of medical education during 2014-2018

Table 11.14. Increment of seats in different teaching institutes

Type/Name	No.
Government Medical College	1256
Private Medical College	1001
Armed Forces Medical College	305
Government Dental College/Unit	0
Private Dental College/Unit	340
Nursing College and/or Institute (Government)	1125
Nursing College and/or Institute (Private)	8500
Medical Assistant Training School (MATS) (Government)	52
Medical Assistant Training School (MATS) (Private)	8670
Institute of Health Technology (IHT) (Government)	166
Institute of Health Technology (IHT) (Private)	1291
Total	22706

6. EXPANSION OF HUMAN RESOURCES IN HEALTH

a. New post creation and recruitment

Table 11.15. New post creation and recruitments for doctors during 2017-2018

Type/Name	No.
BCS (joined)	624
BCS (ongoing)	10000
Adhoc	0
Total	10624

b. New post creation and recruitment of nurses/ midwives during 2014-2018

Table 11.16. New post creation and recruitments for doctors during 2017-2018

Type/Name	No.
Nurse (post created and joined)	15000
Midwife (post created)	3000
Midwife (joined)	1200

7. WONDERFUL DEVELOPMENT IN PUBLIC HEALTH

Table 11.17. Development in key health indicators

Indicator (per 1000 live birth)	2015 (SVRS)	2017 (SVRS)
Under 5 mortality rate	36	31
Neonatal mortality rate	20	17
Maternal mortality rate (per 100000 live birth)	181	172

8. GOLDEN ERA IN PHARMACEUTICAL INDUSTRIES

- Bangladesh now produces 97% of all drugs those are required to meet the demands of the local market in an affordable cost compared to many other countries including her neighbors.
- Revenue from export of drugs produced in Bangladesh has increased from 22471 million BDT in 2016 to 31963 million BDT in 2017.

9. INCREMENT IN BUDGET ALLOCATION IN HEALTH SECTOR

Budget allocation for health sector has been increased in last couple of years. The 4th HPNSP have a budget of 115486 crore taka, which is a big increase from the 3rd Sector program

10. CONTINUOUS ENCOURAGEMENT TO PRIVATE SECTOR AND LOCAL INITIATIVES OF HEALTHCARE

11. HEALTHCARE FOR THE HELPLESS AND THE POOR: HUMANITARIAN AID FOR TREE-MAN, MUKTAMONI, SOFURA-TAHURA AND MORE

- Now famous as “tree man”, Abul Bajandar received complete financial support from the GOB for the treatment of his disease called epidermodysplasia verruciformis.
- Conjoint twins Safura-Tahura were separated through a series of complicated surgical procedure which was fully funded and supported by the GOB.
- Muktamoni was treated for her atypical presentation of hemangioma under GOB support.

12. INITIATIVES REGARDING THE HEALTHCARE OF THE FORCIBLY DISPLACED CITIZENS OF MYANMAR

- Health services provided for the aid of more than one million Rohingya refugees
 - Immunization againsts polio, diphtheria, cholera, measles and rubella.
 - Deworming initiatives.
 - Vitamin A supplementation and other nutritional services.
 - Iron and folic acid supplementation for pregnant women.
 - Normal delivery and newborn care.
 - Mental health support.
 - Malaria control.
 - Detection, treatment and follow-up of HIV/AIDS cases.

b. Resources allocated for those services for one million Rohingya refugees of behalf of GOB in 2017-2018

Table 11.18. Resources for FDMN

Resources	No.
Healthcare center	199
Health post	145
Primary health care center	37
Delivery center	17
Ambulance	21



Figure 11.1. Prime Minister Sheikh Hasina receiving IPS International Achievement Award for the care and aid to the forcibly displaced citizens of Myanmar

13. INTERNATIONAL RECOGNITION IN HEALTH SECTOR IN 2017-2018

- a. IPS International Achievement Award for the care and aid to the forcibly displaced citizens of Myanmar.
- b. Special Distinction Award for Leadership for the care and aid to the forcibly displaced citizens of Myanmar.
- c. Recognition from WHO for achievements in the control of rubella and congenital rubella syndrome.